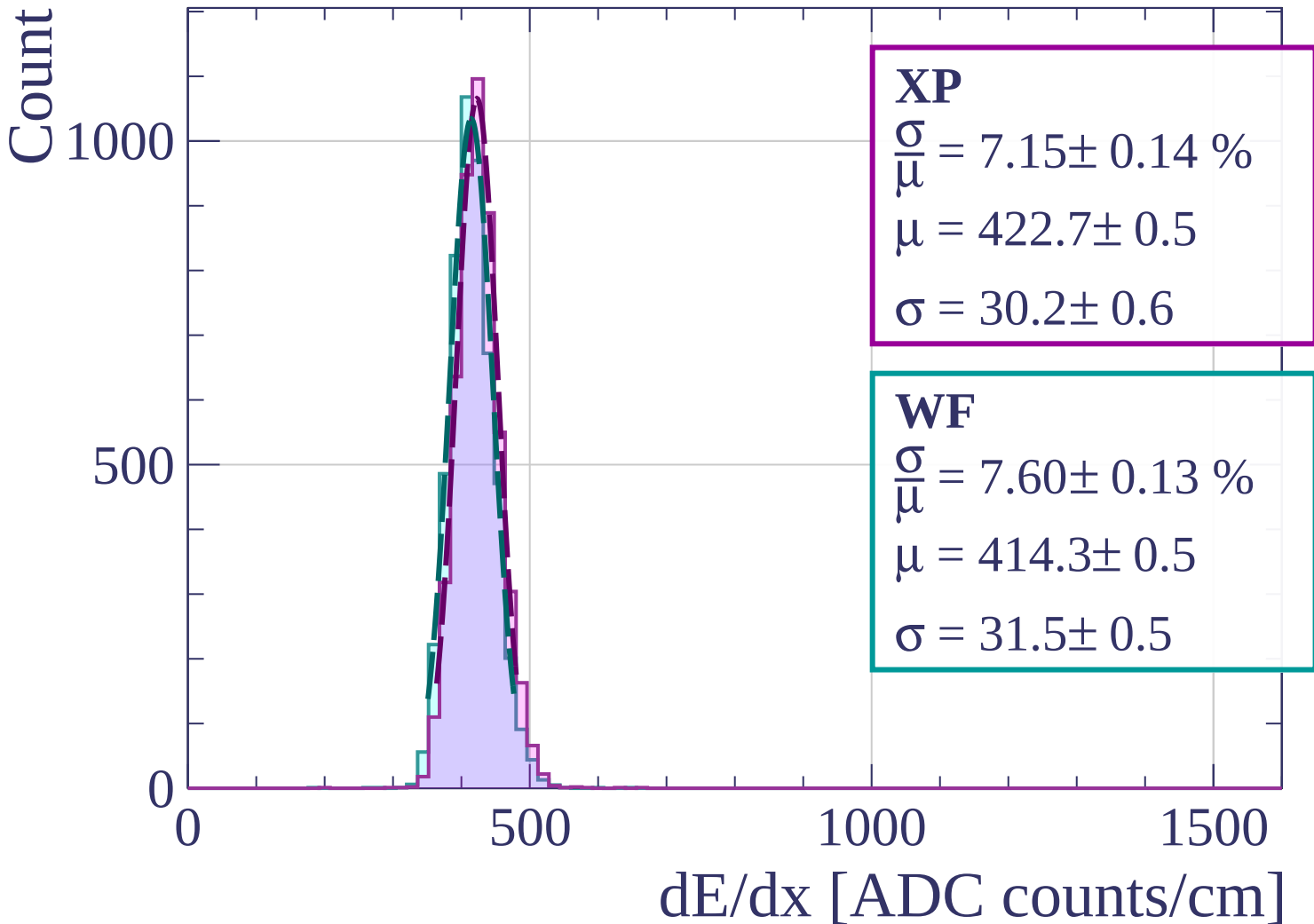
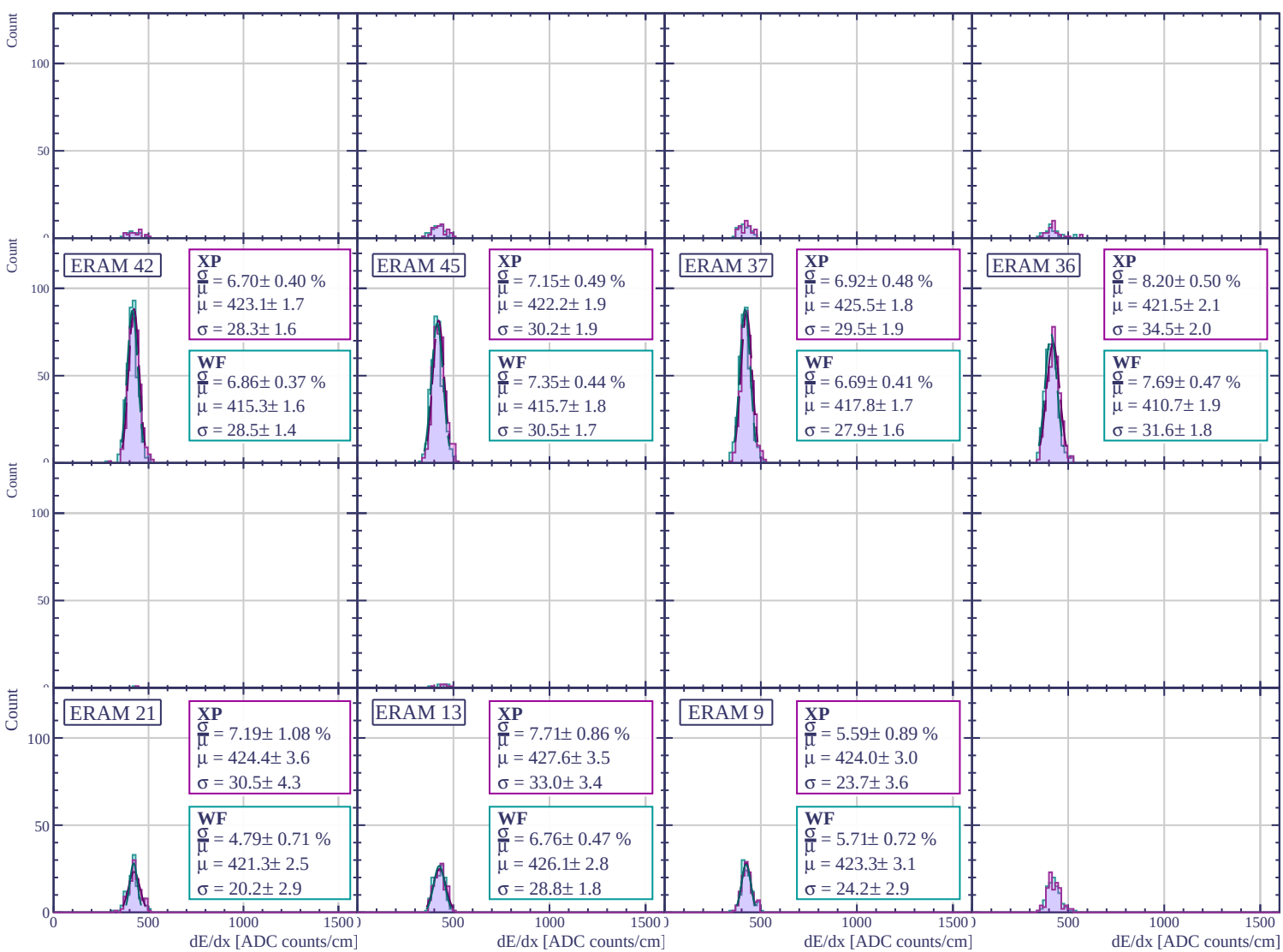
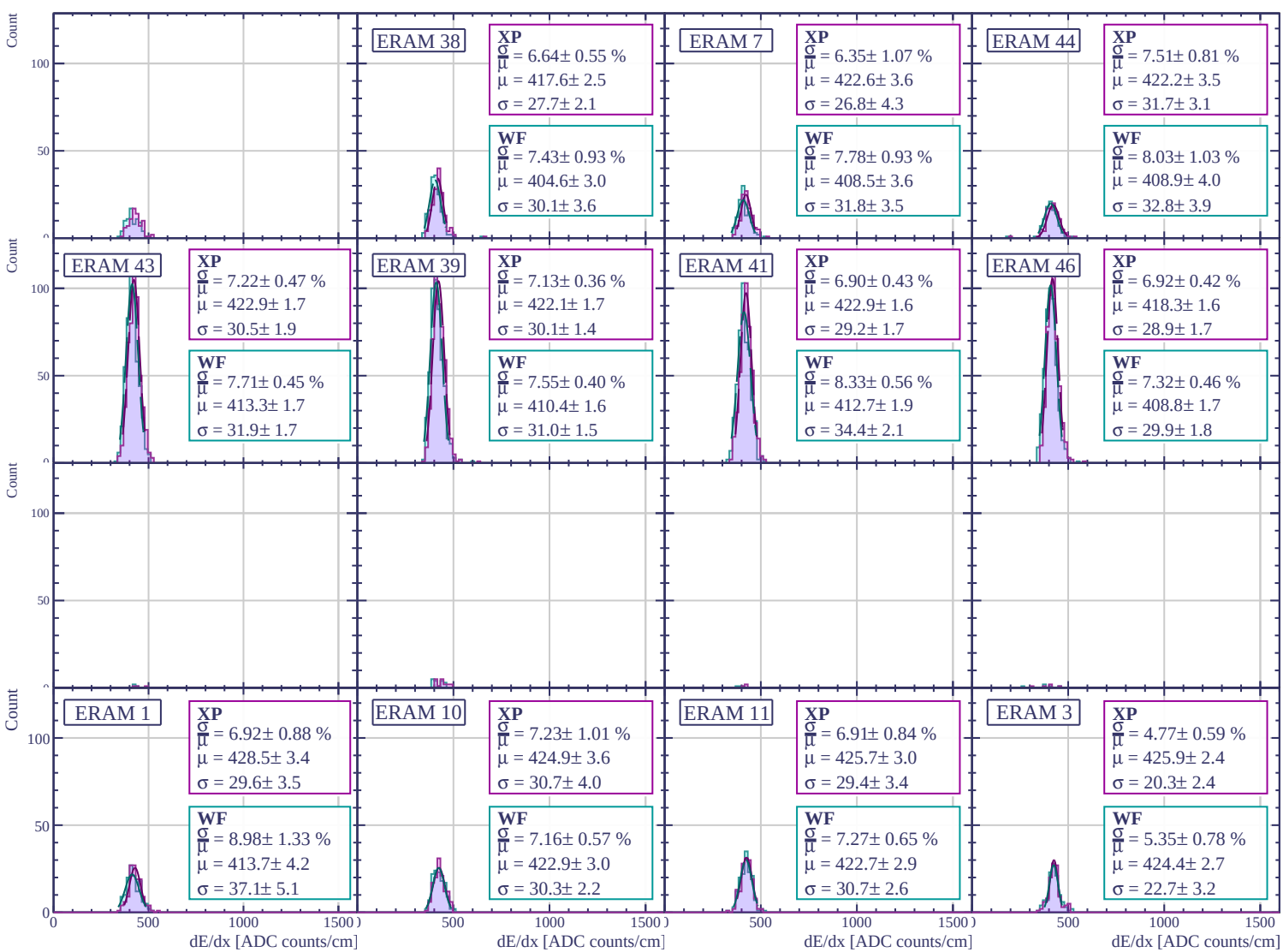
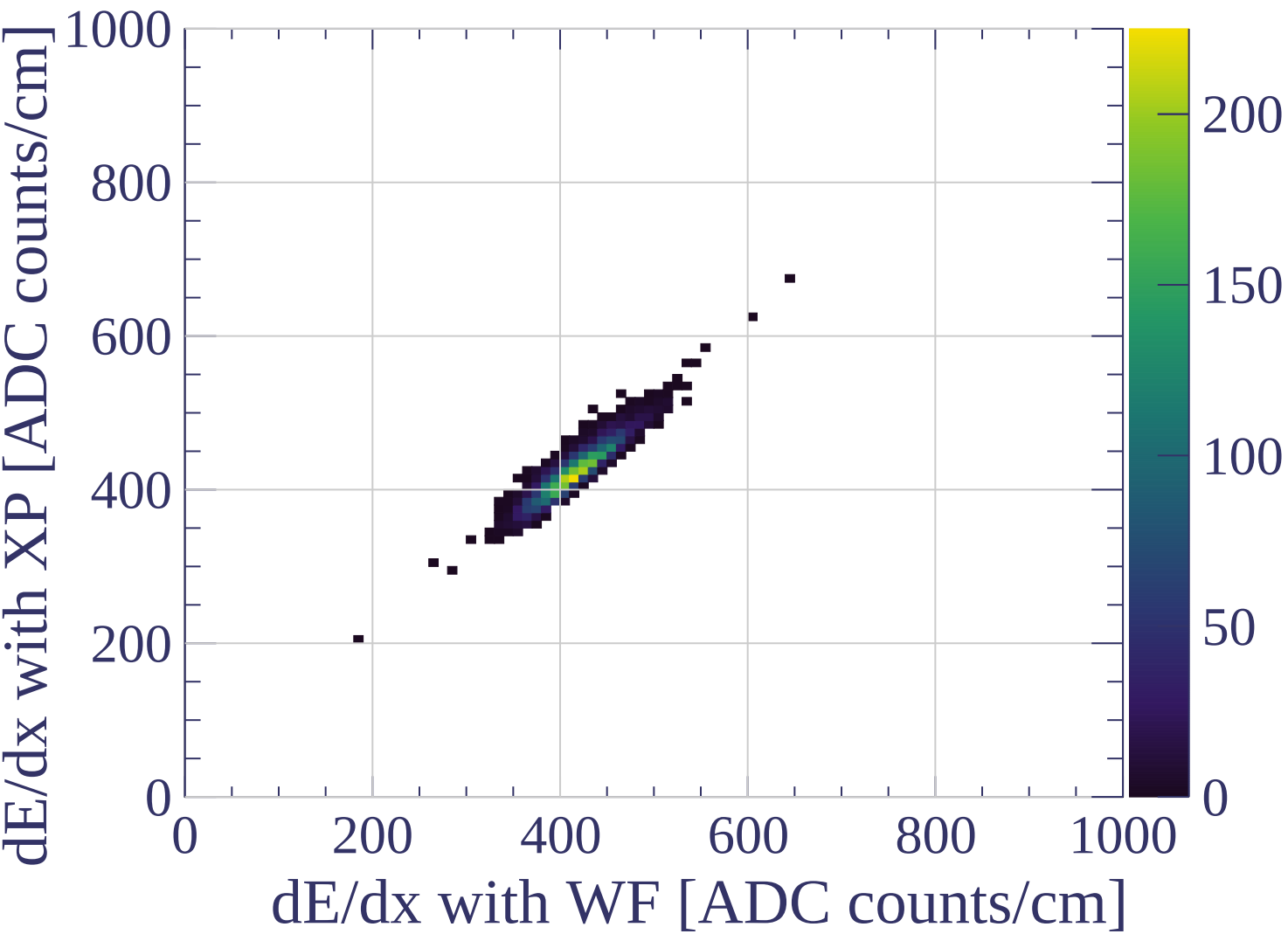
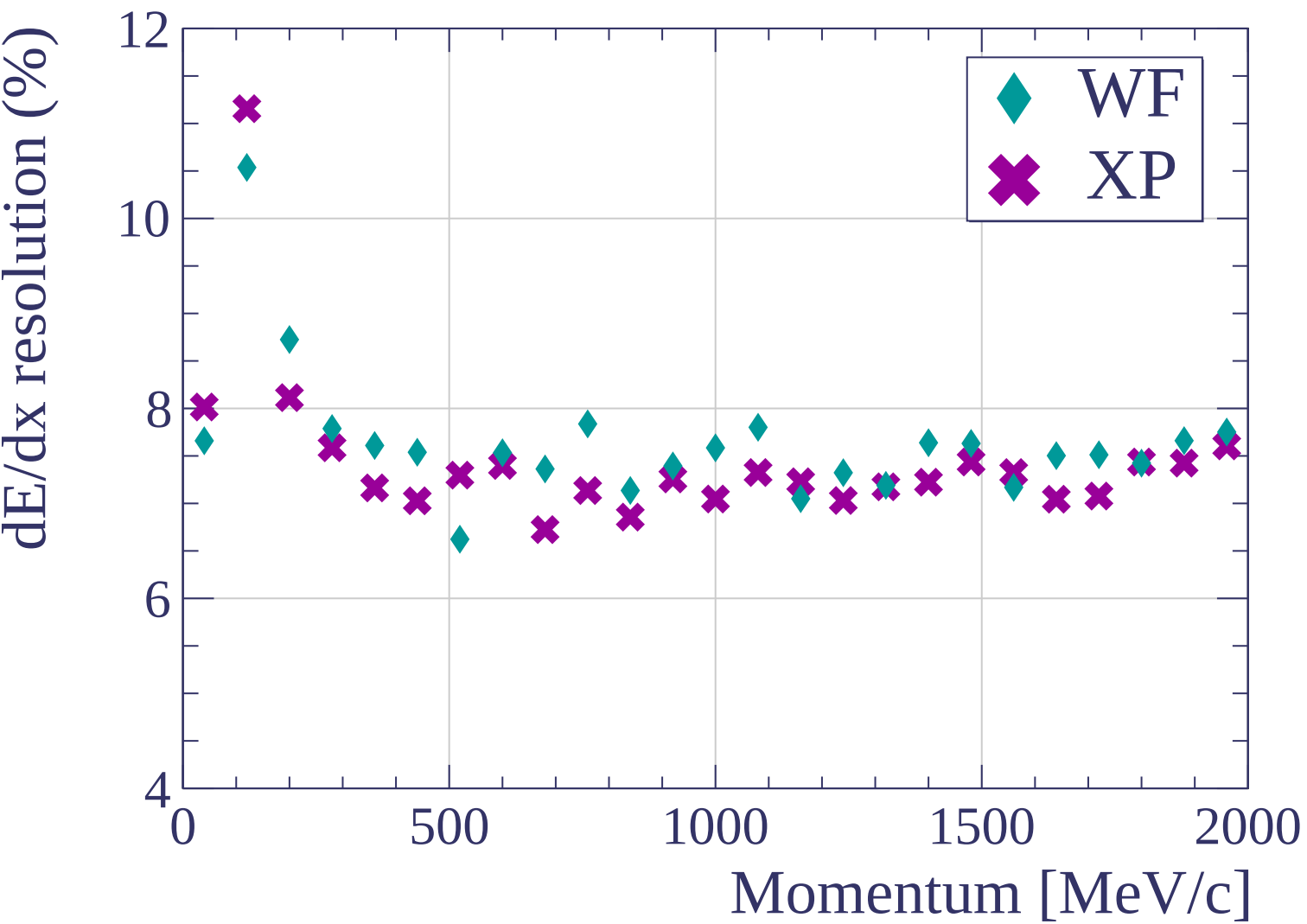


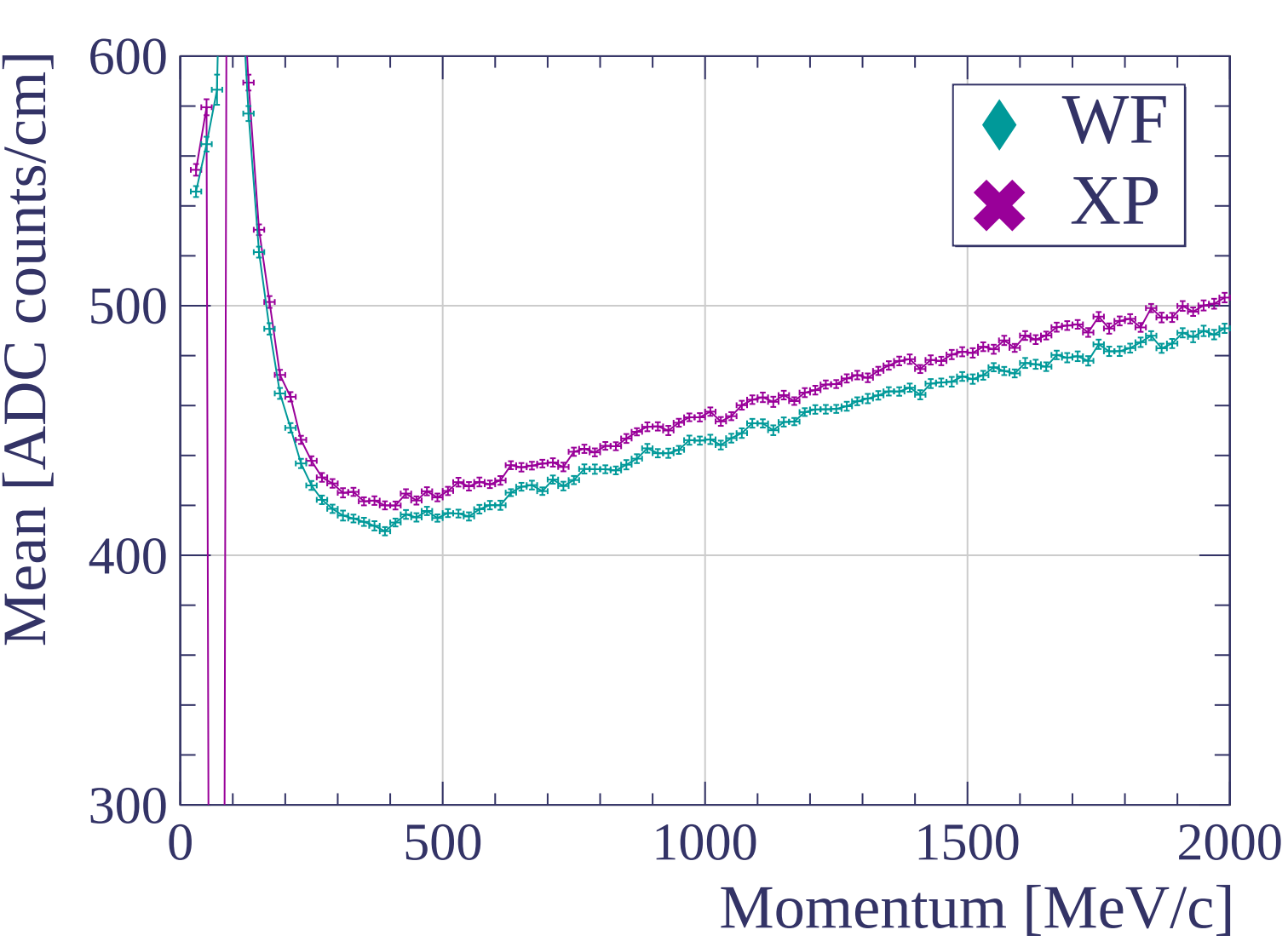


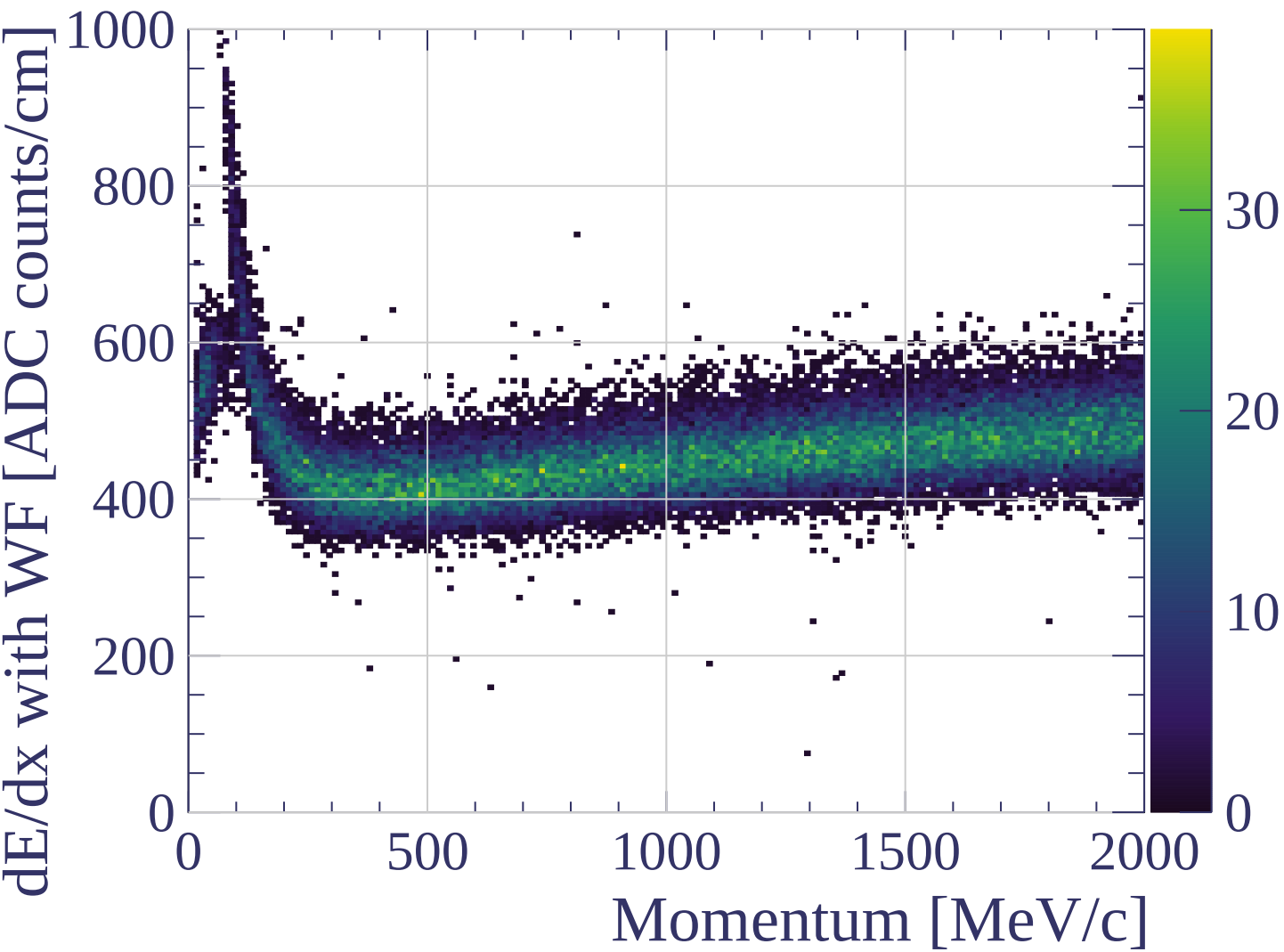


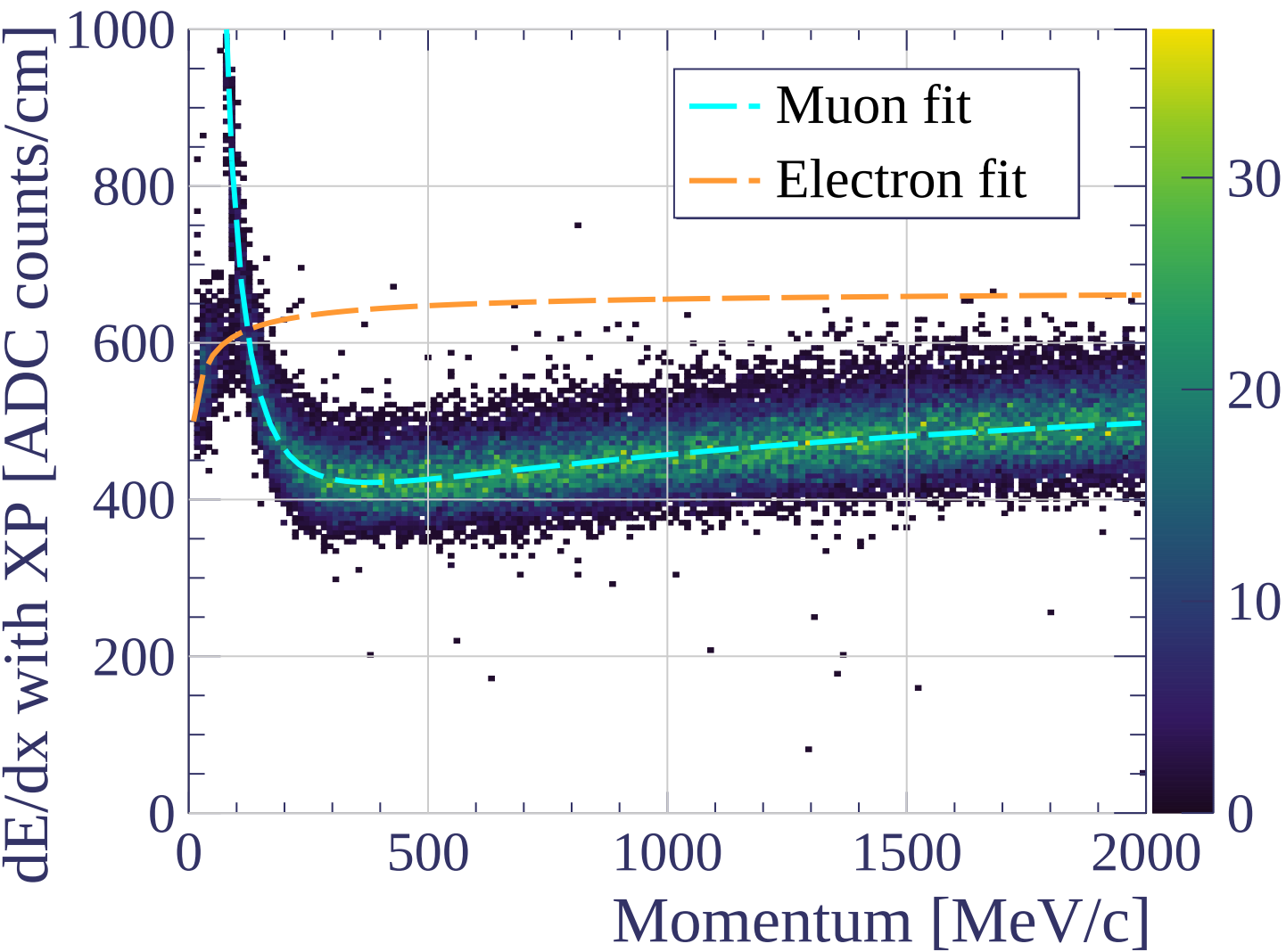


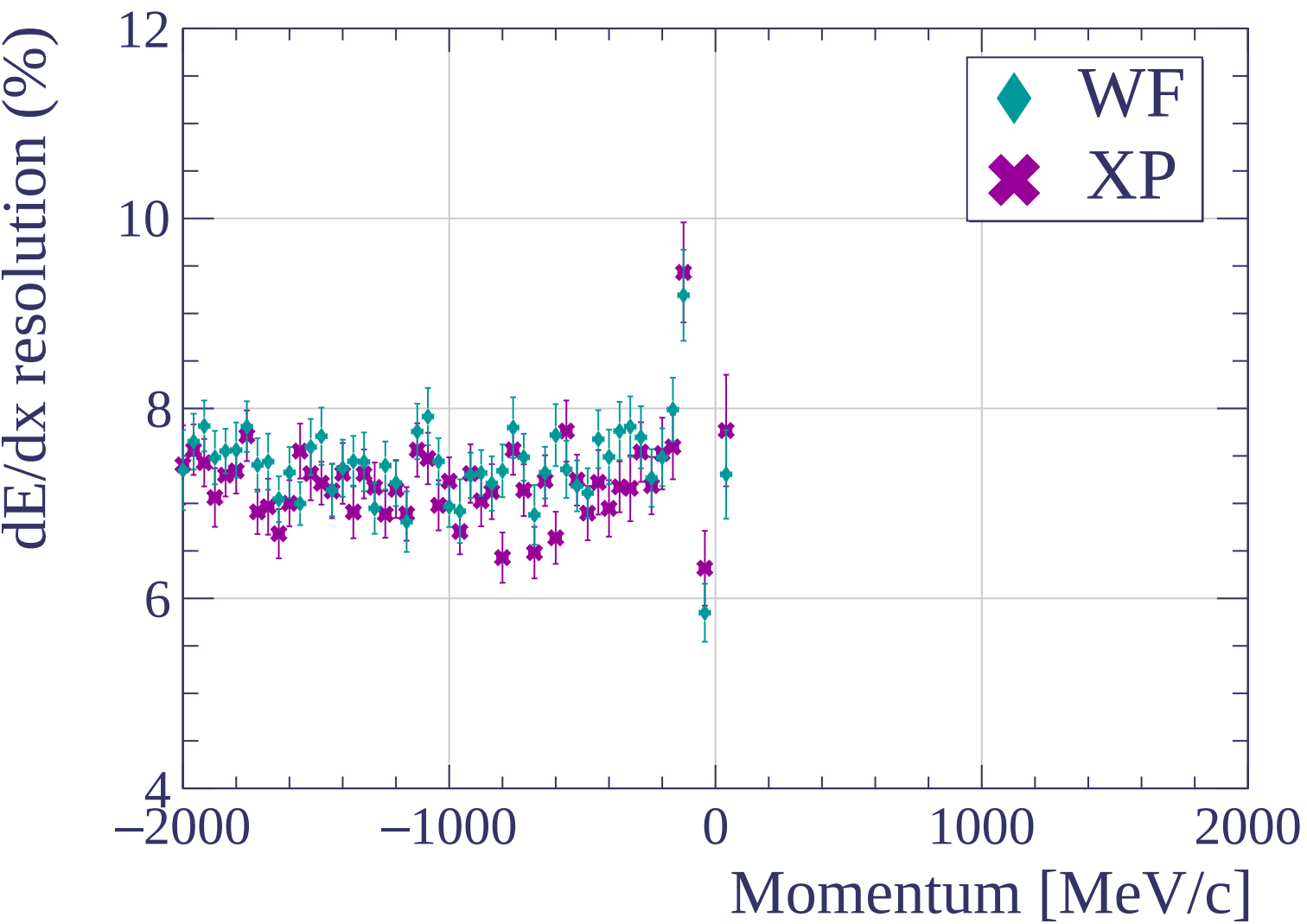


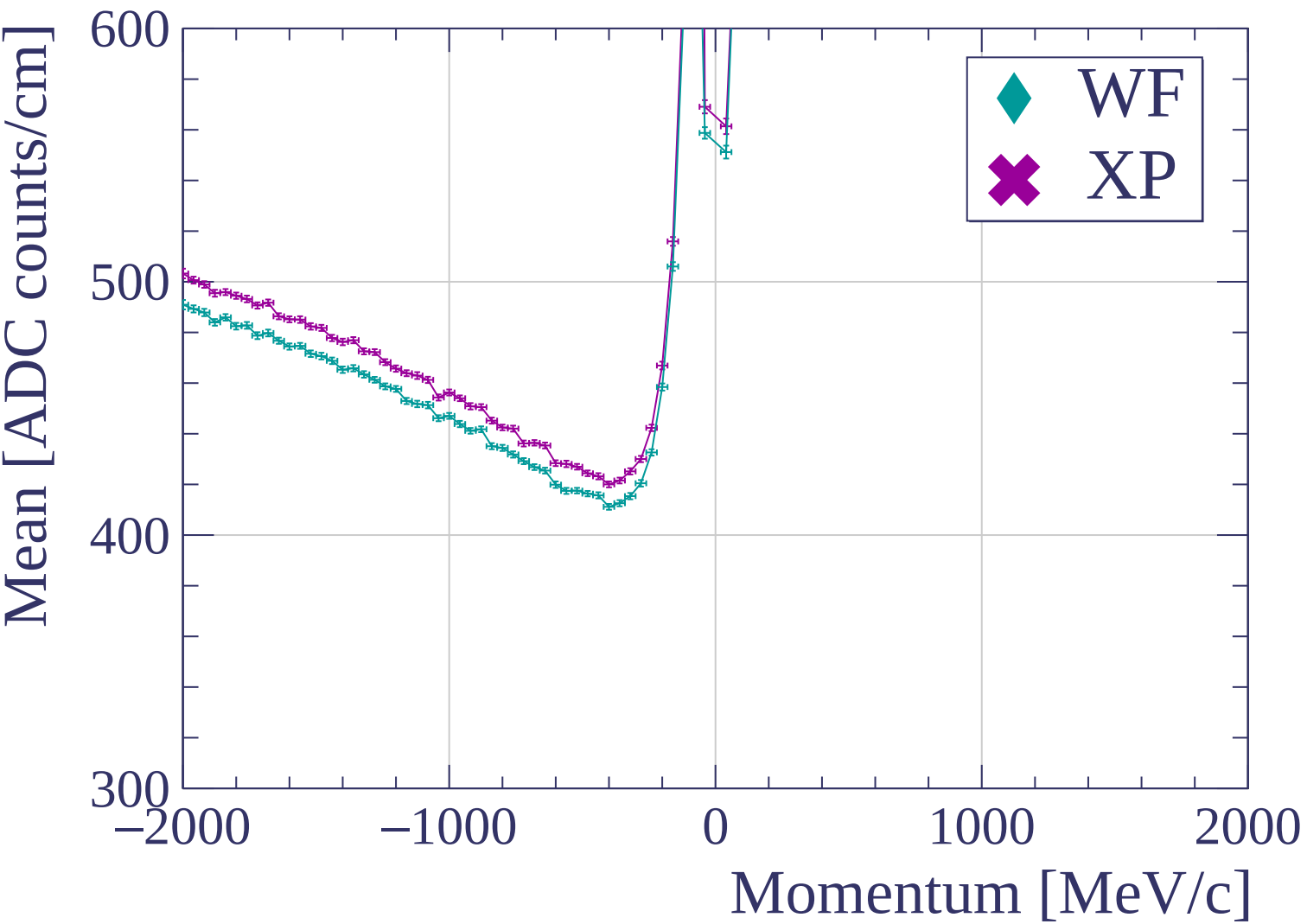


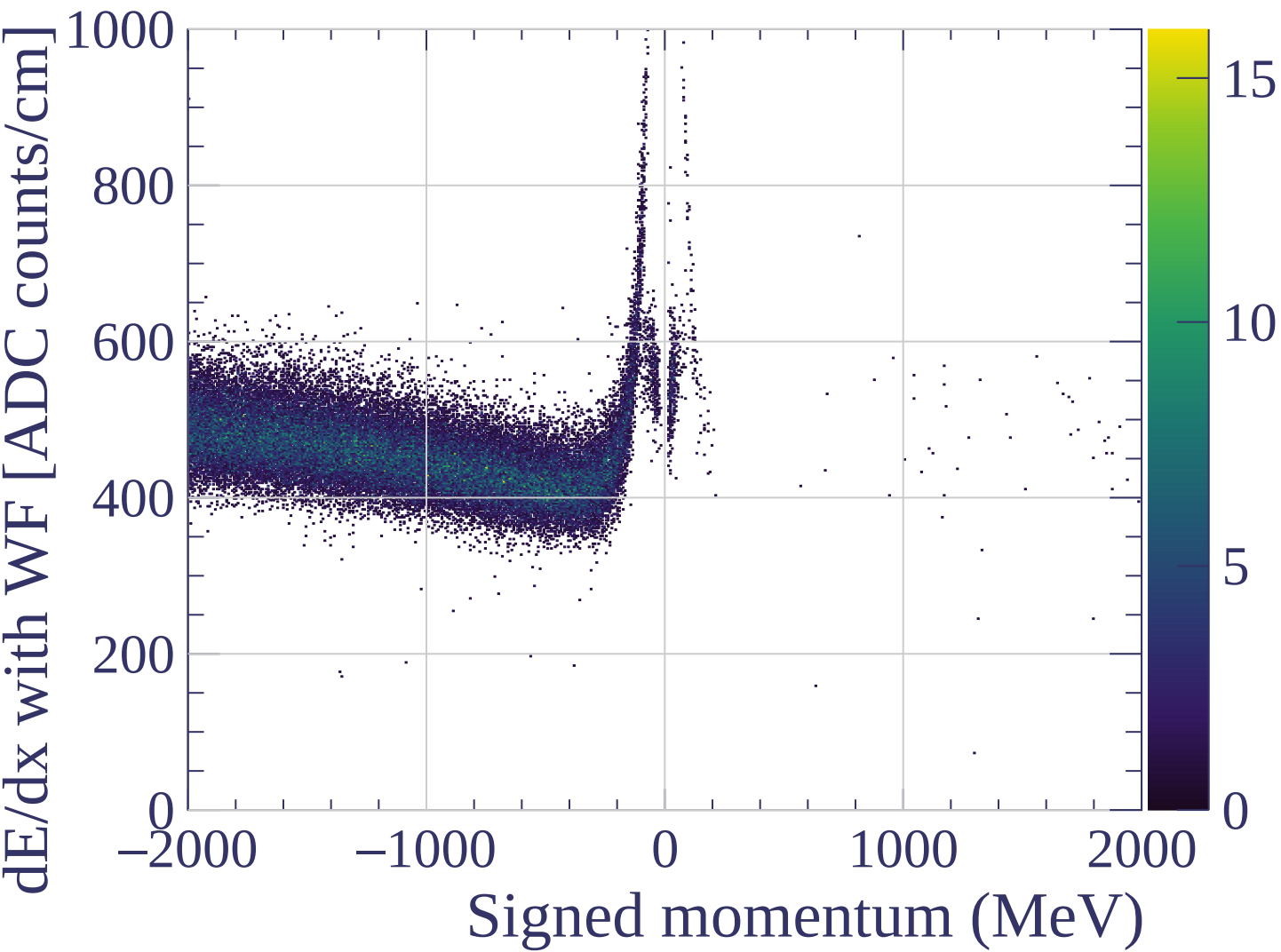


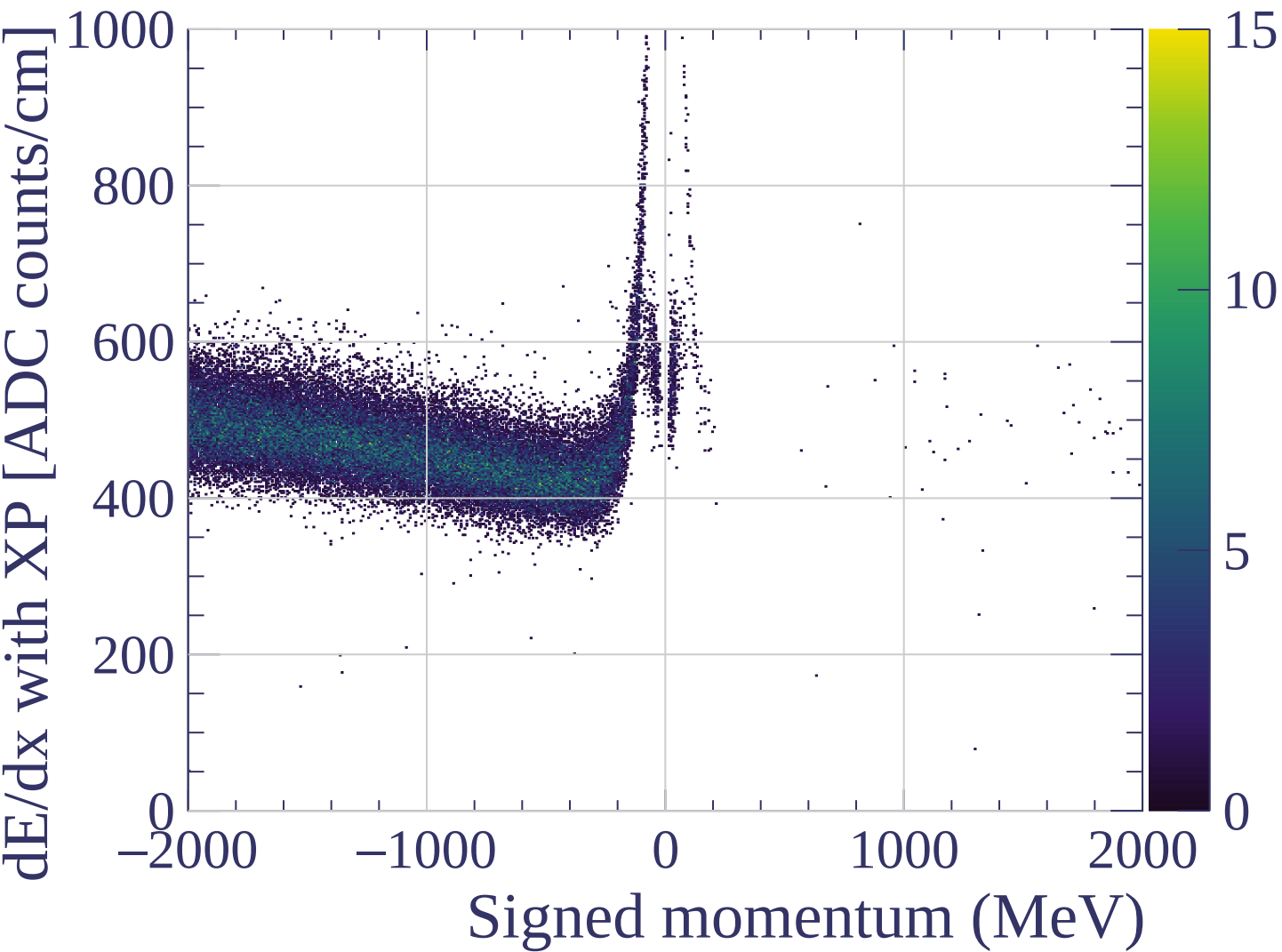




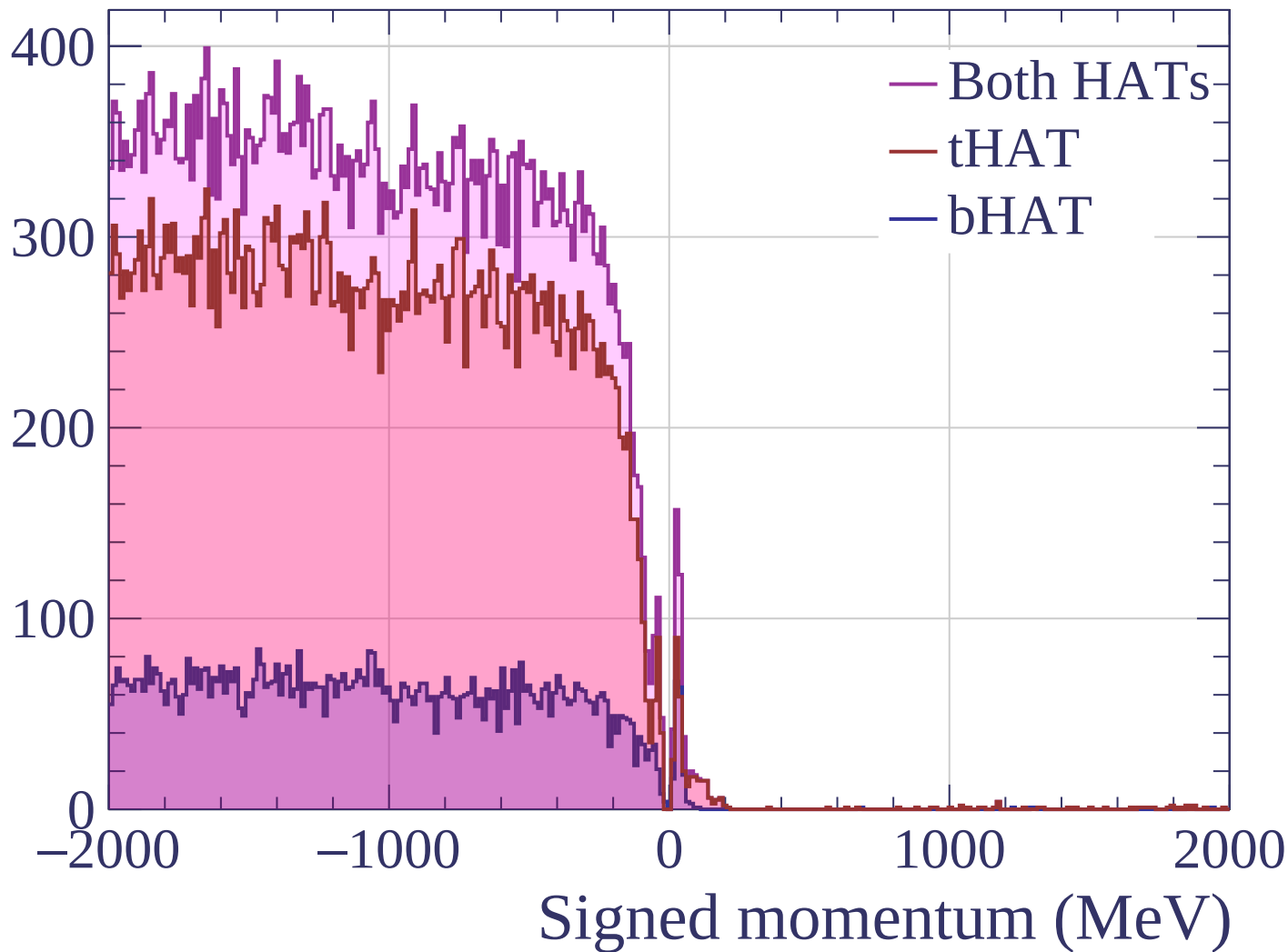


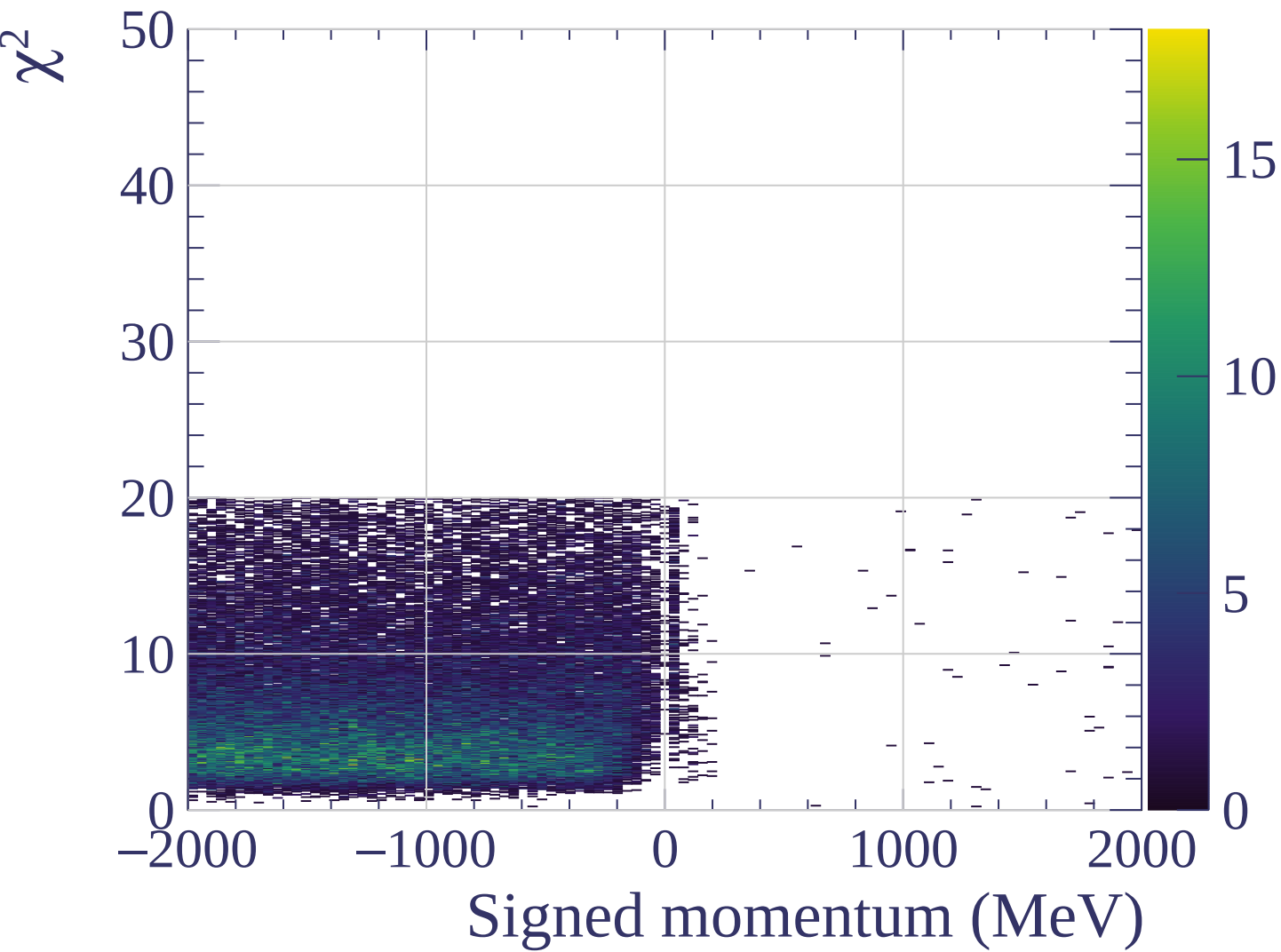


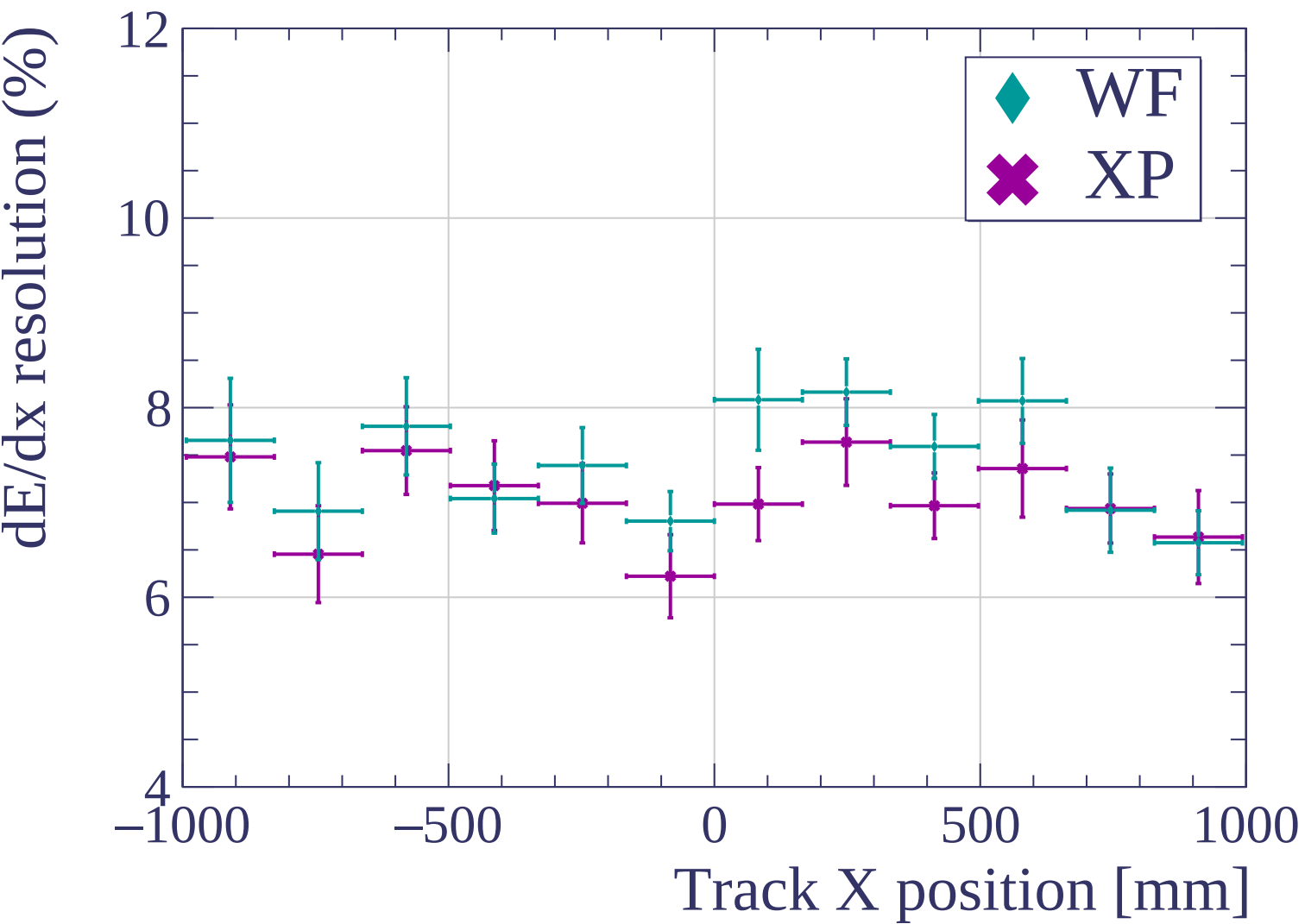


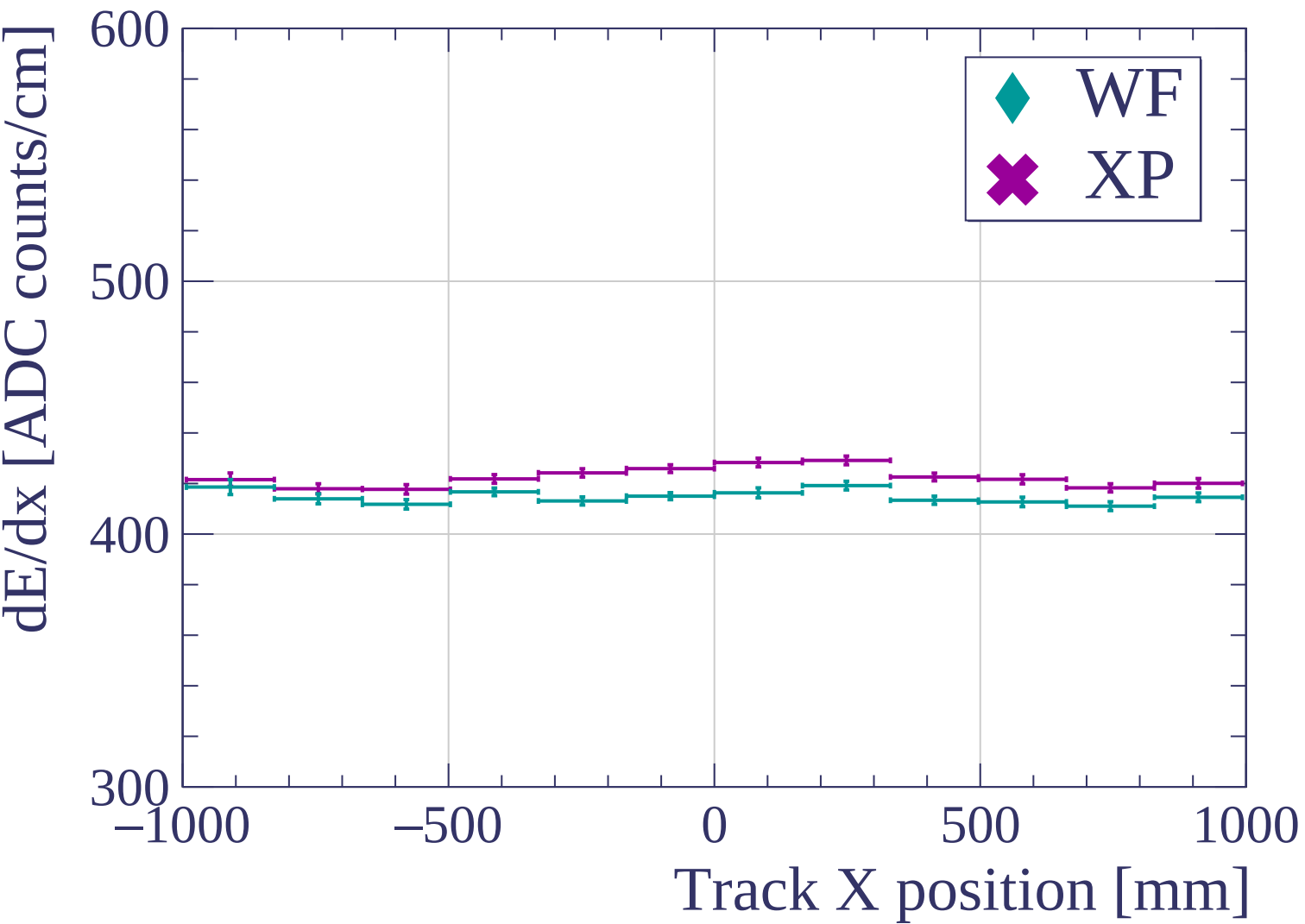


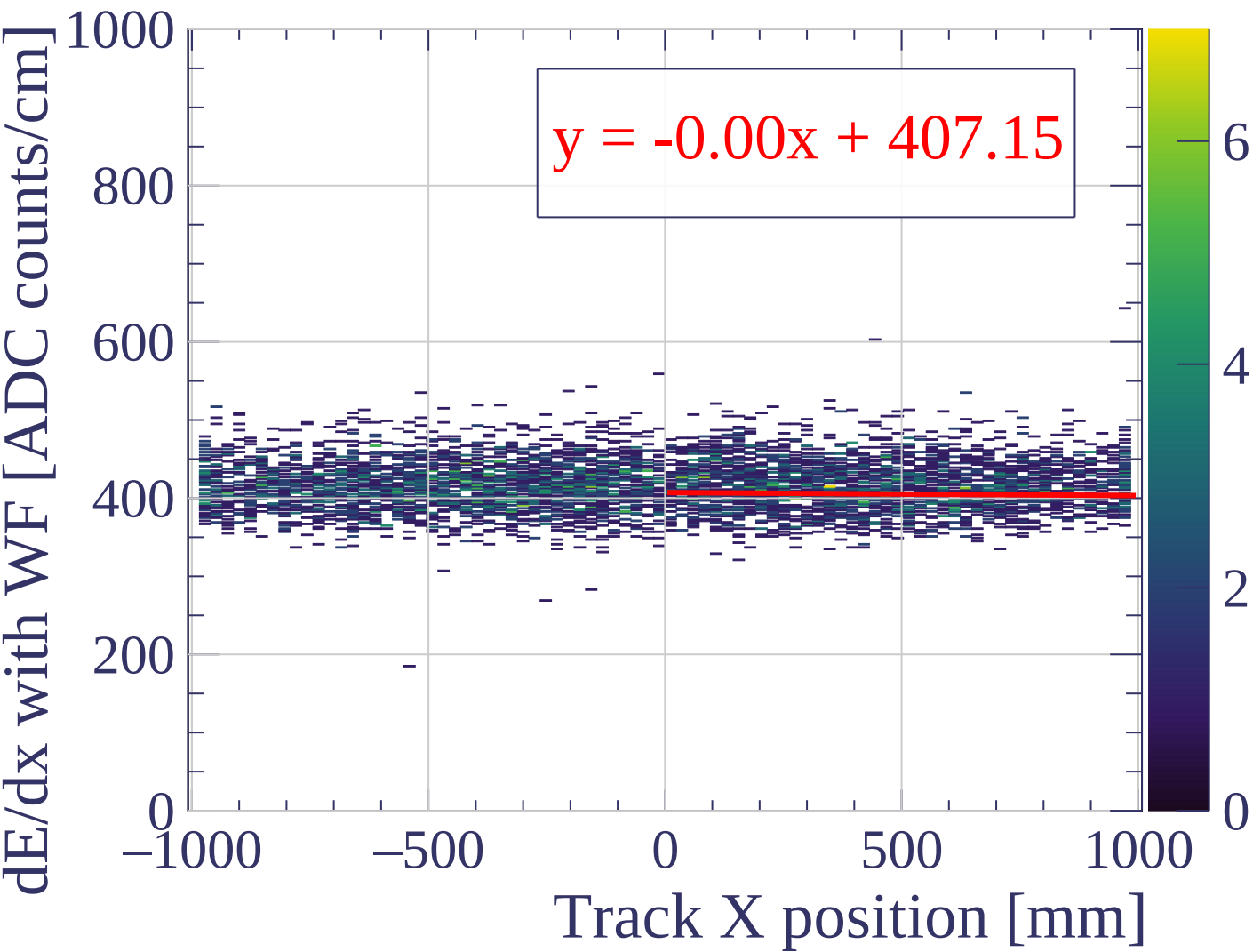
Count

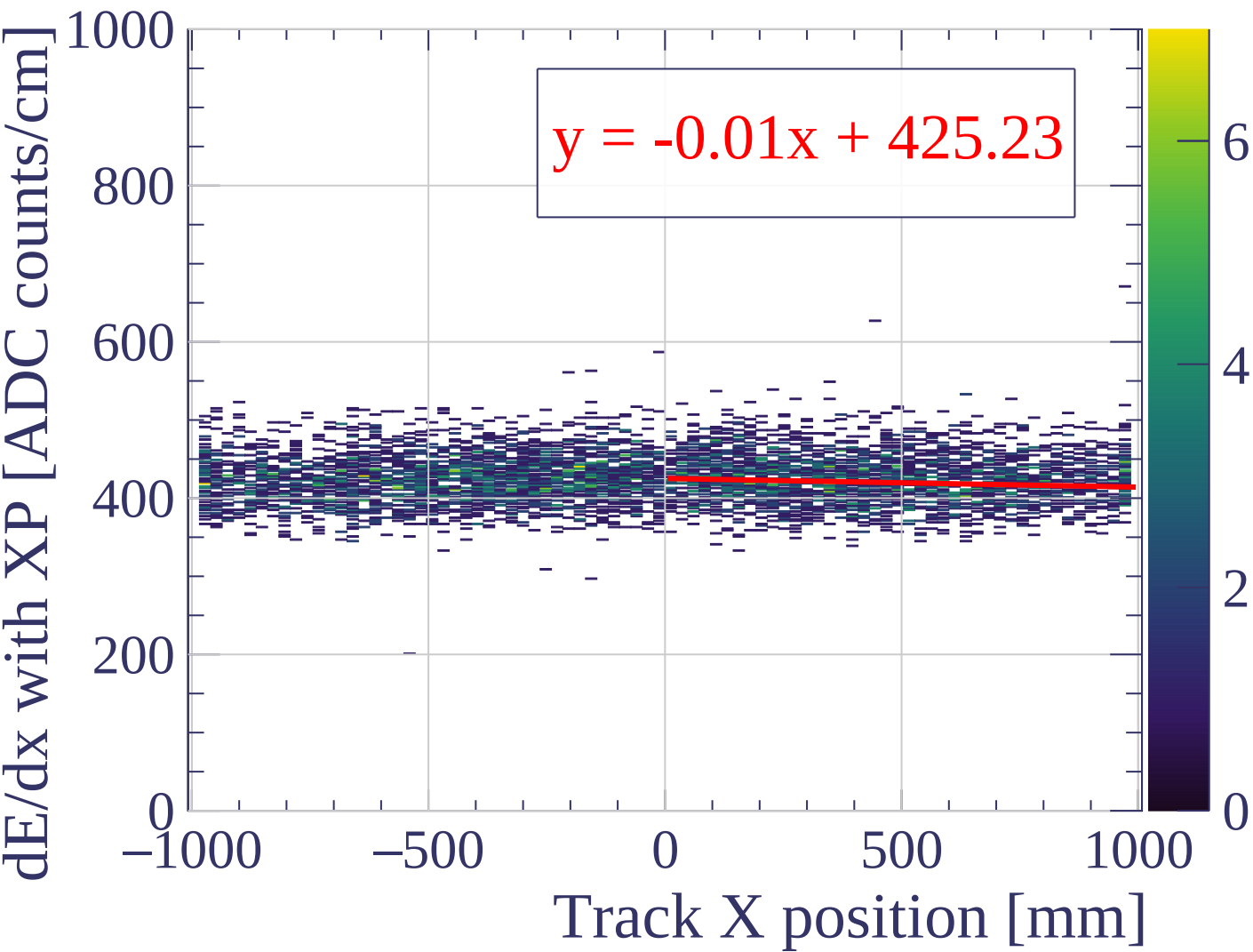


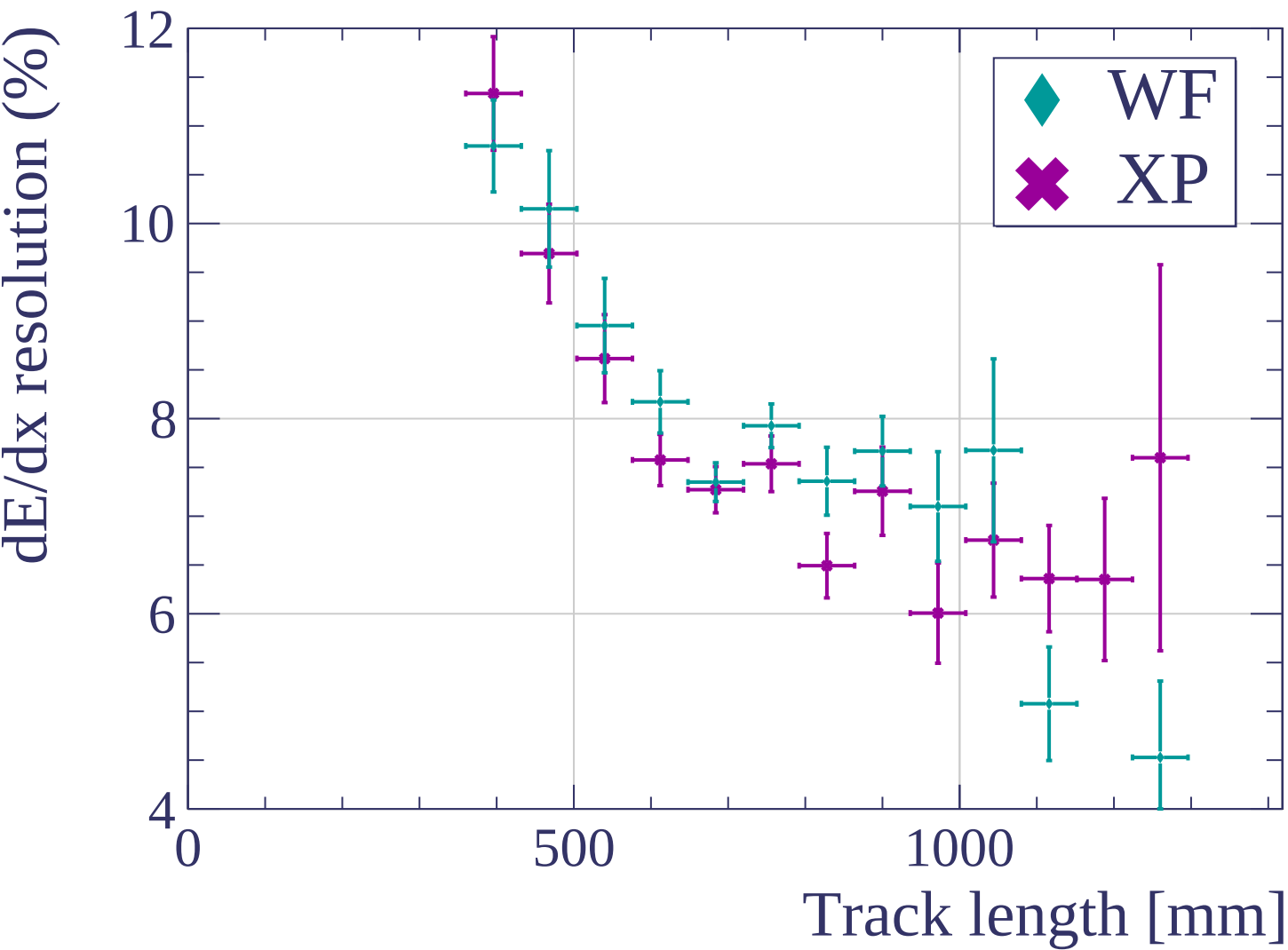


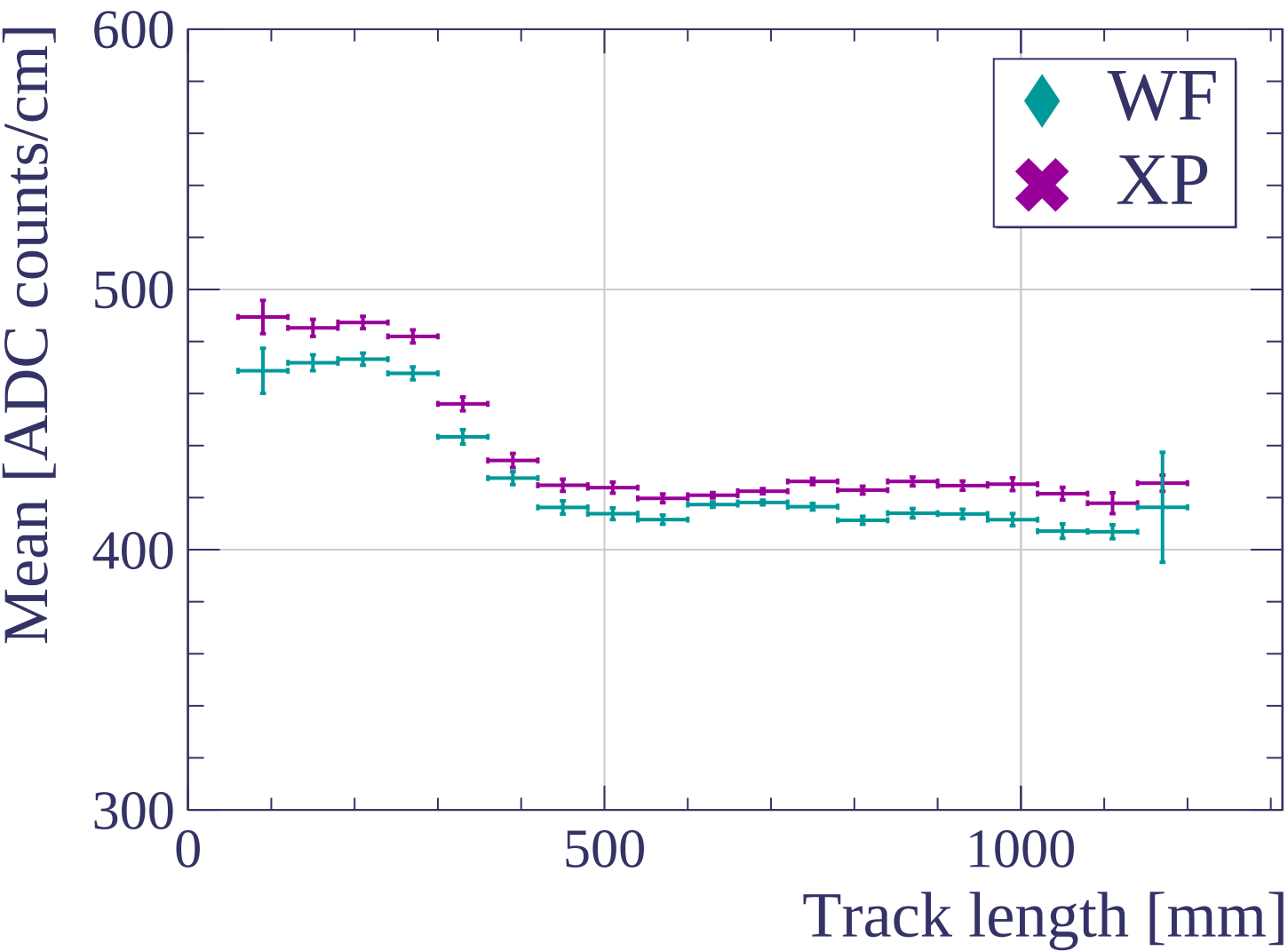


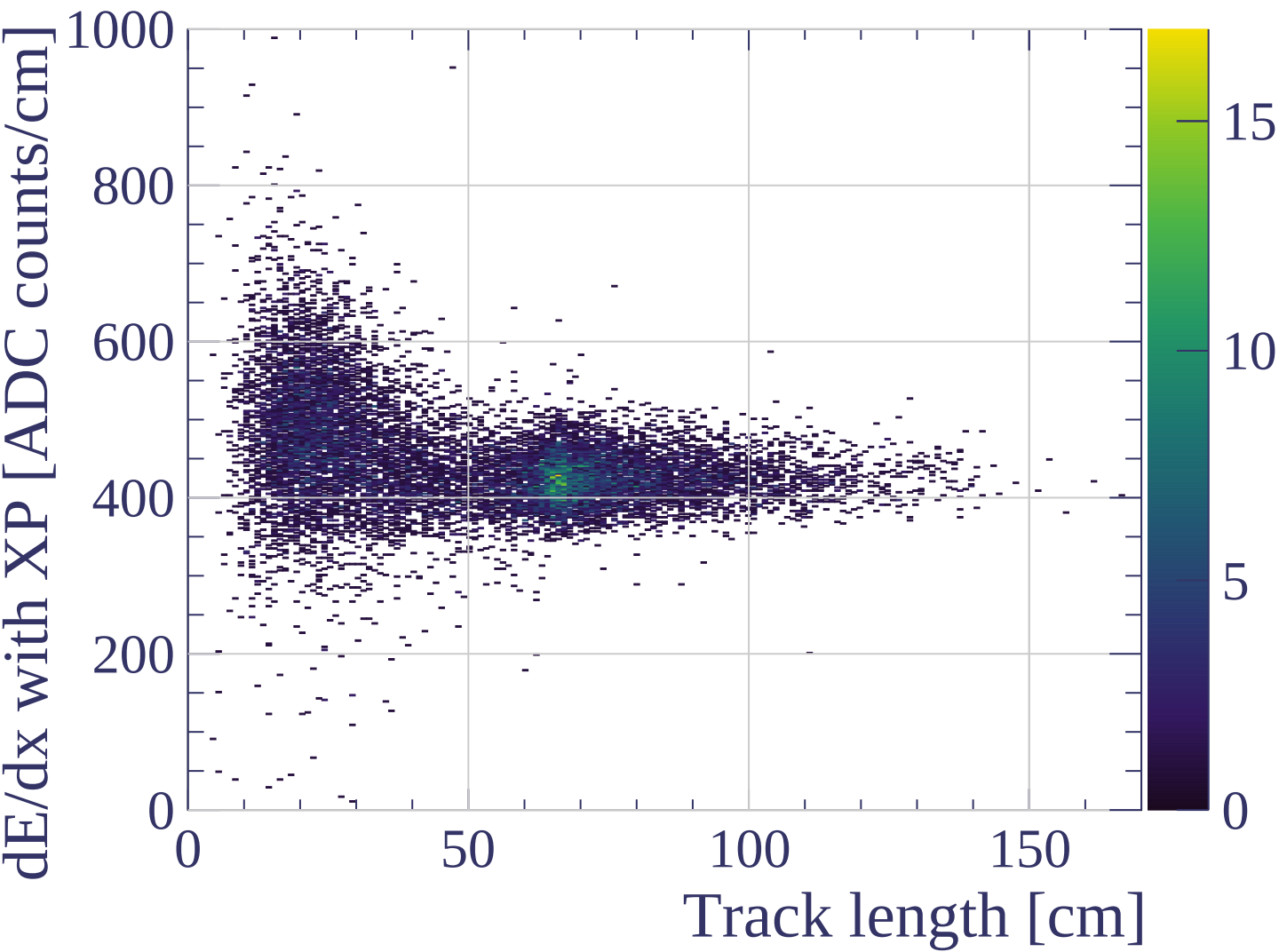


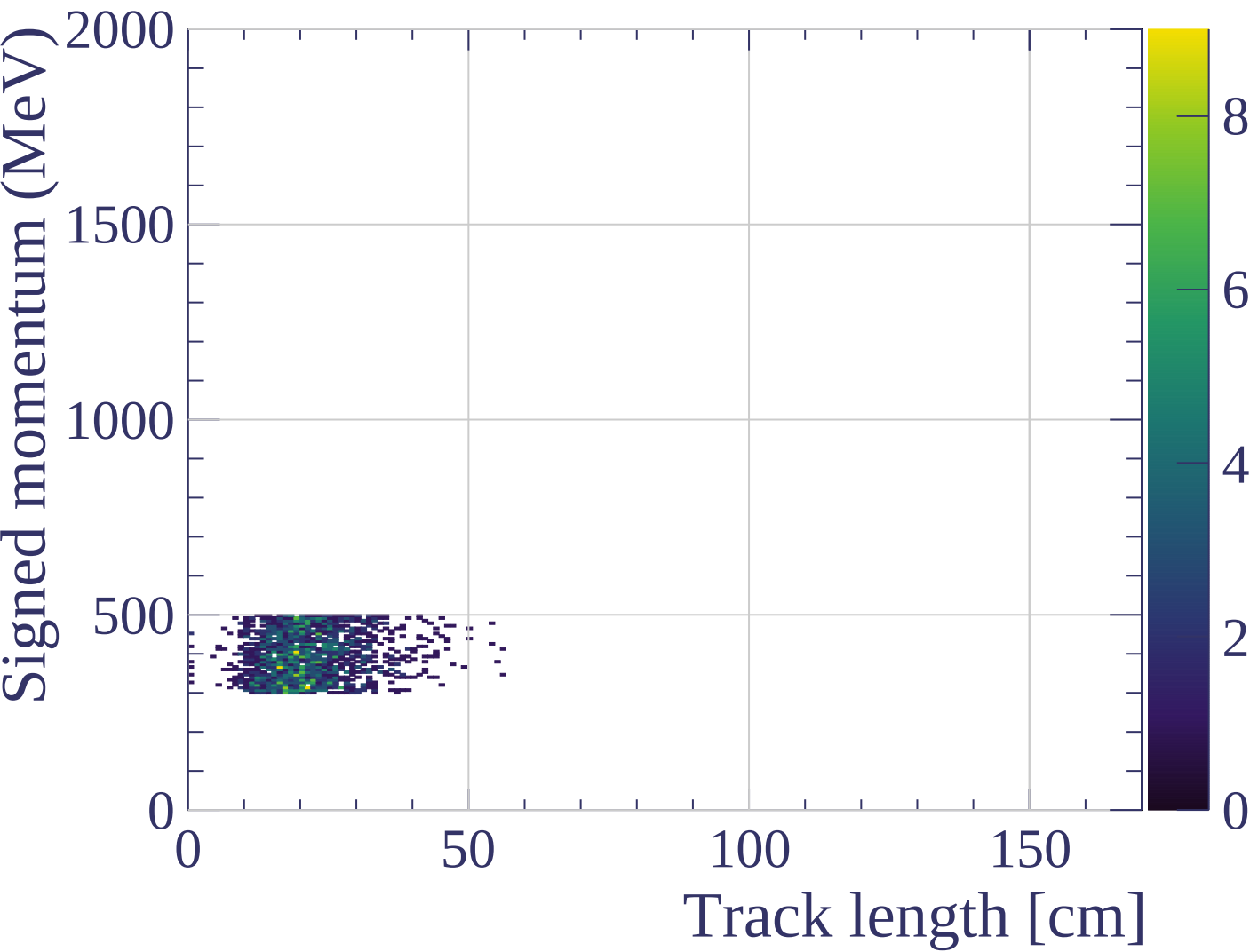


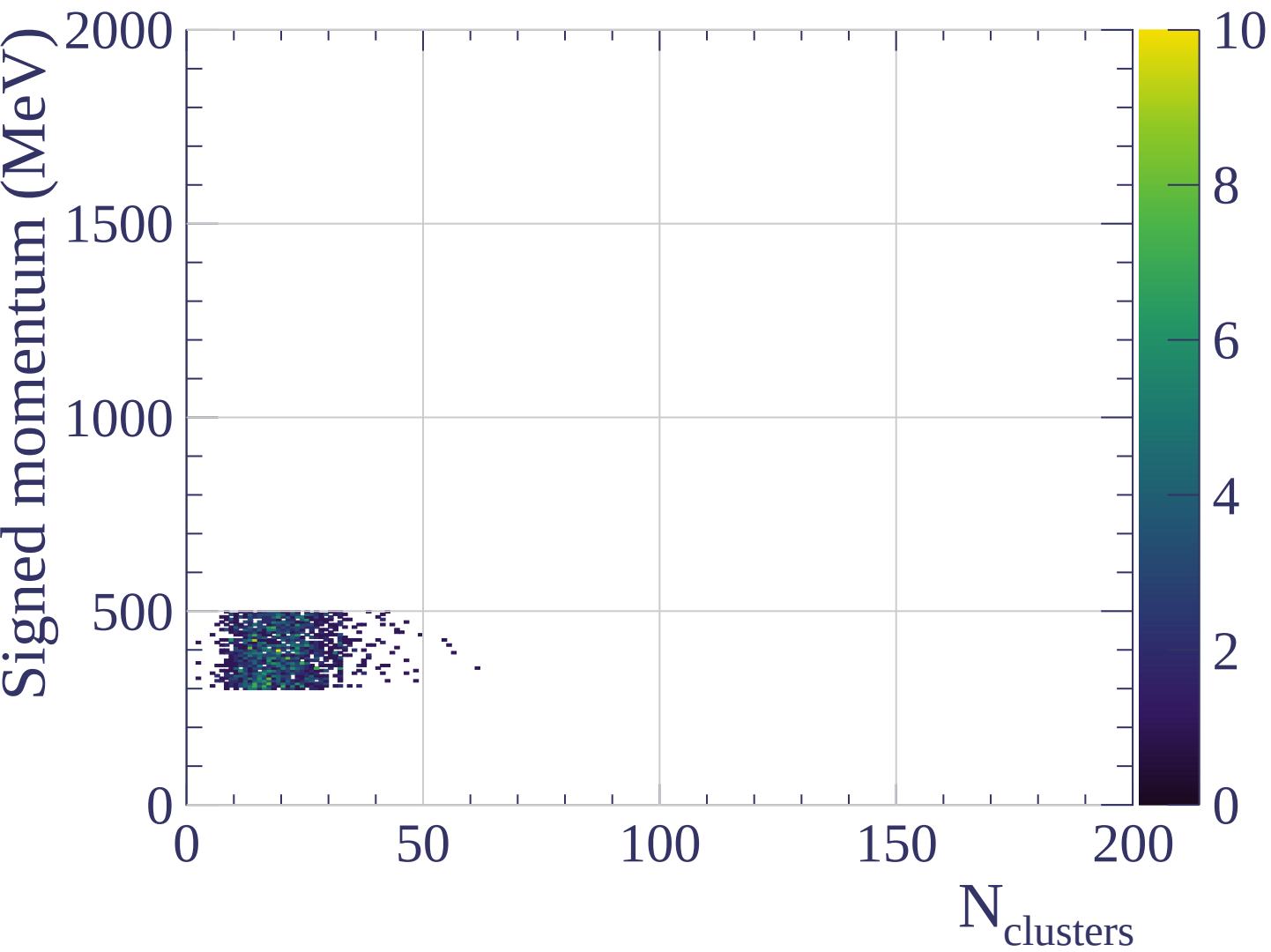


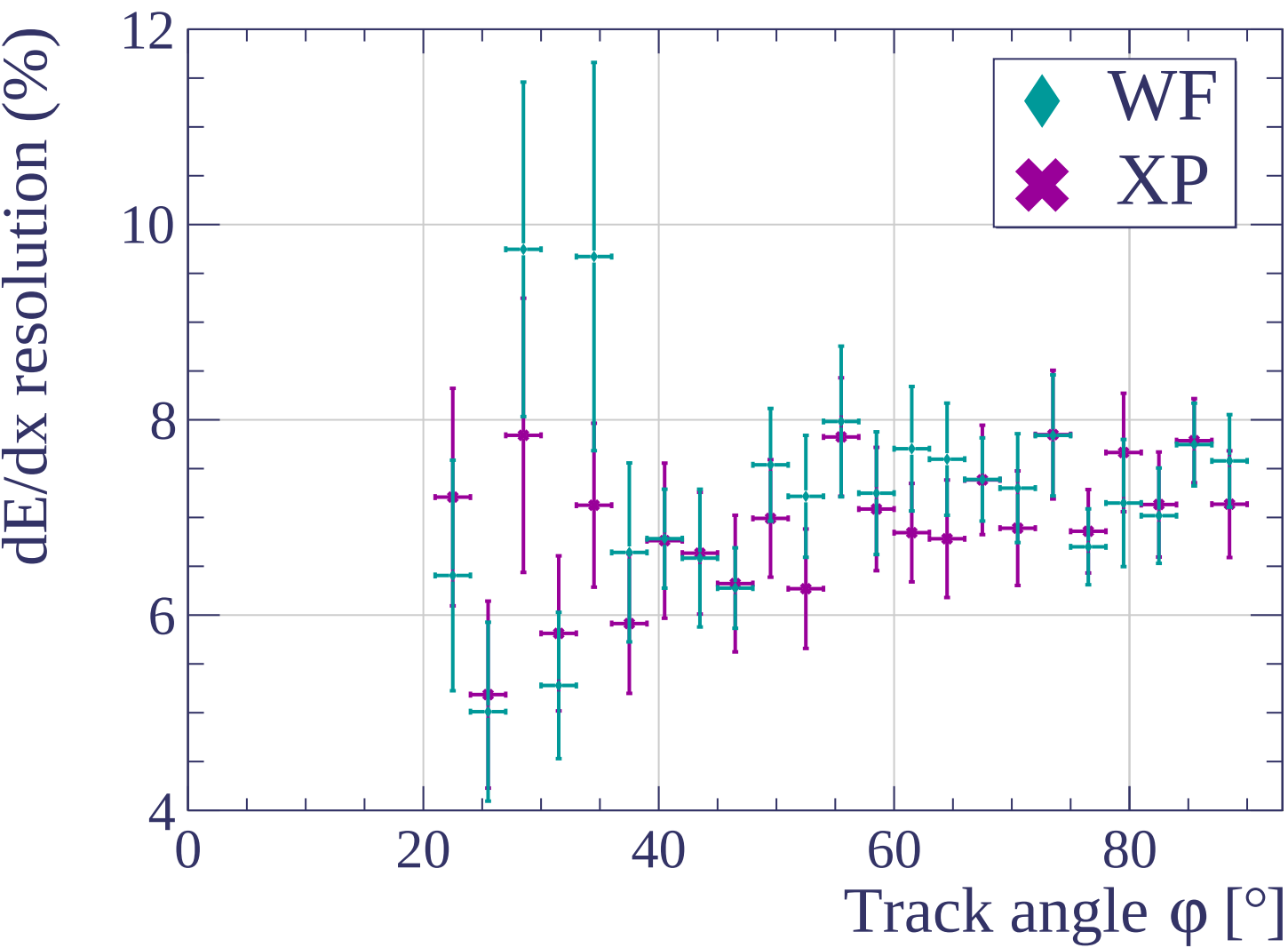


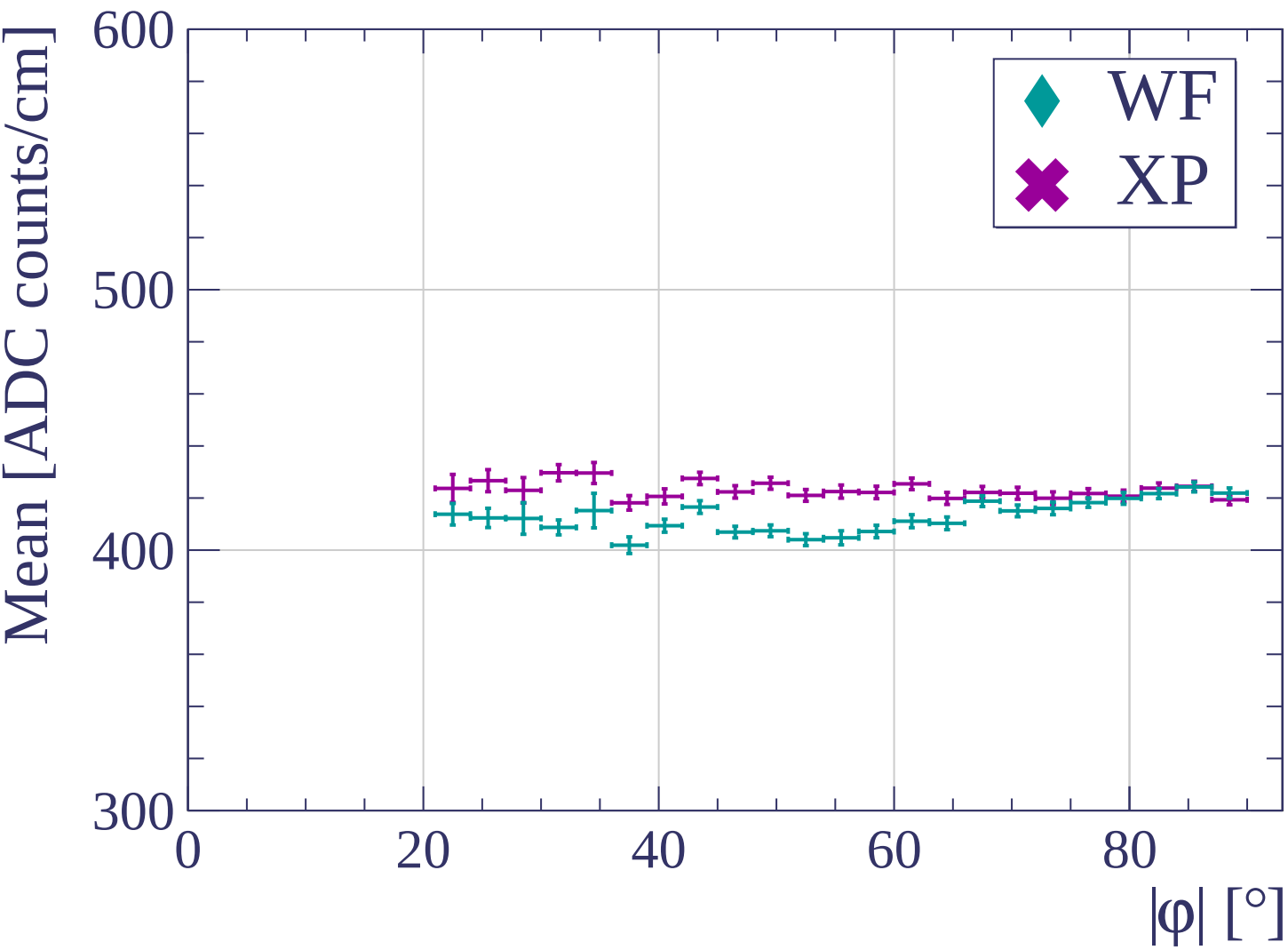


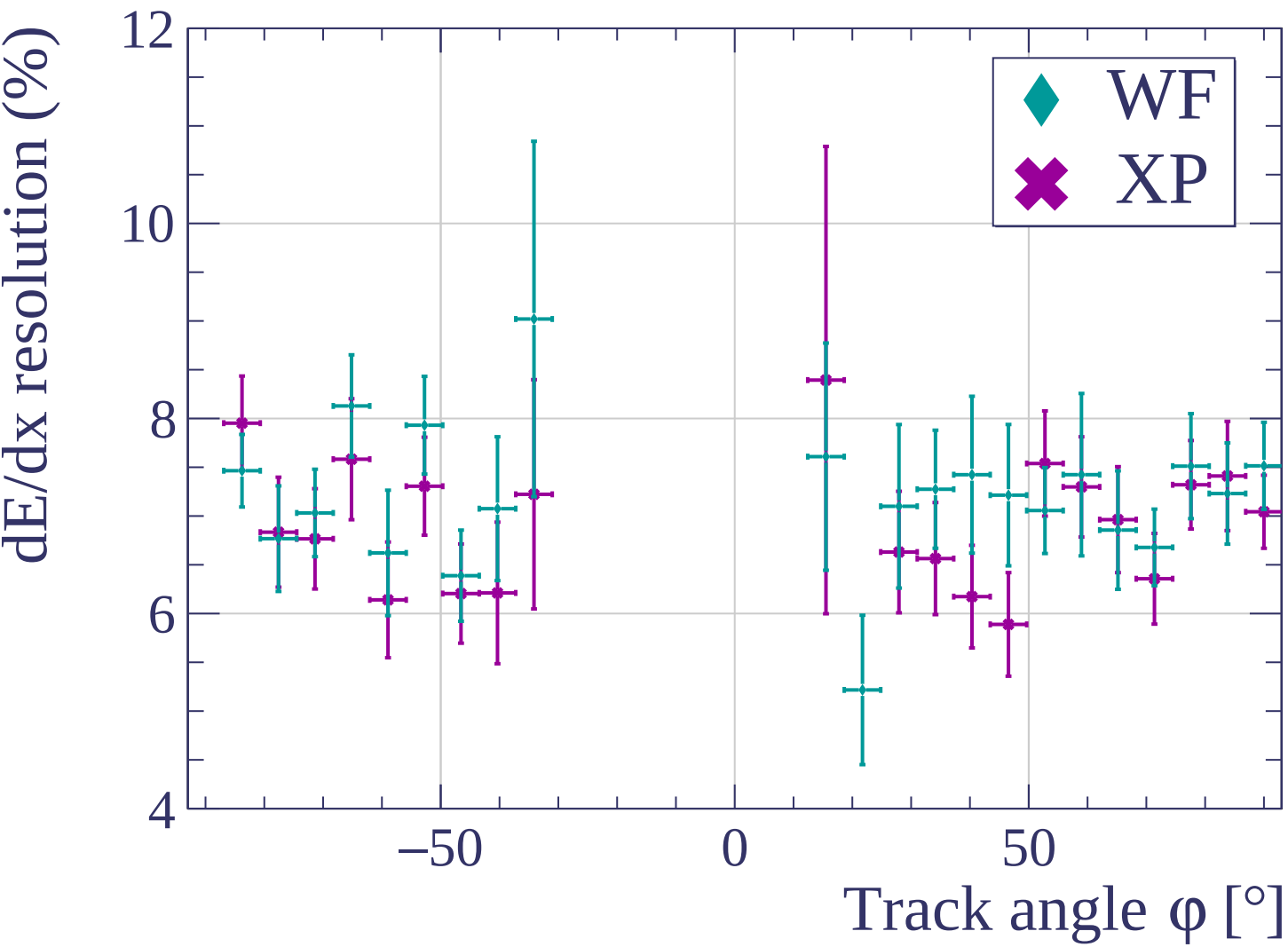


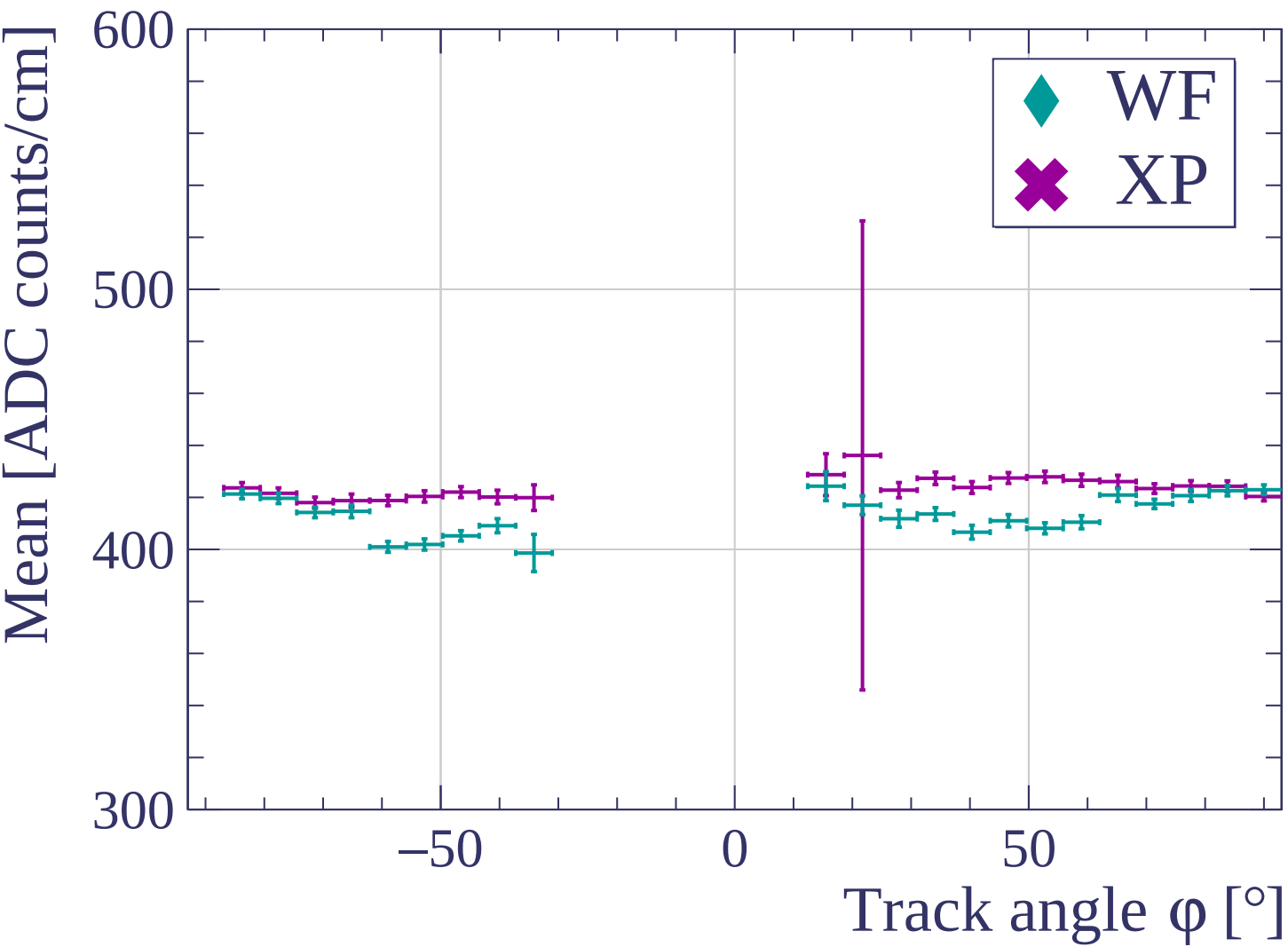


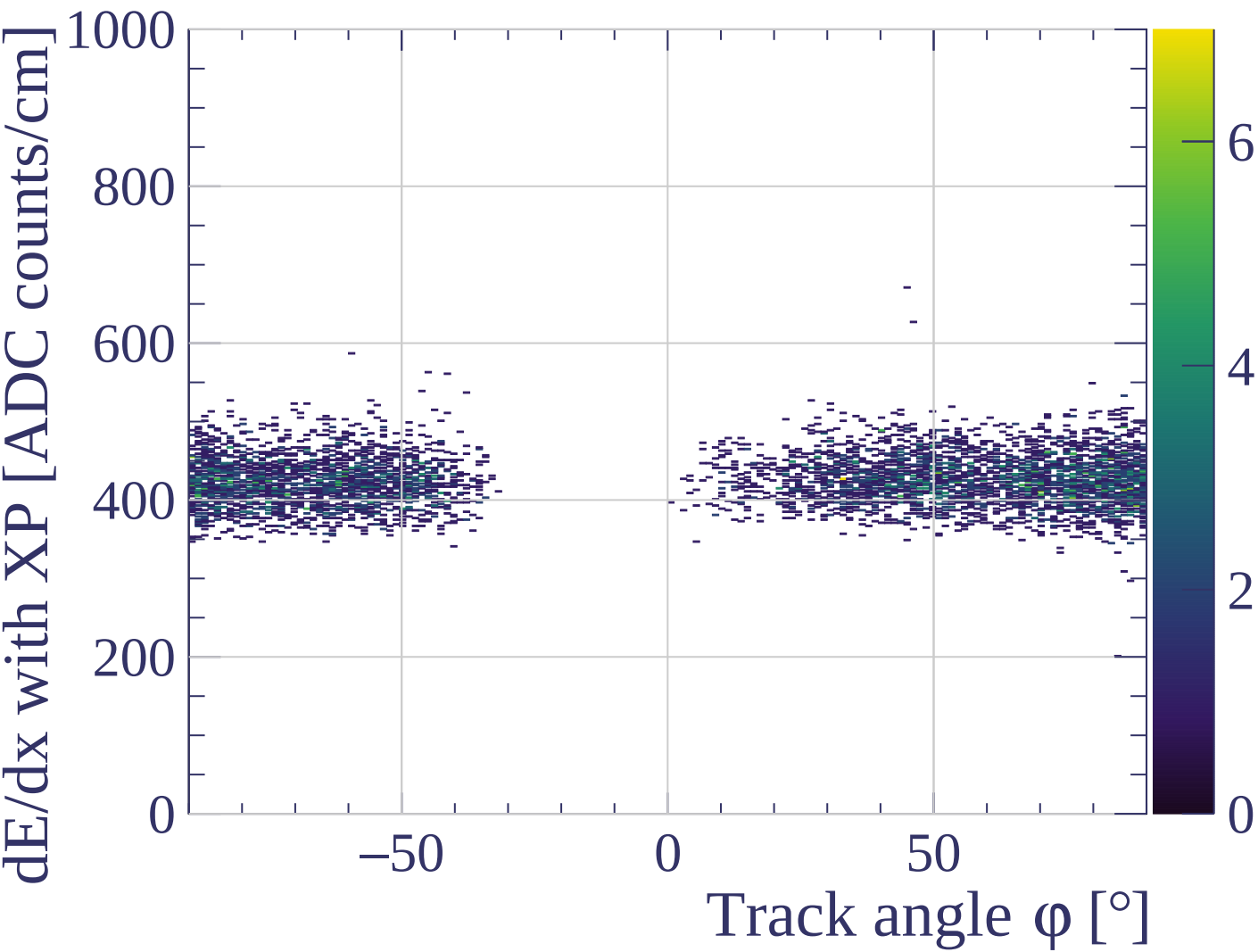




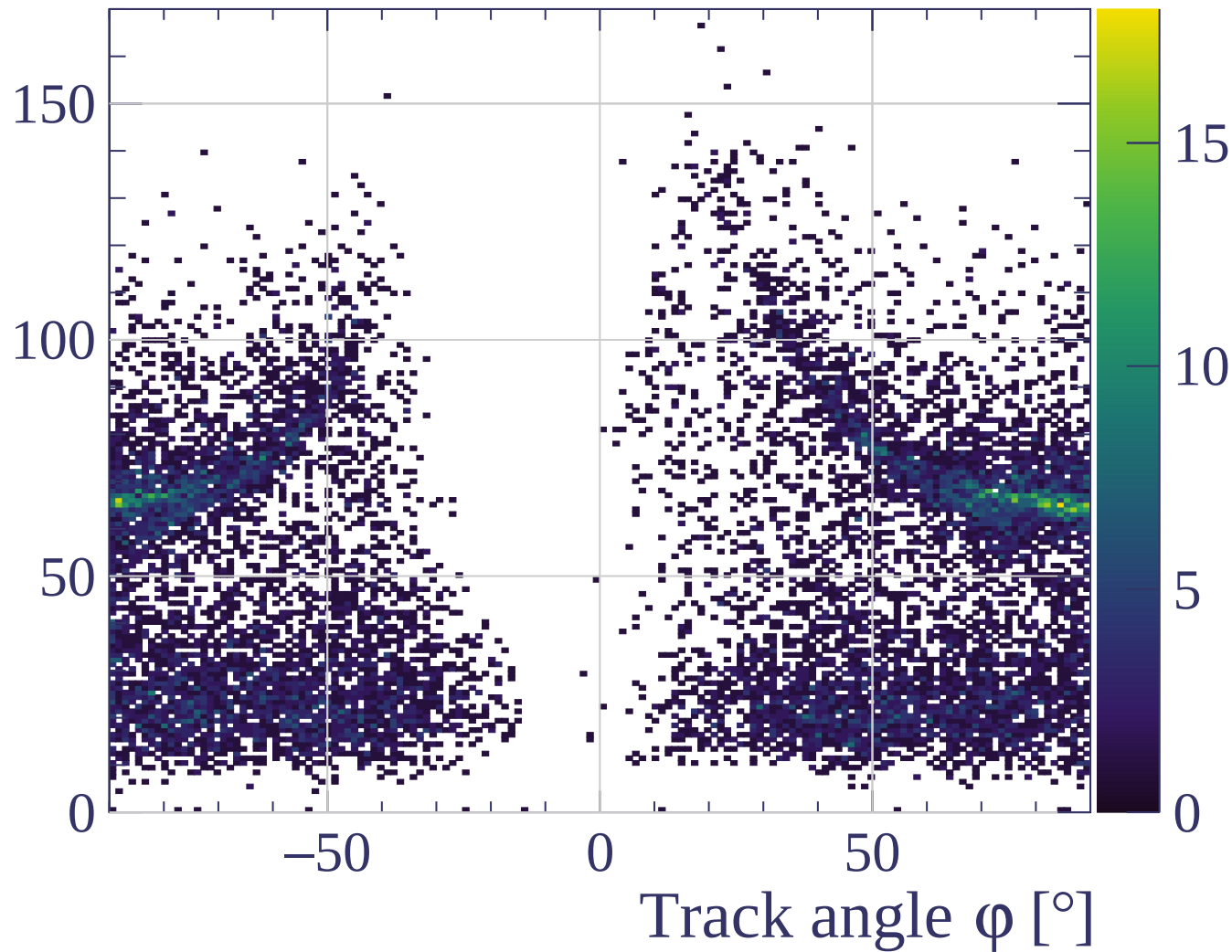


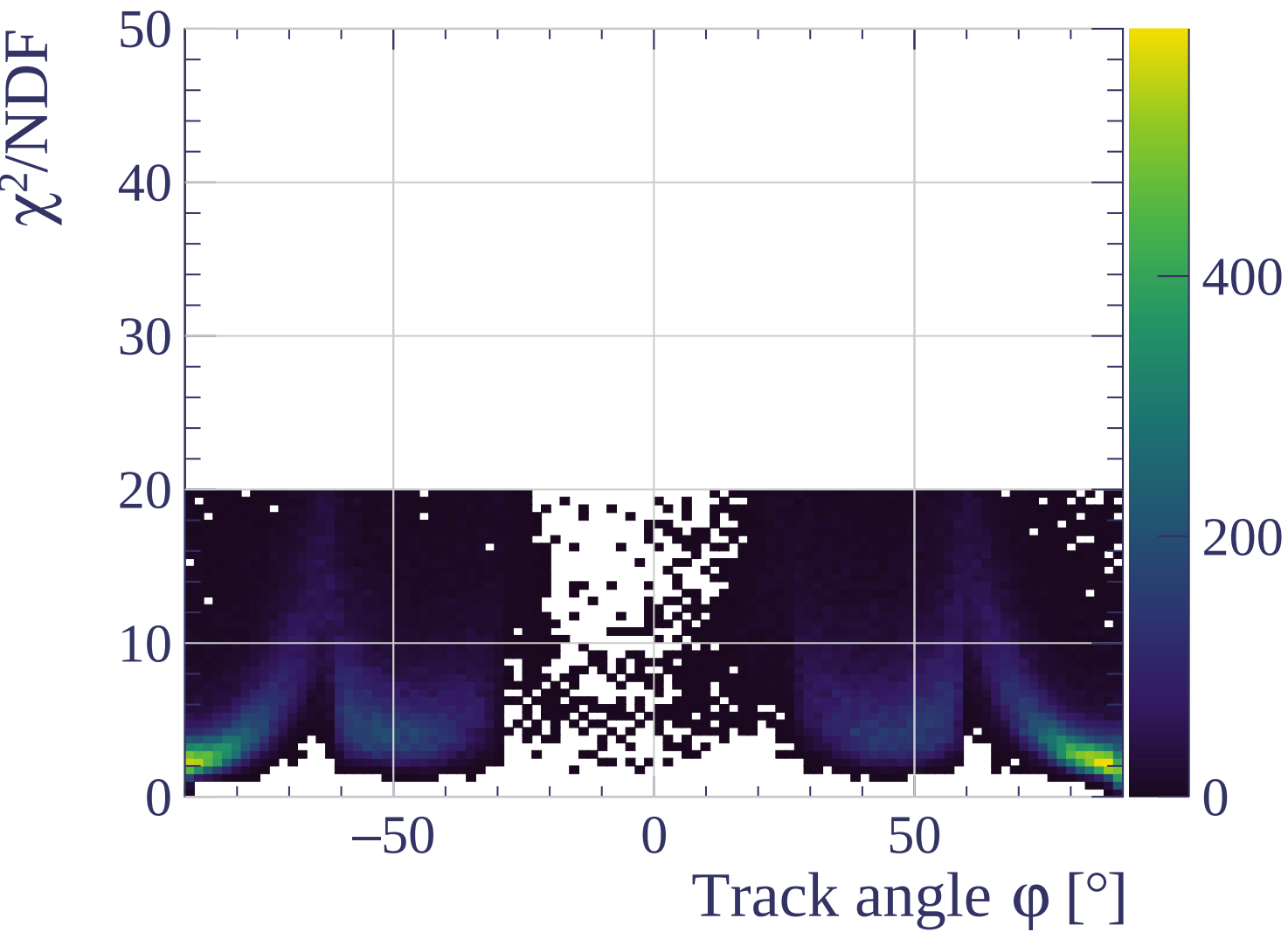


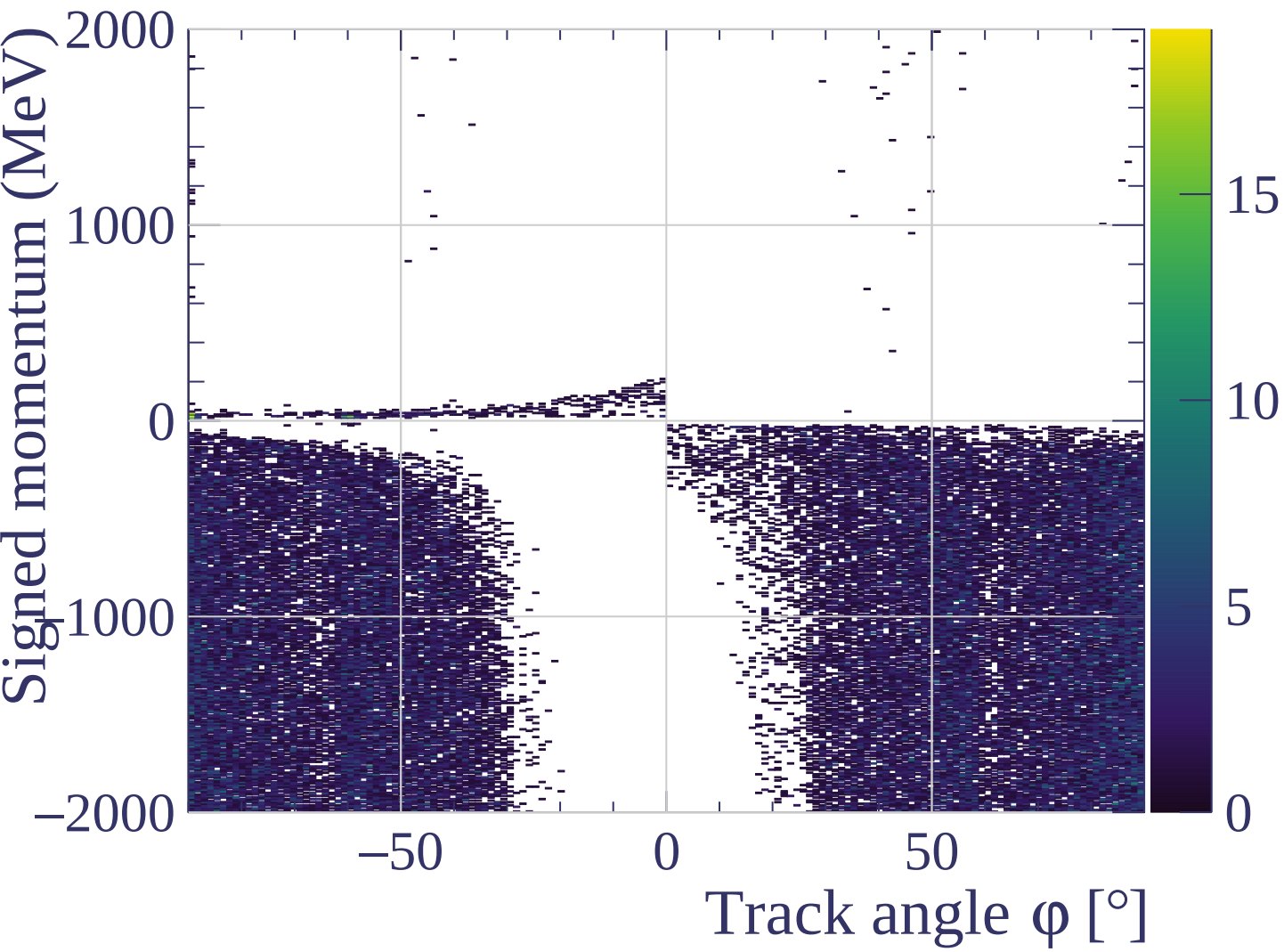




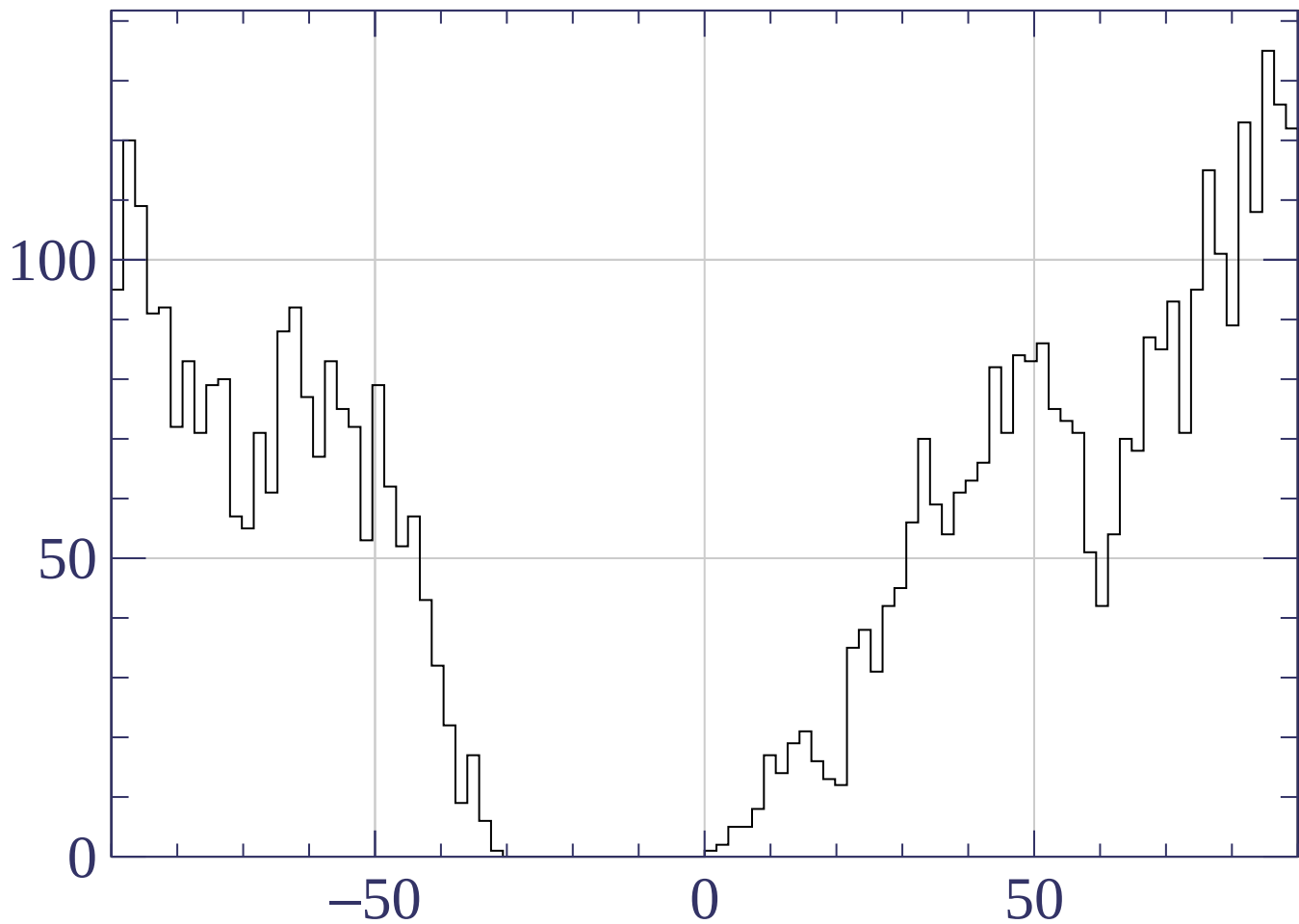
Track length [cm]



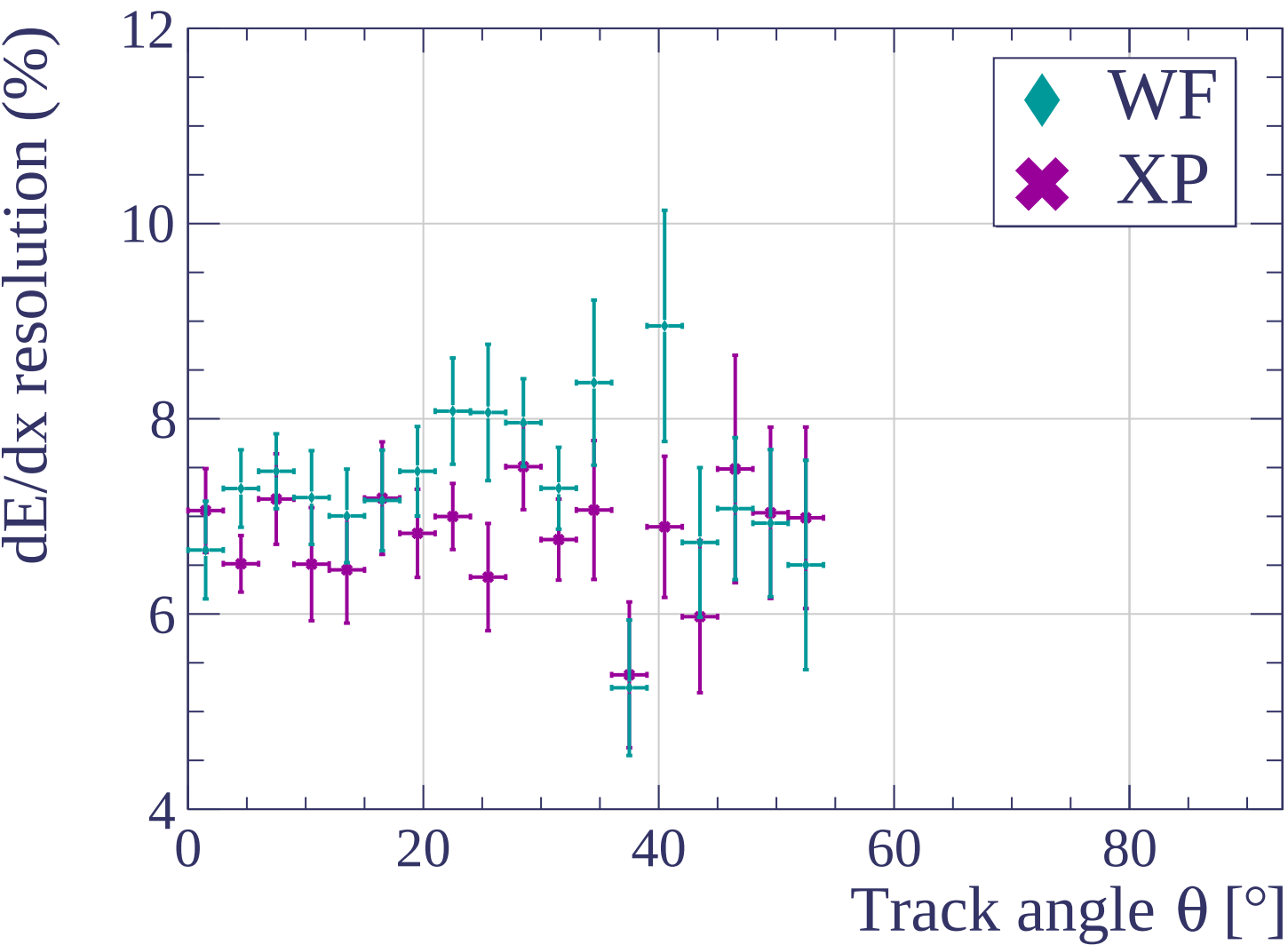


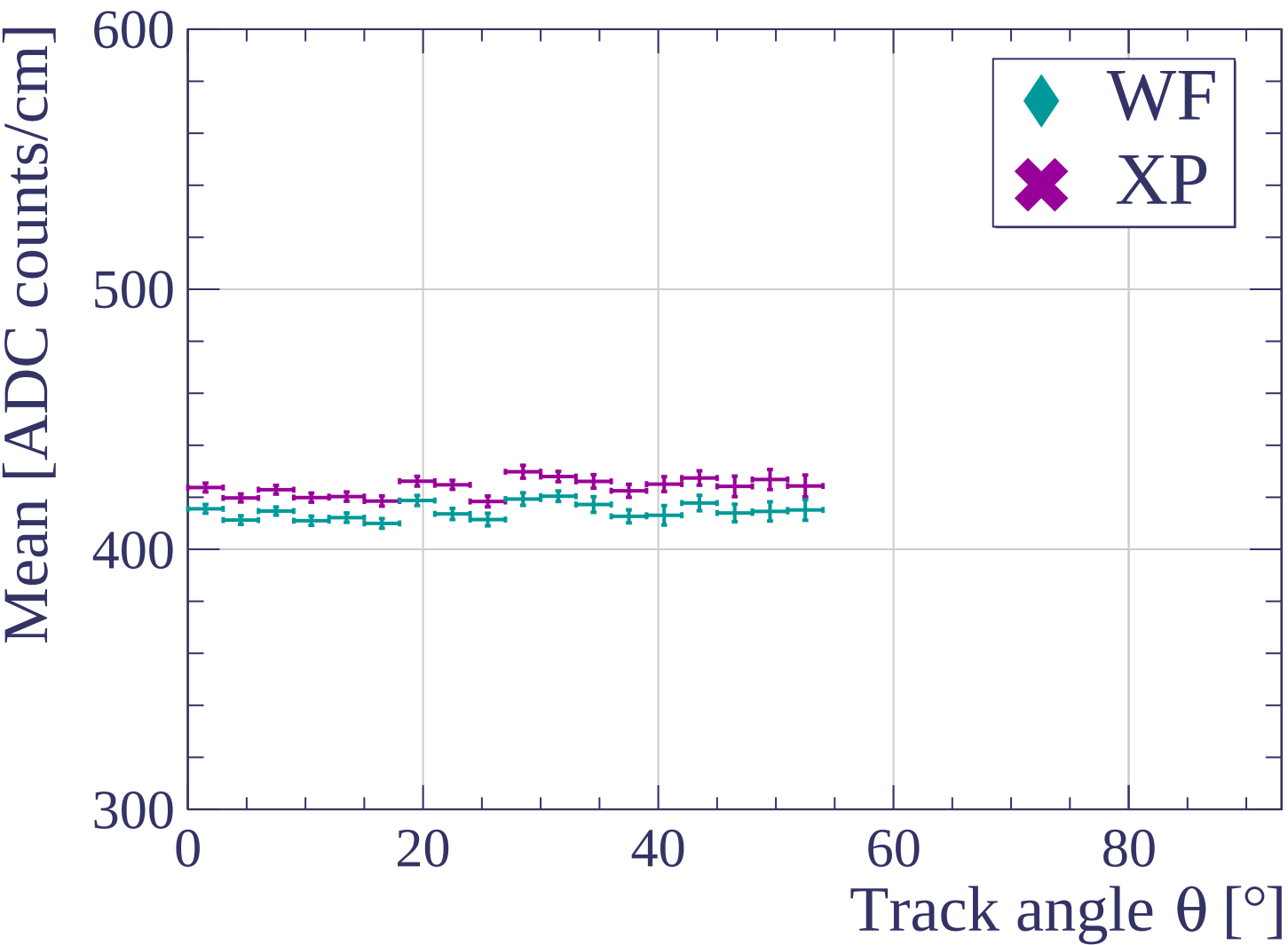


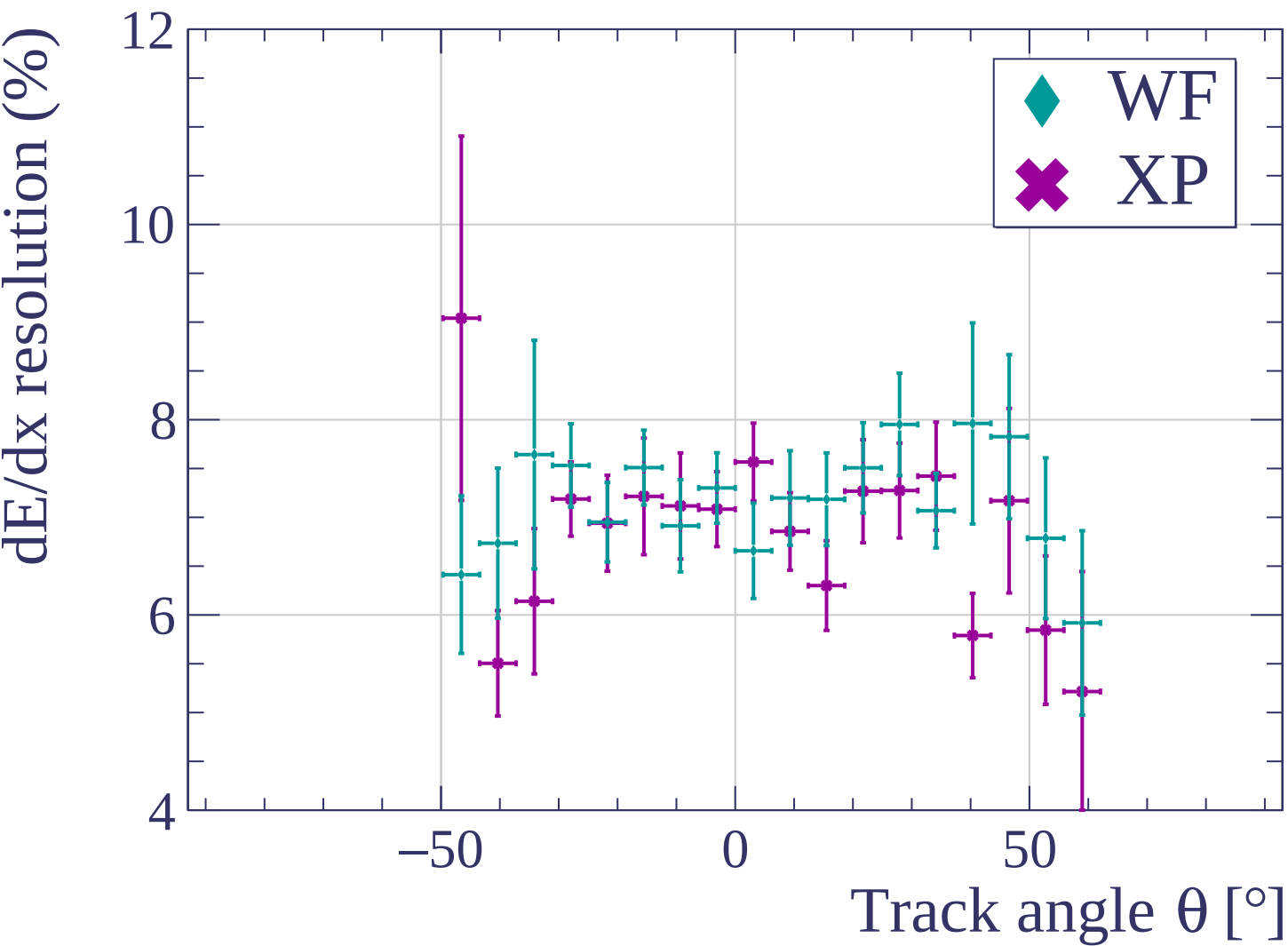
Count

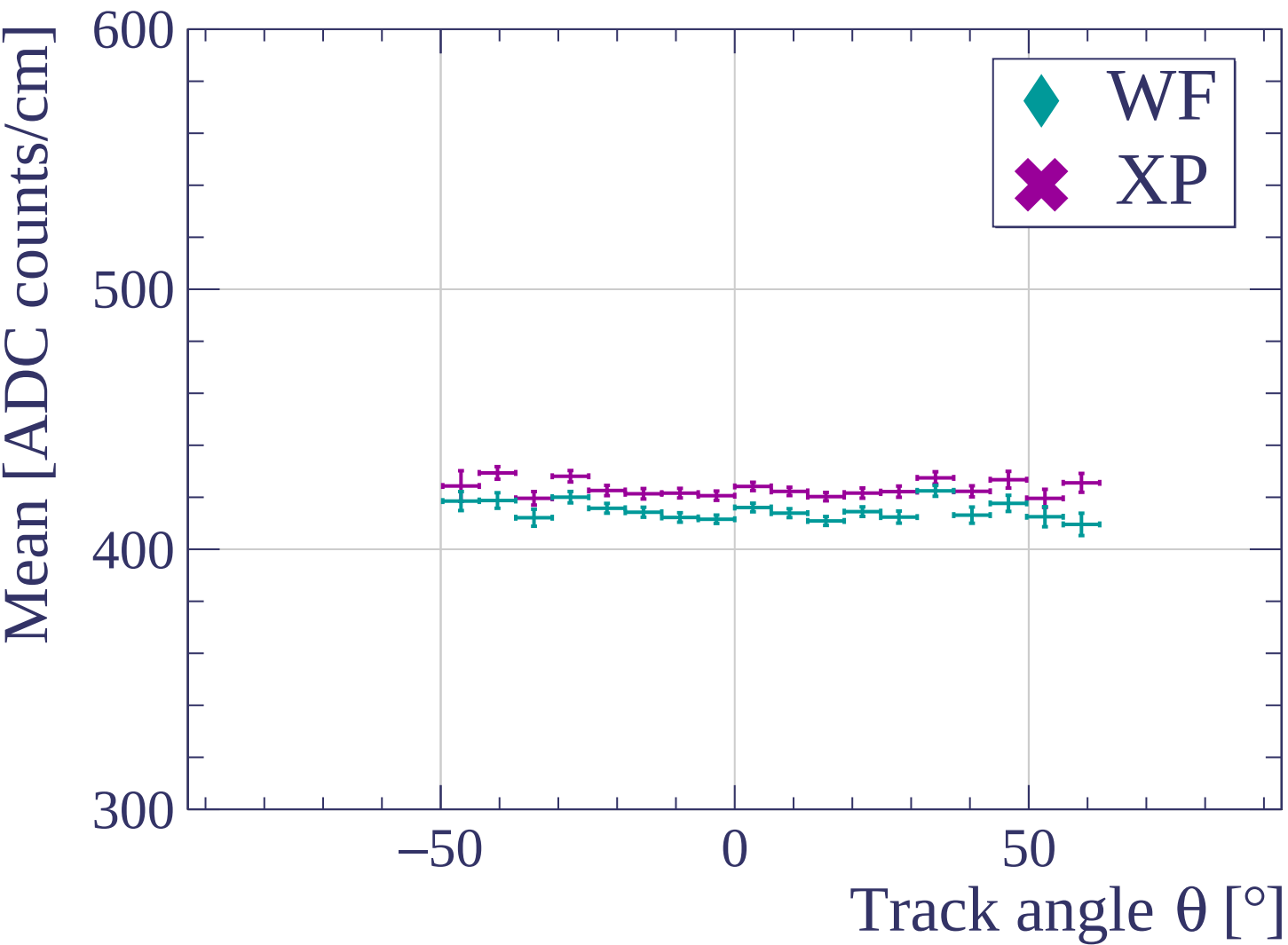


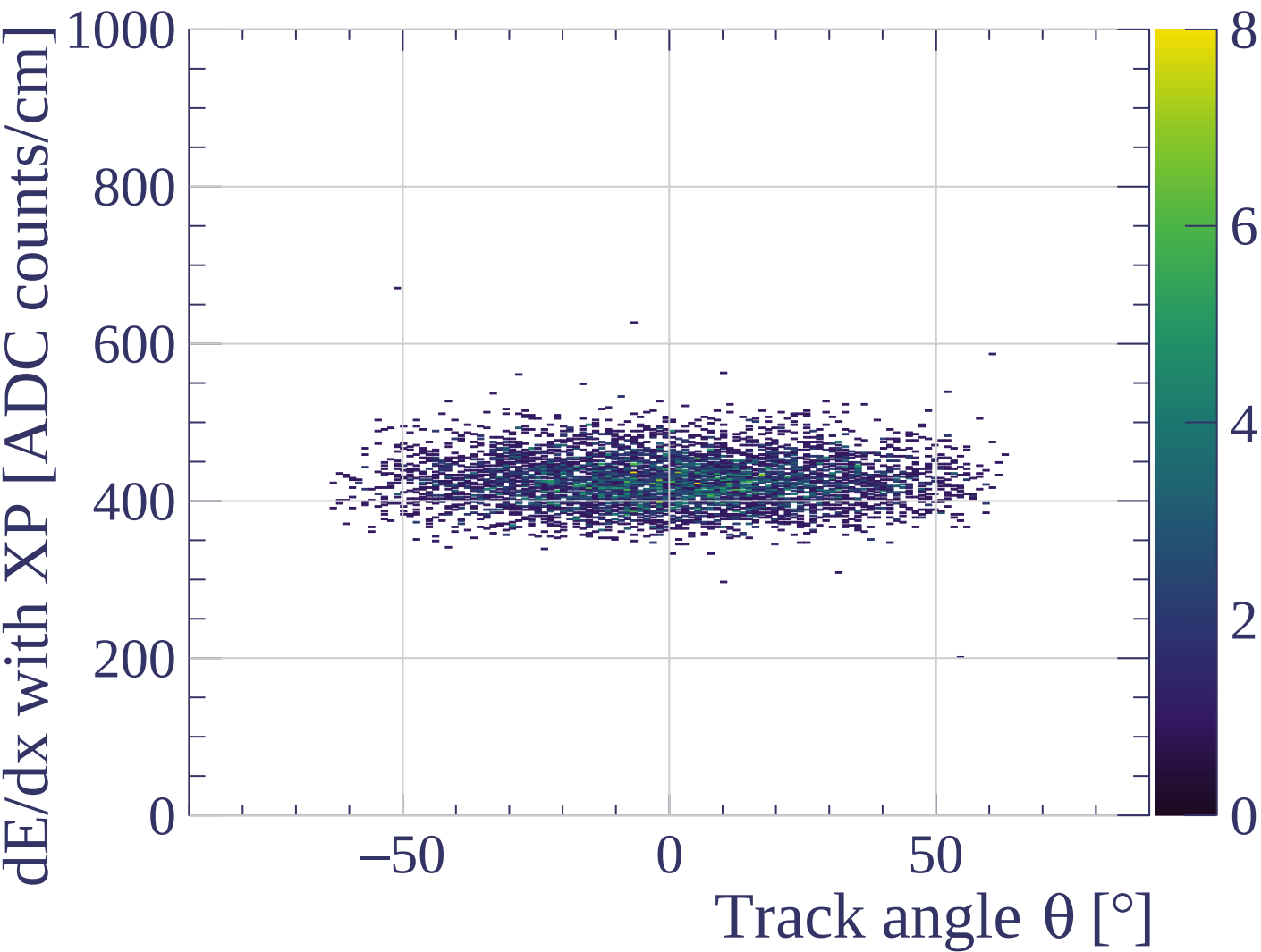
Track angle ϕ [°] angle



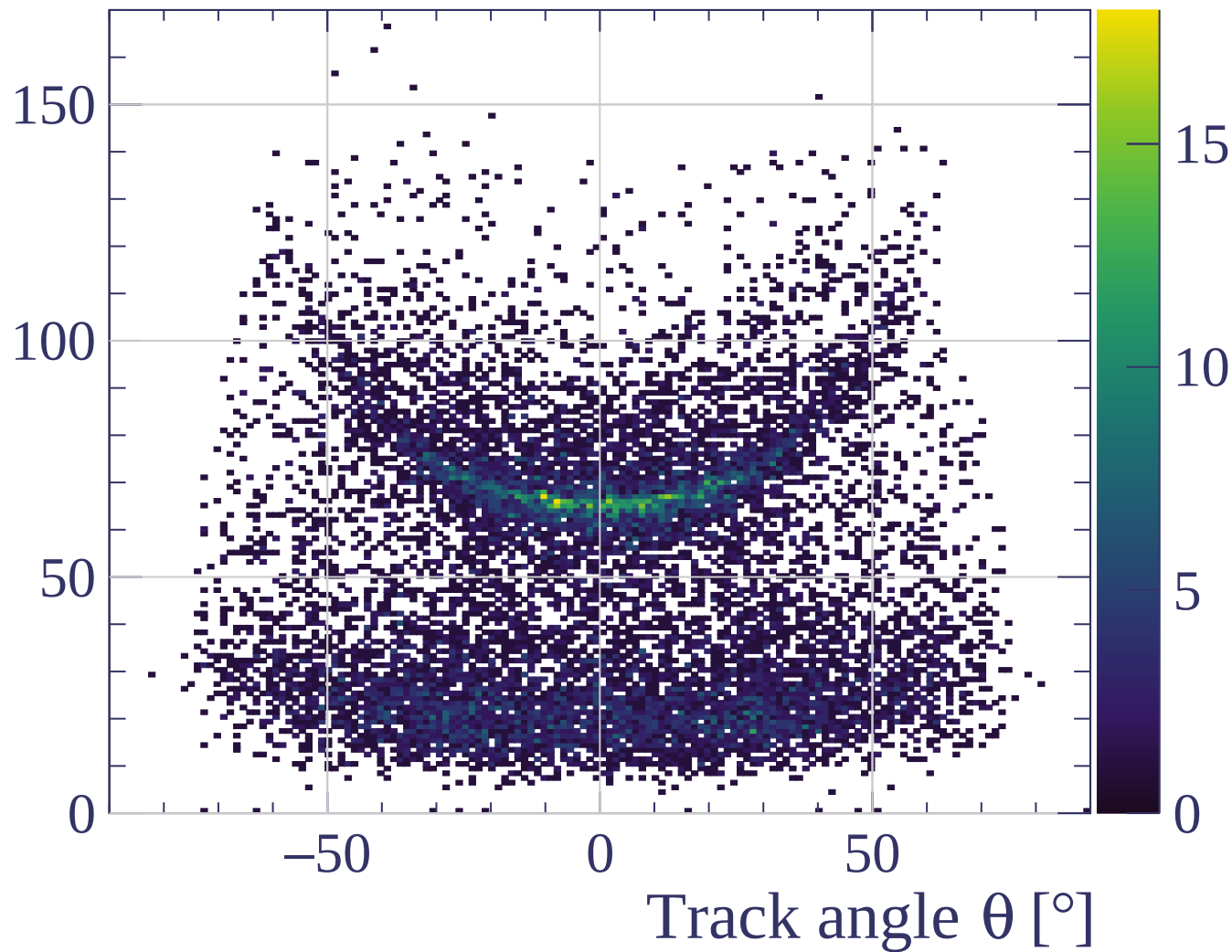


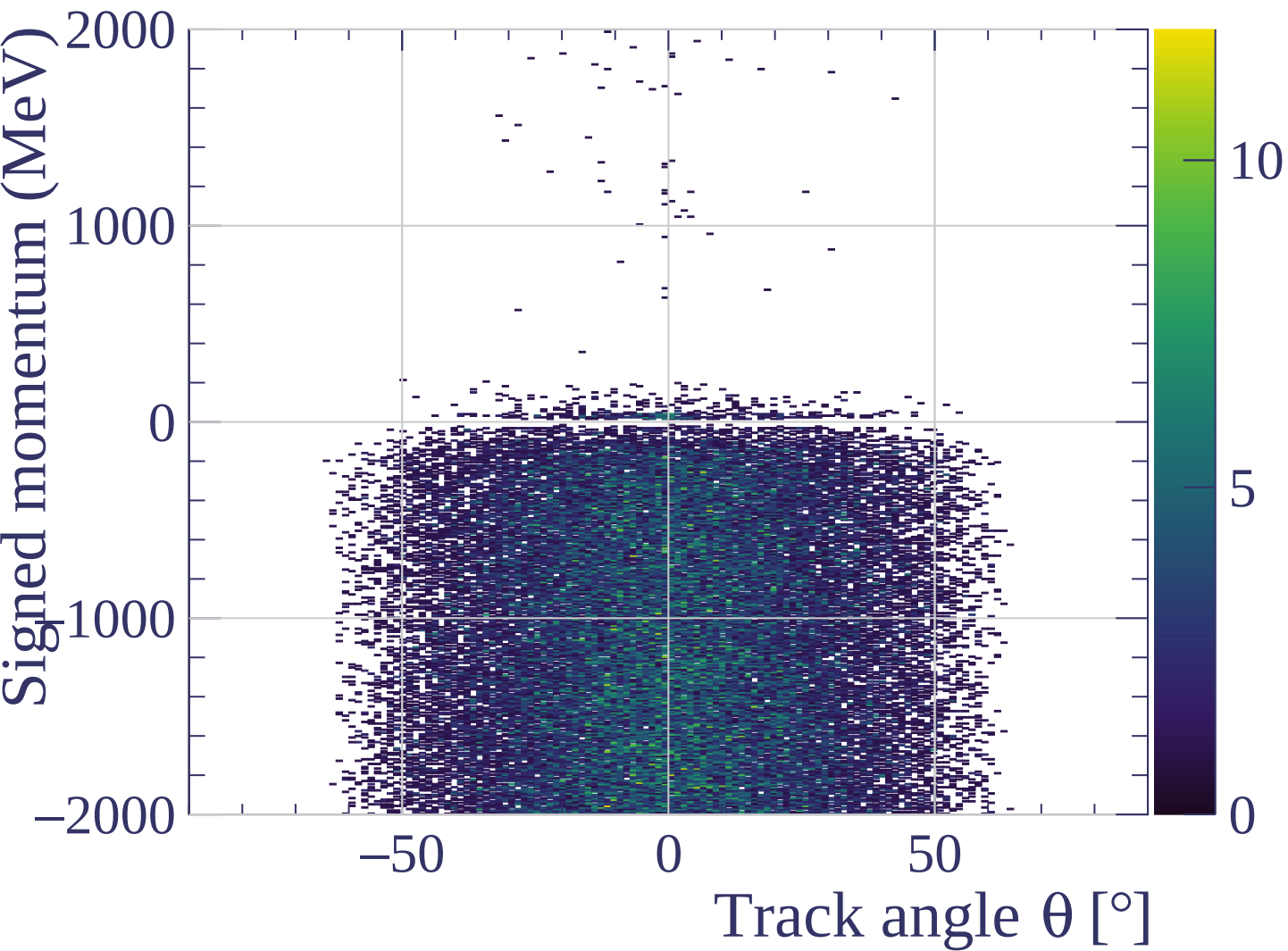




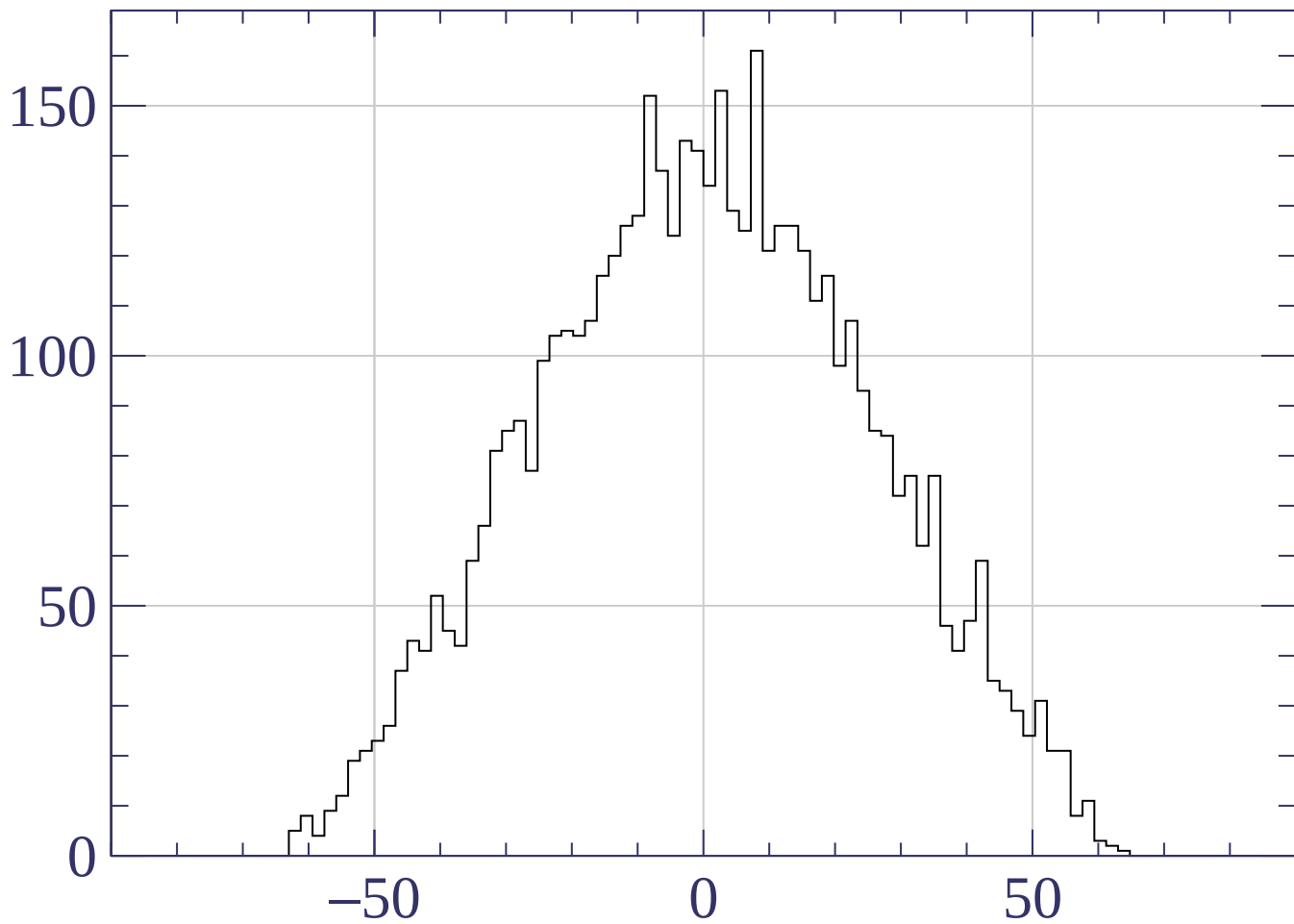


Track length [cm]



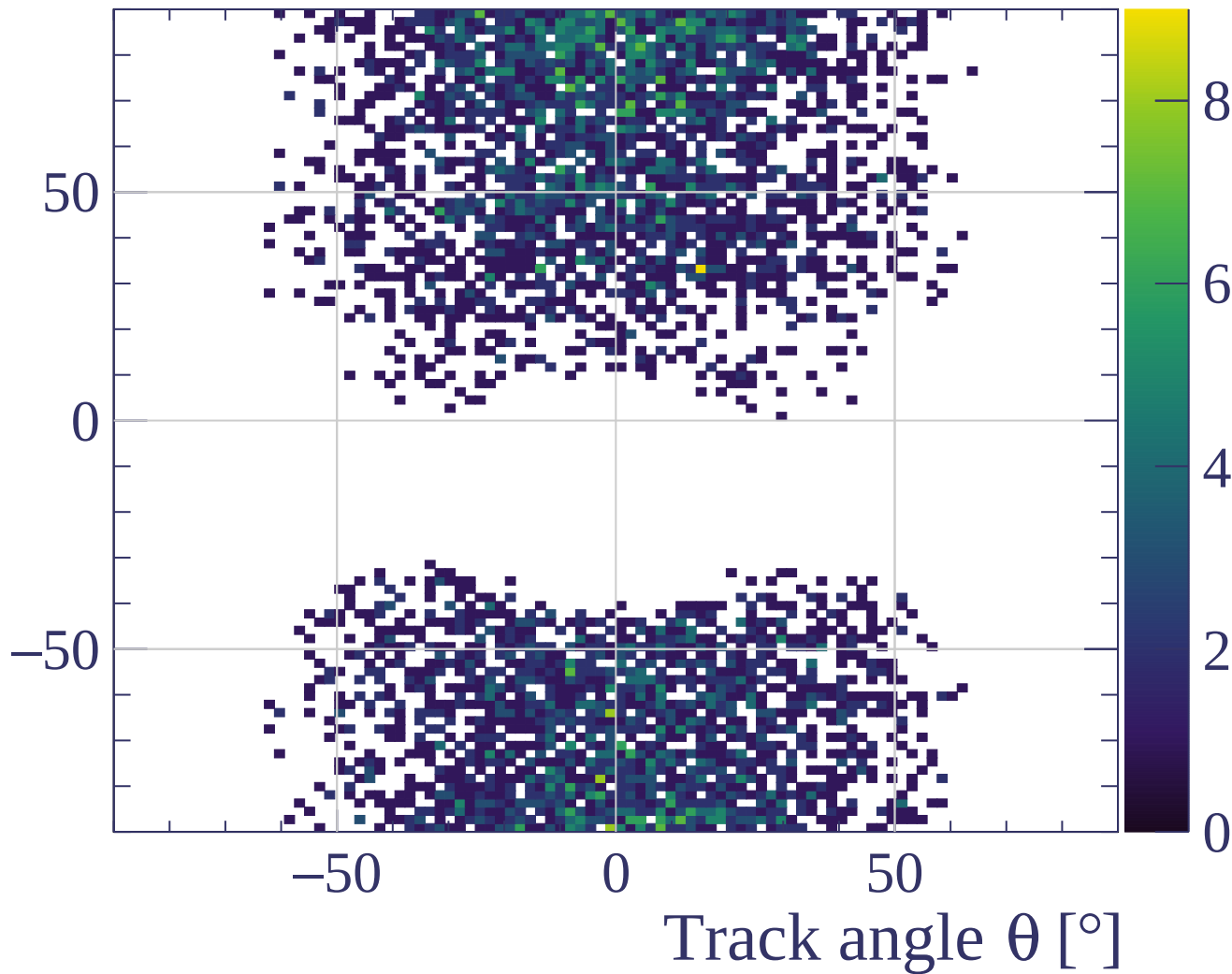


Count

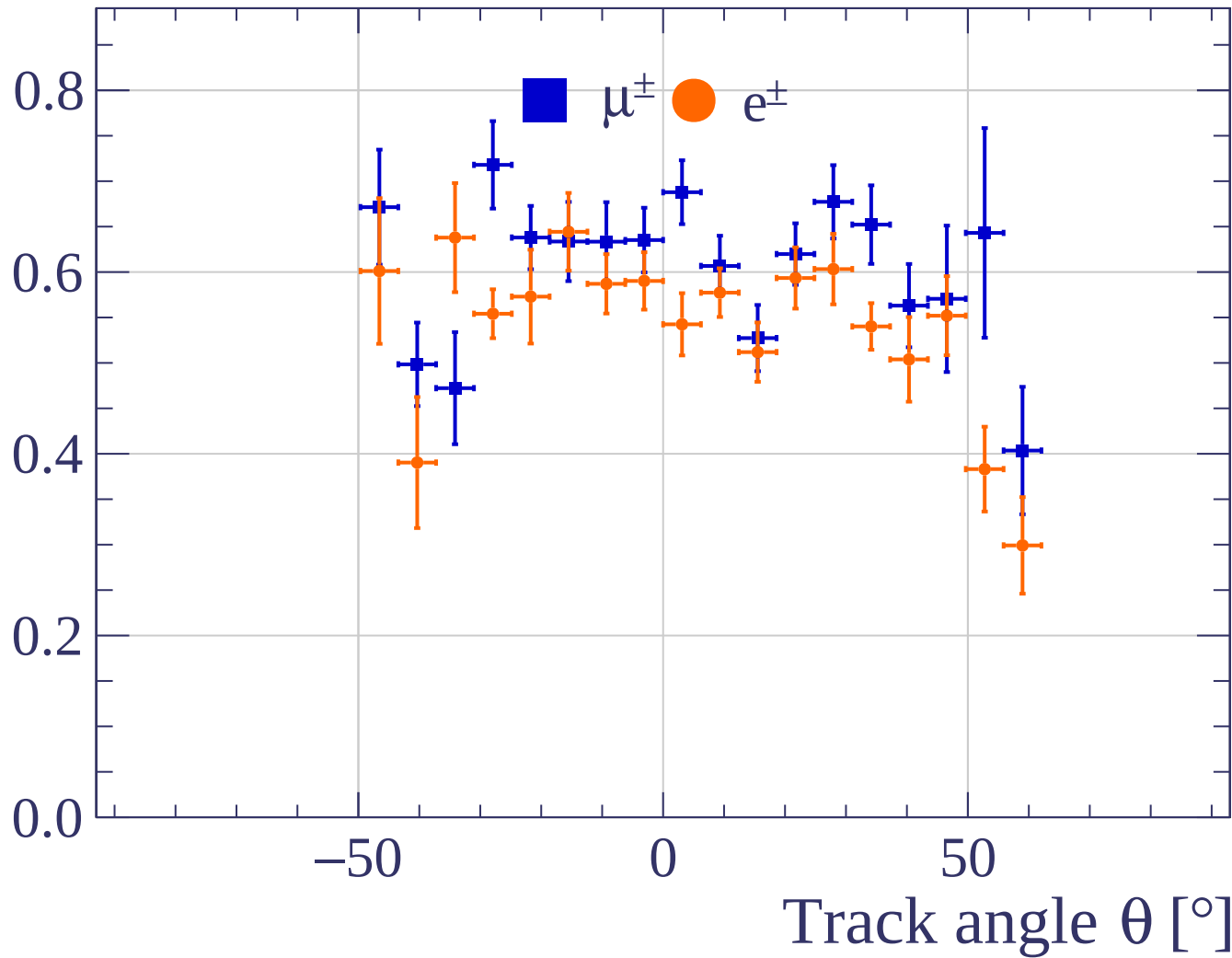


Track angle θ [°]

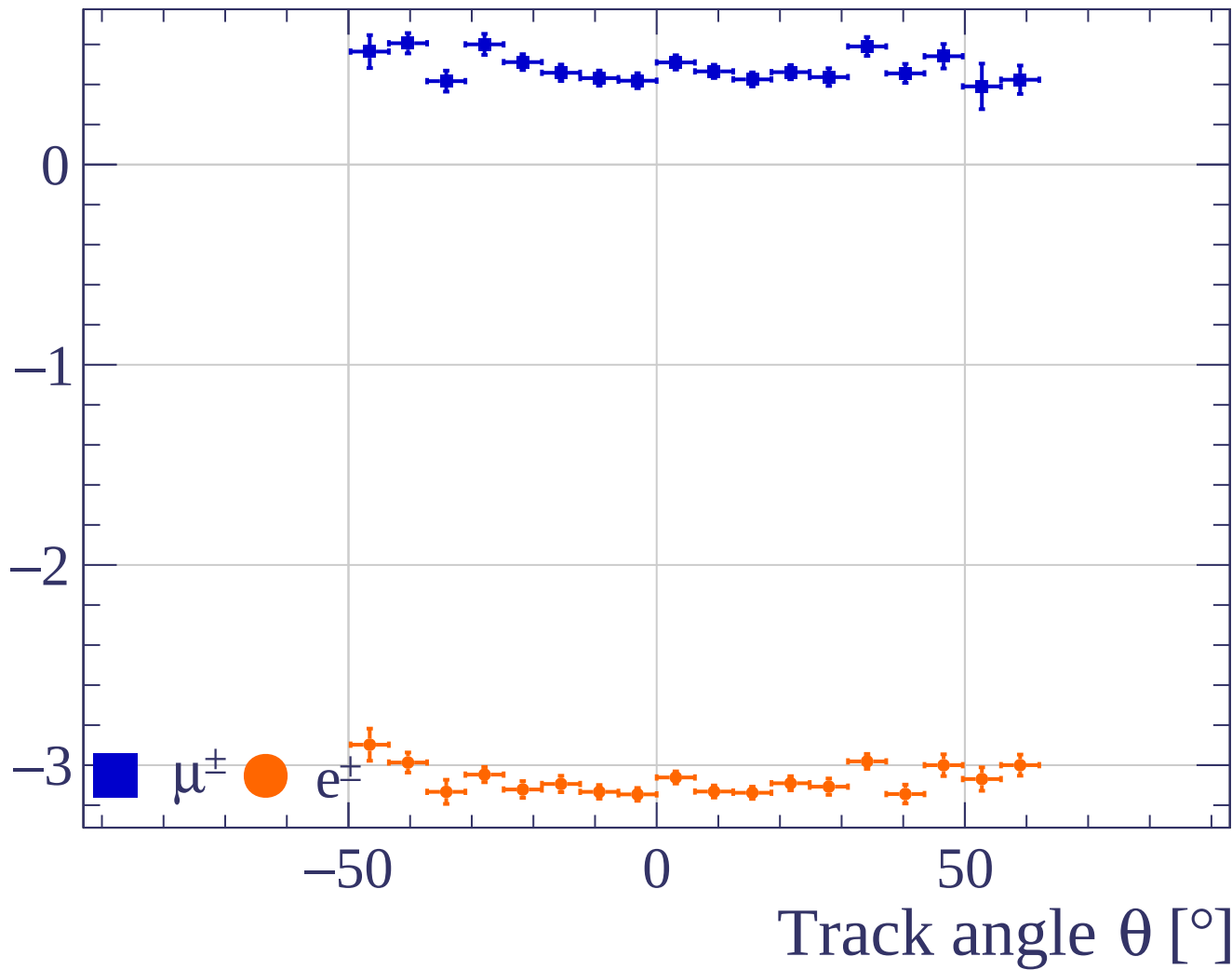
Track angle ϕ [°]



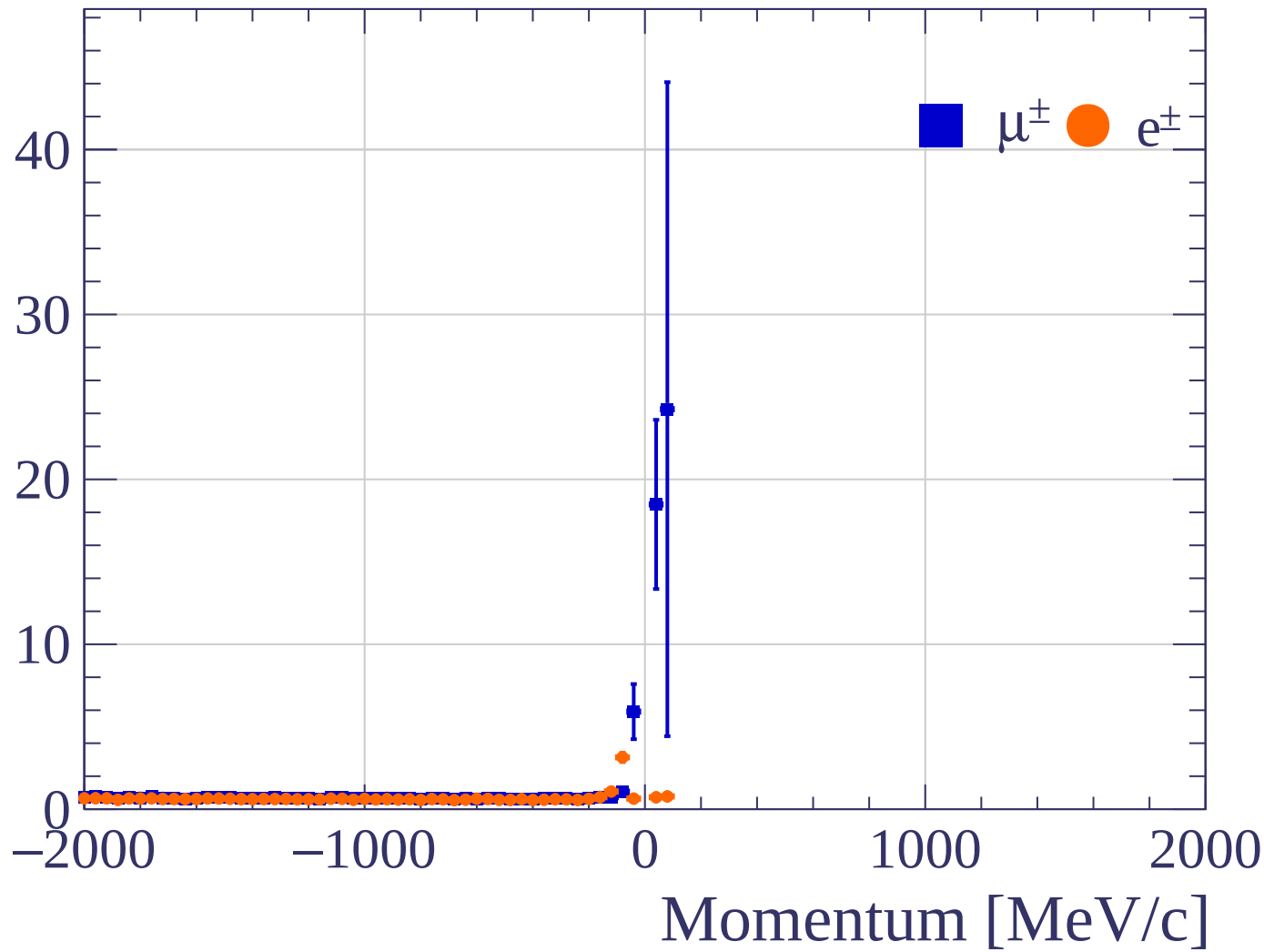
Pulls standard deviation



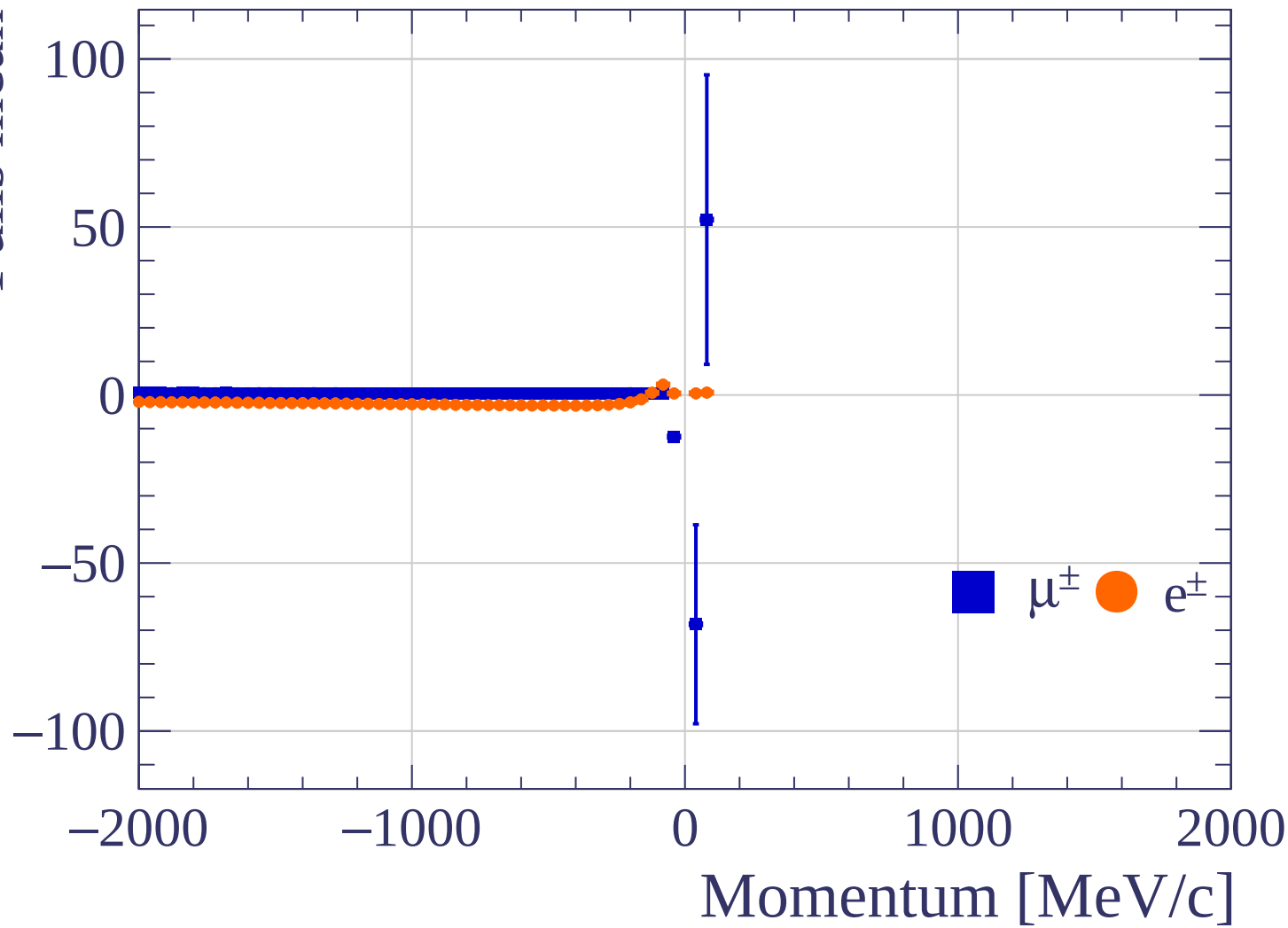
Pulls mean



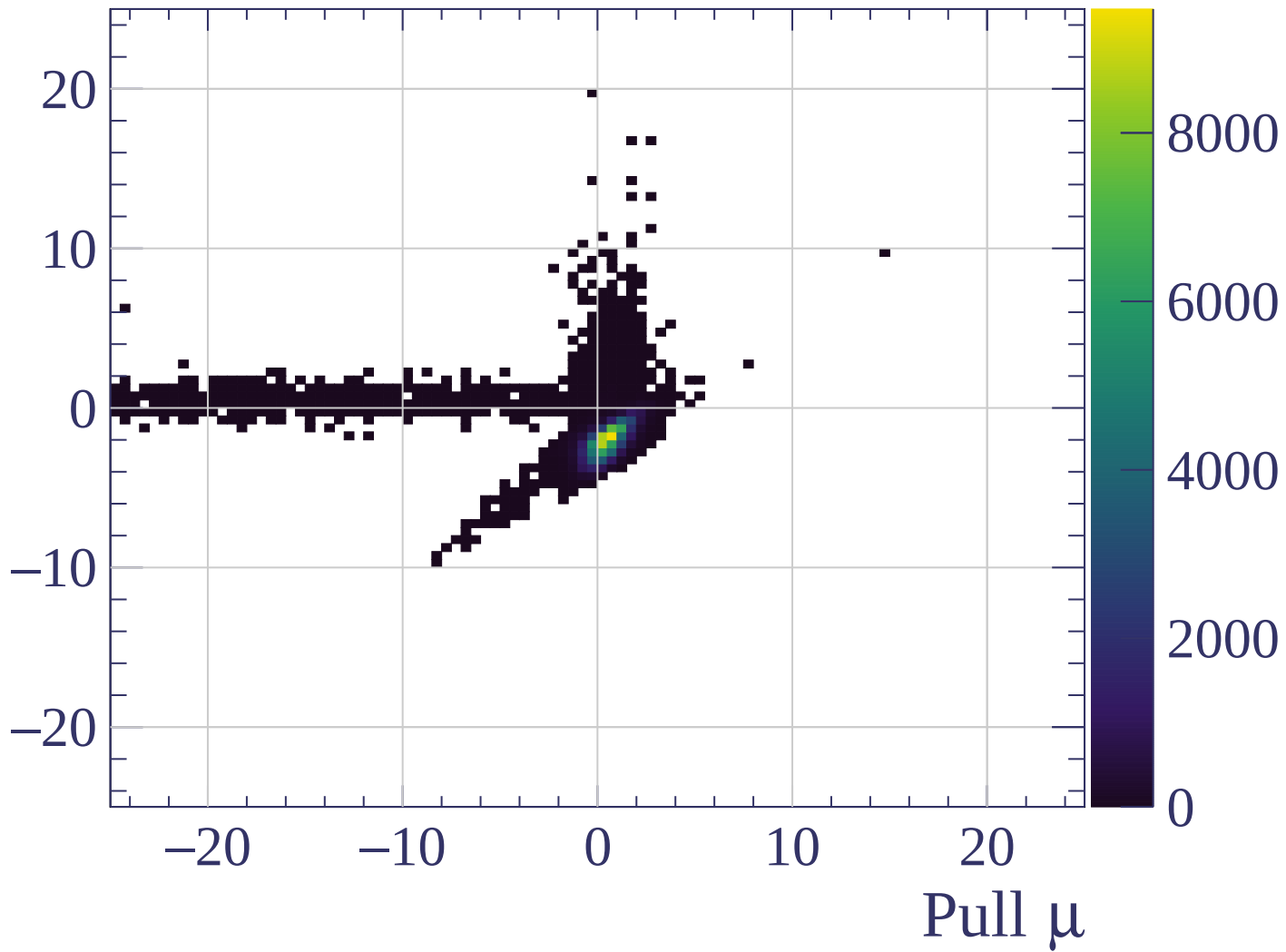
Pulls standard deviation

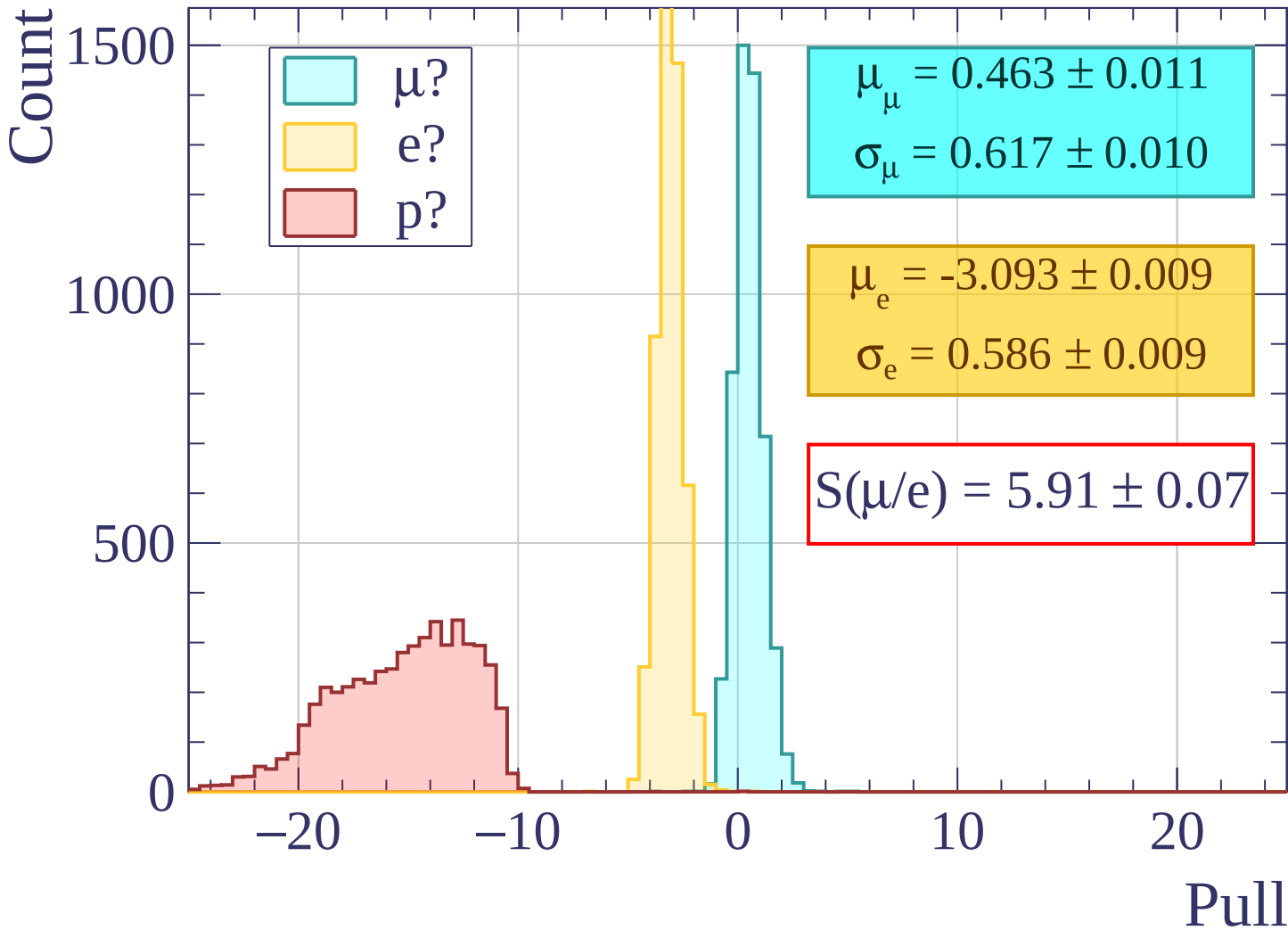


Pulls mean

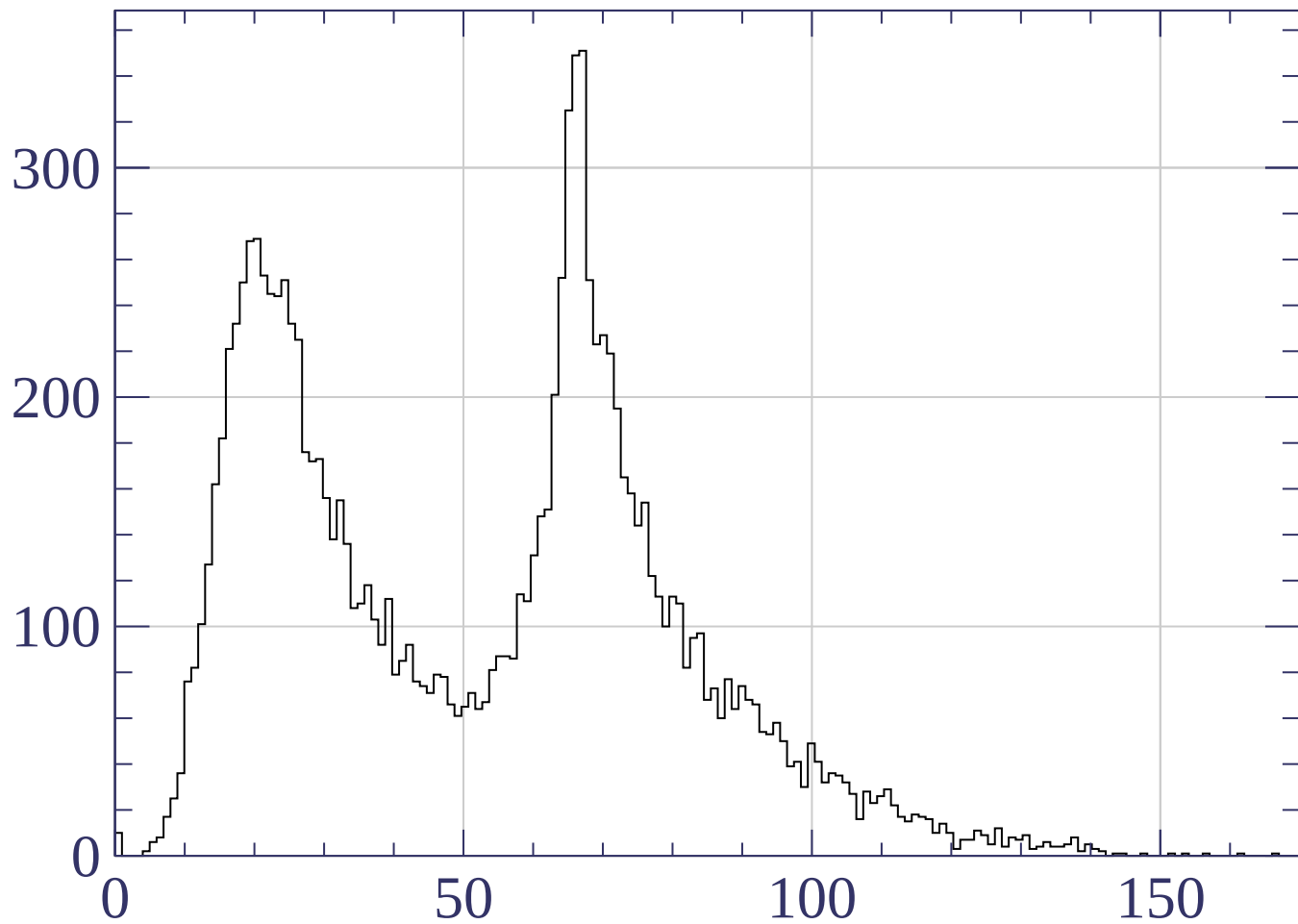


Pull e



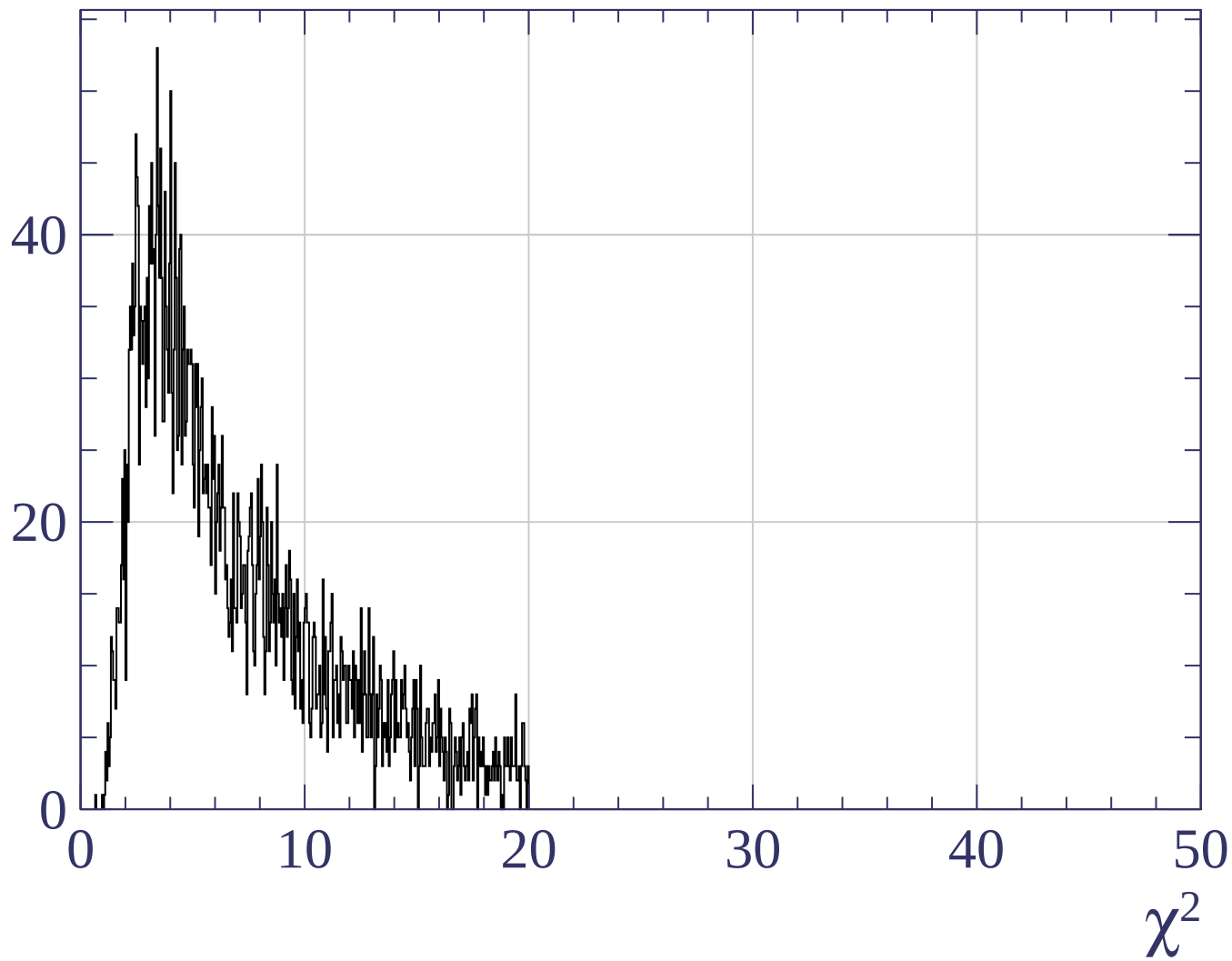


Count

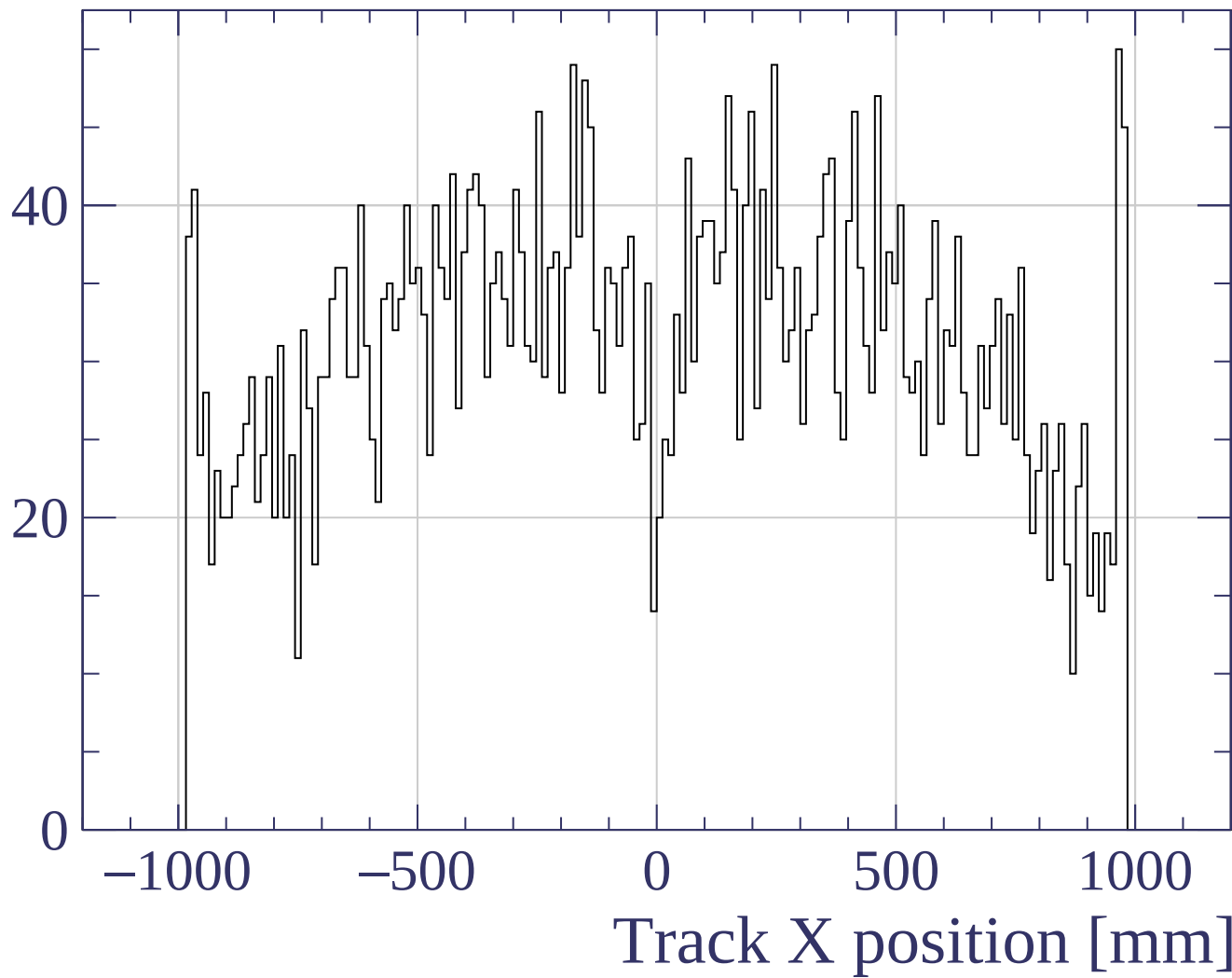


Track length [cm]

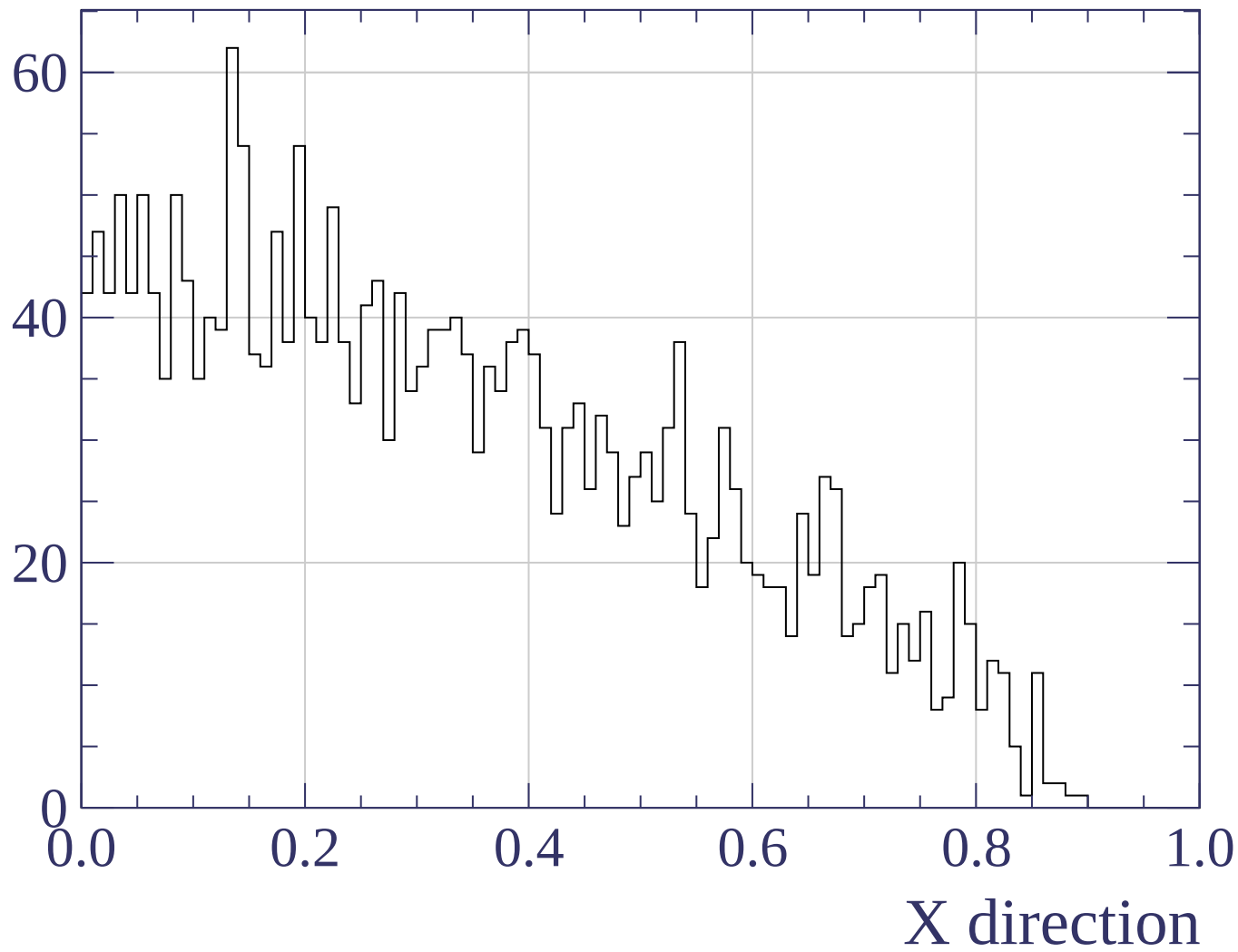
Count



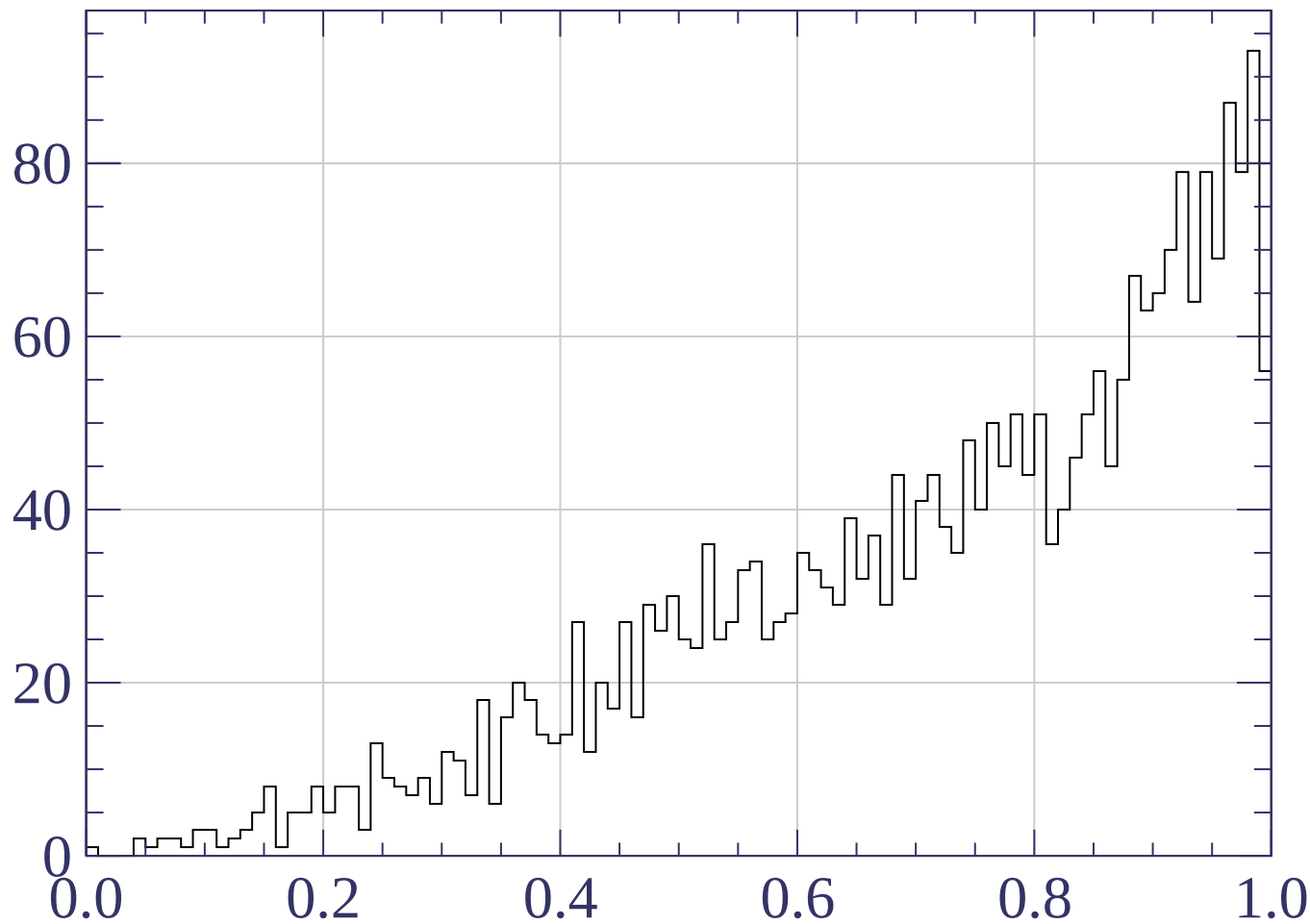
Count



Counts

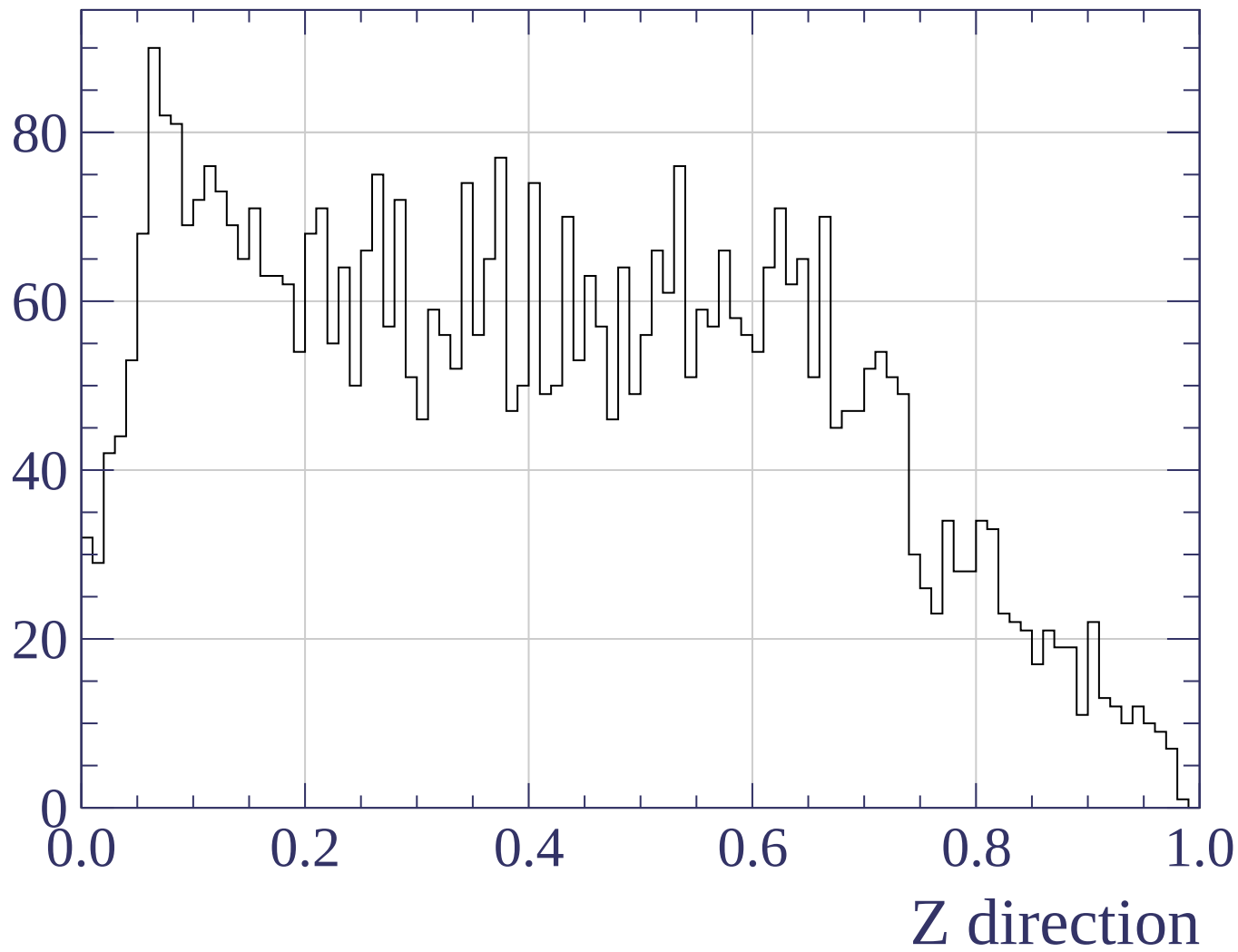


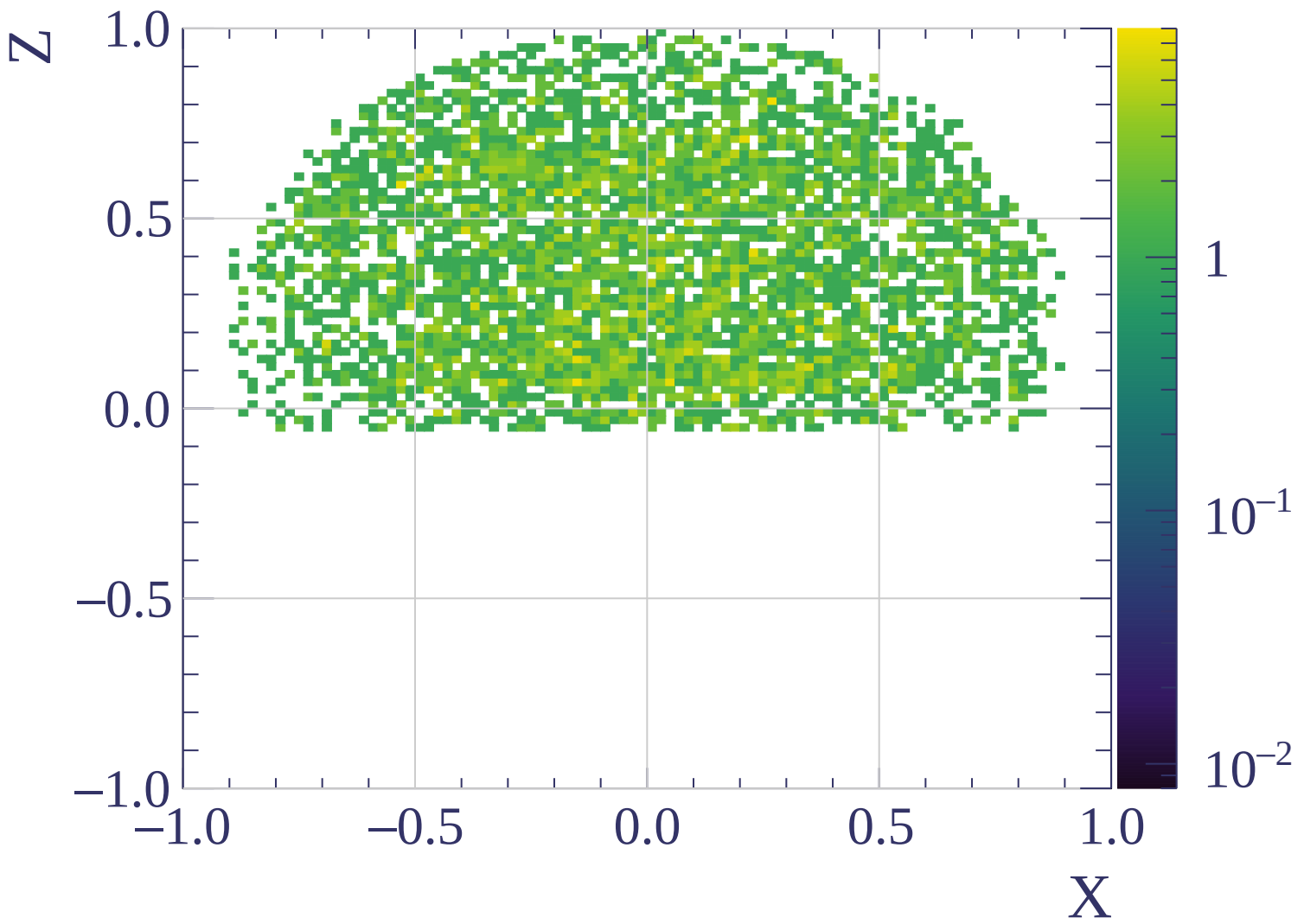
Counts

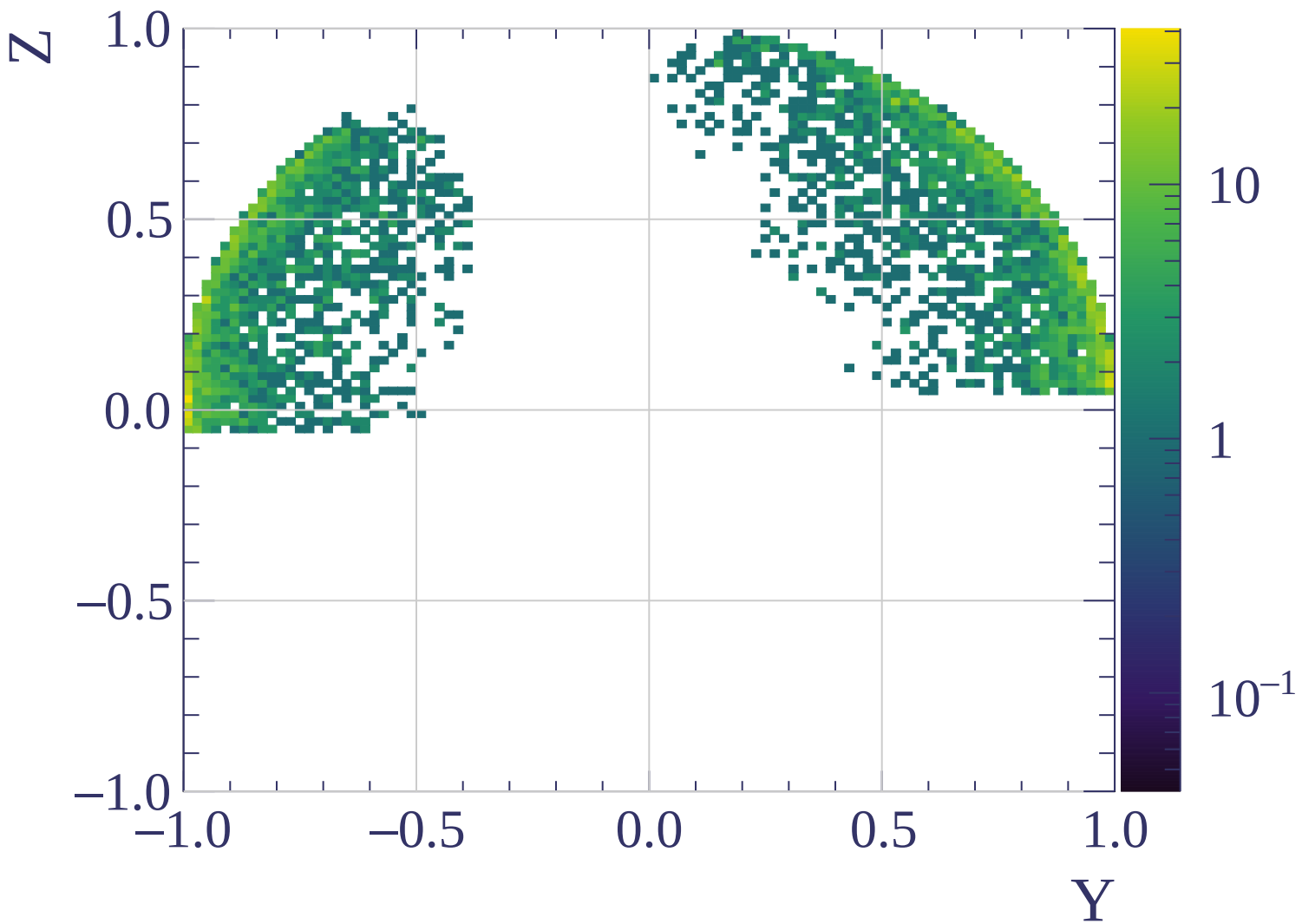


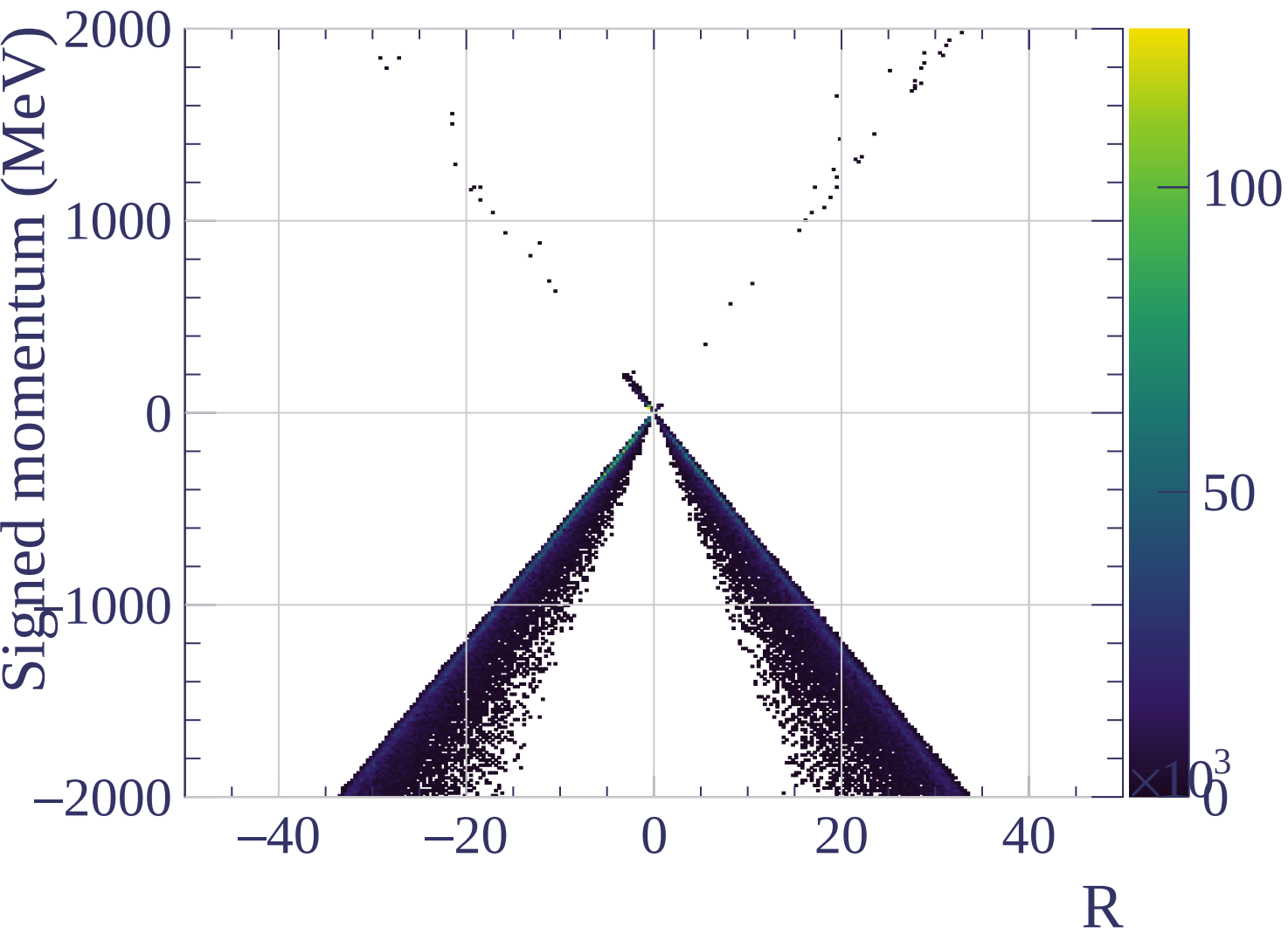
Y direction

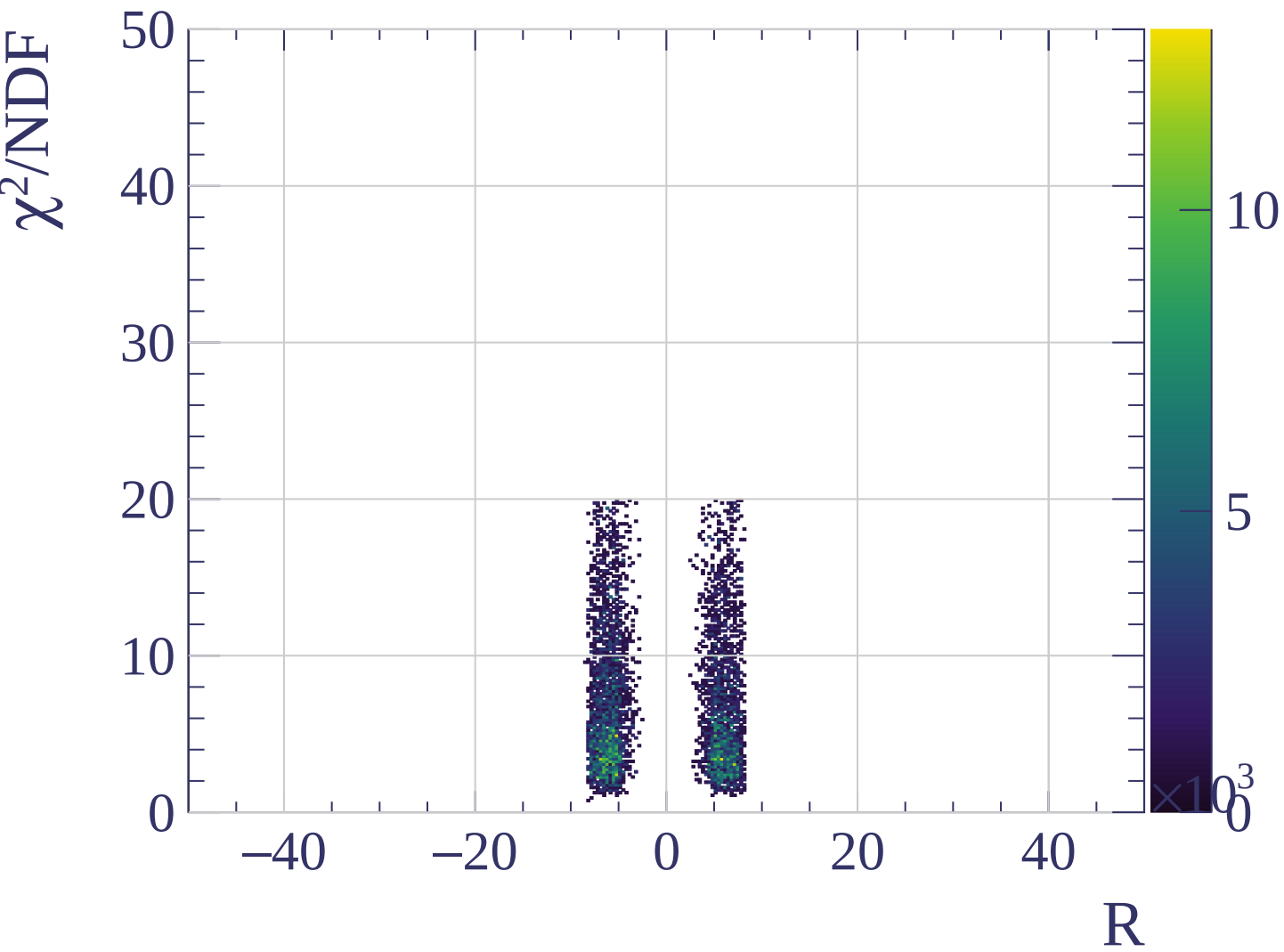
Counts











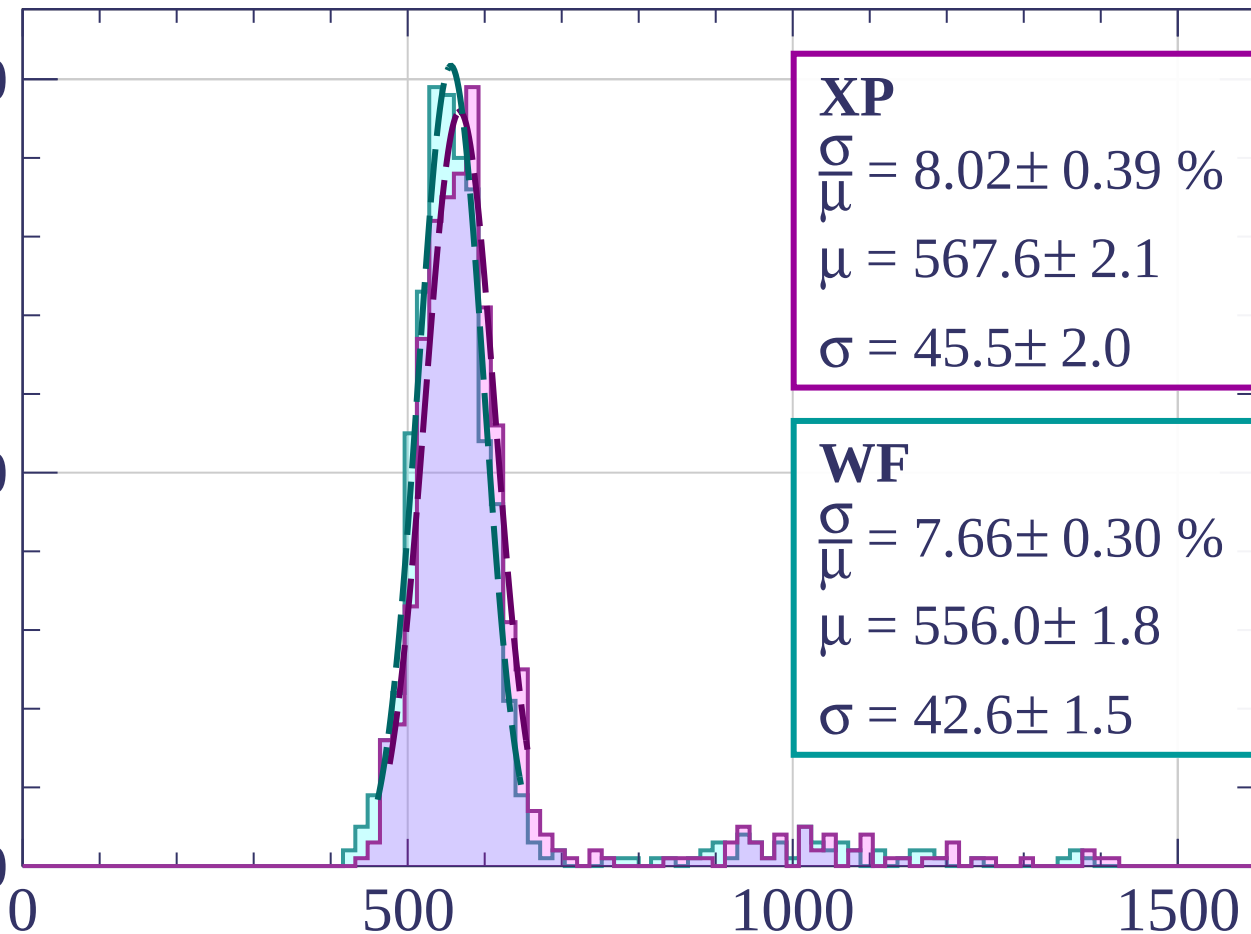
Energy loss | $0 < p < 80$

Count

100

50

0



XP

$$\frac{\sigma}{\mu} = 8.02 \pm 0.39 \%$$

$$\mu = 567.6 \pm 2.1$$

$$\sigma = 45.5 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 7.66 \pm 0.30 \%$$

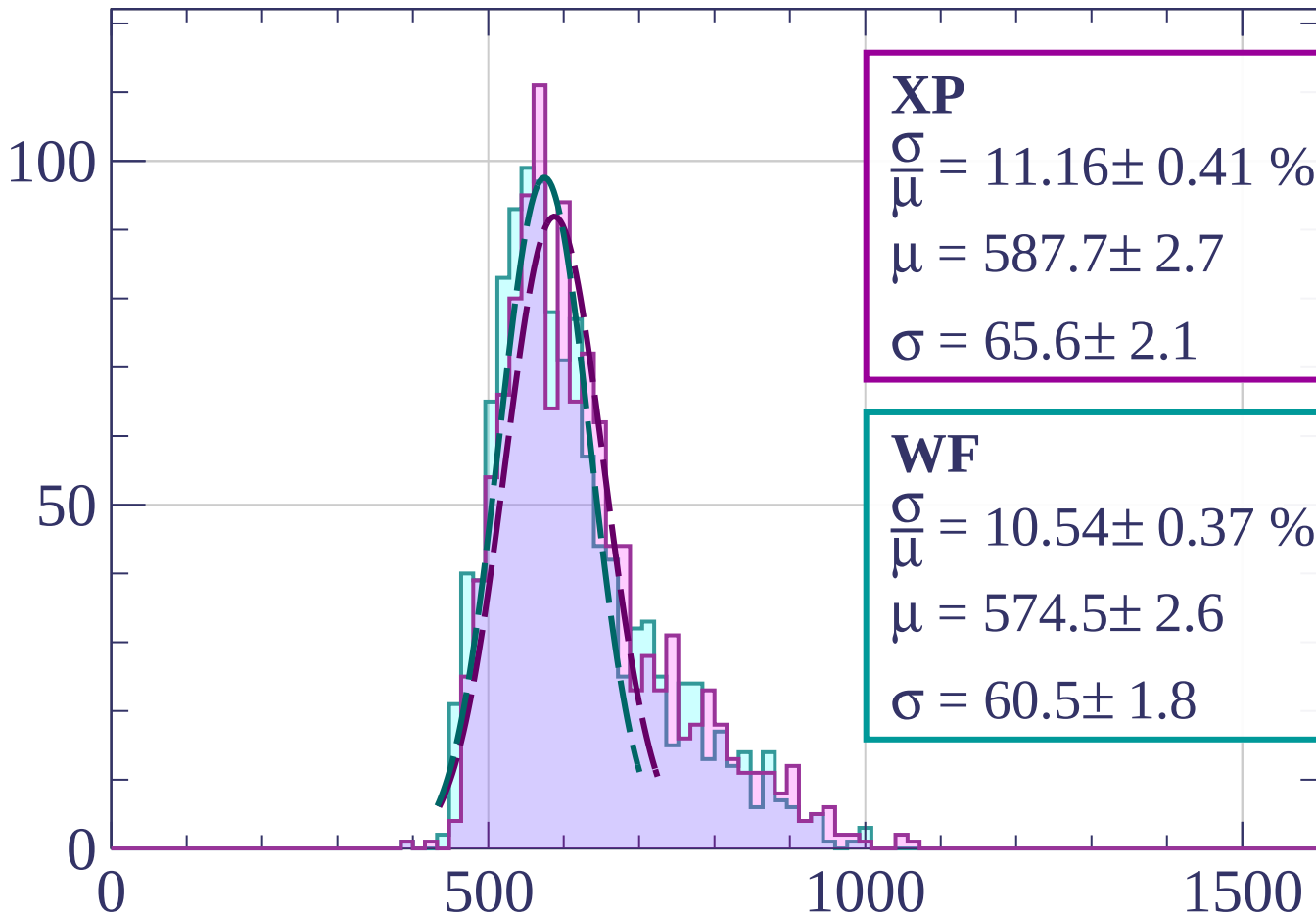
$$\mu = 556.0 \pm 1.8$$

$$\sigma = 42.6 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $80 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 11.16 \pm 0.41 \%$$

$$\mu = 587.7 \pm 2.7$$

$$\sigma = 65.6 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 10.54 \pm 0.37 \%$$

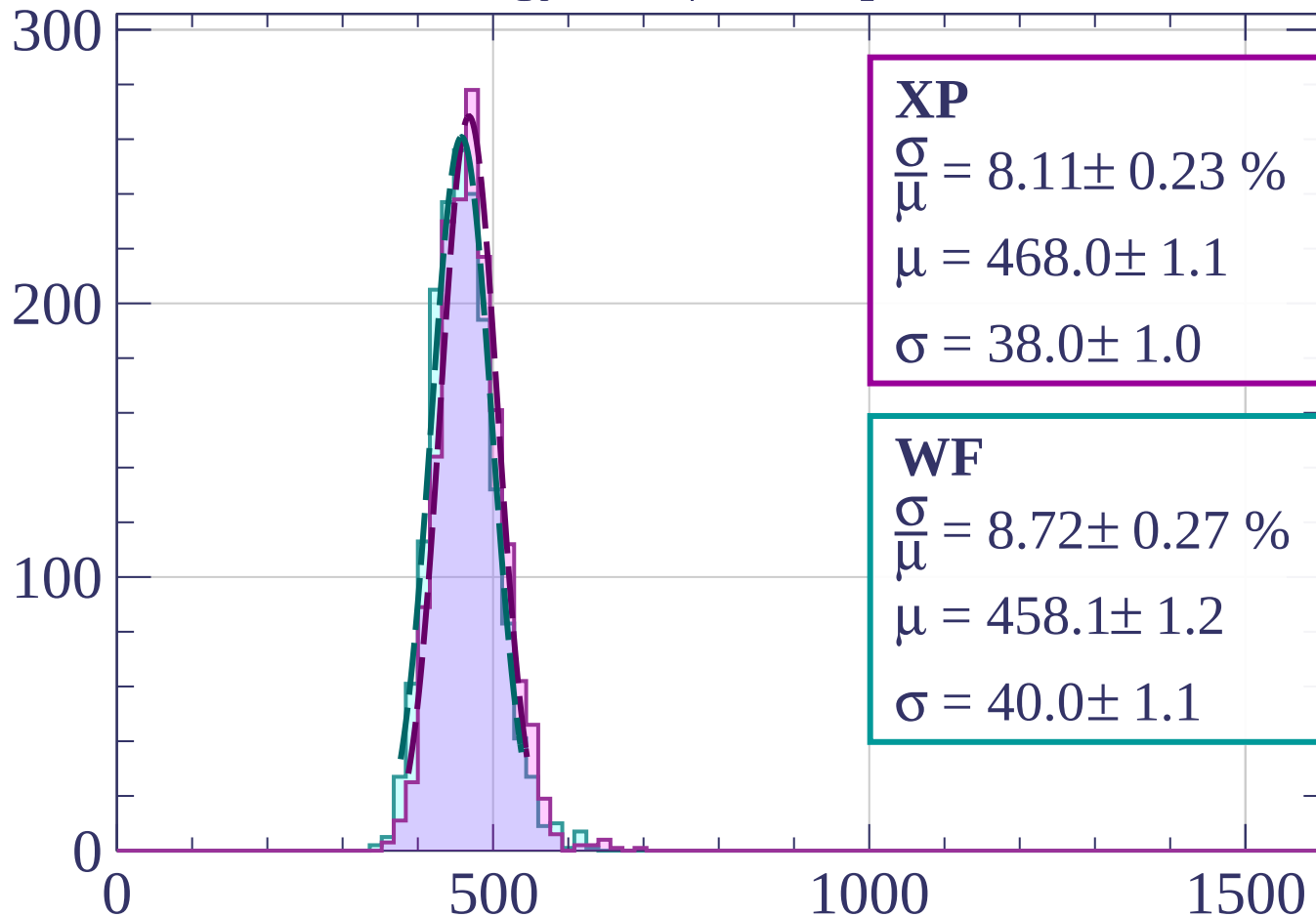
$$\mu = 574.5 \pm 2.6$$

$$\sigma = 60.5 \pm 1.8$$

dE/dx [ADC counts/cm]

Energy loss | $160 < p < 240$

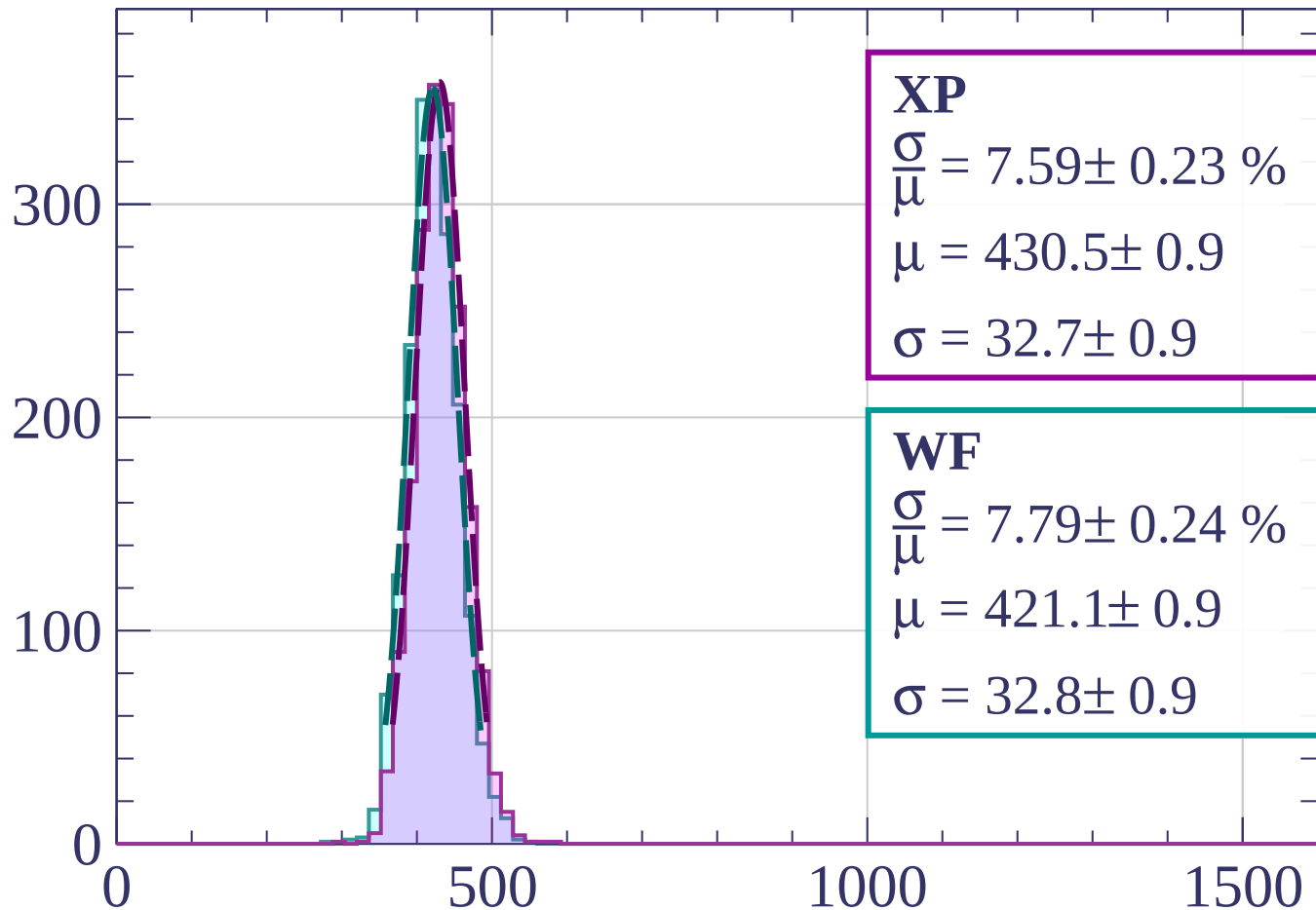
Count



dE/dx [ADC counts/cm]

Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 7.59 \pm 0.23 \%$$

$$\mu = 430.5 \pm 0.9$$

$$\sigma = 32.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.79 \pm 0.24 \%$$

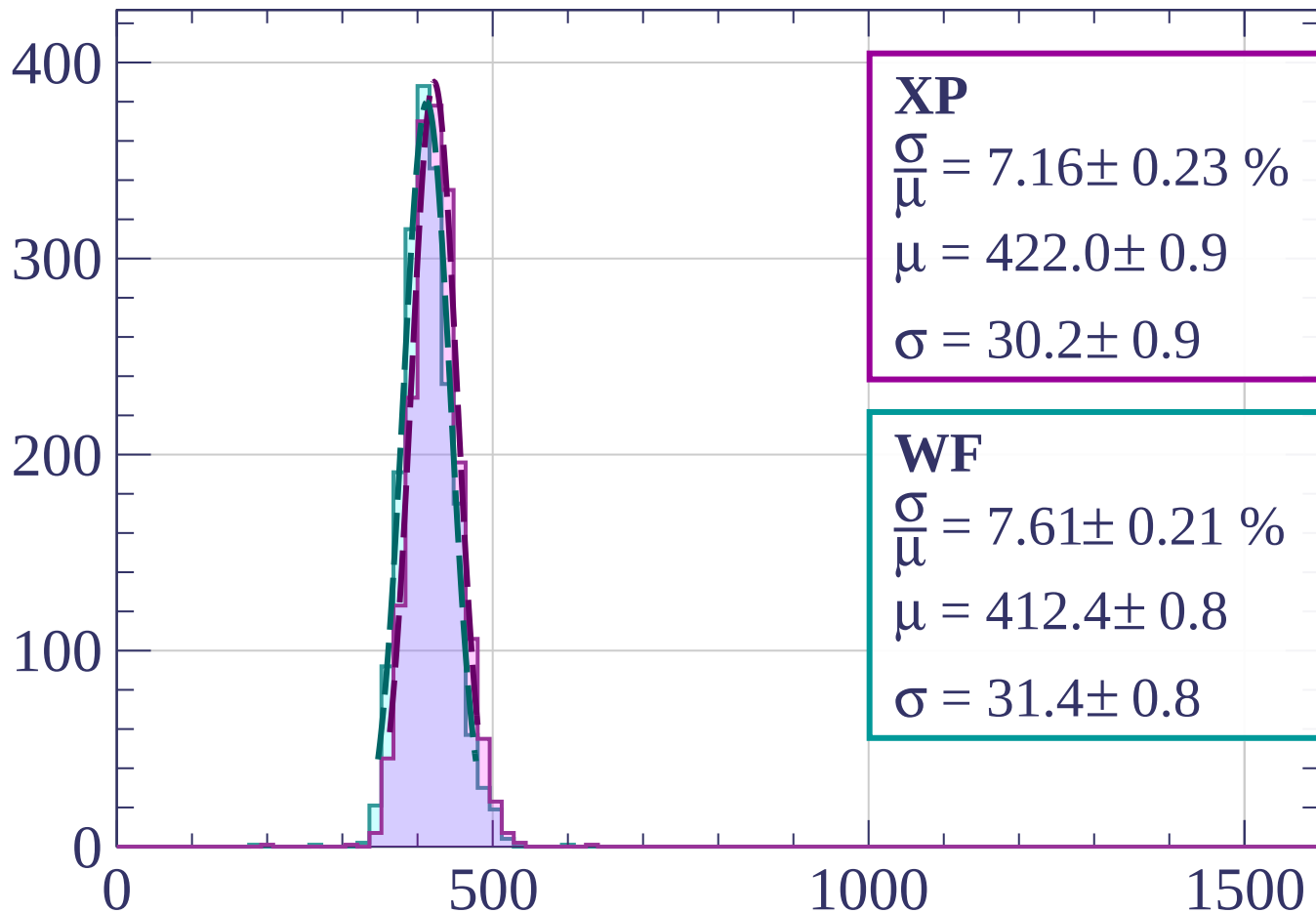
$$\mu = 421.1 \pm 0.9$$

$$\sigma = 32.8 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $320 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 7.16 \pm 0.23 \%$$

$$\mu = 422.0 \pm 0.9$$

$$\sigma = 30.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.61 \pm 0.21 \%$$

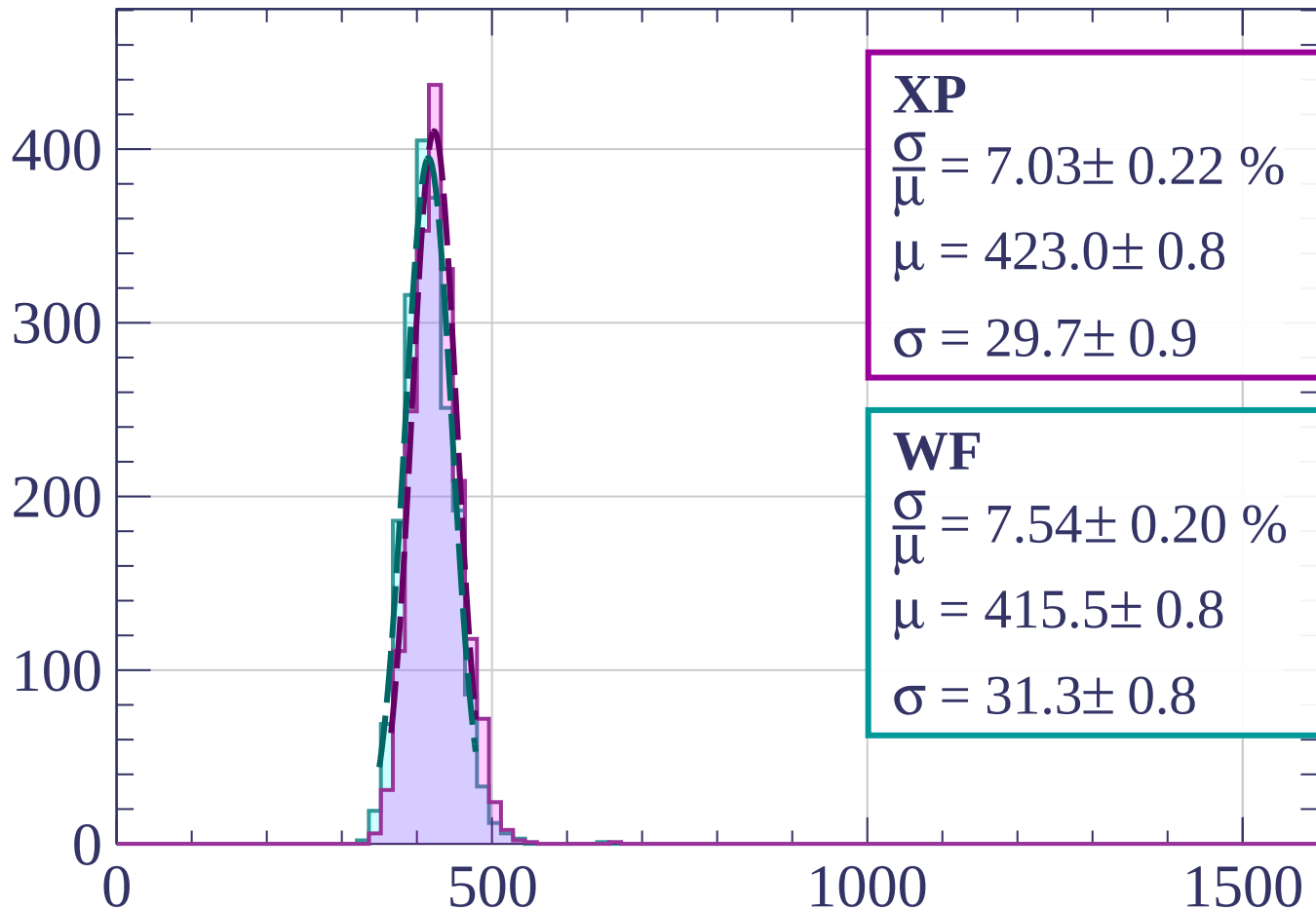
$$\mu = 412.4 \pm 0.8$$

$$\sigma = 31.4 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 7.03 \pm 0.22 \%$$

$$\mu = 423.0 \pm 0.8$$

$$\sigma = 29.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.54 \pm 0.20 \%$$

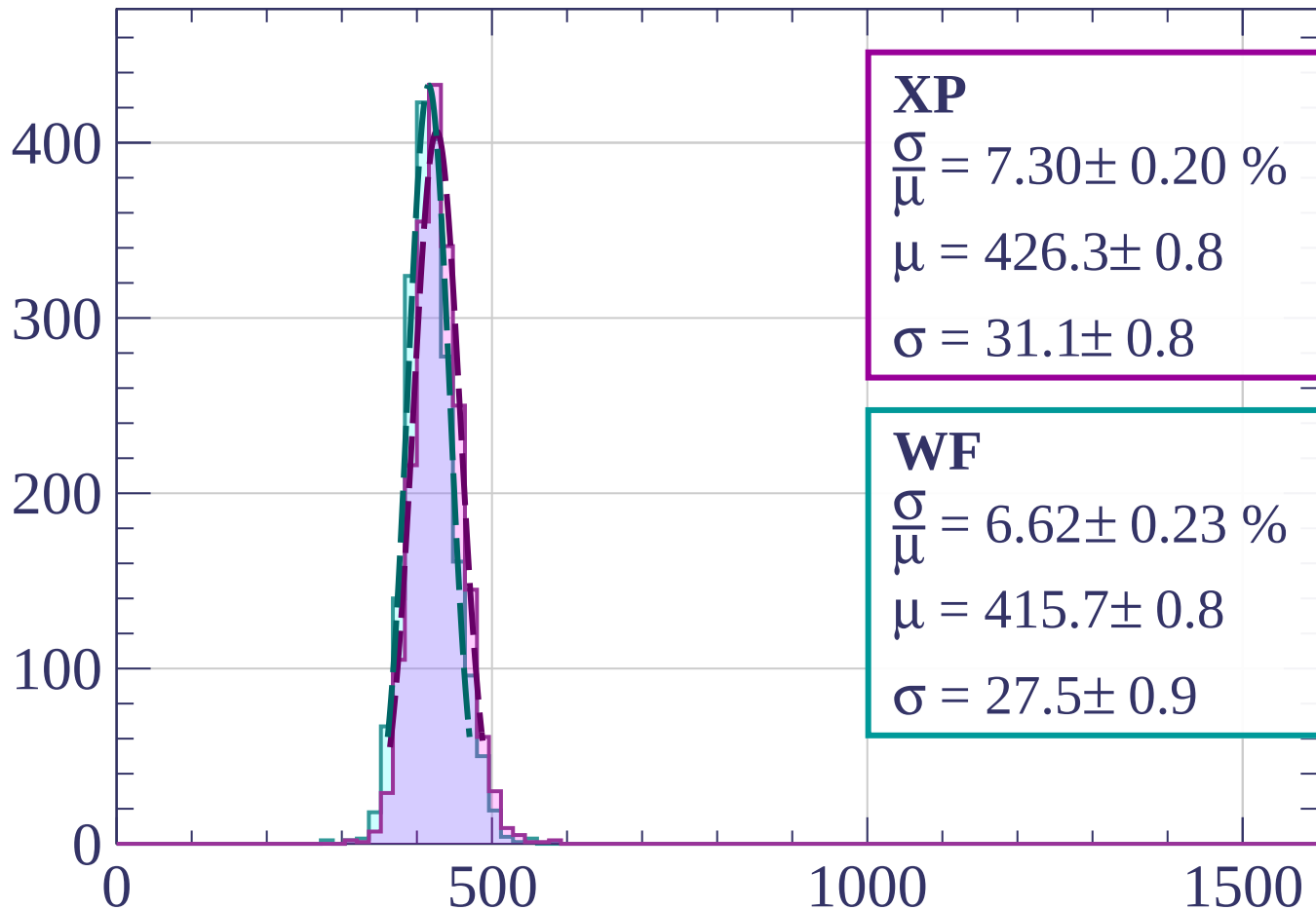
$$\mu = 415.5 \pm 0.8$$

$$\sigma = 31.3 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 7.30 \pm 0.20 \%$$

$$\mu = 426.3 \pm 0.8$$

$$\sigma = 31.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.62 \pm 0.23 \%$$

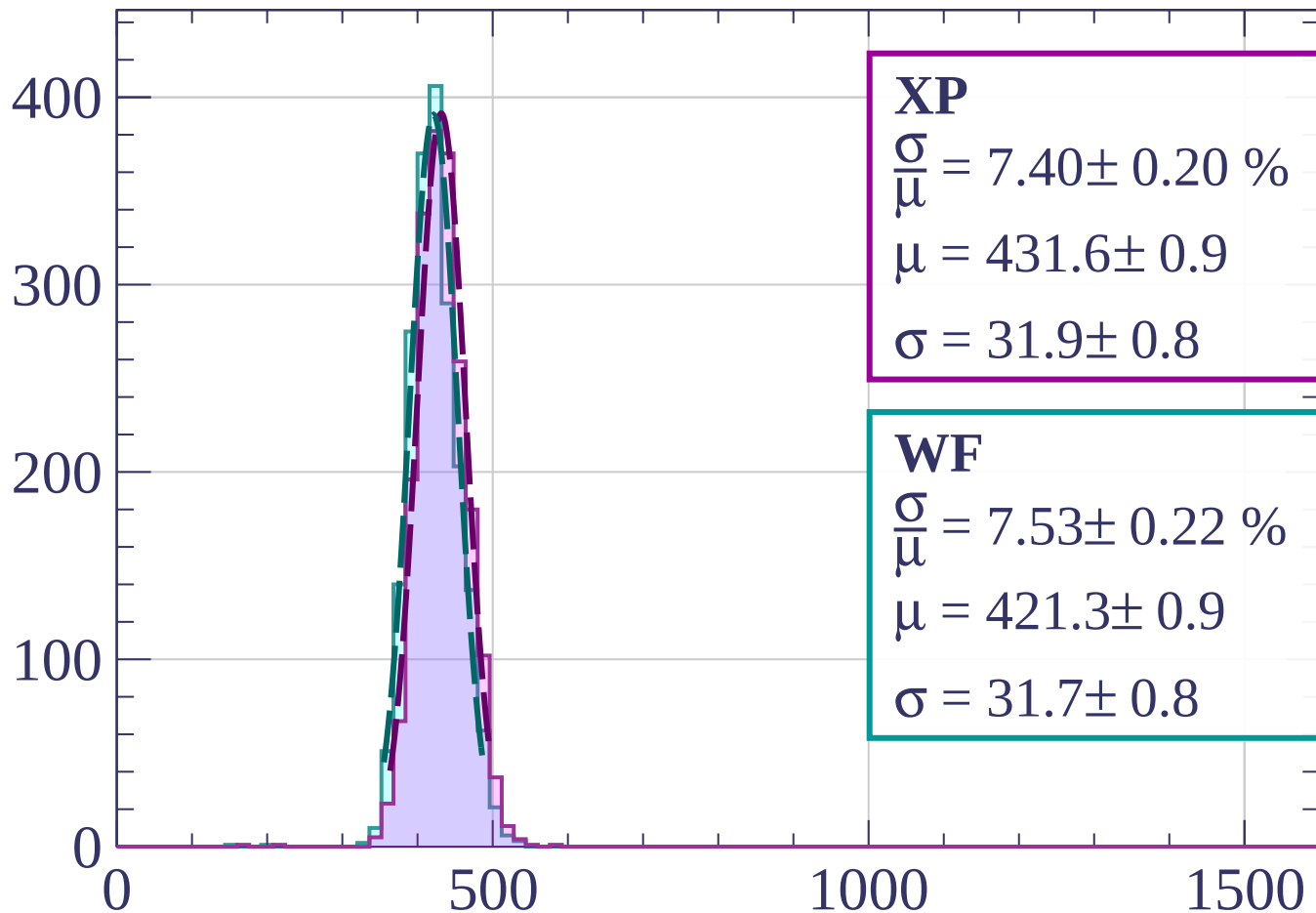
$$\mu = 415.7 \pm 0.8$$

$$\sigma = 27.5 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.20 \%$$

$$\mu = 431.6 \pm 0.9$$

$$\sigma = 31.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.53 \pm 0.22 \%$$

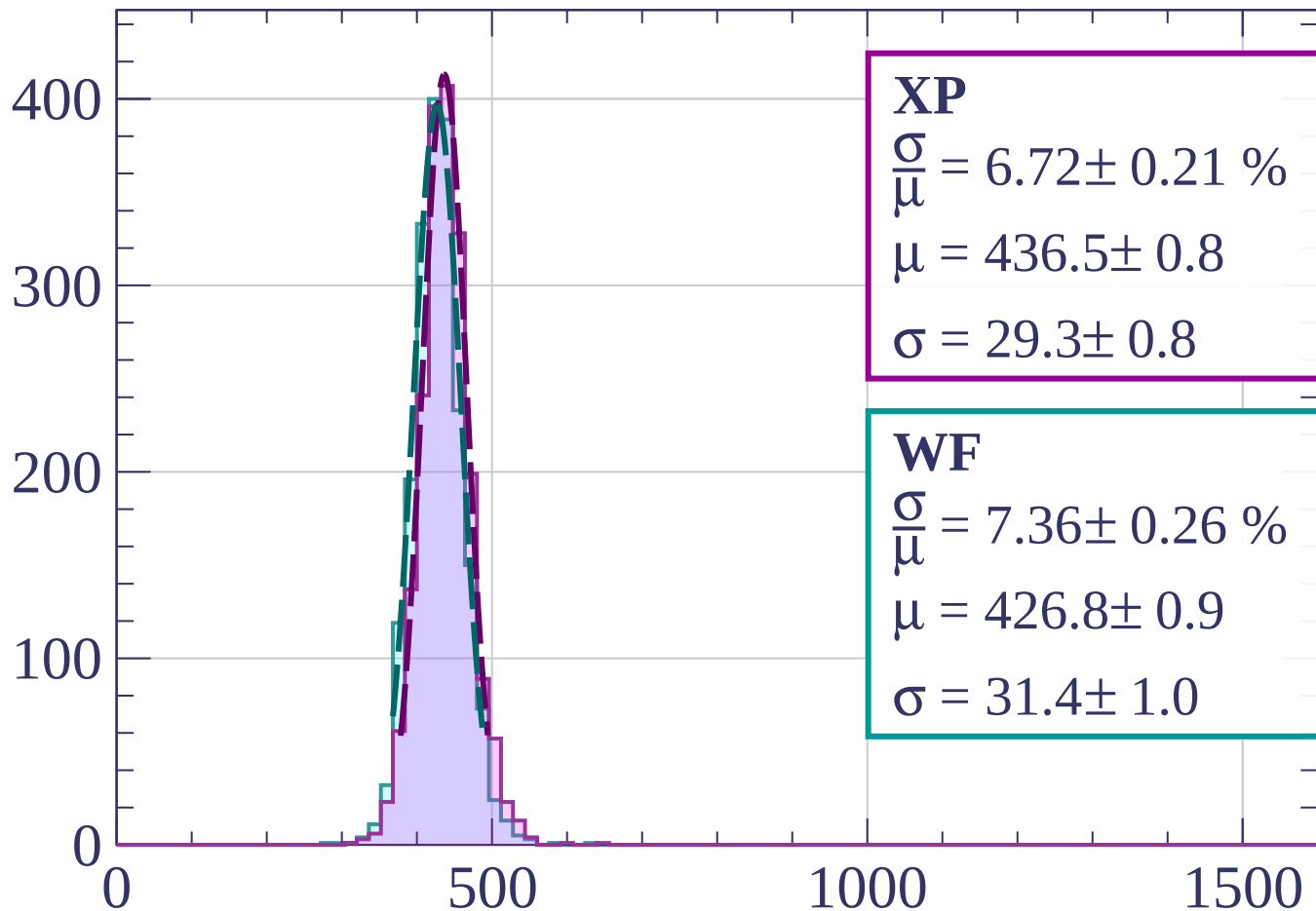
$$\mu = 421.3 \pm 0.9$$

$$\sigma = 31.7 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 6.72 \pm 0.21 \%$$

$$\mu = 436.5 \pm 0.8$$

$$\sigma = 29.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.36 \pm 0.26 \%$$

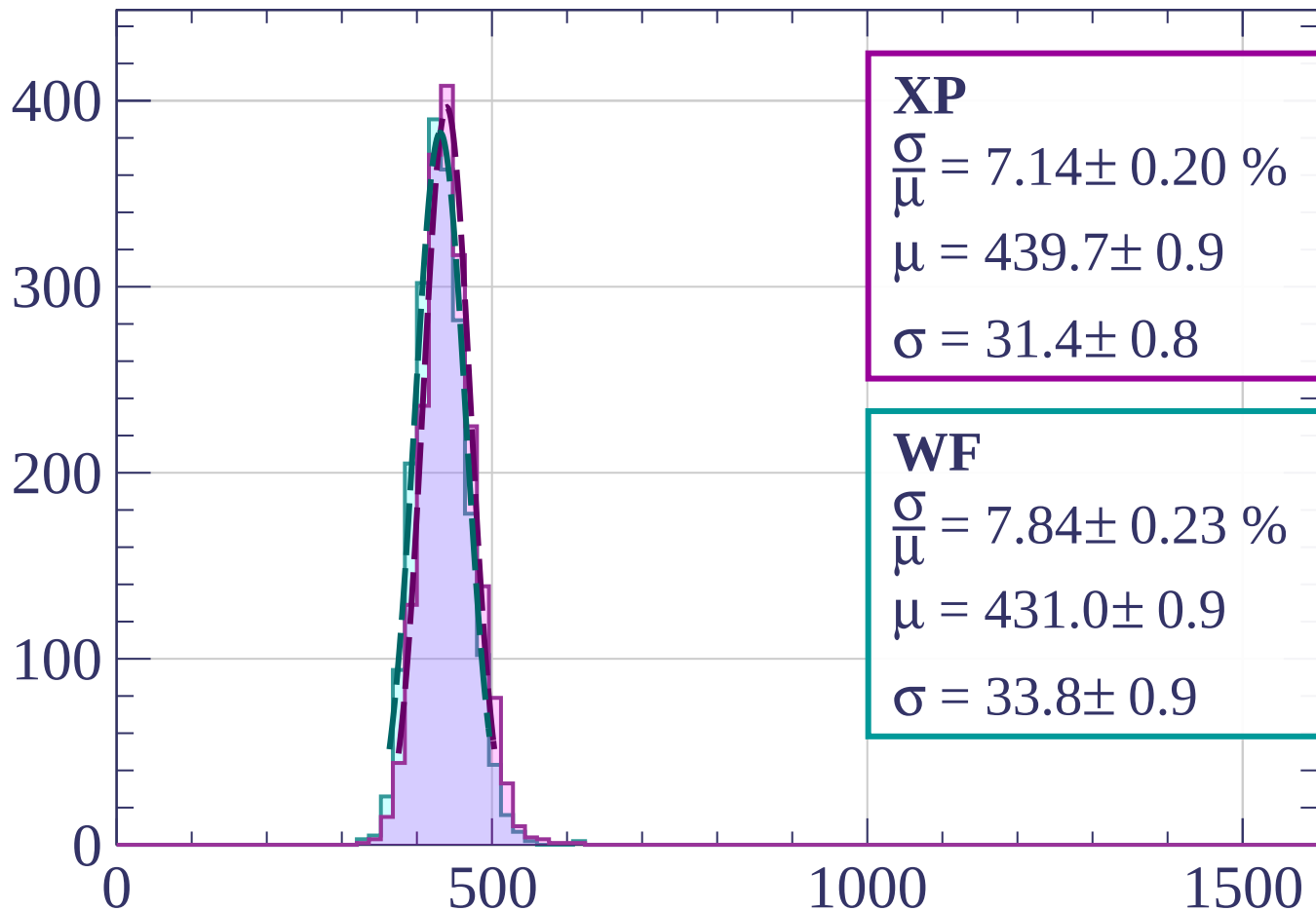
$$\mu = 426.8 \pm 0.9$$

$$\sigma = 31.4 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $720 < p < 800$

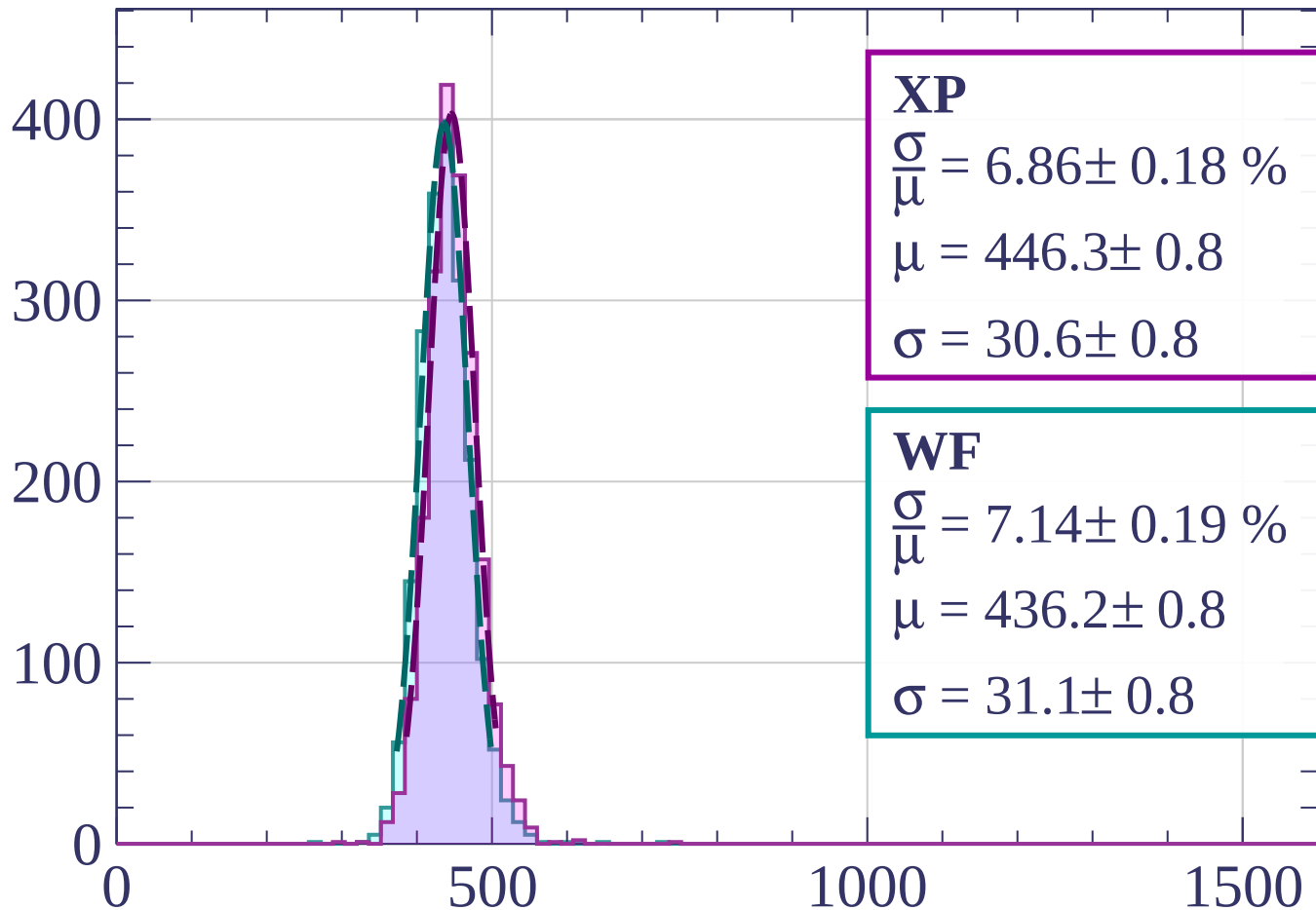
Count



dE/dx [ADC counts/cm]

Energy loss | $800 < p < 880$

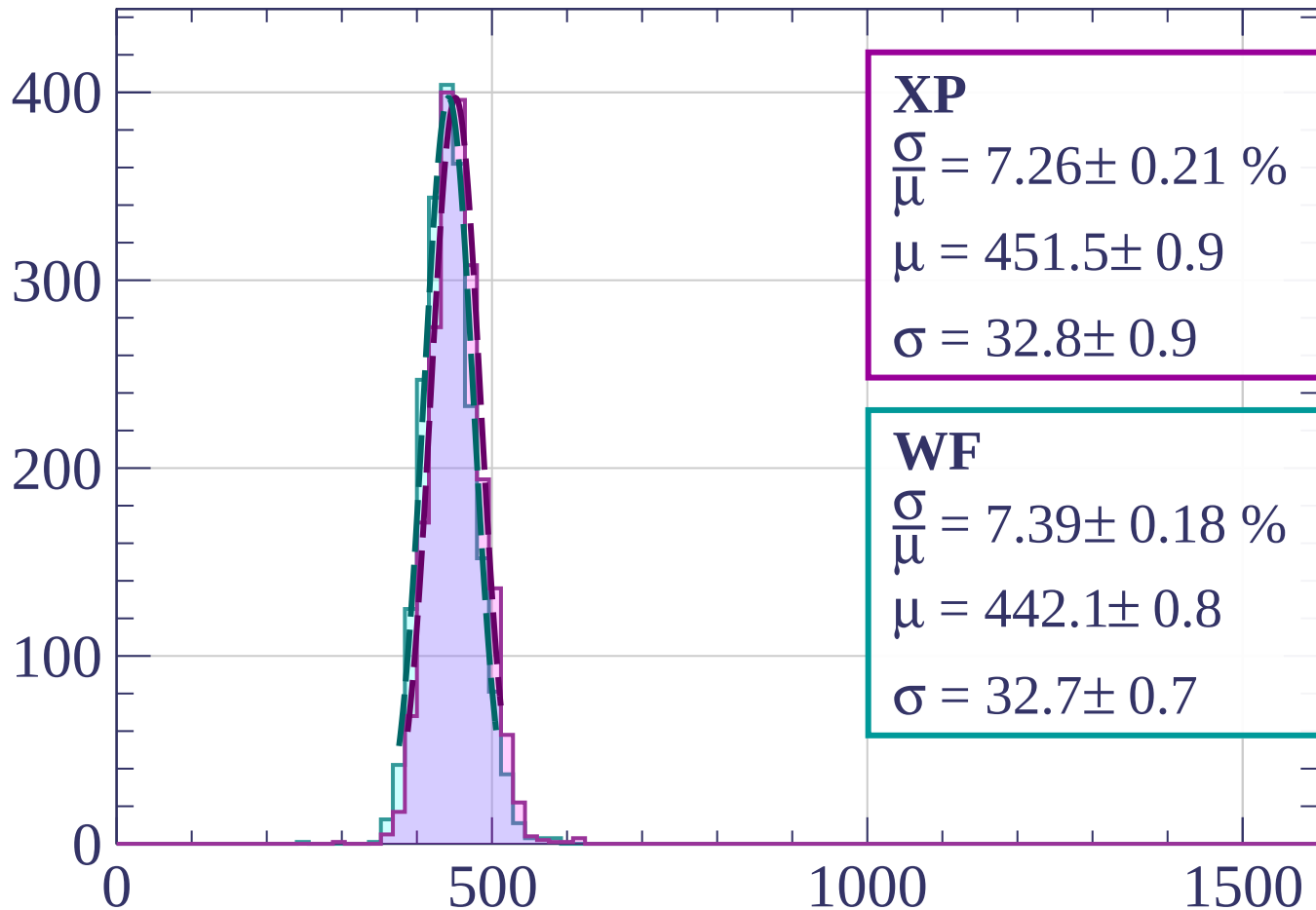
Count



dE/dx [ADC counts/cm]

Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 7.26 \pm 0.21 \%$$

$$\mu = 451.5 \pm 0.9$$

$$\sigma = 32.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.39 \pm 0.18 \%$$

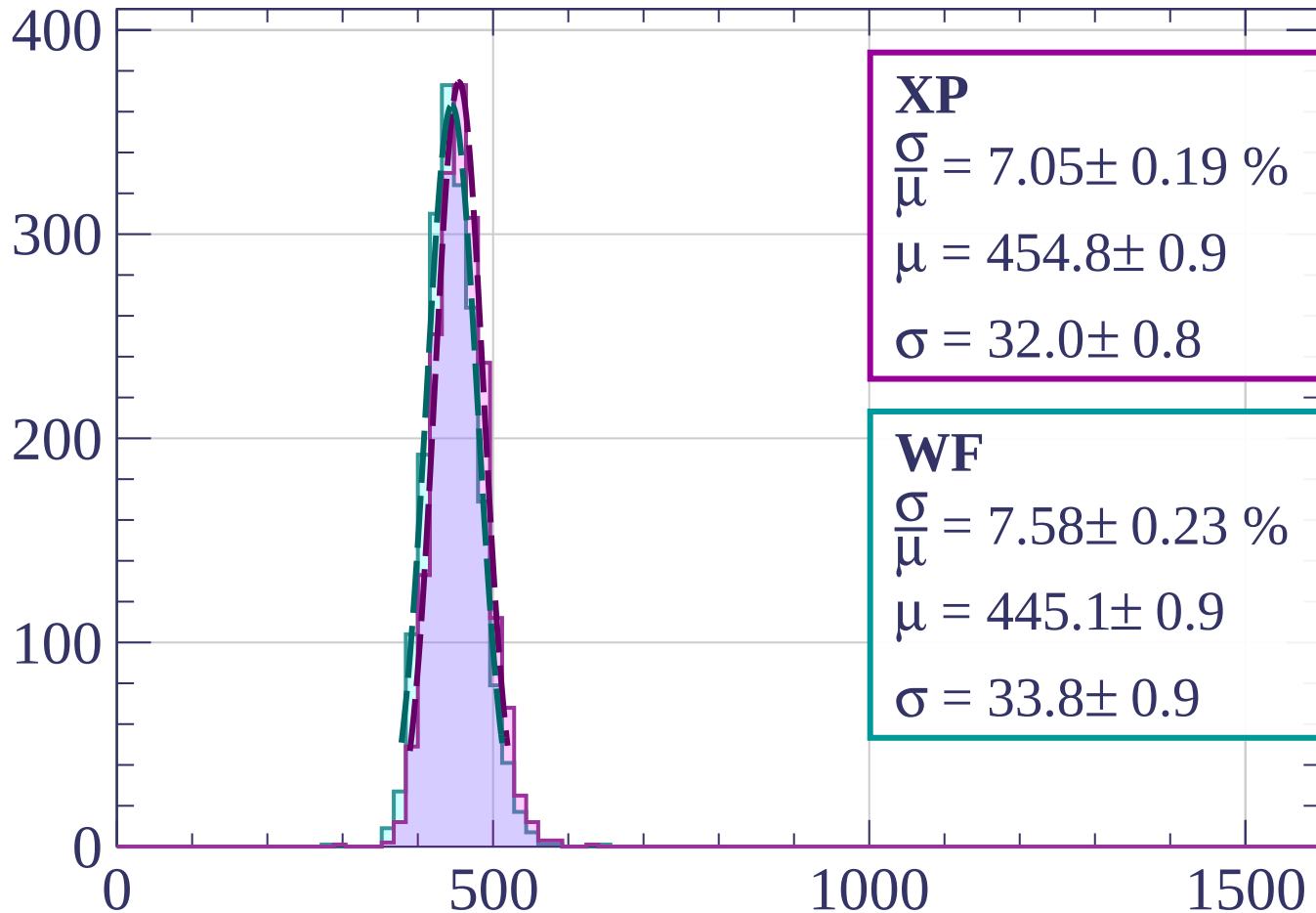
$$\mu = 442.1 \pm 0.8$$

$$\sigma = 32.7 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $960 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 7.05 \pm 0.19 \%$$

$$\mu = 454.8 \pm 0.9$$

$$\sigma = 32.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.58 \pm 0.23 \%$$

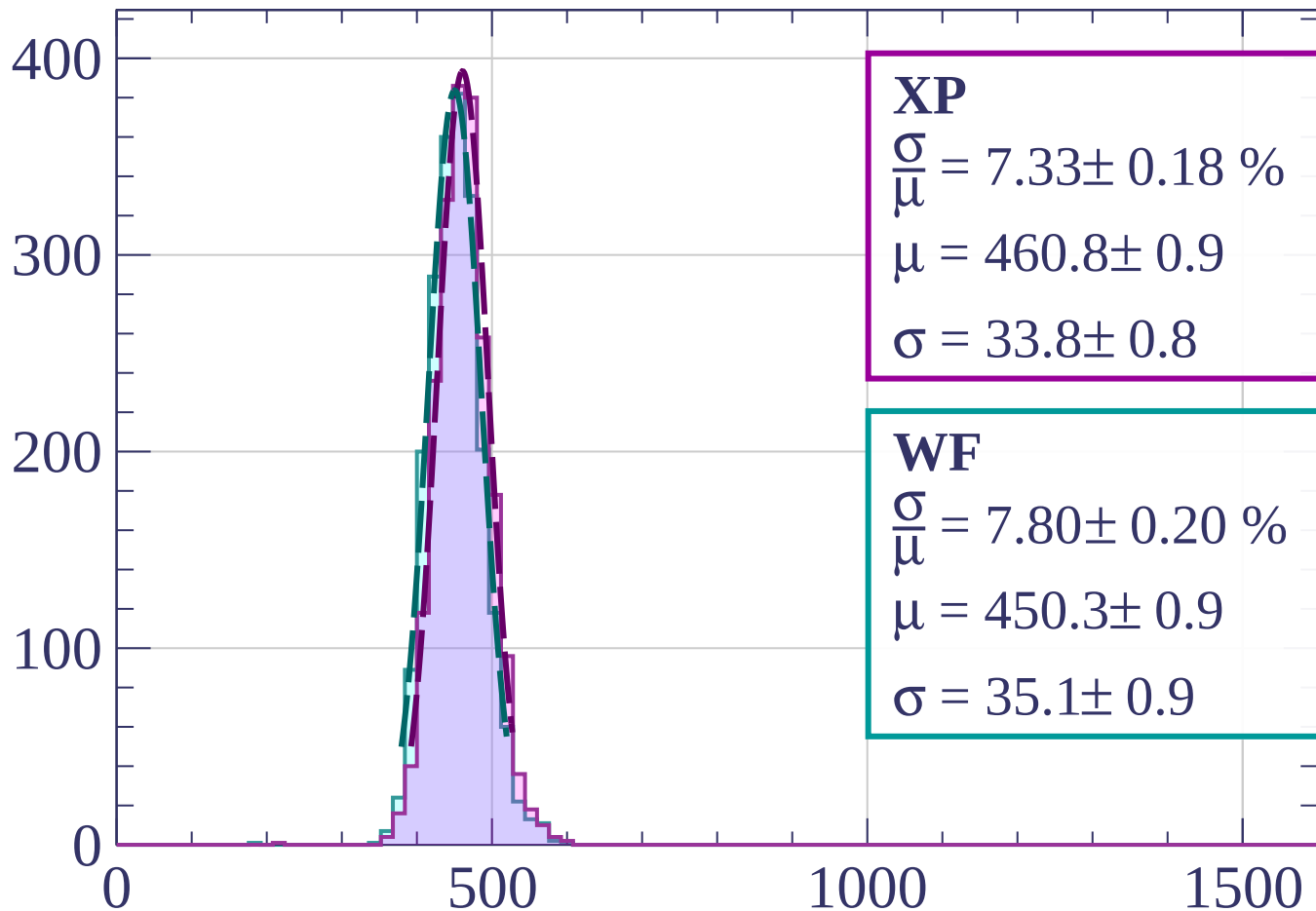
$$\mu = 445.1 \pm 0.9$$

$$\sigma = 33.8 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 7.33 \pm 0.18 \%$$

$$\mu = 460.8 \pm 0.9$$

$$\sigma = 33.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.80 \pm 0.20 \%$$

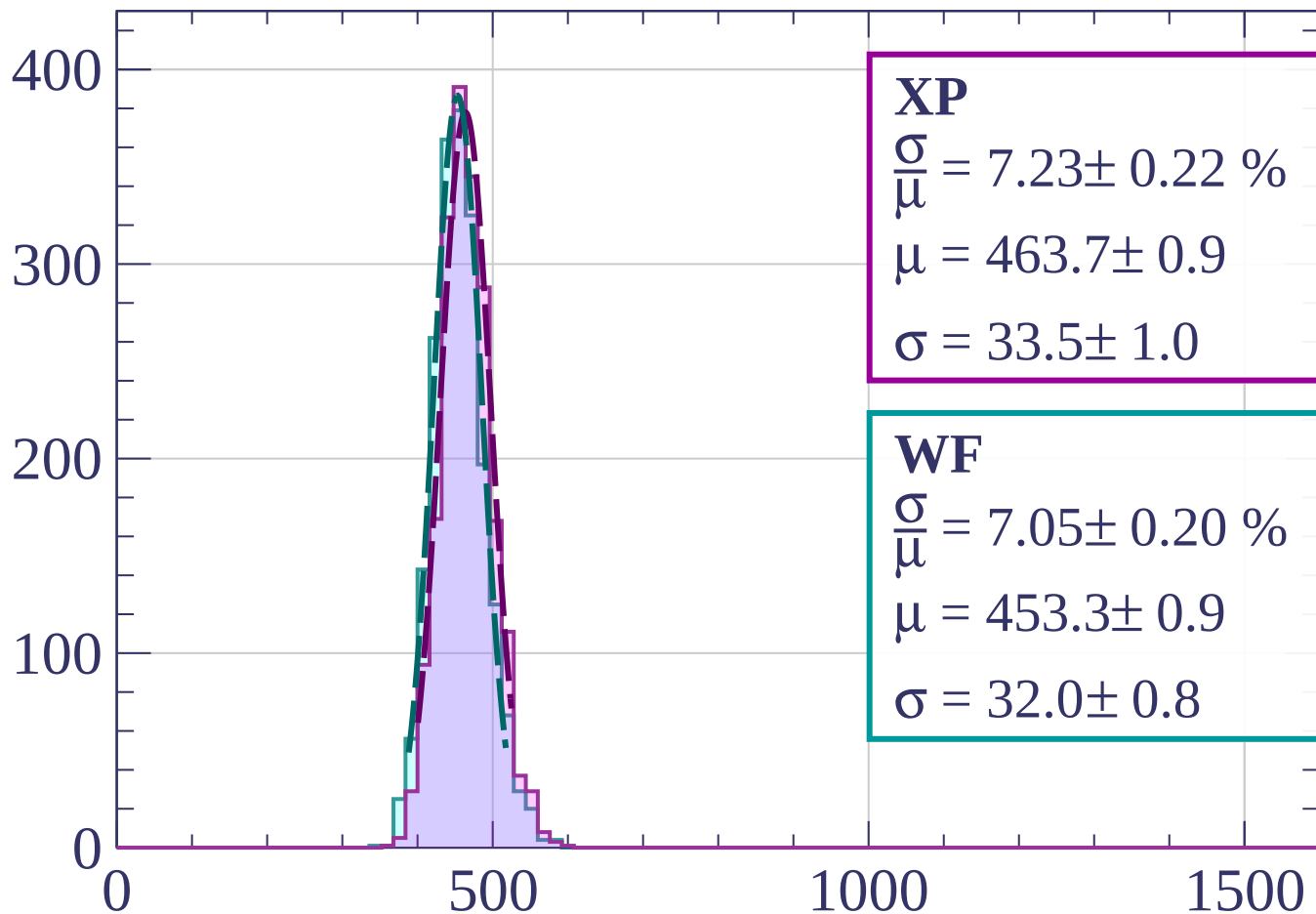
$$\mu = 450.3 \pm 0.9$$

$$\sigma = 35.1 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1200$

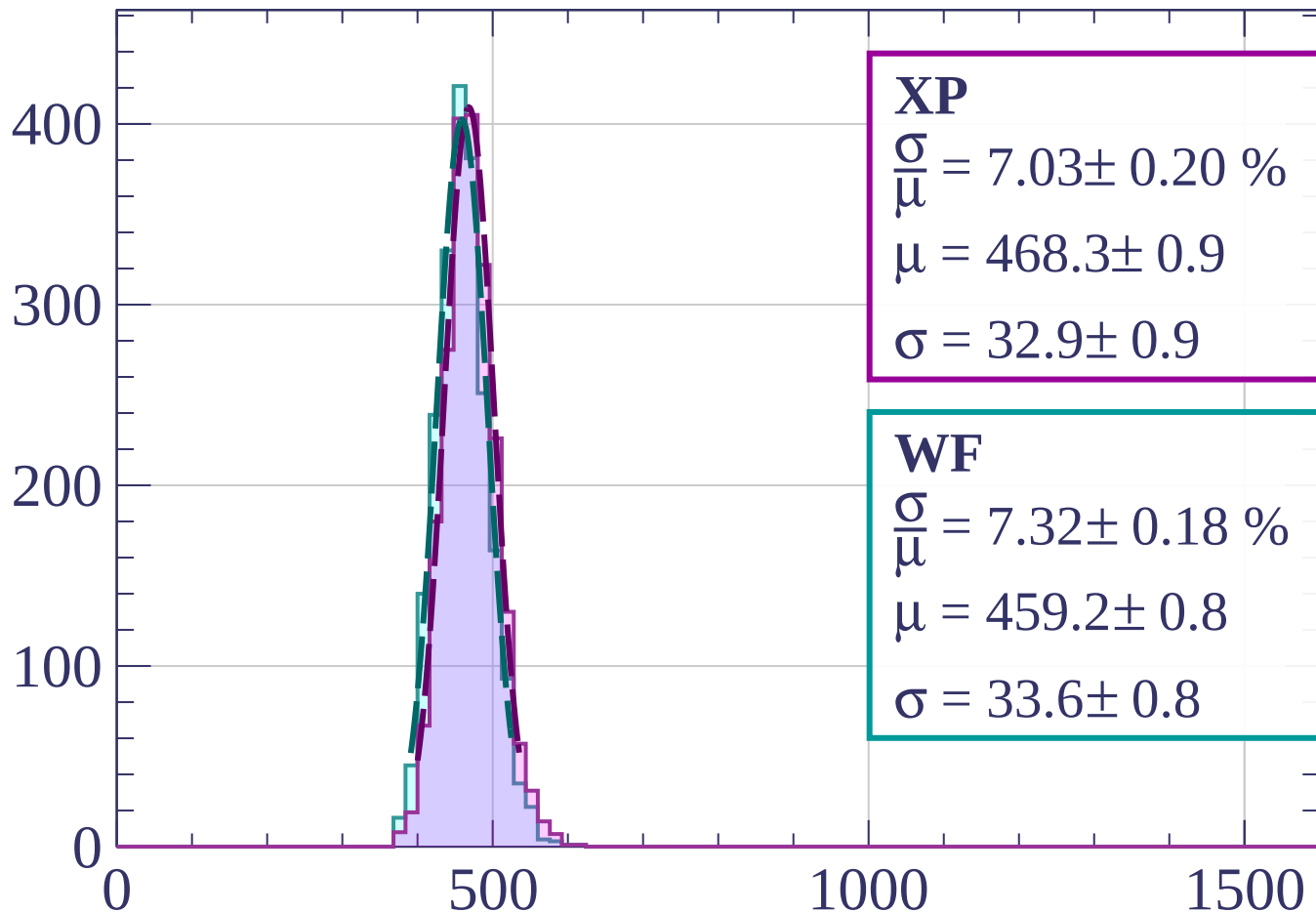
Count



dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1280$

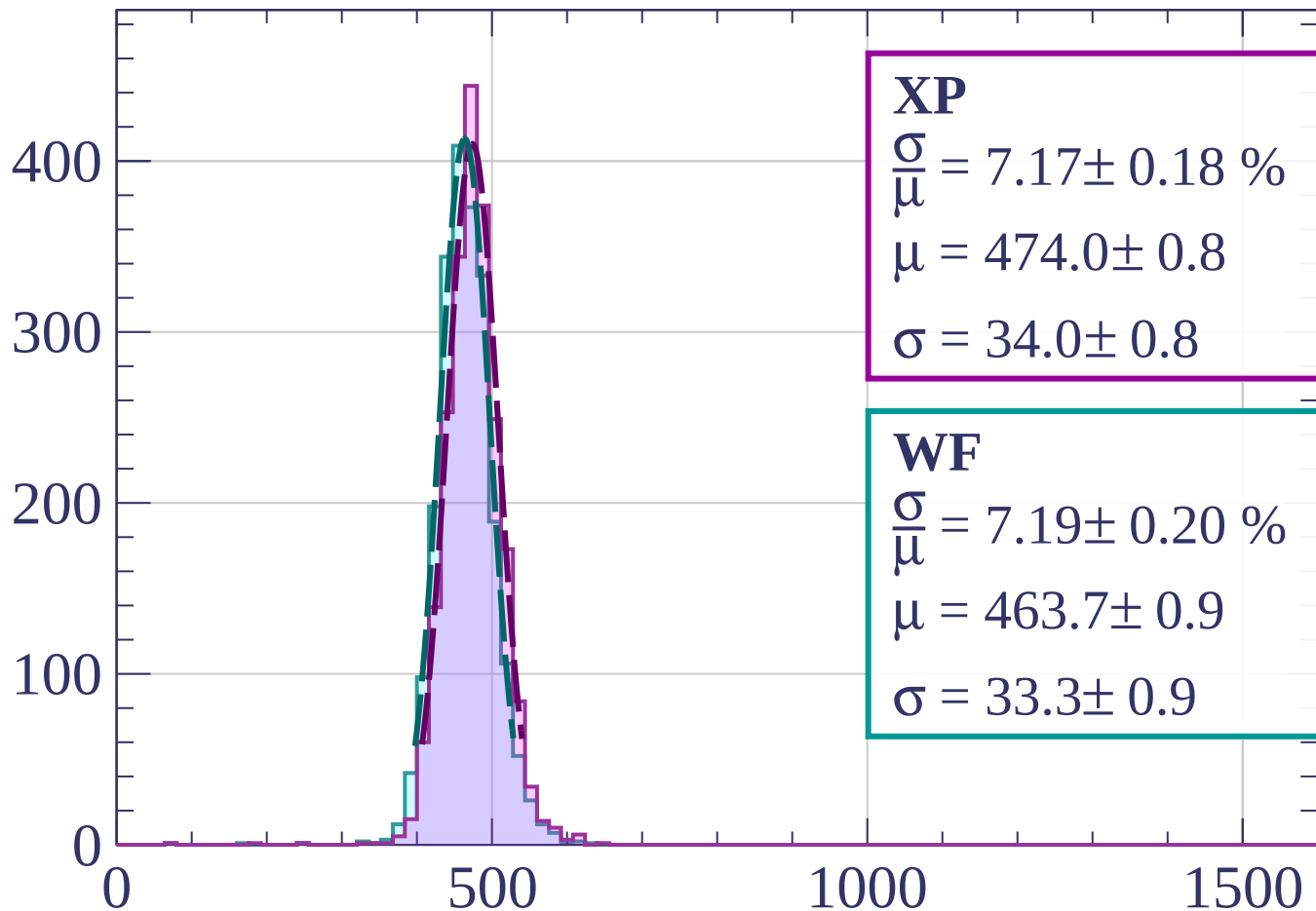
Count



dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 7.17 \pm 0.18 \%$$

$$\mu = 474.0 \pm 0.8$$

$$\sigma = 34.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.19 \pm 0.20 \%$$

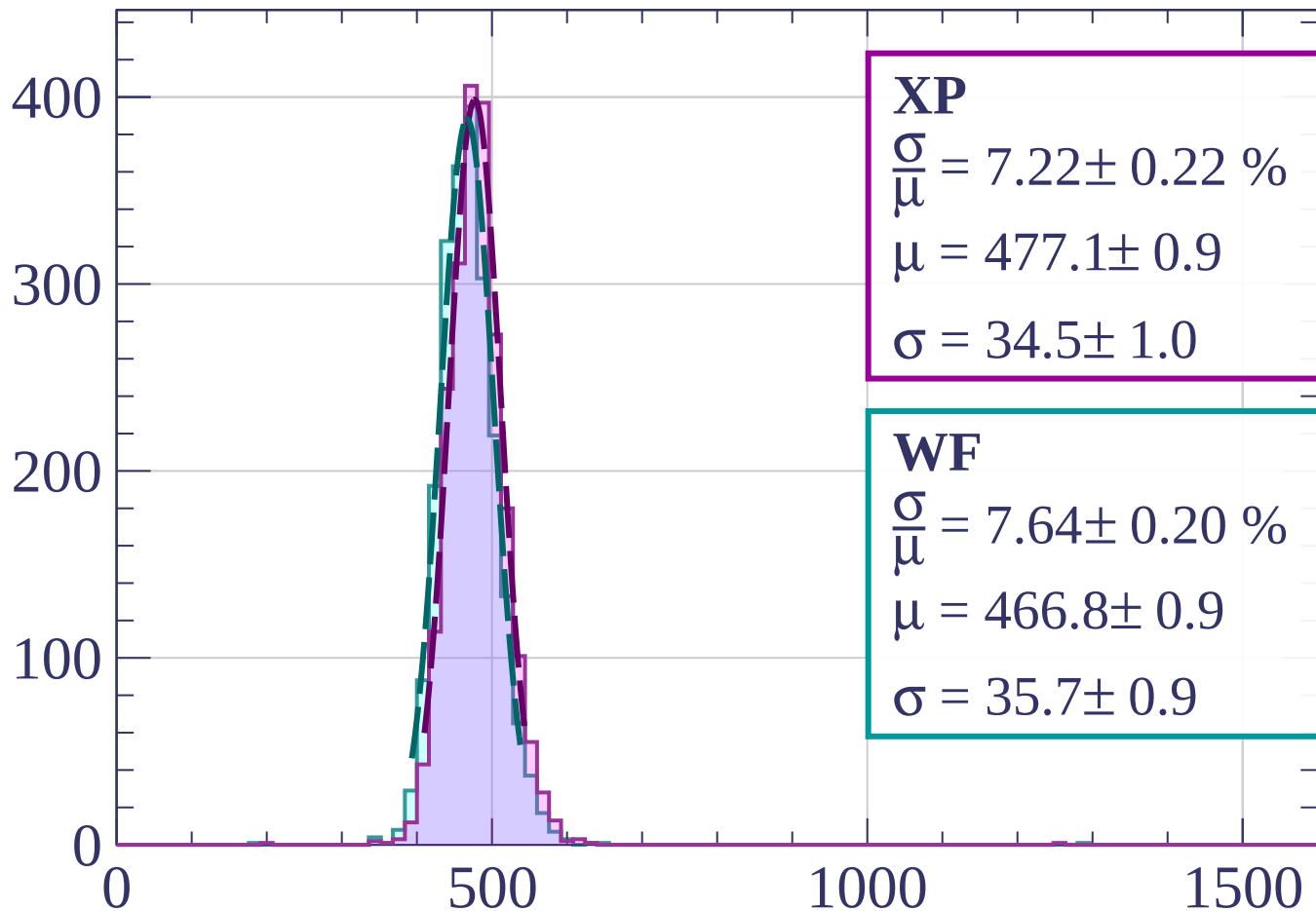
$$\mu = 463.7 \pm 0.9$$

$$\sigma = 33.3 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 7.22 \pm 0.22 \%$$

$$\mu = 477.1 \pm 0.9$$

$$\sigma = 34.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.64 \pm 0.20 \%$$

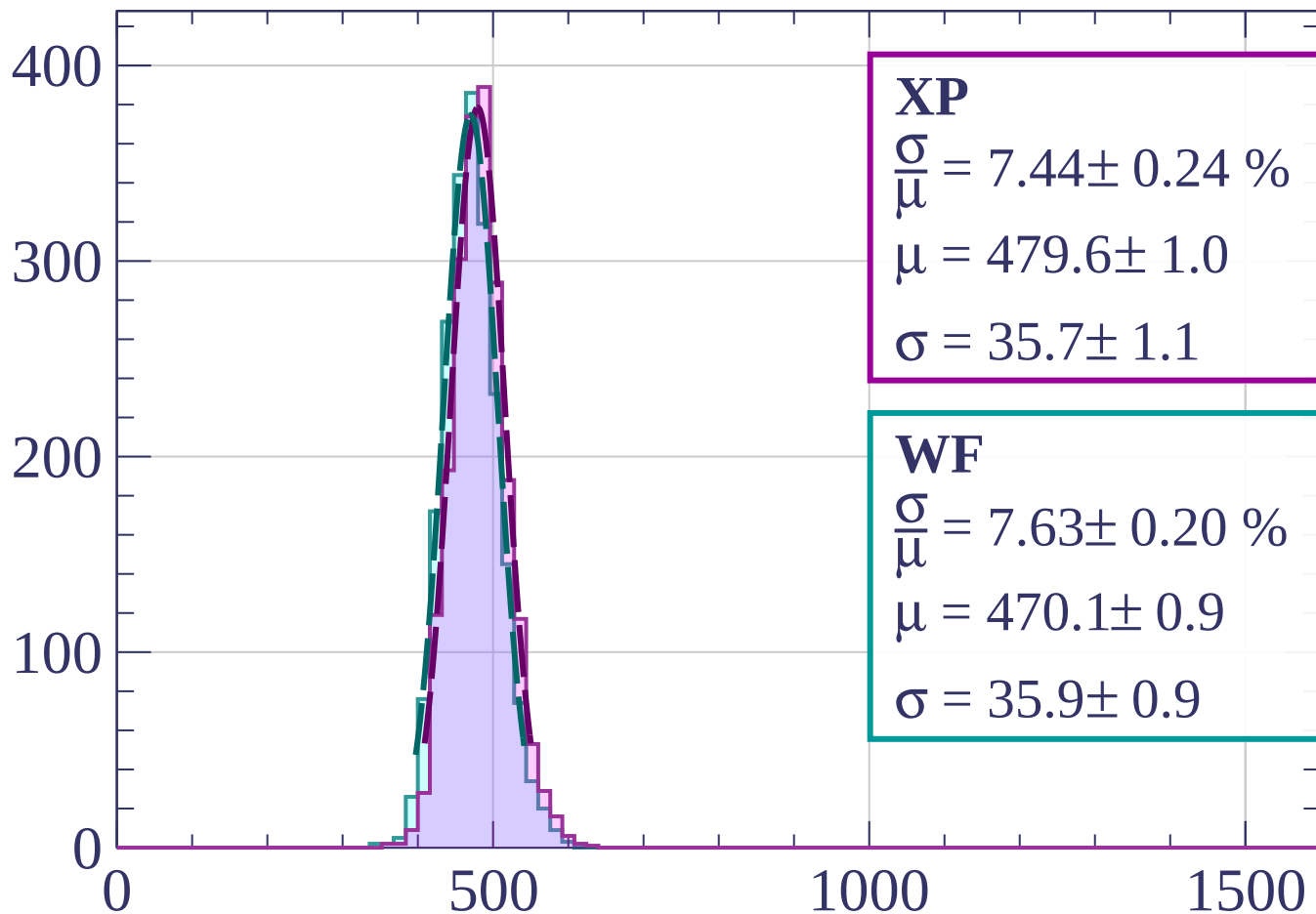
$$\mu = 466.8 \pm 0.9$$

$$\sigma = 35.7 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 7.44 \pm 0.24 \%$$

$$\mu = 479.6 \pm 1.0$$

$$\sigma = 35.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.63 \pm 0.20 \%$$

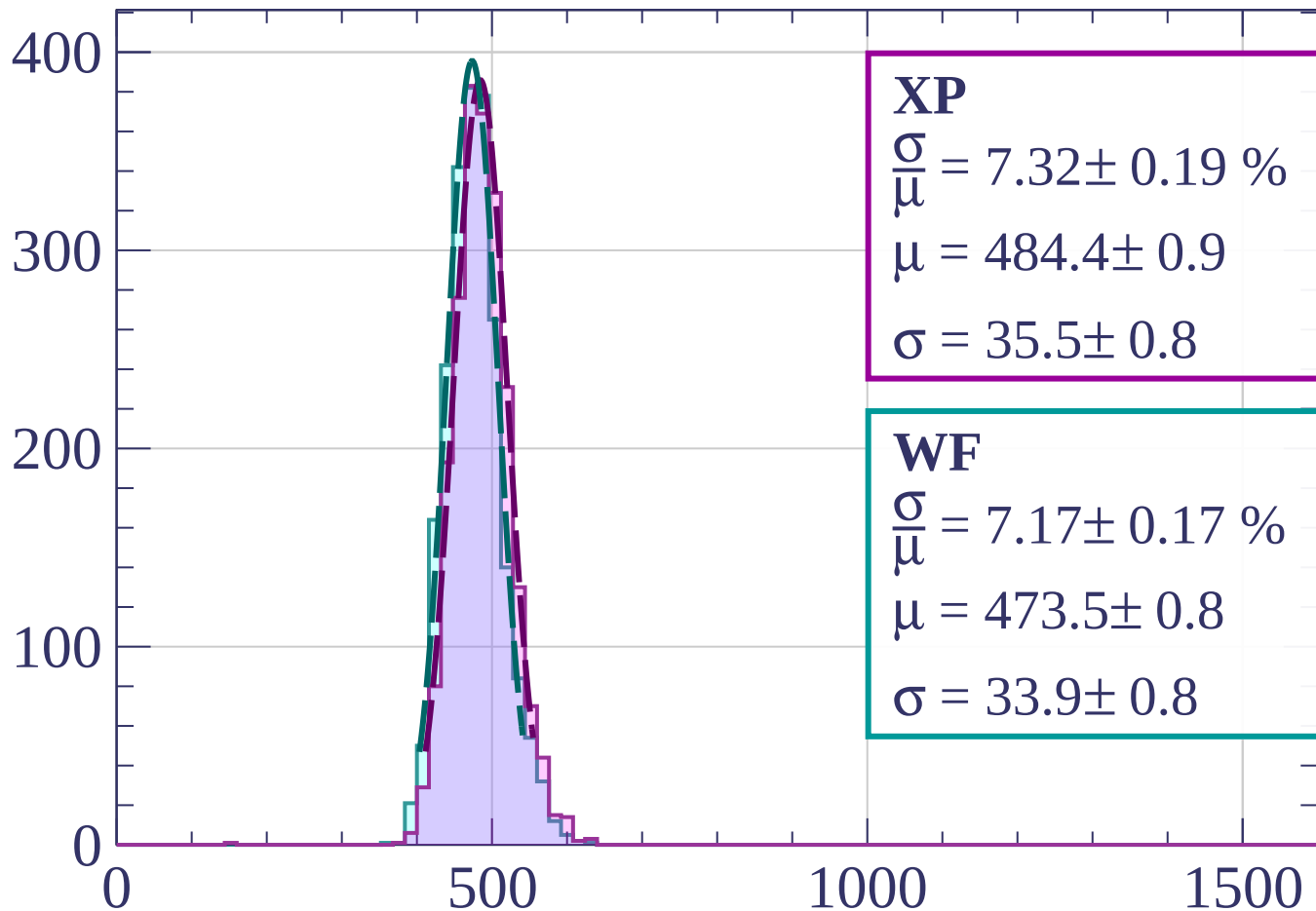
$$\mu = 470.1 \pm 0.9$$

$$\sigma = 35.9 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.32 \pm 0.19 \%$$

$$\mu = 484.4 \pm 0.9$$

$$\sigma = 35.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.17 \pm 0.17 \%$$

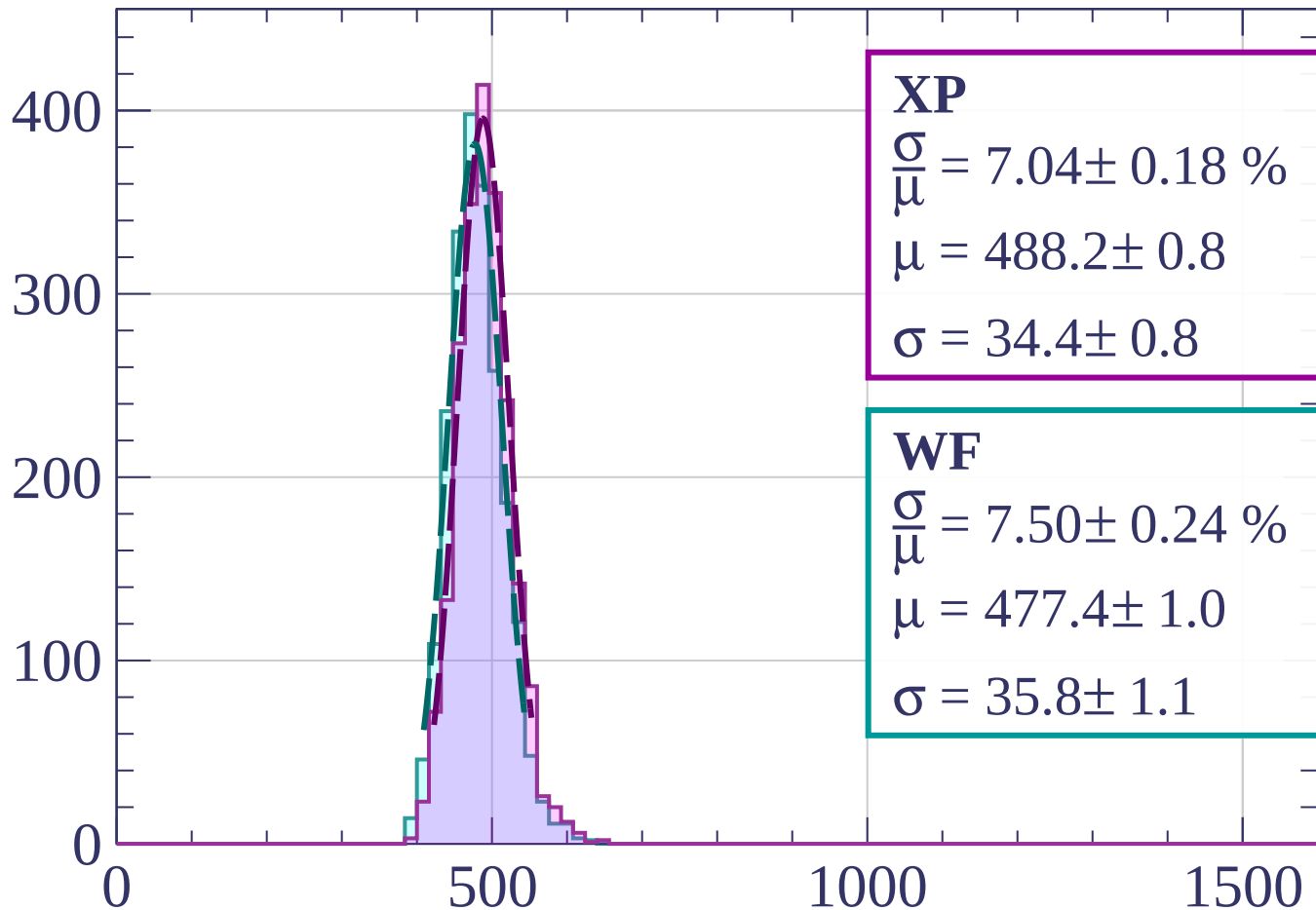
$$\mu = 473.5 \pm 0.8$$

$$\sigma = 33.9 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 7.04 \pm 0.18 \%$$

$$\mu = 488.2 \pm 0.8$$

$$\sigma = 34.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.50 \pm 0.24 \%$$

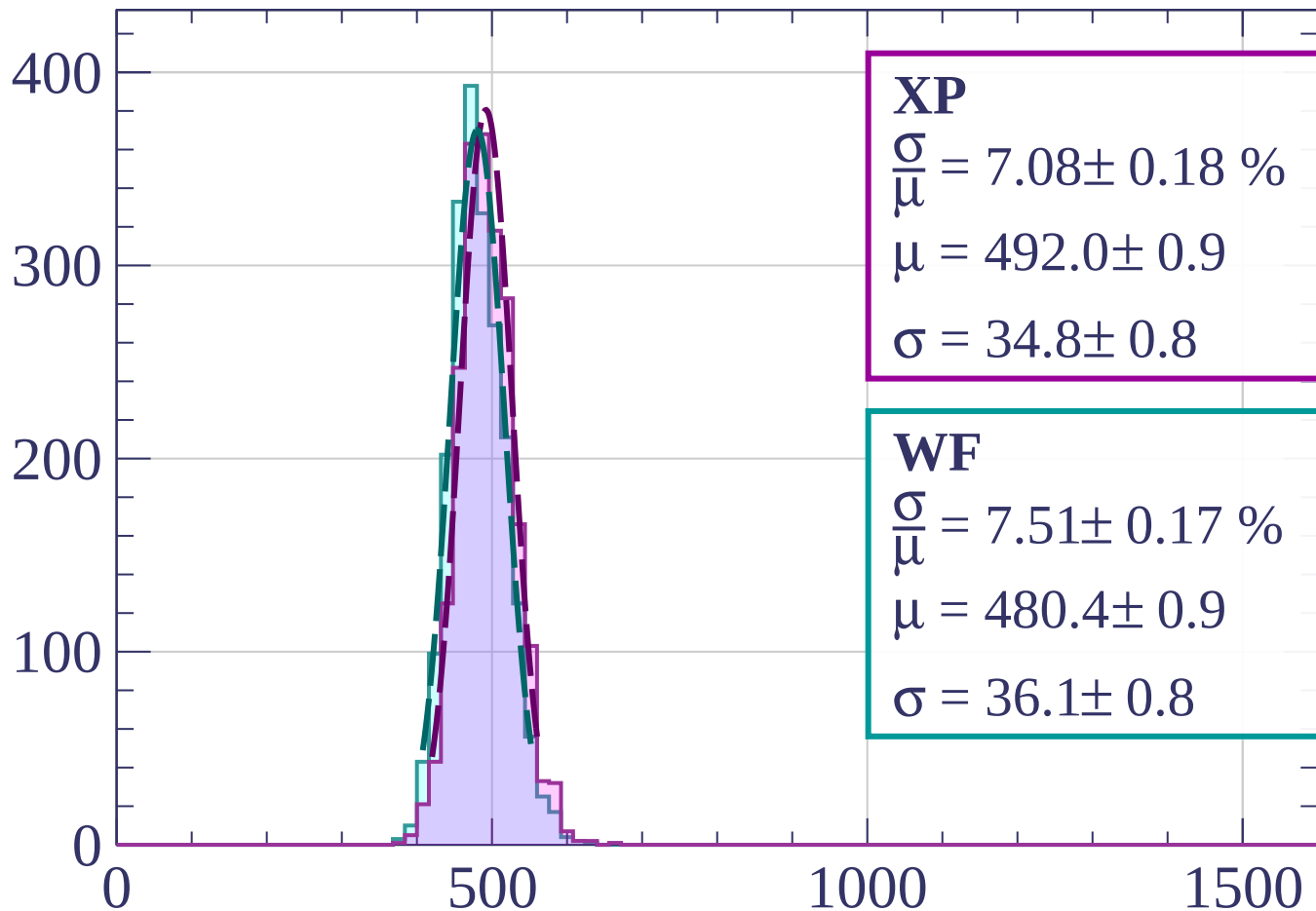
$$\mu = 477.4 \pm 1.0$$

$$\sigma = 35.8 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $1680 < p < 1760$

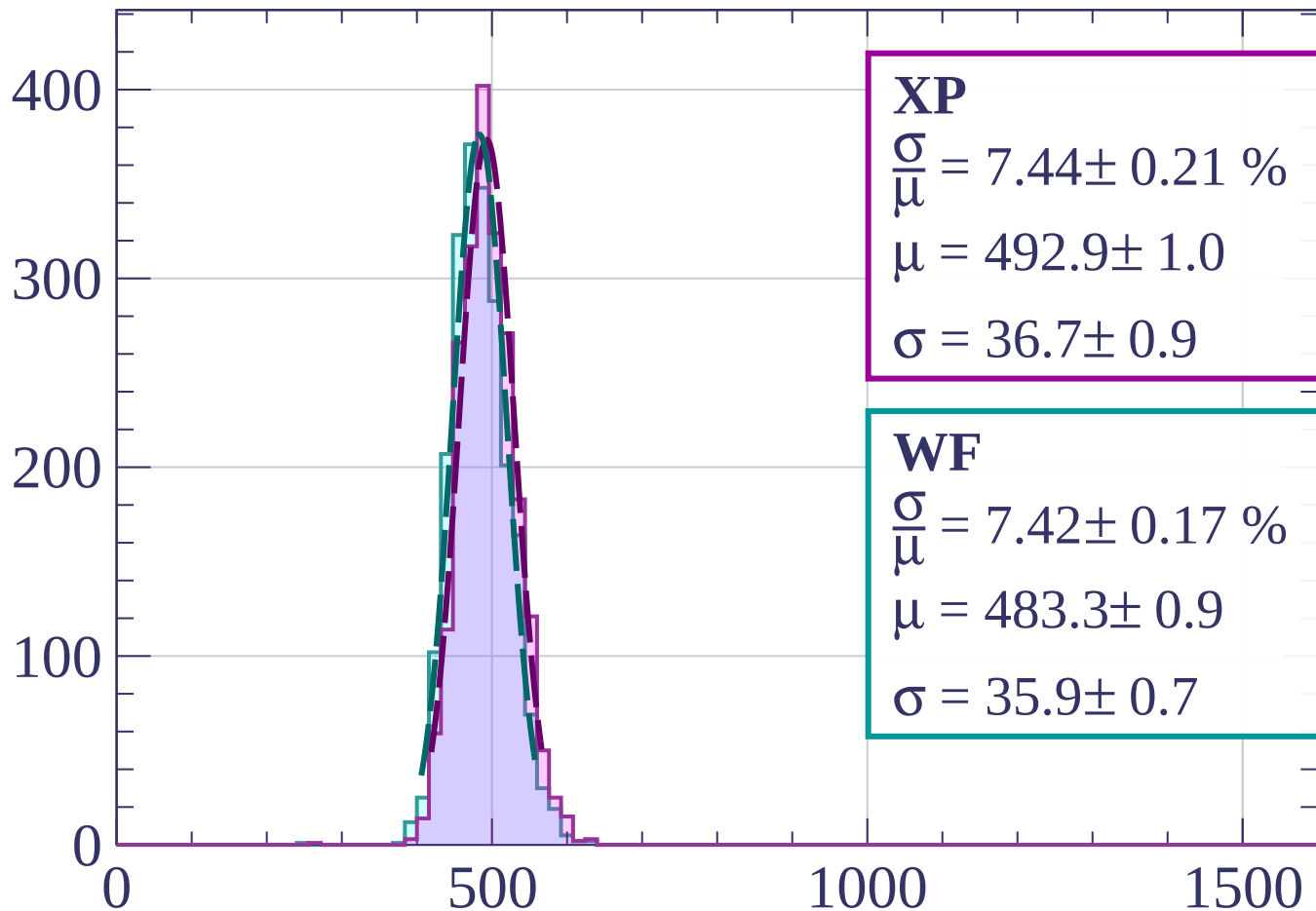
Count



dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 7.44 \pm 0.21 \%$$

$$\mu = 492.9 \pm 1.0$$

$$\sigma = 36.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.42 \pm 0.17 \%$$

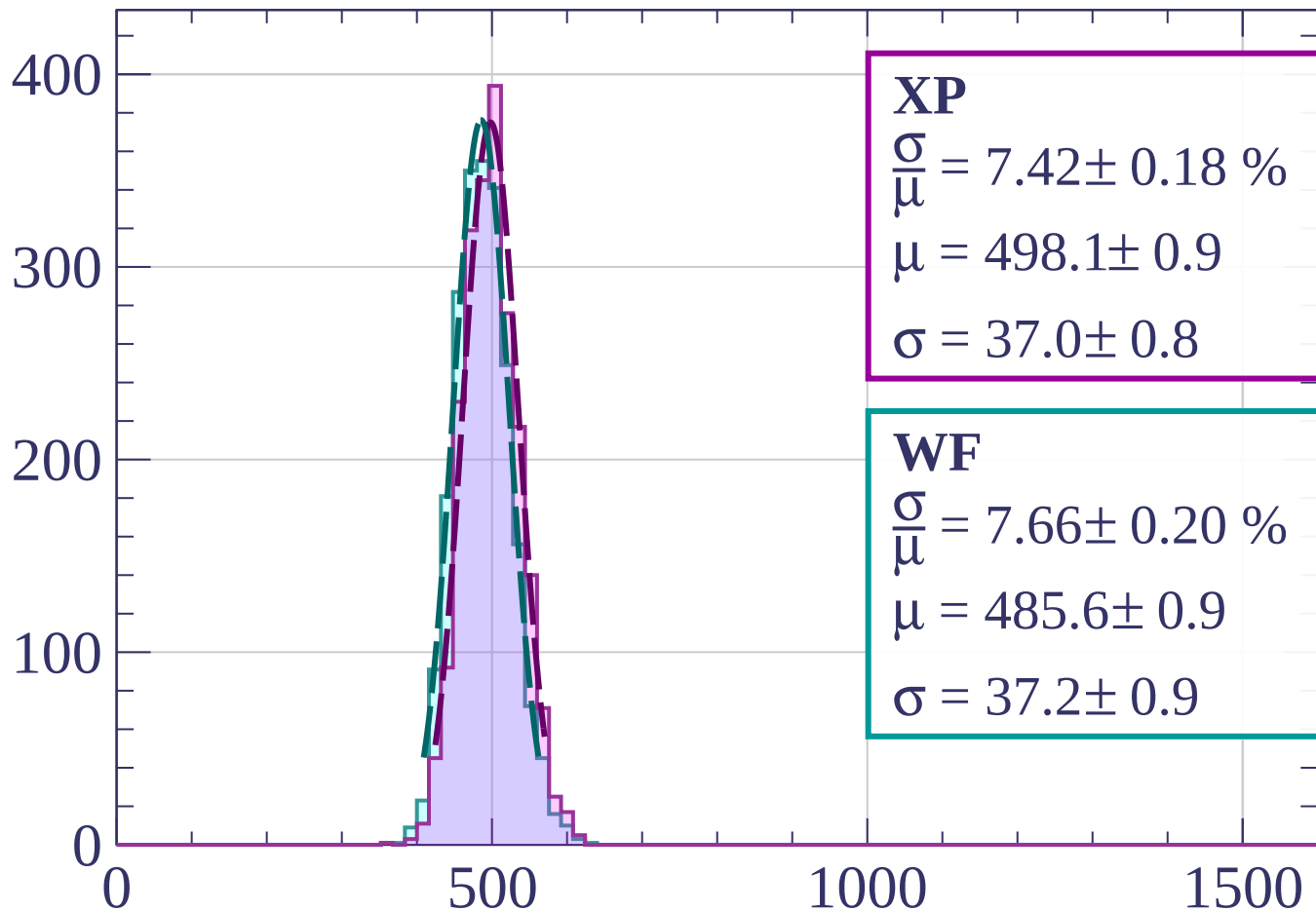
$$\mu = 483.3 \pm 0.9$$

$$\sigma = 35.9 \pm 0.7$$

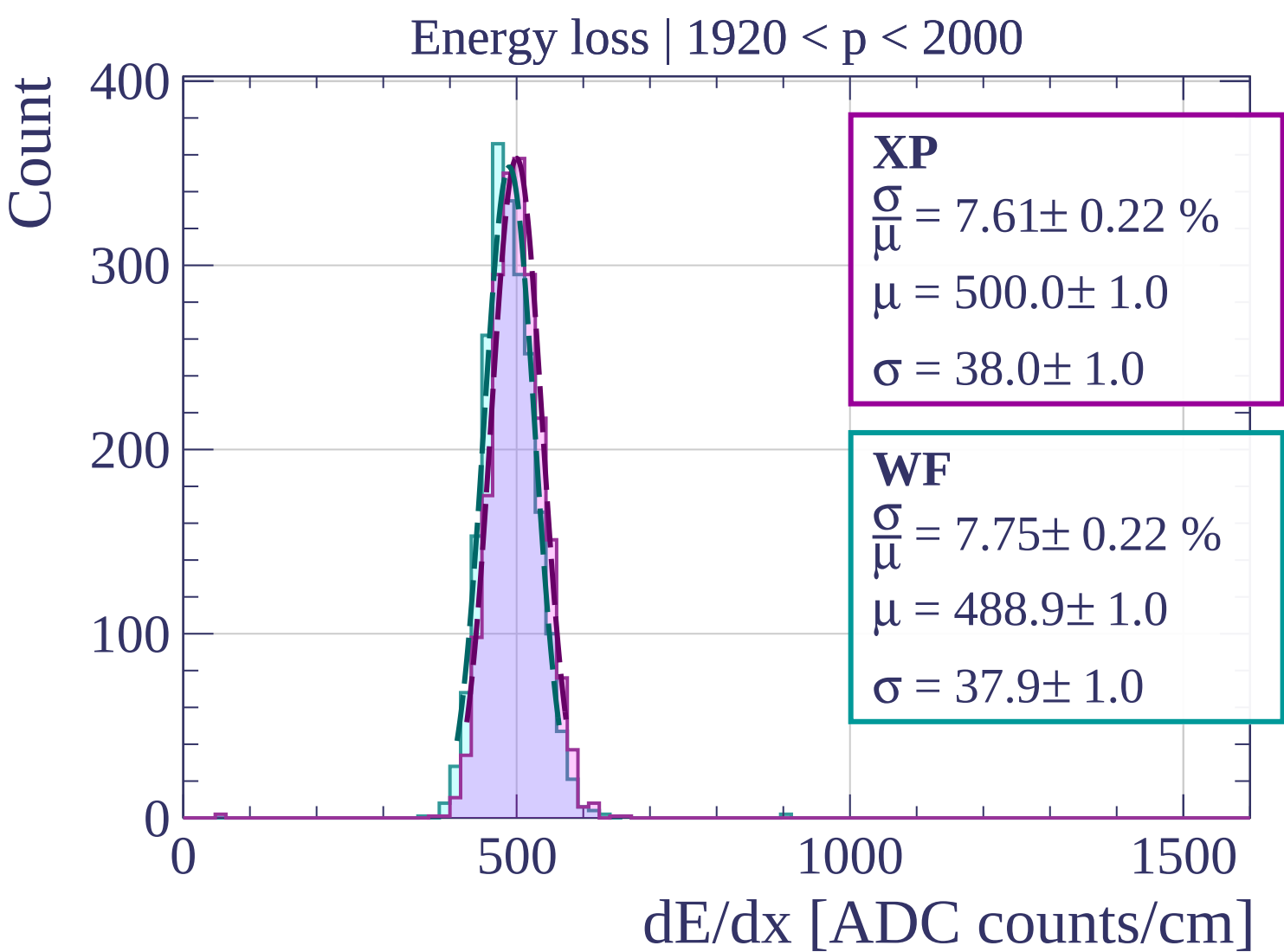
dE/dx [ADC counts/cm]

Energy loss | $1840 < p < 1920$

Count

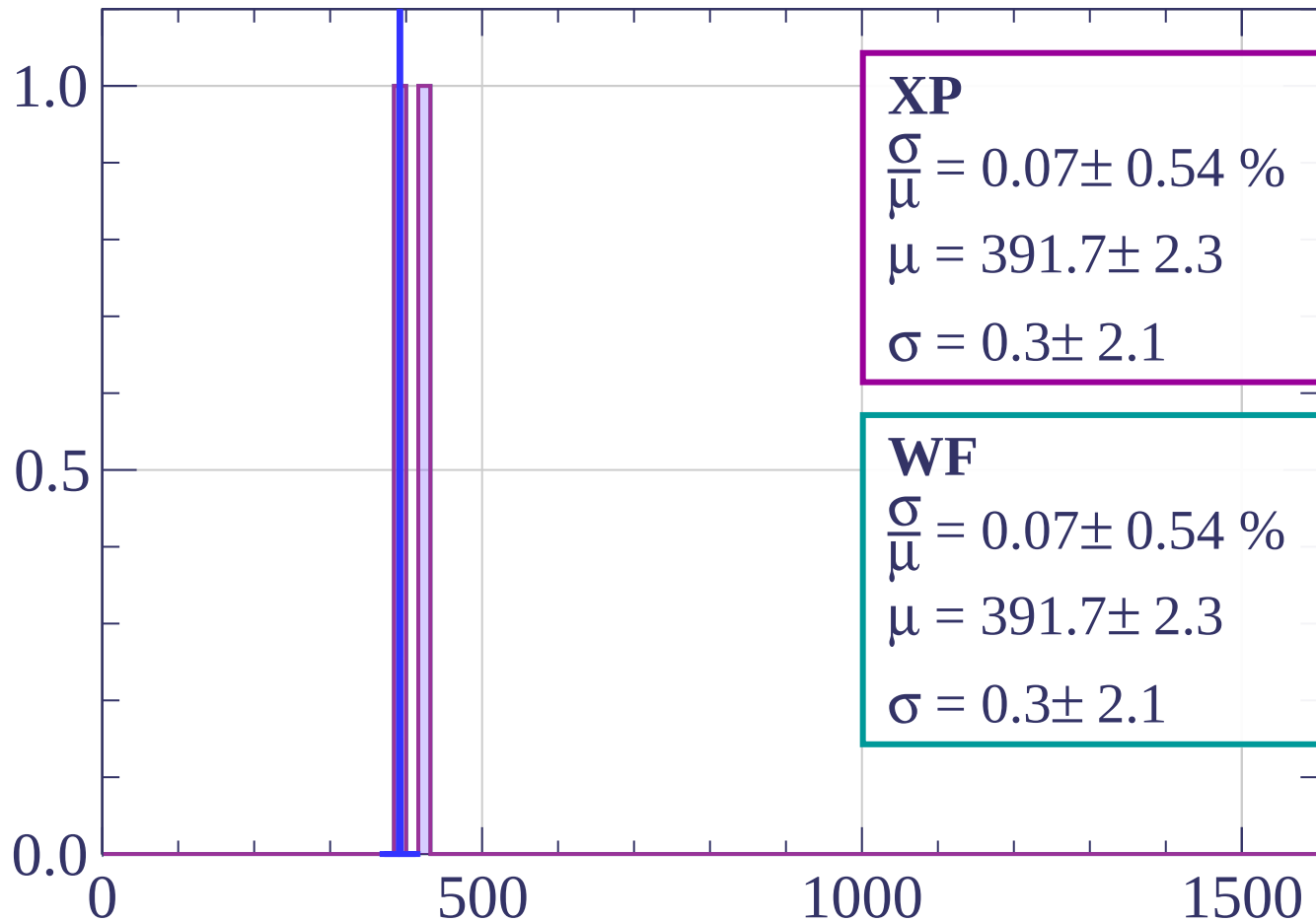


dE/dx [ADC counts/cm]



Count

Energy loss | $0 < \phi < 3$



XP

$$\frac{\sigma}{\mu} = 0.07 \pm 0.54 \%$$

$$\mu = 391.7 \pm 2.3$$

$$\sigma = 0.3 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 0.07 \pm 0.54 \%$$

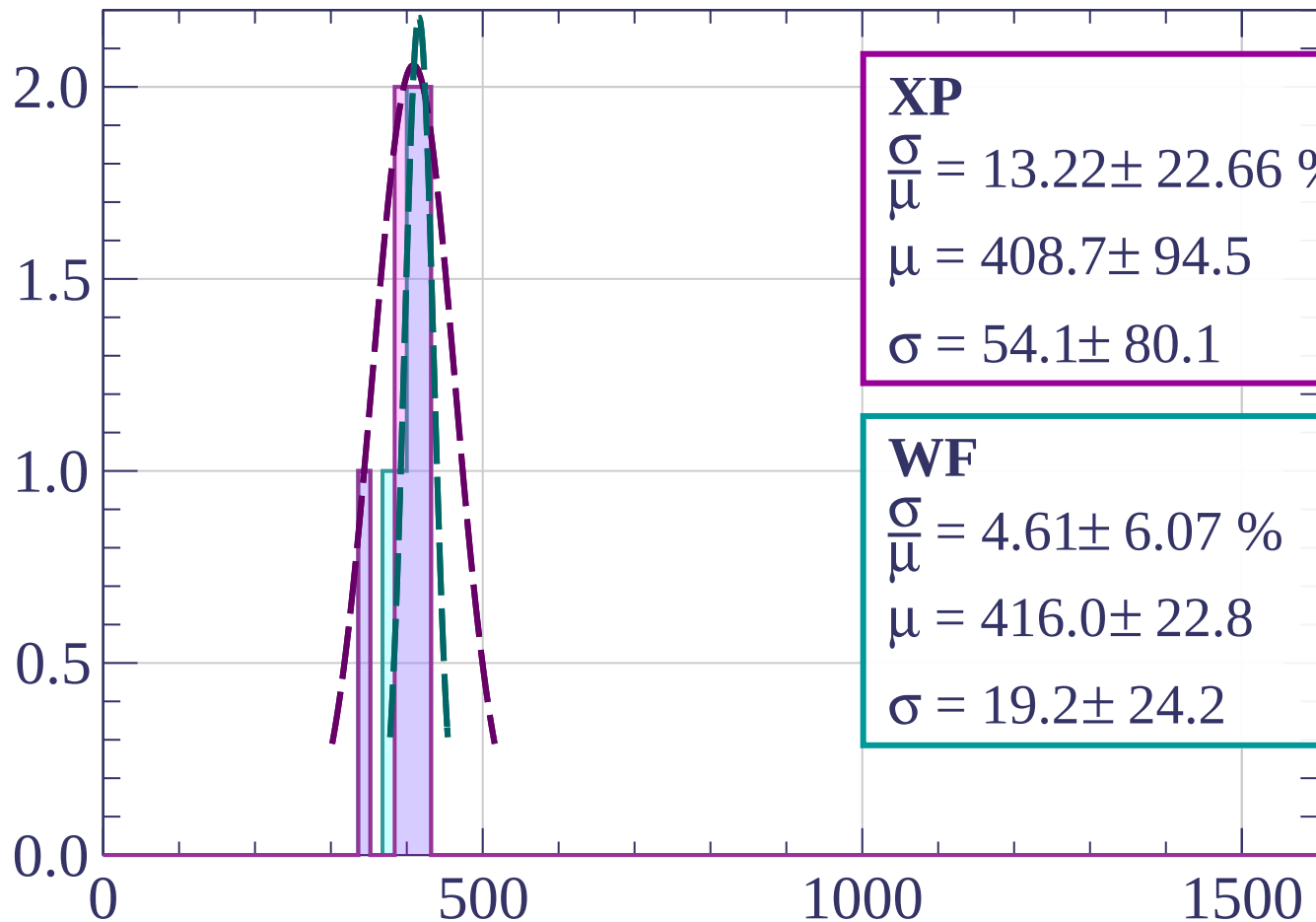
$$\mu = 391.7 \pm 2.3$$

$$\sigma = 0.3 \pm 2.1$$

dE/dx [ADC counts/cm]

Count

Energy loss | $3 < \phi < 6$



XP

$$\frac{\sigma}{\mu} = 13.22 \pm 22.66 \%$$

$$\mu = 408.7 \pm 94.5$$

$$\sigma = 54.1 \pm 80.1$$

WF

$$\frac{\sigma}{\mu} = 4.61 \pm 6.07 \%$$

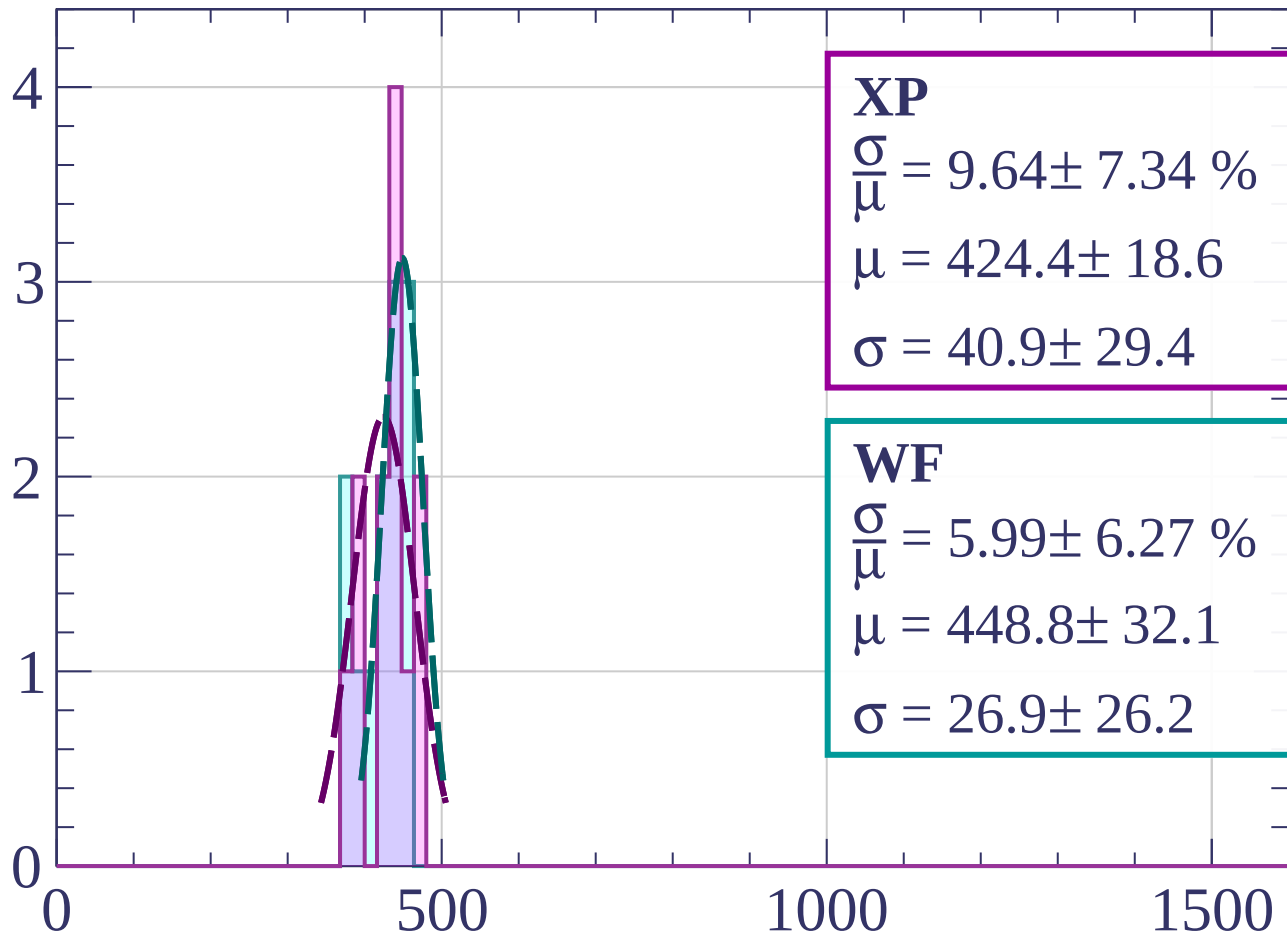
$$\mu = 416.0 \pm 22.8$$

$$\sigma = 19.2 \pm 24.2$$

dE/dx [ADC counts/cm]

Count

Energy loss | $6 < \phi < 9$



XP

$$\frac{\sigma}{\mu} = 9.64 \pm 7.34 \%$$

$$\mu = 424.4 \pm 18.6$$

$$\sigma = 40.9 \pm 29.4$$

WF

$$\frac{\sigma}{\mu} = 5.99 \pm 6.27 \%$$

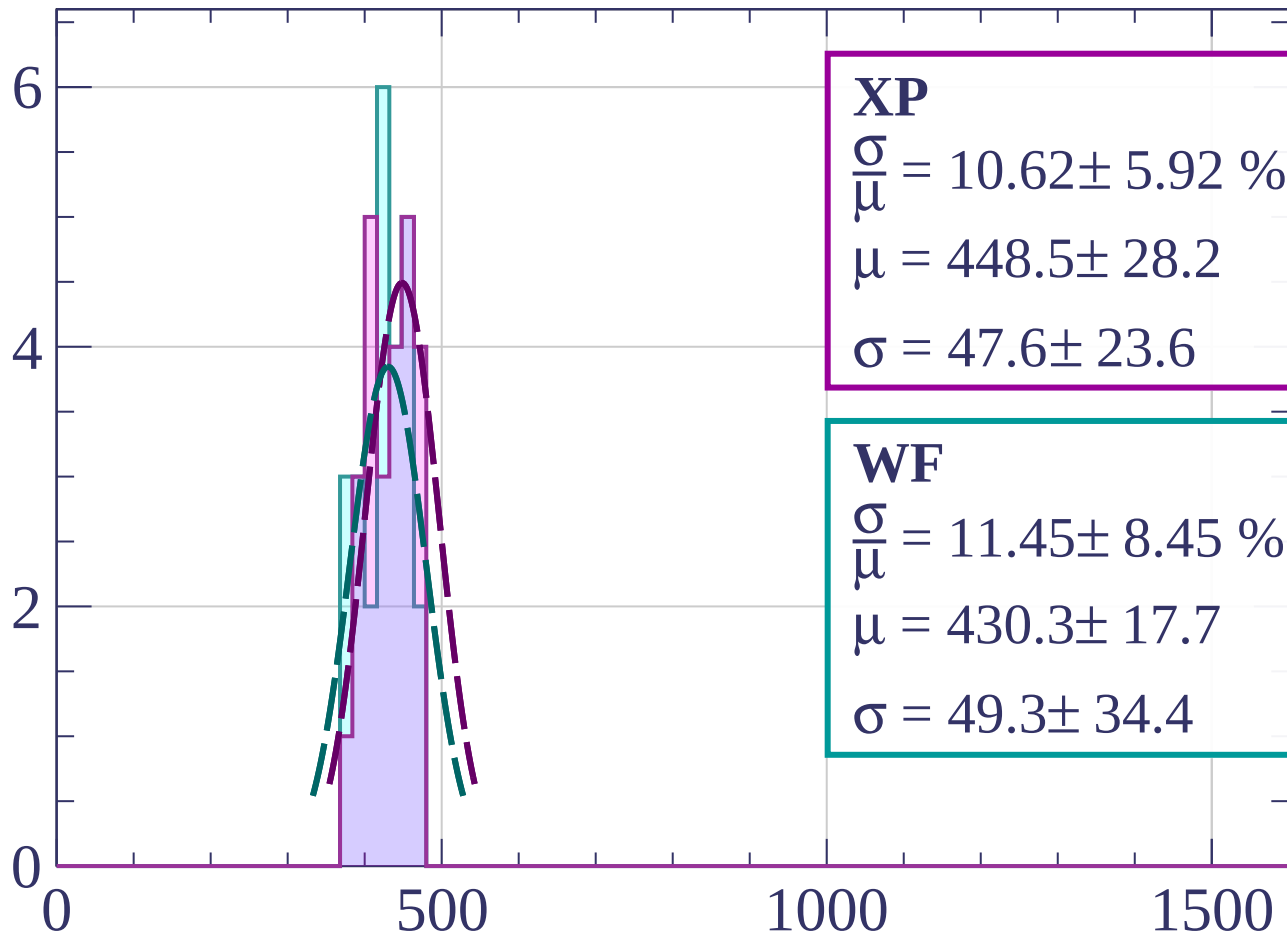
$$\mu = 448.8 \pm 32.1$$

$$\sigma = 26.9 \pm 26.2$$

dE/dx [ADC counts/cm]

Energy loss | $9 < \phi < 12$

Count



XP

$$\frac{\sigma}{\mu} = 10.62 \pm 5.92 \%$$

$$\mu = 448.5 \pm 28.2$$

$$\sigma = 47.6 \pm 23.6$$

WF

$$\frac{\sigma}{\mu} = 11.45 \pm 8.45 \%$$

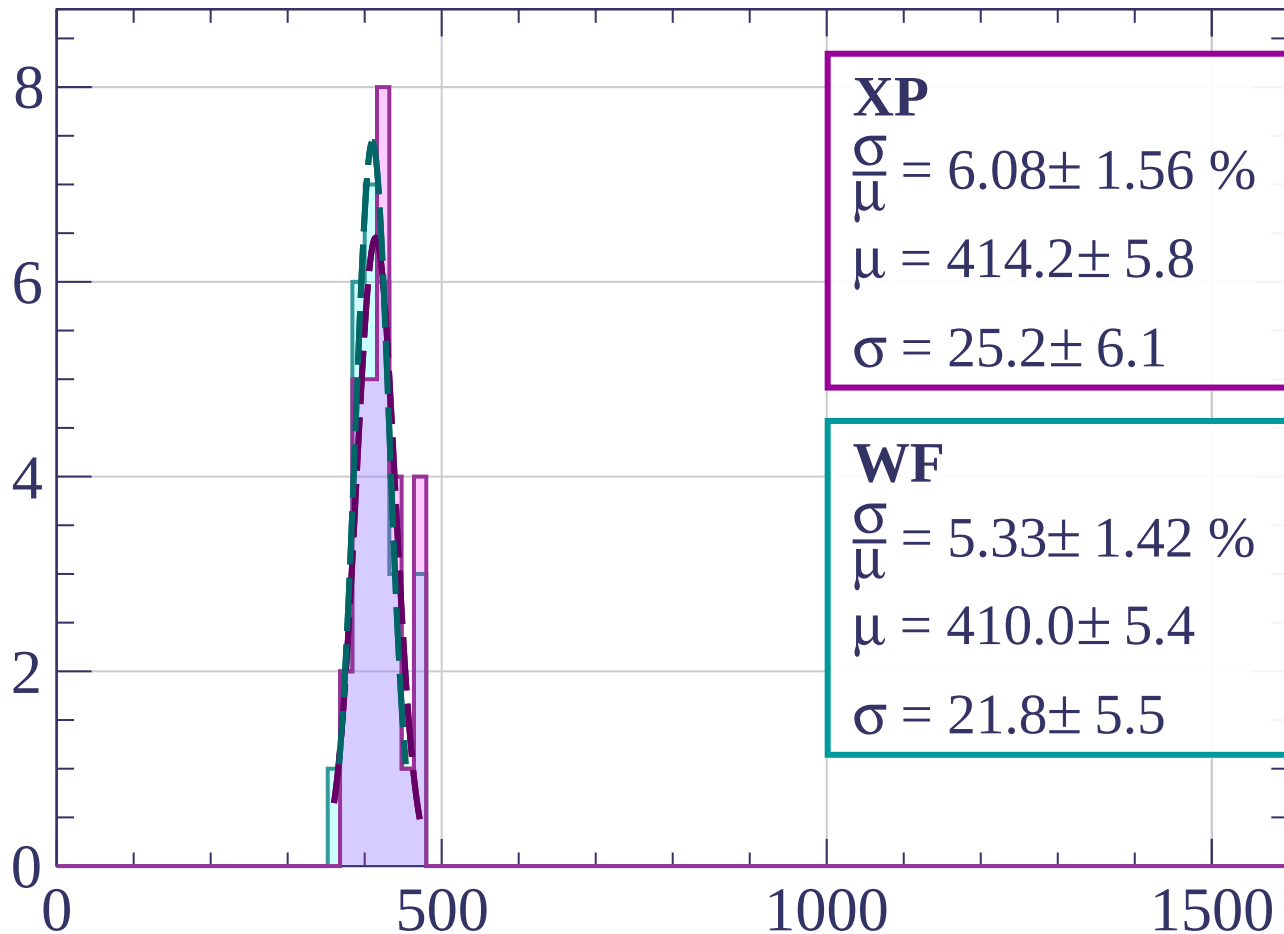
$$\mu = 430.3 \pm 17.7$$

$$\sigma = 49.3 \pm 34.4$$

dE/dx [ADC counts/cm]

Energy loss | $12 < \phi < 15$

Count



XP

$$\frac{\sigma}{\mu} = 6.08 \pm 1.56 \%$$

$$\mu = 414.2 \pm 5.8$$

$$\sigma = 25.2 \pm 6.1$$

WF

$$\frac{\sigma}{\mu} = 5.33 \pm 1.42 \%$$

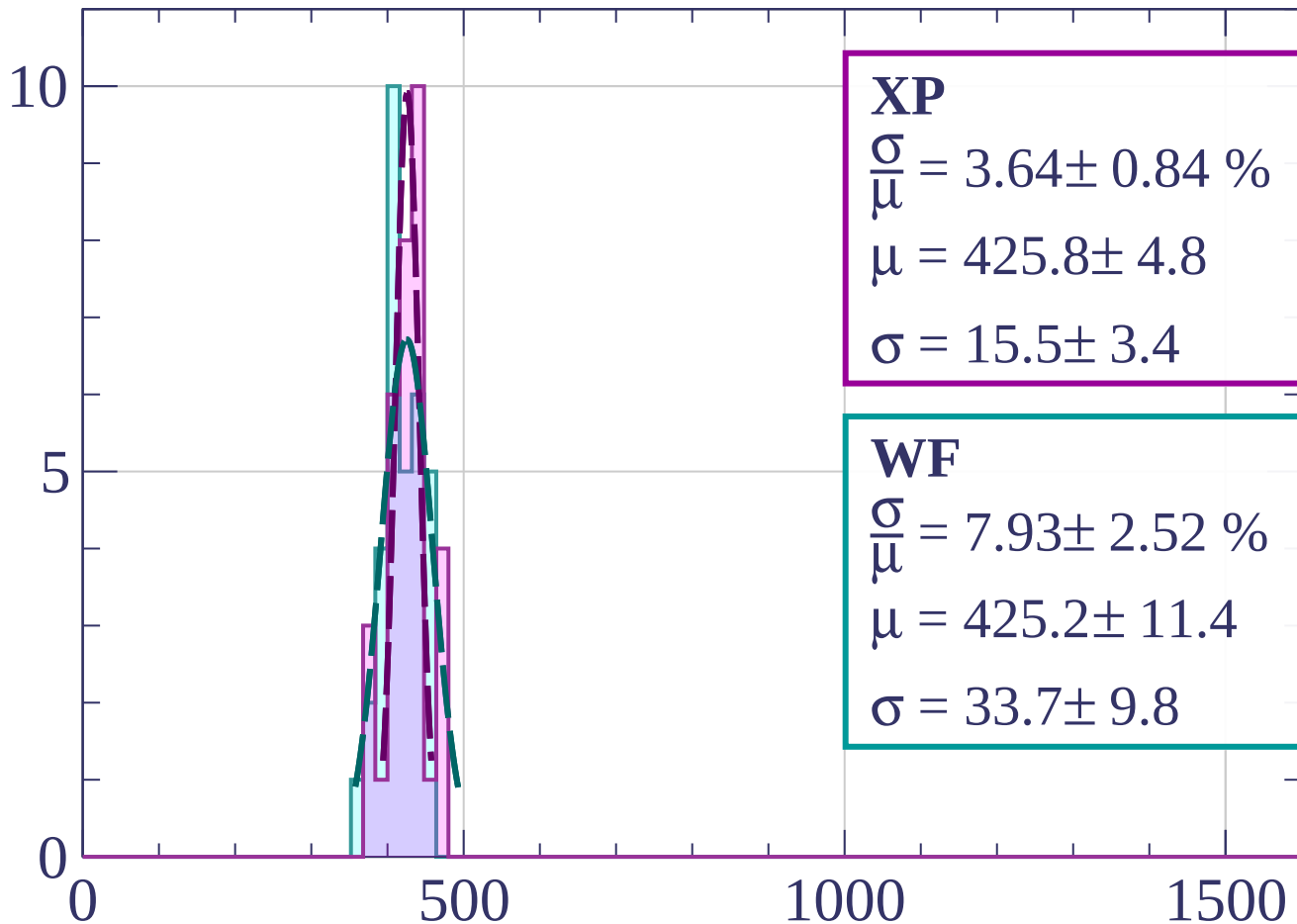
$$\mu = 410.0 \pm 5.4$$

$$\sigma = 21.8 \pm 5.5$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \phi < 18$

Count



XP

$$\frac{\sigma}{\mu} = 3.64 \pm 0.84 \%$$

$$\mu = 425.8 \pm 4.8$$

$$\sigma = 15.5 \pm 3.4$$

WF

$$\frac{\sigma}{\mu} = 7.93 \pm 2.52 \%$$

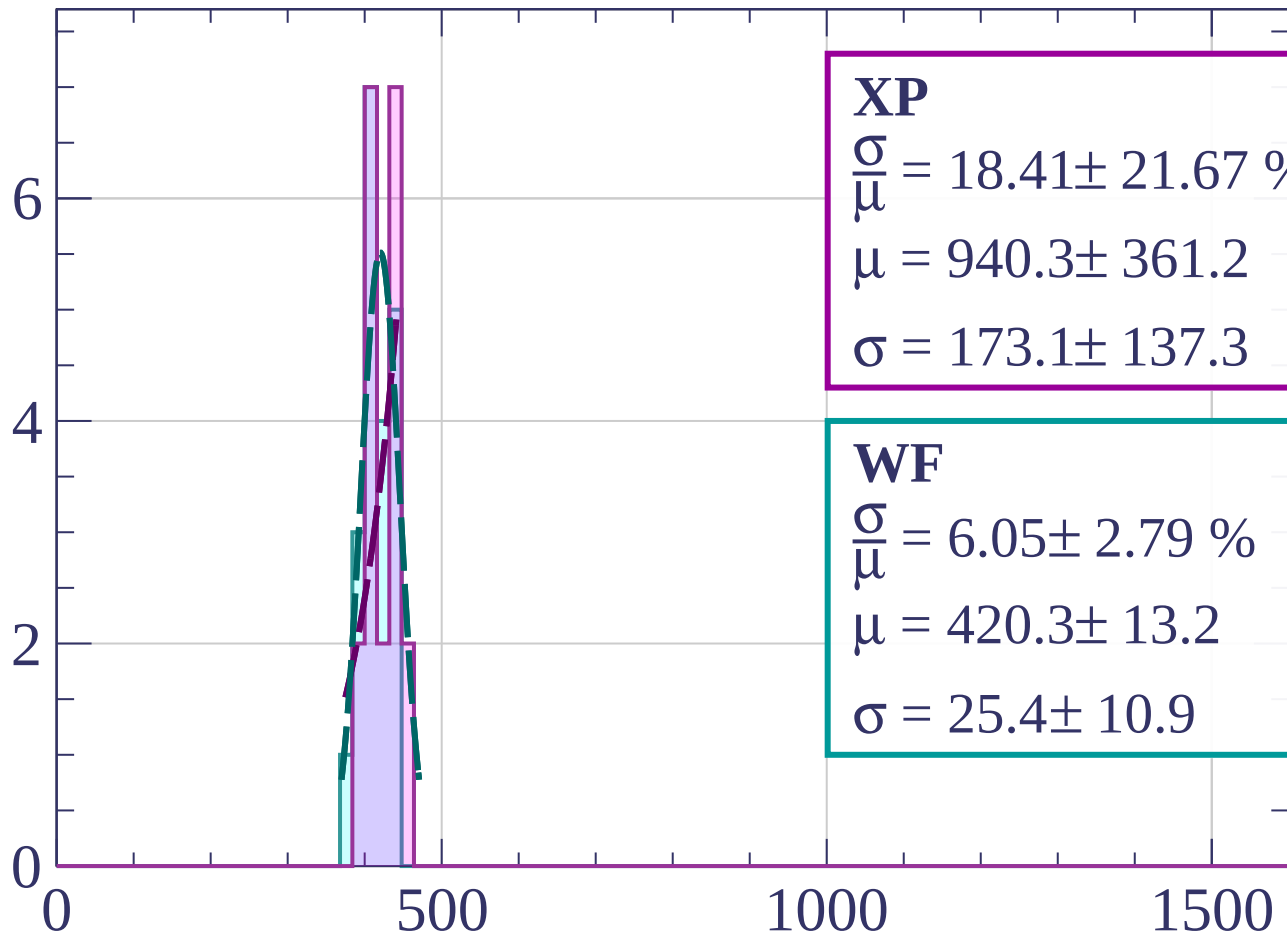
$$\mu = 425.2 \pm 11.4$$

$$\sigma = 33.7 \pm 9.8$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \phi < 21$

Count



XP

$$\frac{\sigma}{\mu} = 18.41 \pm 21.67 \%$$

$$\mu = 940.3 \pm 361.2$$

$$\sigma = 173.1 \pm 137.3$$

WF

$$\frac{\sigma}{\mu} = 6.05 \pm 2.79 \%$$

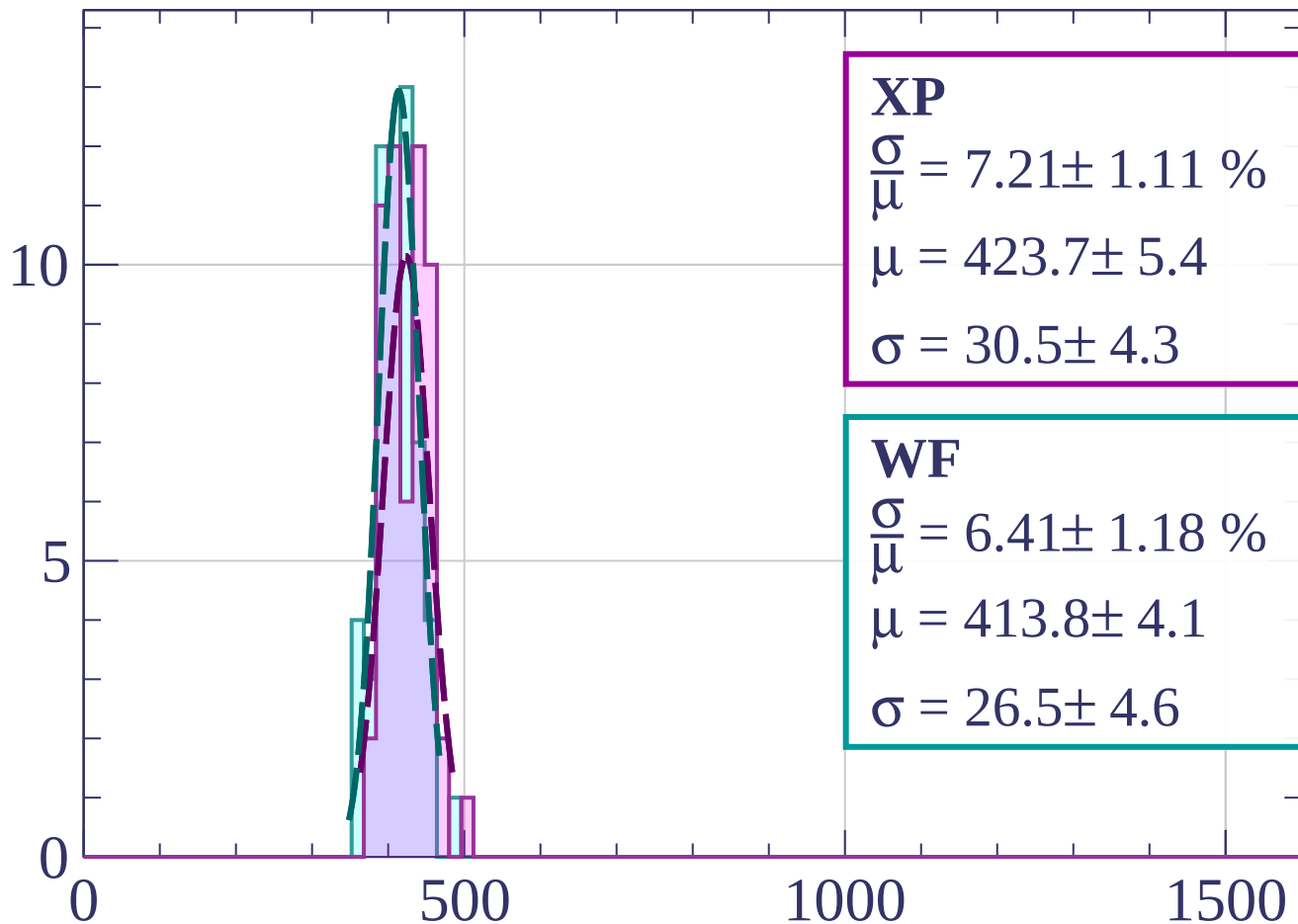
$$\mu = 420.3 \pm 13.2$$

$$\sigma = 25.4 \pm 10.9$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 7.21 \pm 1.11 \%$$

$$\mu = 423.7 \pm 5.4$$

$$\sigma = 30.5 \pm 4.3$$

WF

$$\frac{\sigma}{\mu} = 6.41 \pm 1.18 \%$$

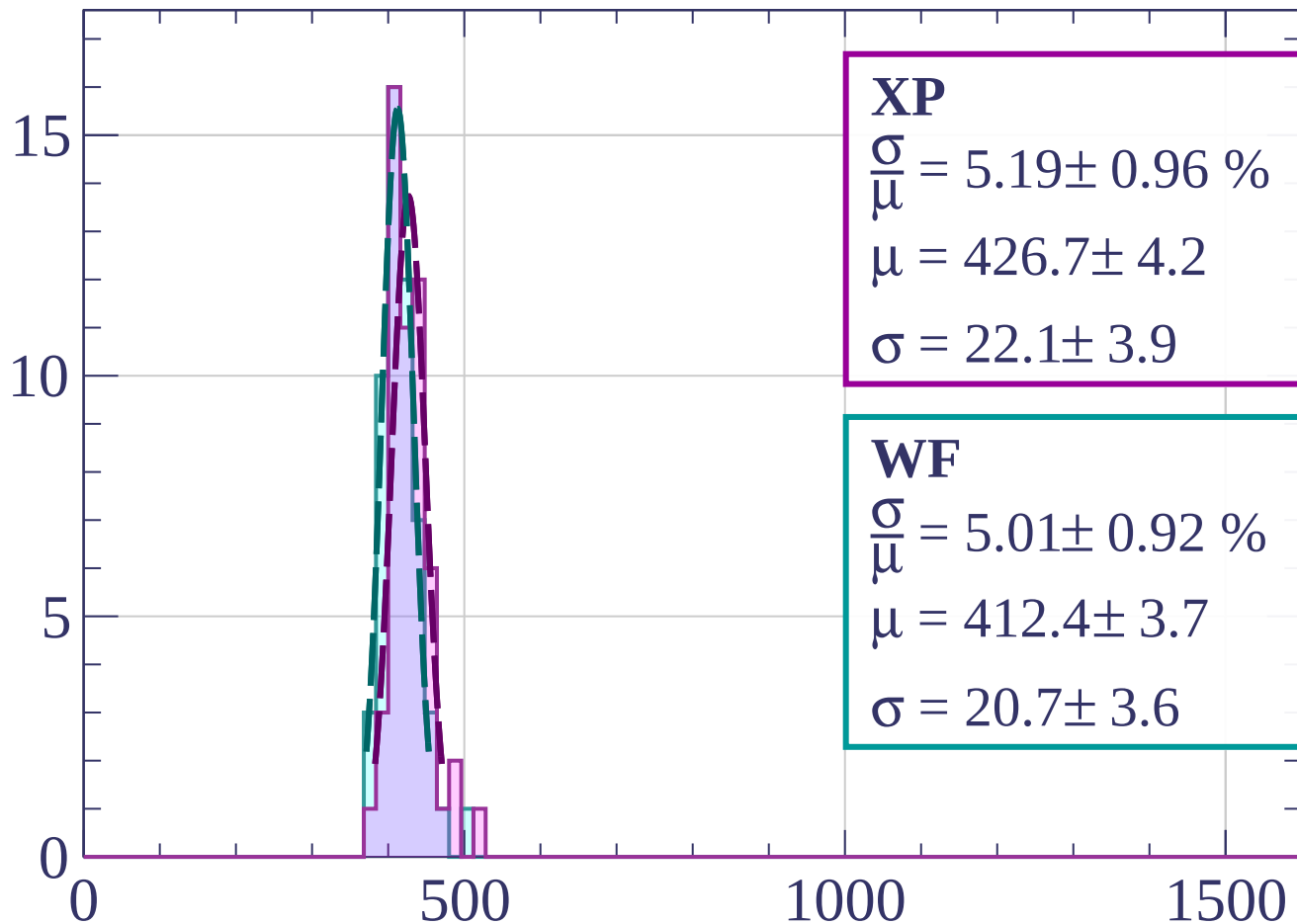
$$\mu = 413.8 \pm 4.1$$

$$\sigma = 26.5 \pm 4.6$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \phi < 27$

Count



XP

$$\frac{\sigma}{\mu} = 5.19 \pm 0.96 \%$$

$$\mu = 426.7 \pm 4.2$$

$$\sigma = 22.1 \pm 3.9$$

WF

$$\frac{\sigma}{\mu} = 5.01 \pm 0.92 \%$$

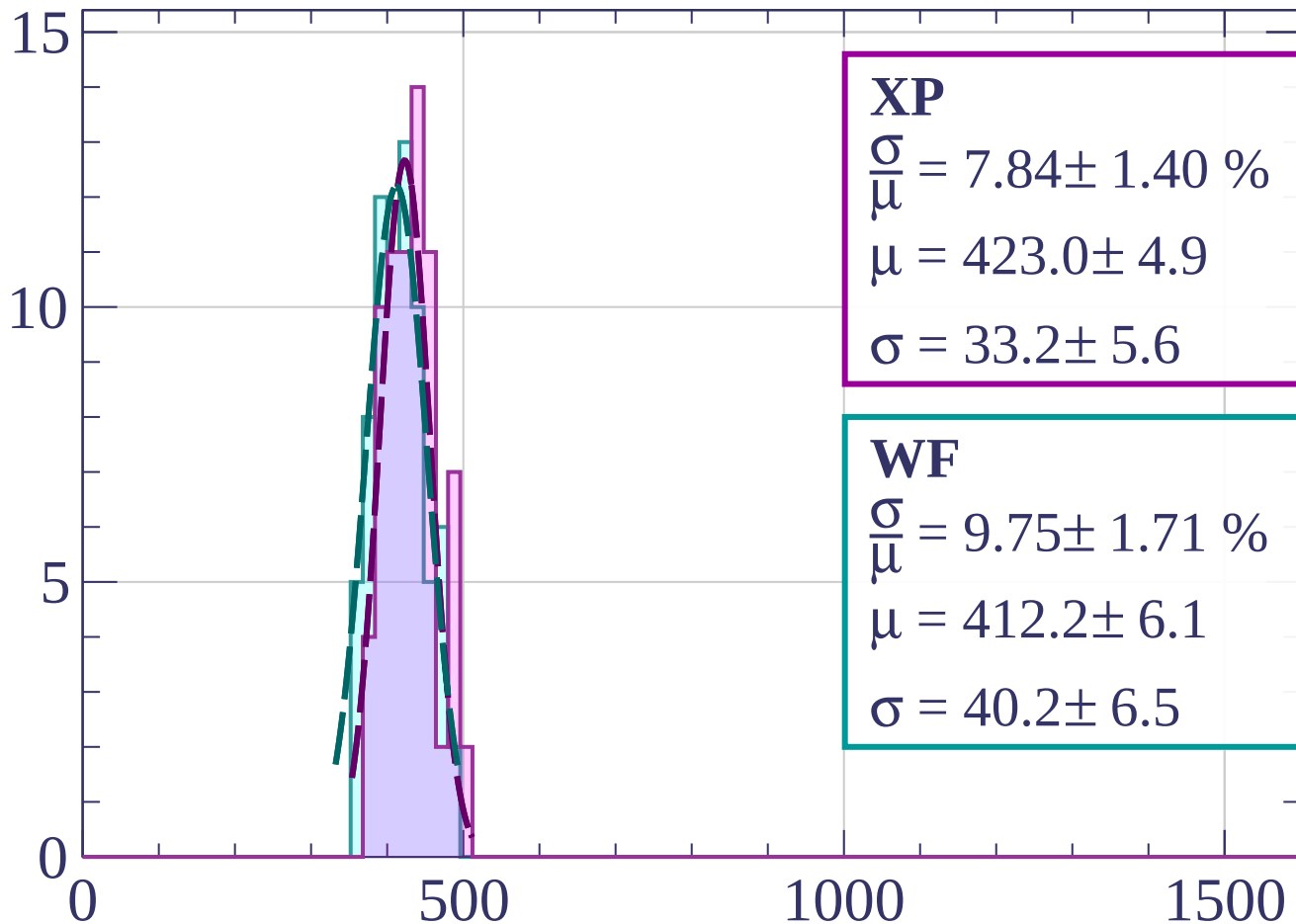
$$\mu = 412.4 \pm 3.7$$

$$\sigma = 20.7 \pm 3.6$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \phi < 30$

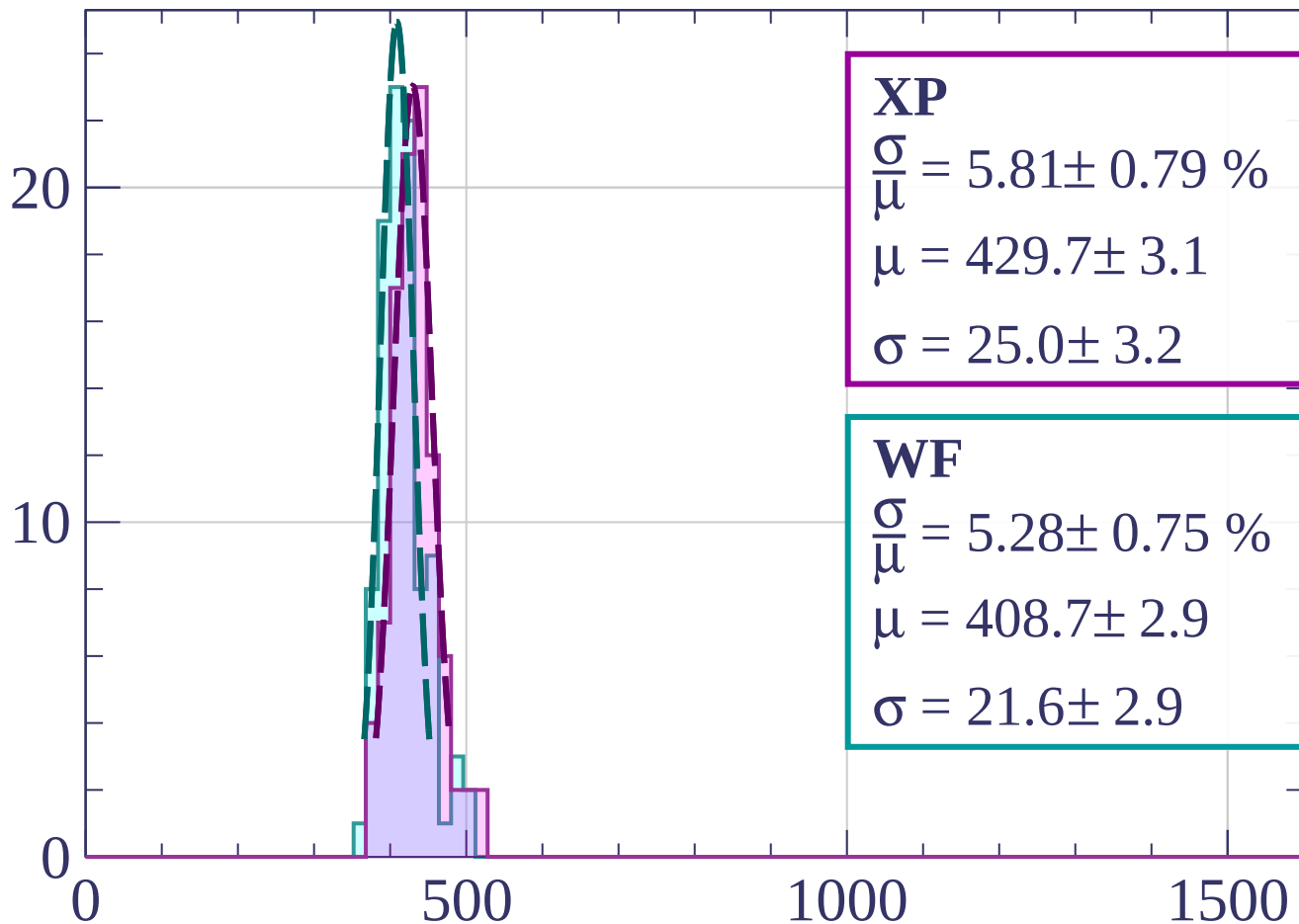
Count



dE/dx [ADC counts/cm]

Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 5.81 \pm 0.79 \%$$

$$\mu = 429.7 \pm 3.1$$

$$\sigma = 25.0 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 5.28 \pm 0.75 \%$$

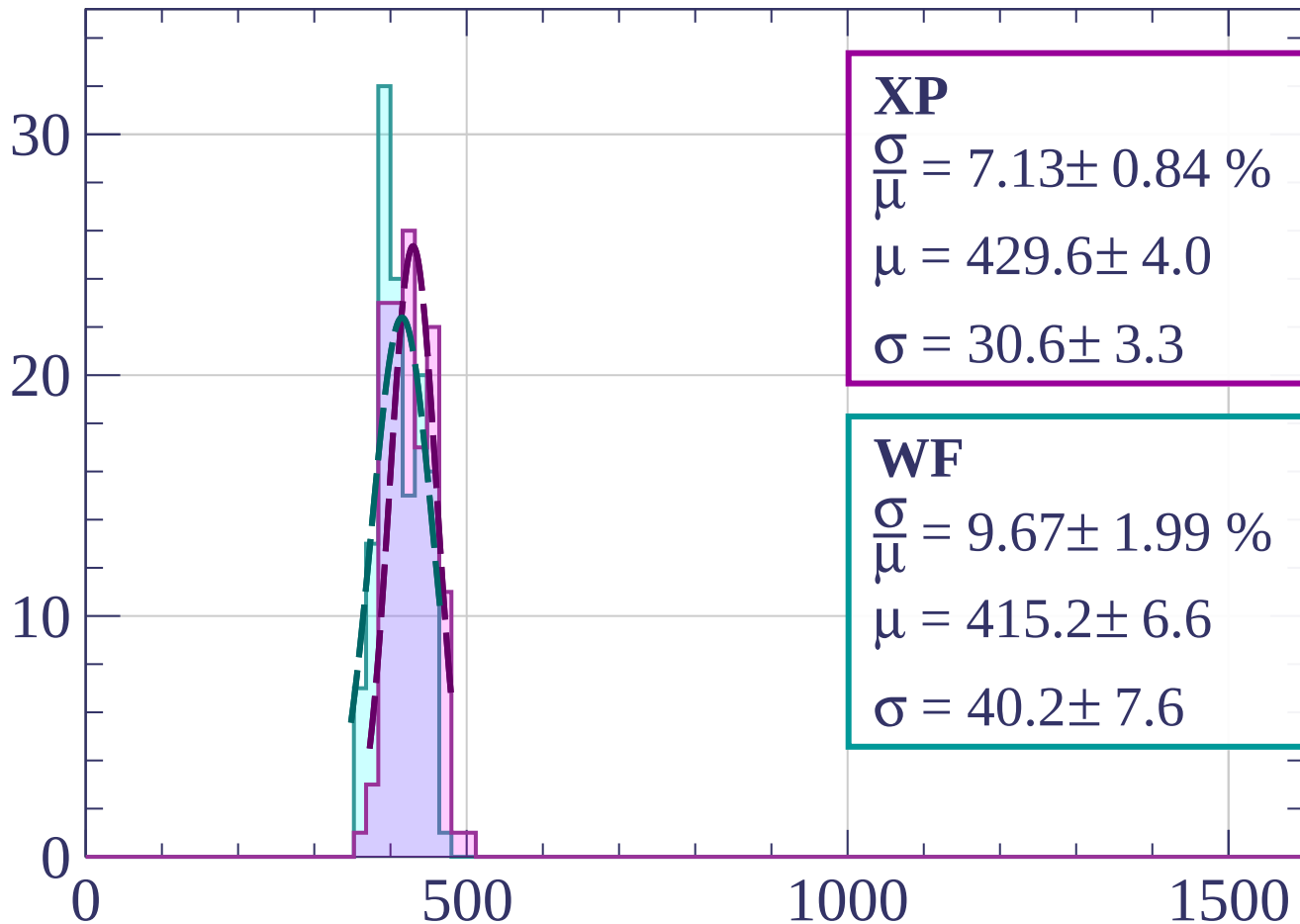
$$\mu = 408.7 \pm 2.9$$

$$\sigma = 21.6 \pm 2.9$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \phi < 36$

Count



XP

$$\frac{\sigma}{\mu} = 7.13 \pm 0.84 \%$$

$$\mu = 429.6 \pm 4.0$$

$$\sigma = 30.6 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 9.67 \pm 1.99 \%$$

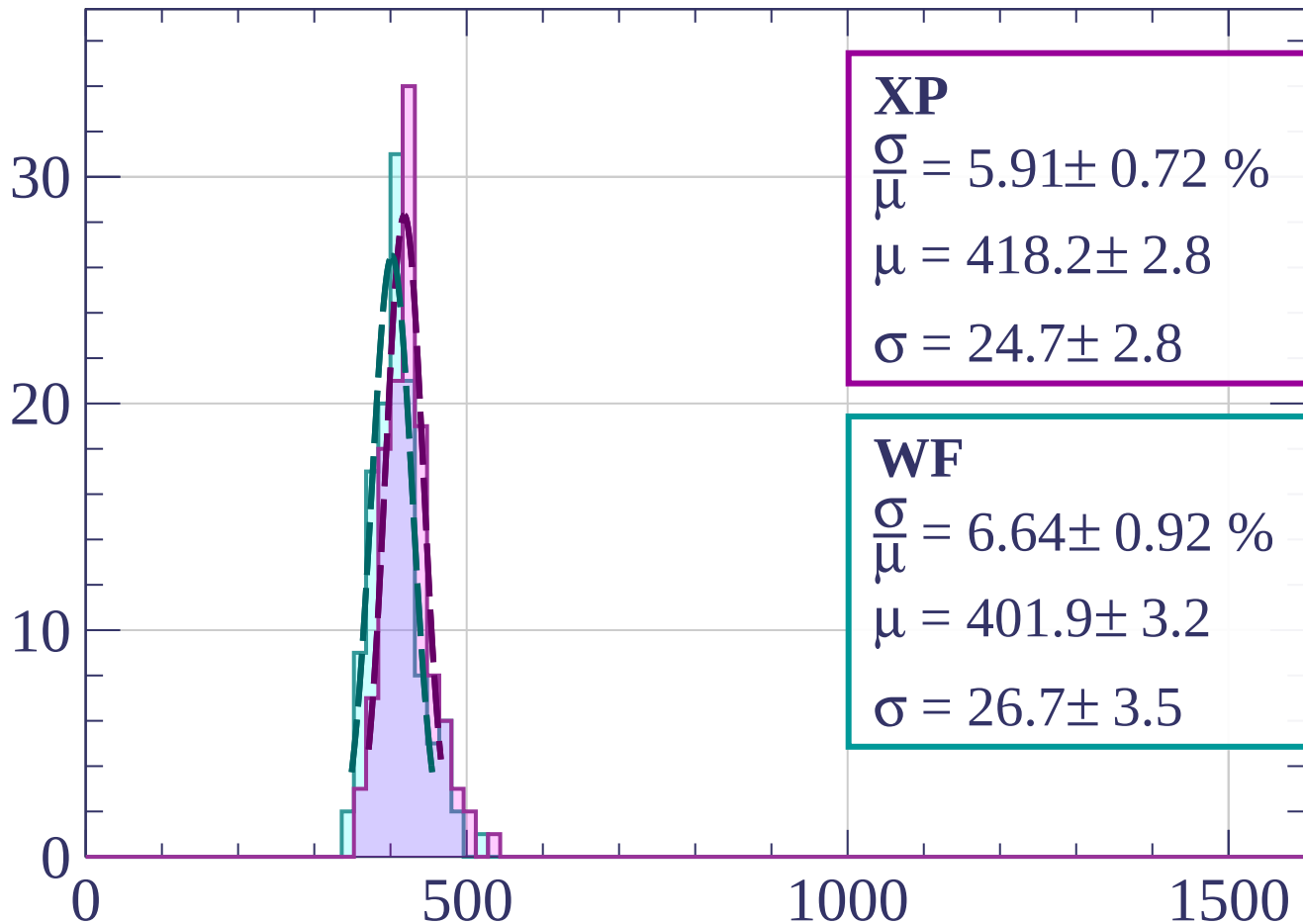
$$\mu = 415.2 \pm 6.6$$

$$\sigma = 40.2 \pm 7.6$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \phi < 39$

Count



XP

$$\frac{\sigma}{\mu} = 5.91 \pm 0.72 \%$$

$$\mu = 418.2 \pm 2.8$$

$$\sigma = 24.7 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 6.64 \pm 0.92 \%$$

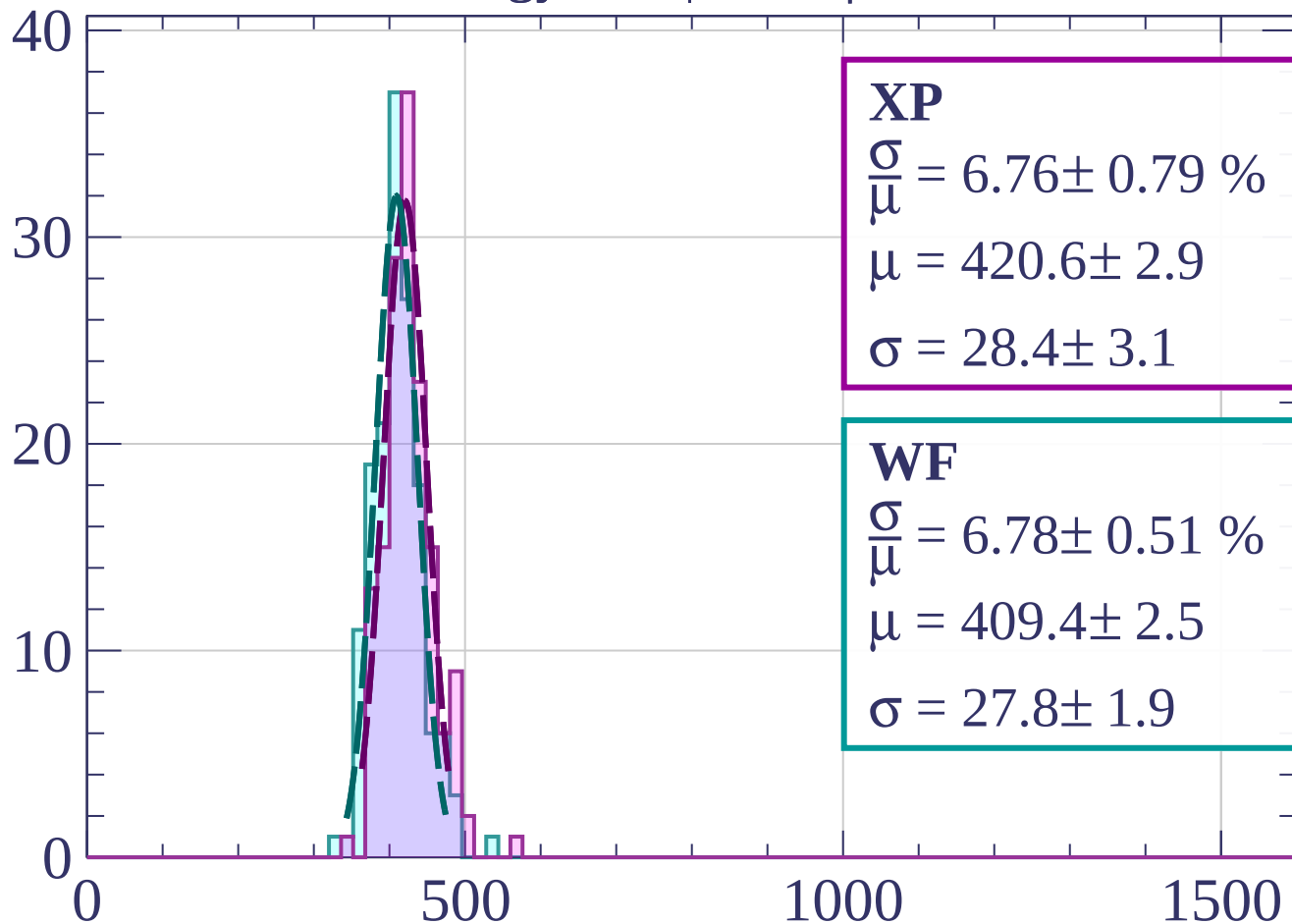
$$\mu = 401.9 \pm 3.2$$

$$\sigma = 26.7 \pm 3.5$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \phi < 42$

Count



XP

$$\frac{\sigma}{\mu} = 6.76 \pm 0.79 \%$$

$$\mu = 420.6 \pm 2.9$$

$$\sigma = 28.4 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = 6.78 \pm 0.51 \%$$

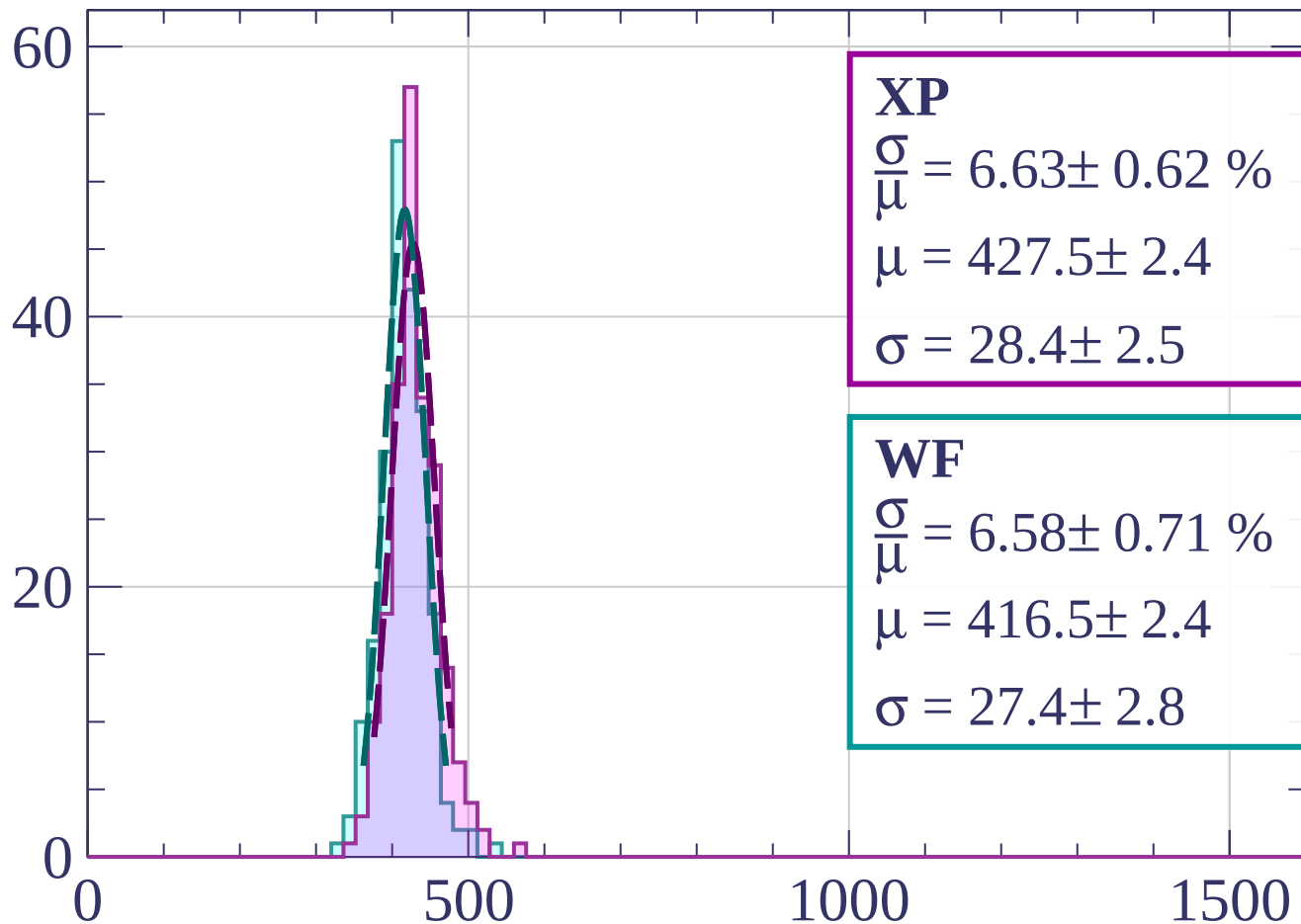
$$\mu = 409.4 \pm 2.5$$

$$\sigma = 27.8 \pm 1.9$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \phi < 45$

Count



XP

$$\frac{\sigma}{\mu} = 6.63 \pm 0.62 \%$$

$$\mu = 427.5 \pm 2.4$$

$$\sigma = 28.4 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 6.58 \pm 0.71 \%$$

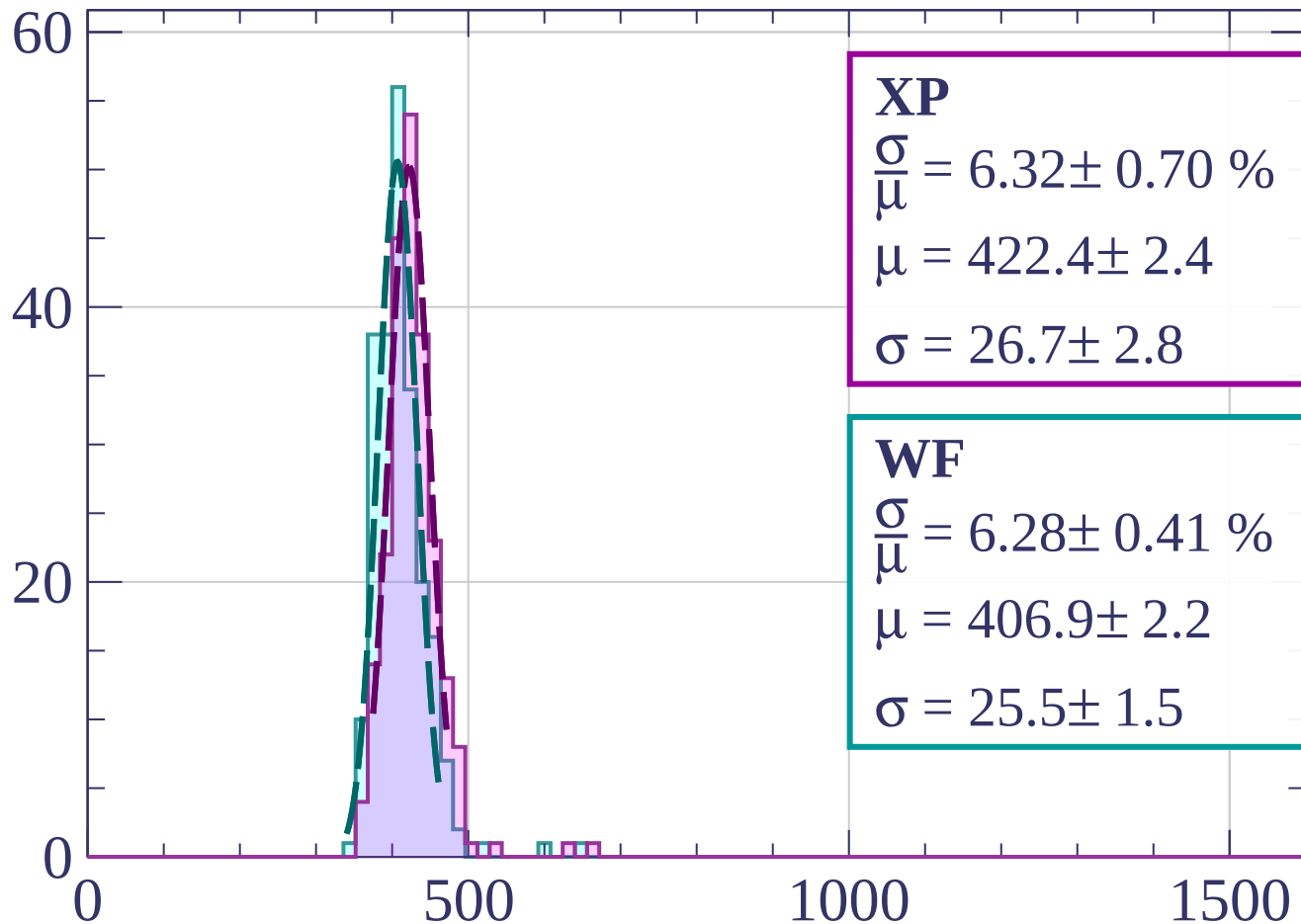
$$\mu = 416.5 \pm 2.4$$

$$\sigma = 27.4 \pm 2.8$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 6.32 \pm 0.70 \%$$

$$\mu = 422.4 \pm 2.4$$

$$\sigma = 26.7 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 6.28 \pm 0.41 \%$$

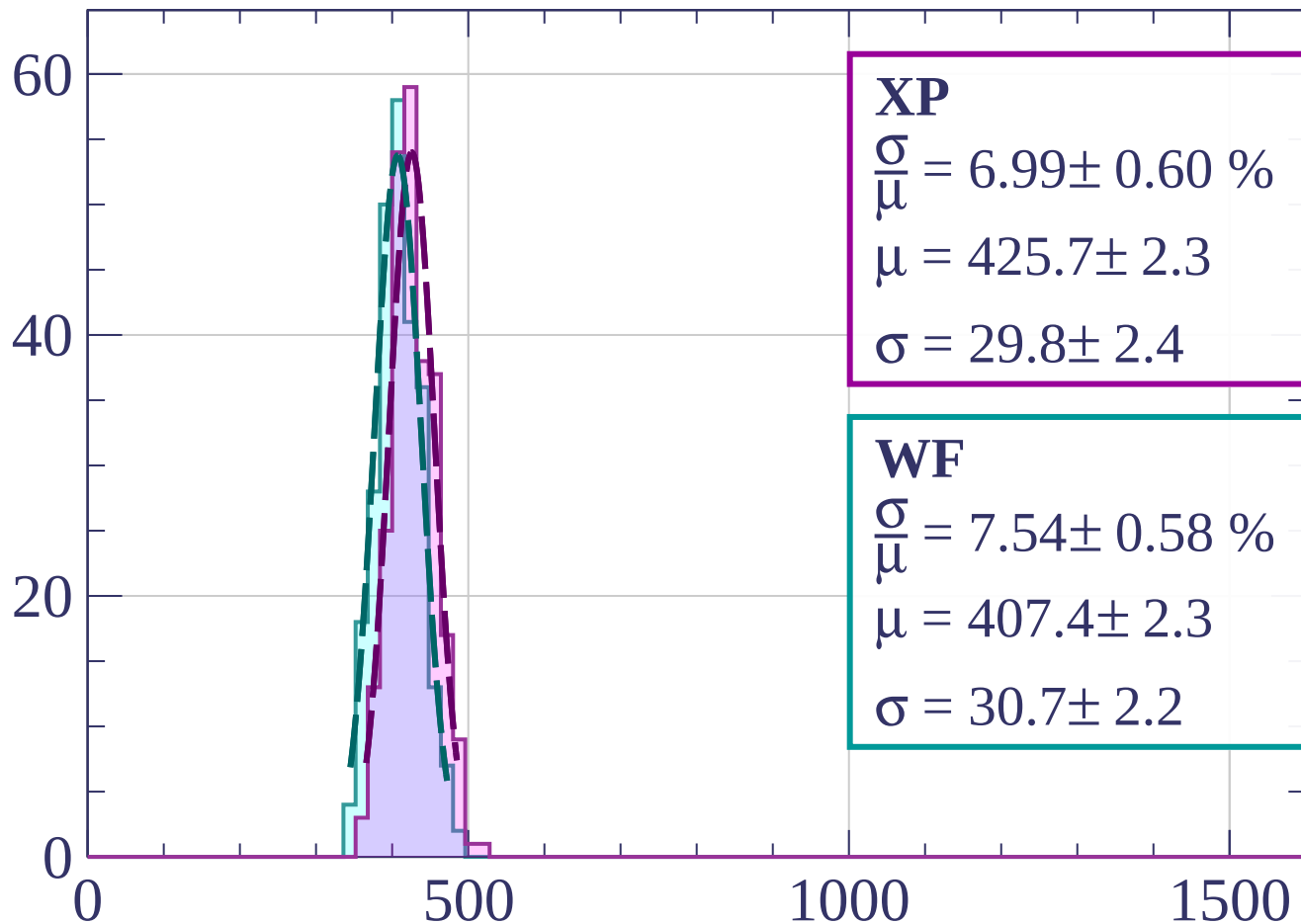
$$\mu = 406.9 \pm 2.2$$

$$\sigma = 25.5 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $48 < \phi < 51$

Count



XP

$$\frac{\sigma}{\mu} = 6.99 \pm 0.60 \%$$

$$\mu = 425.7 \pm 2.3$$

$$\sigma = 29.8 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 7.54 \pm 0.58 \%$$

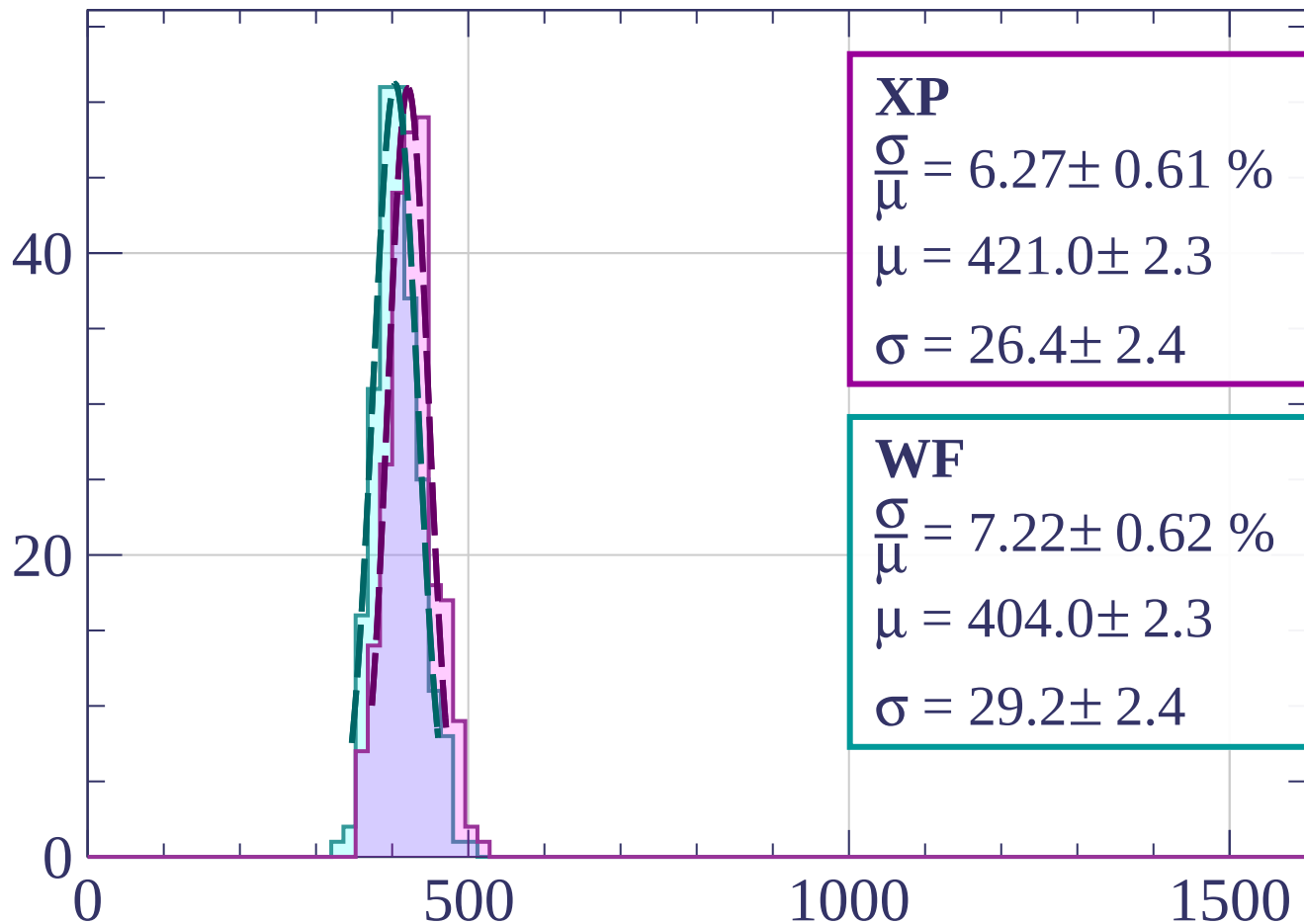
$$\mu = 407.4 \pm 2.3$$

$$\sigma = 30.7 \pm 2.2$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 6.27 \pm 0.61 \%$$

$$\mu = 421.0 \pm 2.3$$

$$\sigma = 26.4 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 7.22 \pm 0.62 \%$$

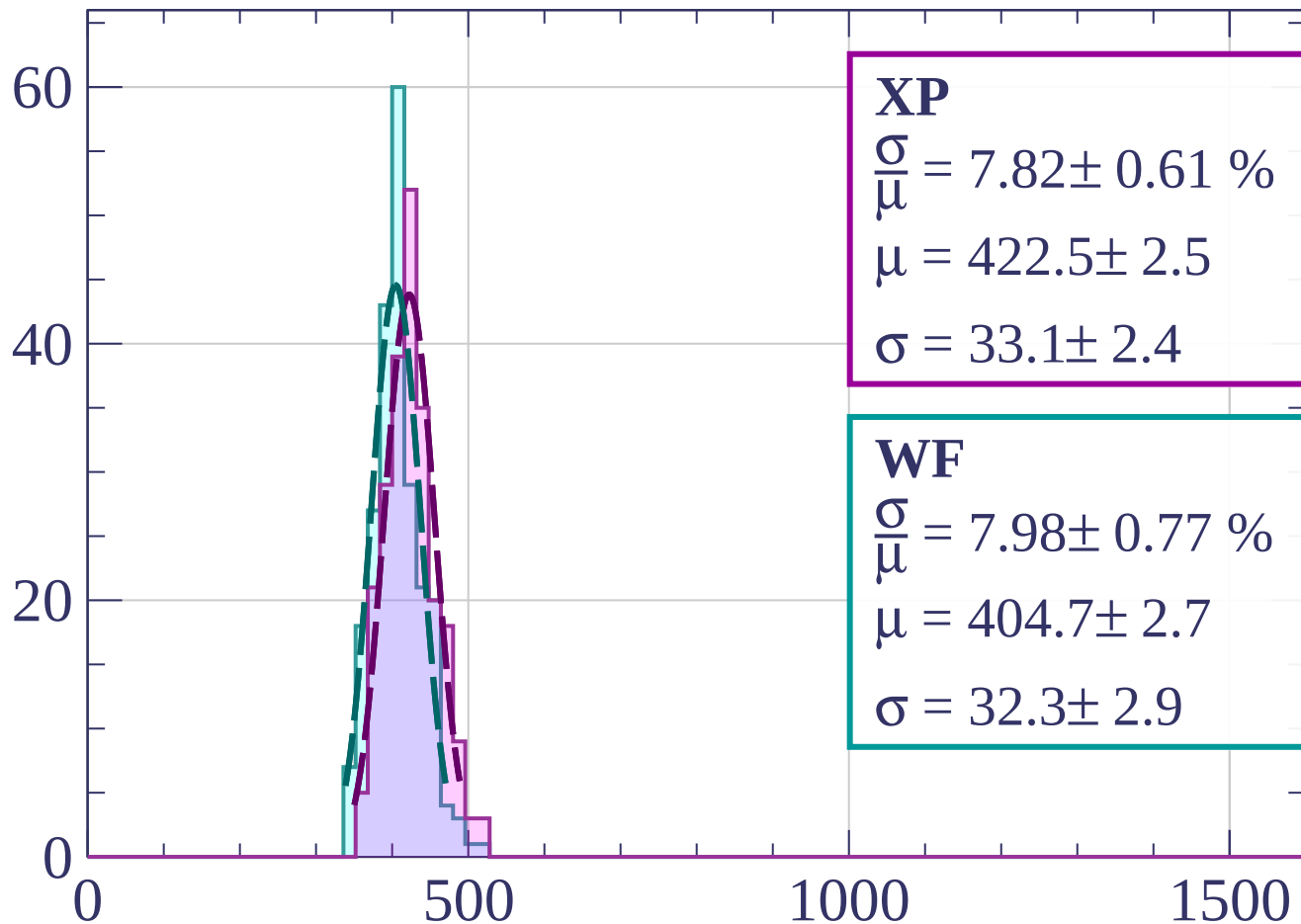
$$\mu = 404.0 \pm 2.3$$

$$\sigma = 29.2 \pm 2.4$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \phi < 57$

Count



XP

$$\frac{\sigma}{\mu} = 7.82 \pm 0.61 \%$$

$$\mu = 422.5 \pm 2.5$$

$$\sigma = 33.1 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 7.98 \pm 0.77 \%$$

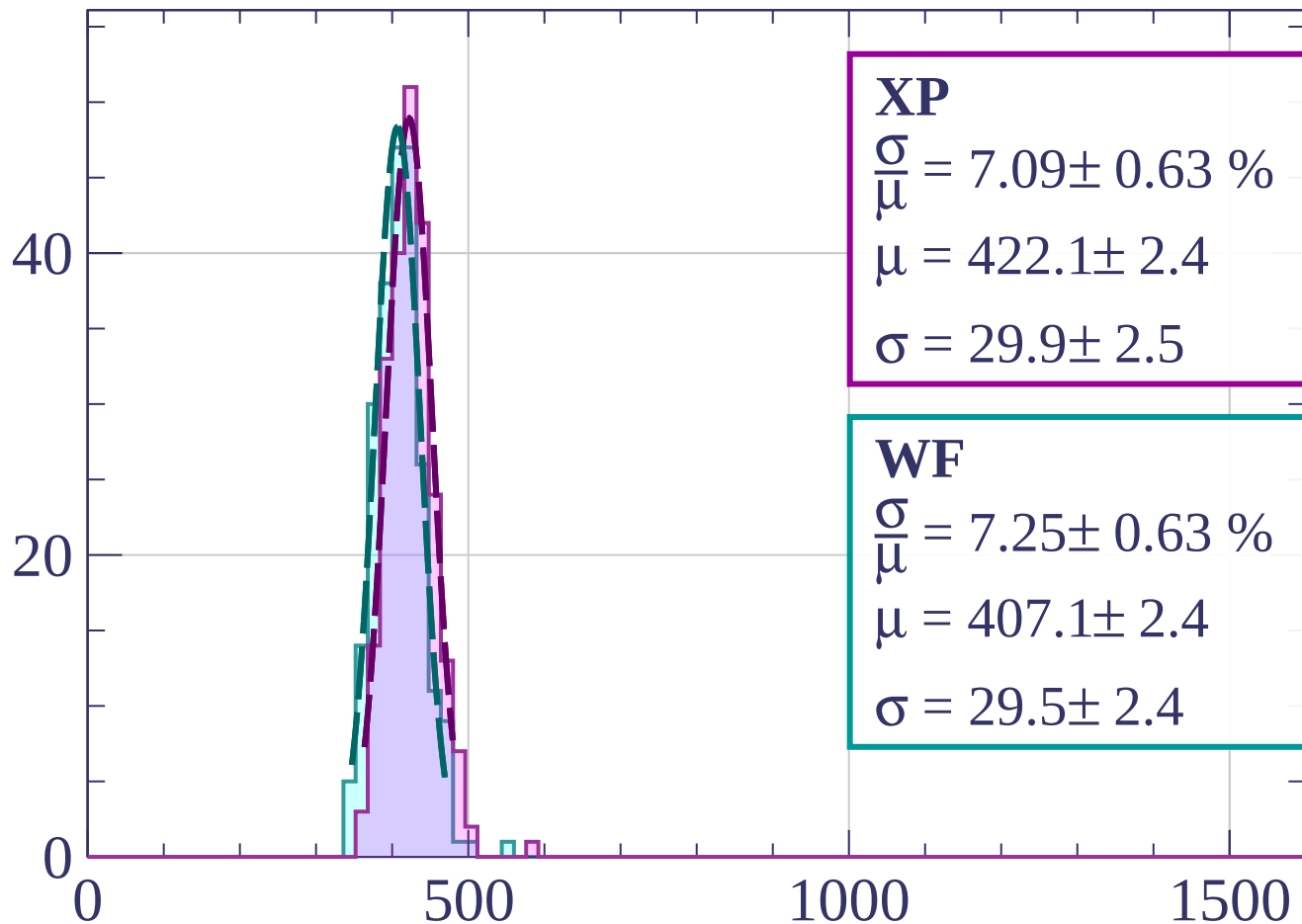
$$\mu = 404.7 \pm 2.7$$

$$\sigma = 32.3 \pm 2.9$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \phi < 60$

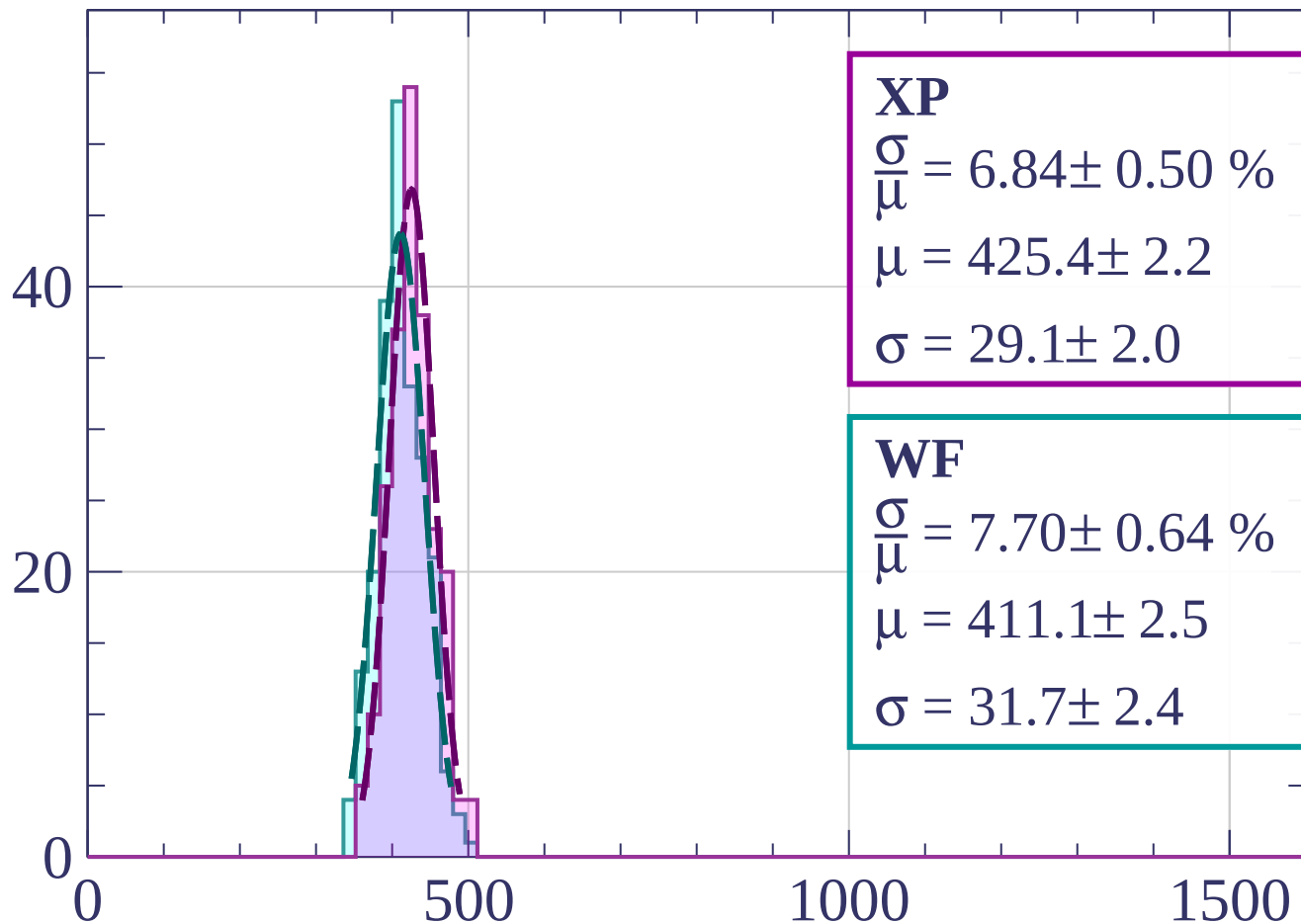
Count



dE/dx [ADC counts/cm]

Energy loss | $60 < \phi < 63$

Count



XP

$$\frac{\sigma}{\mu} = 6.84 \pm 0.50 \%$$

$$\mu = 425.4 \pm 2.2$$

$$\sigma = 29.1 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 7.70 \pm 0.64 \%$$

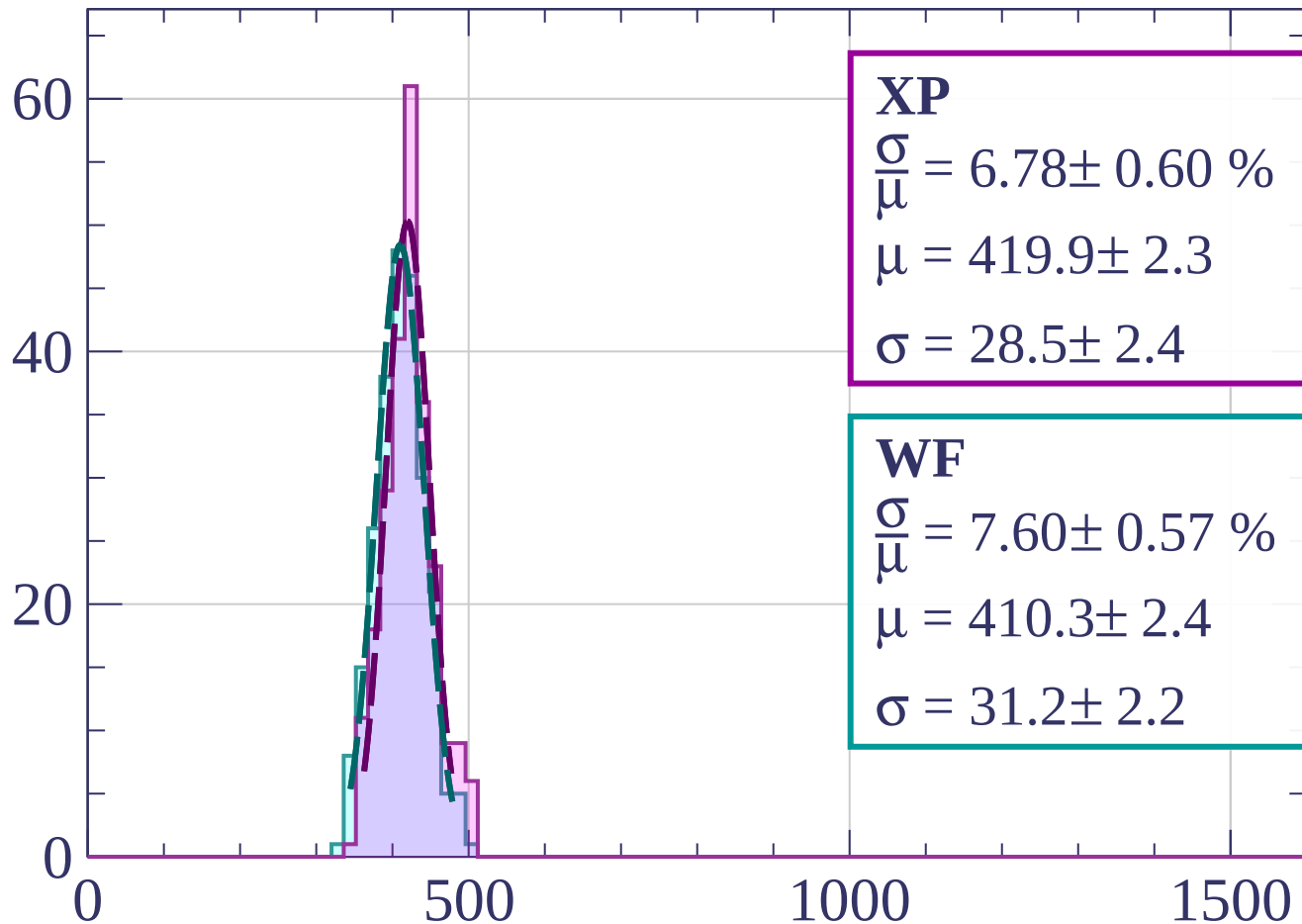
$$\mu = 411.1 \pm 2.5$$

$$\sigma = 31.7 \pm 2.4$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \phi < 66$

Count



XP

$$\frac{\sigma}{\mu} = 6.78 \pm 0.60 \%$$

$$\mu = 419.9 \pm 2.3$$

$$\sigma = 28.5 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 7.60 \pm 0.57 \%$$

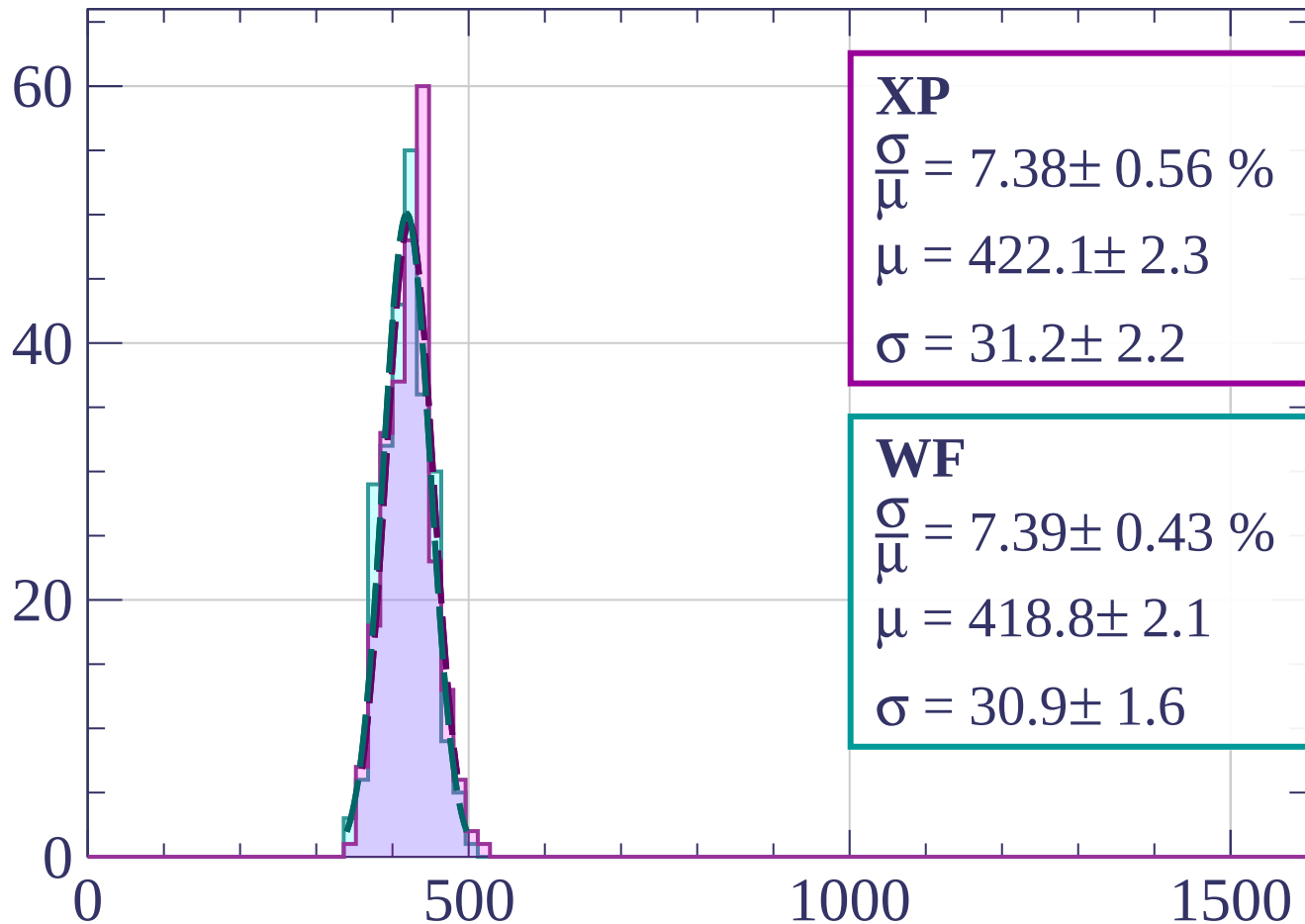
$$\mu = 410.3 \pm 2.4$$

$$\sigma = 31.2 \pm 2.2$$

dE/dx [ADC counts/cm]

Energy loss | $66 < \phi < 69$

Count



XP

$$\frac{\sigma}{\mu} = 7.38 \pm 0.56 \%$$

$$\mu = 422.1 \pm 2.3$$

$$\sigma = 31.2 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 7.39 \pm 0.43 \%$$

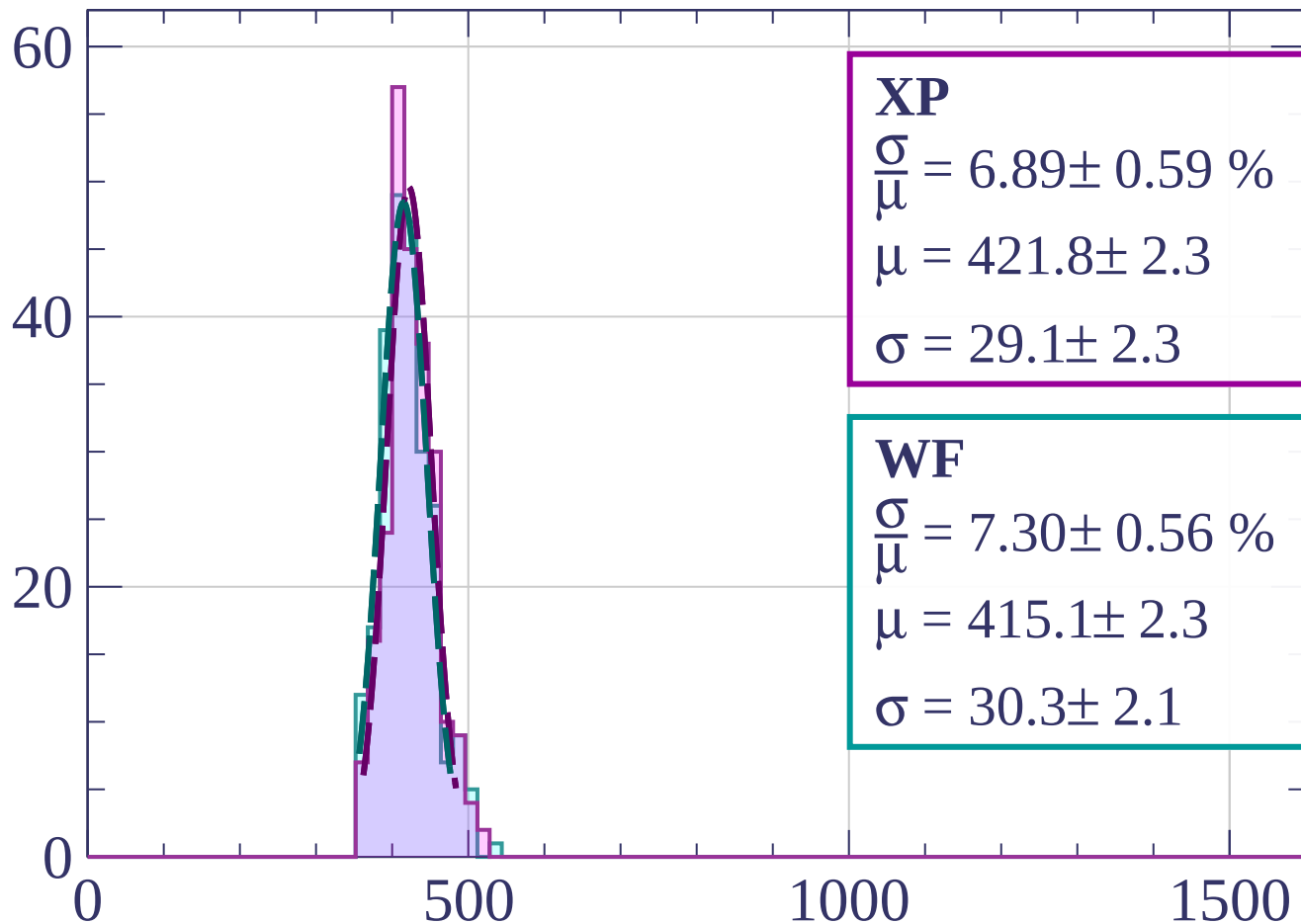
$$\mu = 418.8 \pm 2.1$$

$$\sigma = 30.9 \pm 1.6$$

dE/dx [ADC counts/cm]

Energy loss | $69 < \phi < 72$

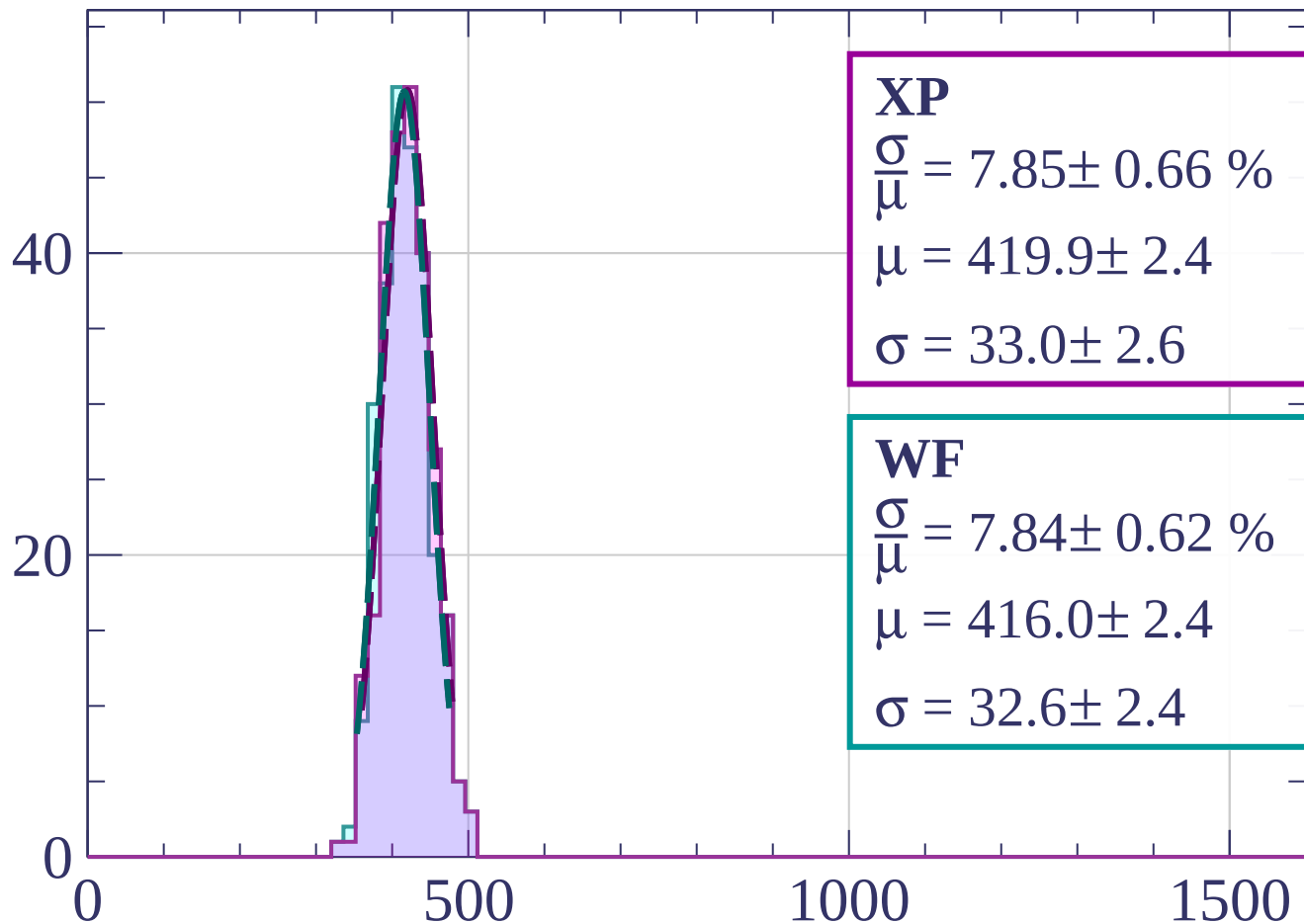
Count



dE/dx [ADC counts/cm]

Energy loss | $72 < \phi < 75$

Count



XP

$$\frac{\sigma}{\mu} = 7.85 \pm 0.66 \%$$

$$\mu = 419.9 \pm 2.4$$

$$\sigma = 33.0 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 7.84 \pm 0.62 \%$$

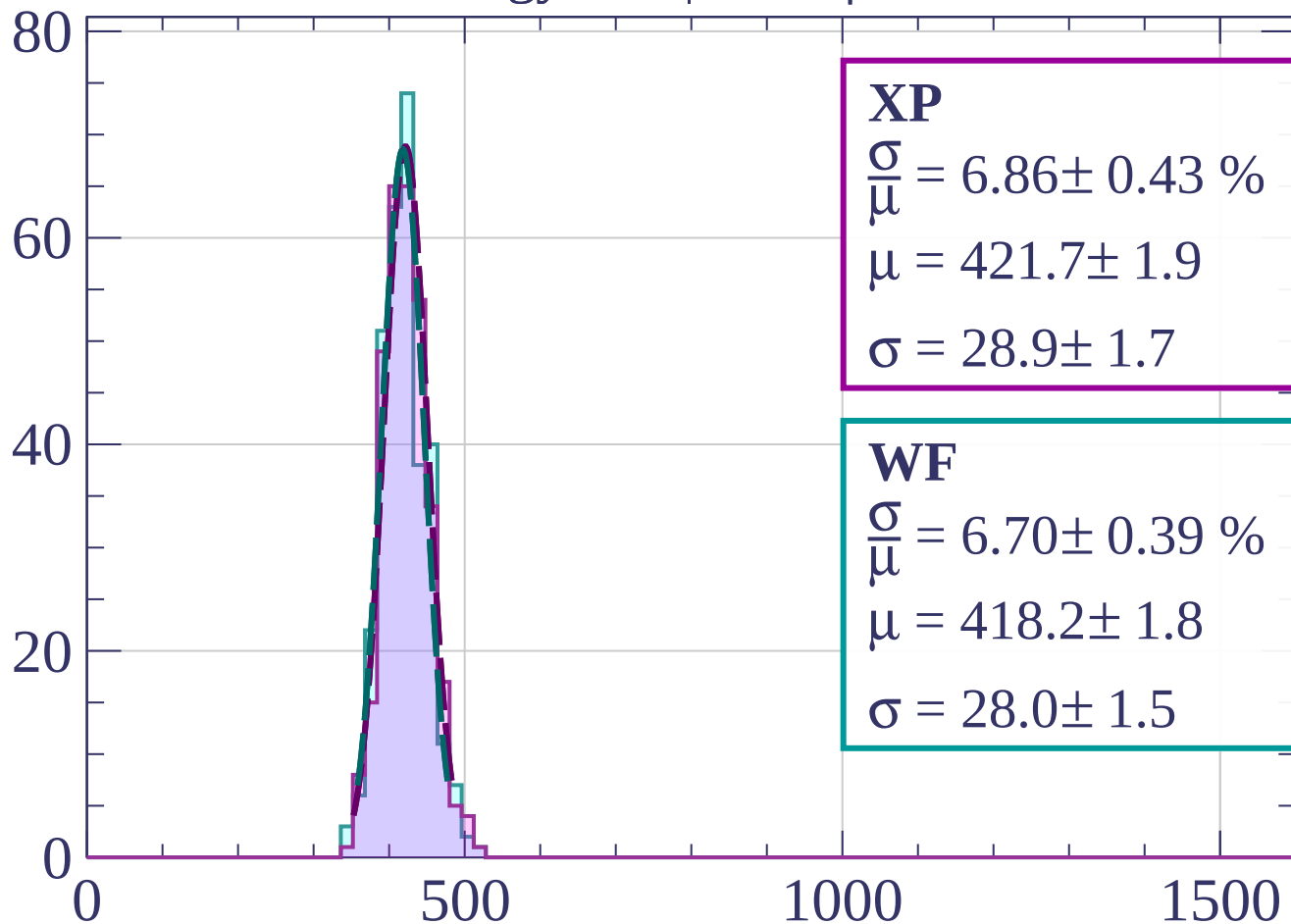
$$\mu = 416.0 \pm 2.4$$

$$\sigma = 32.6 \pm 2.4$$

dE/dx [ADC counts/cm]

Energy loss | $75 < \phi < 78$

Count



XP

$$\frac{\sigma}{\mu} = 6.86 \pm 0.43 \%$$

$$\mu = 421.7 \pm 1.9$$

$$\sigma = 28.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.70 \pm 0.39 \%$$

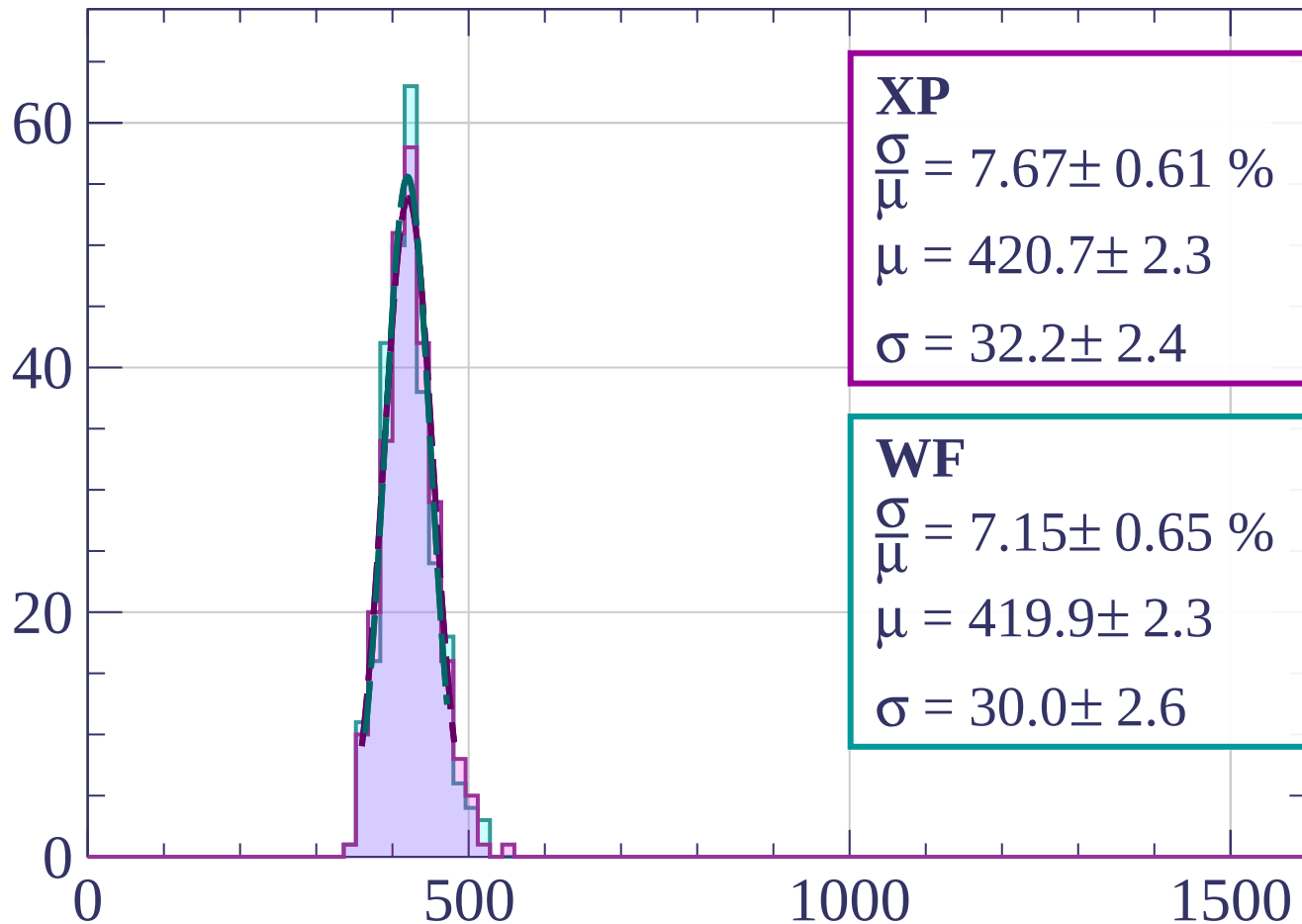
$$\mu = 418.2 \pm 1.8$$

$$\sigma = 28.0 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $78 < \phi < 81$

Count



XP

$$\frac{\sigma}{\mu} = 7.67 \pm 0.61 \%$$

$$\mu = 420.7 \pm 2.3$$

$$\sigma = 32.2 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 7.15 \pm 0.65 \%$$

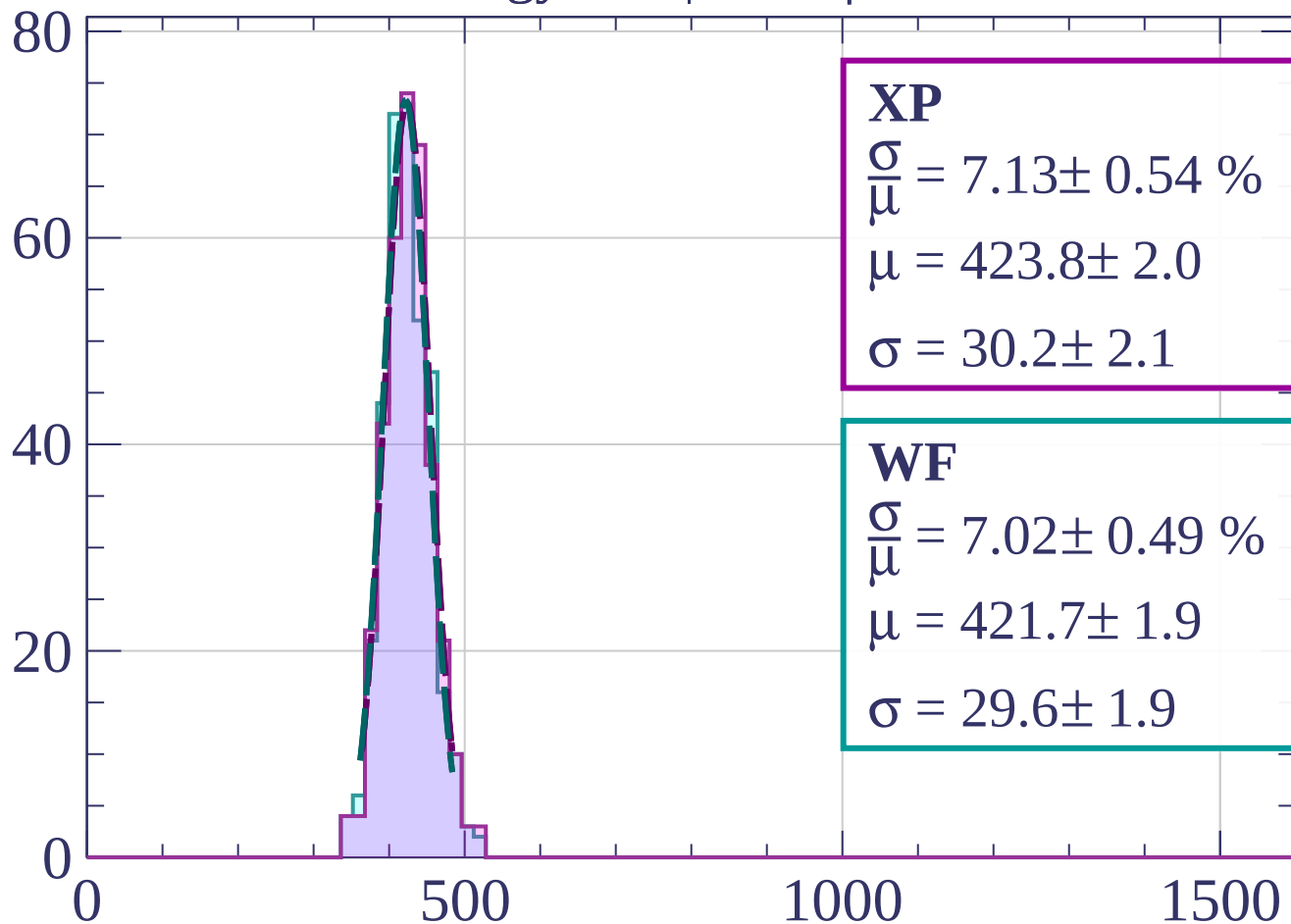
$$\mu = 419.9 \pm 2.3$$

$$\sigma = 30.0 \pm 2.6$$

dE/dx [ADC counts/cm]

Energy loss | $81 < \phi < 84$

Count



XP

$$\frac{\sigma}{\mu} = 7.13 \pm 0.54 \%$$

$$\mu = 423.8 \pm 2.0$$

$$\sigma = 30.2 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 7.02 \pm 0.49 \%$$

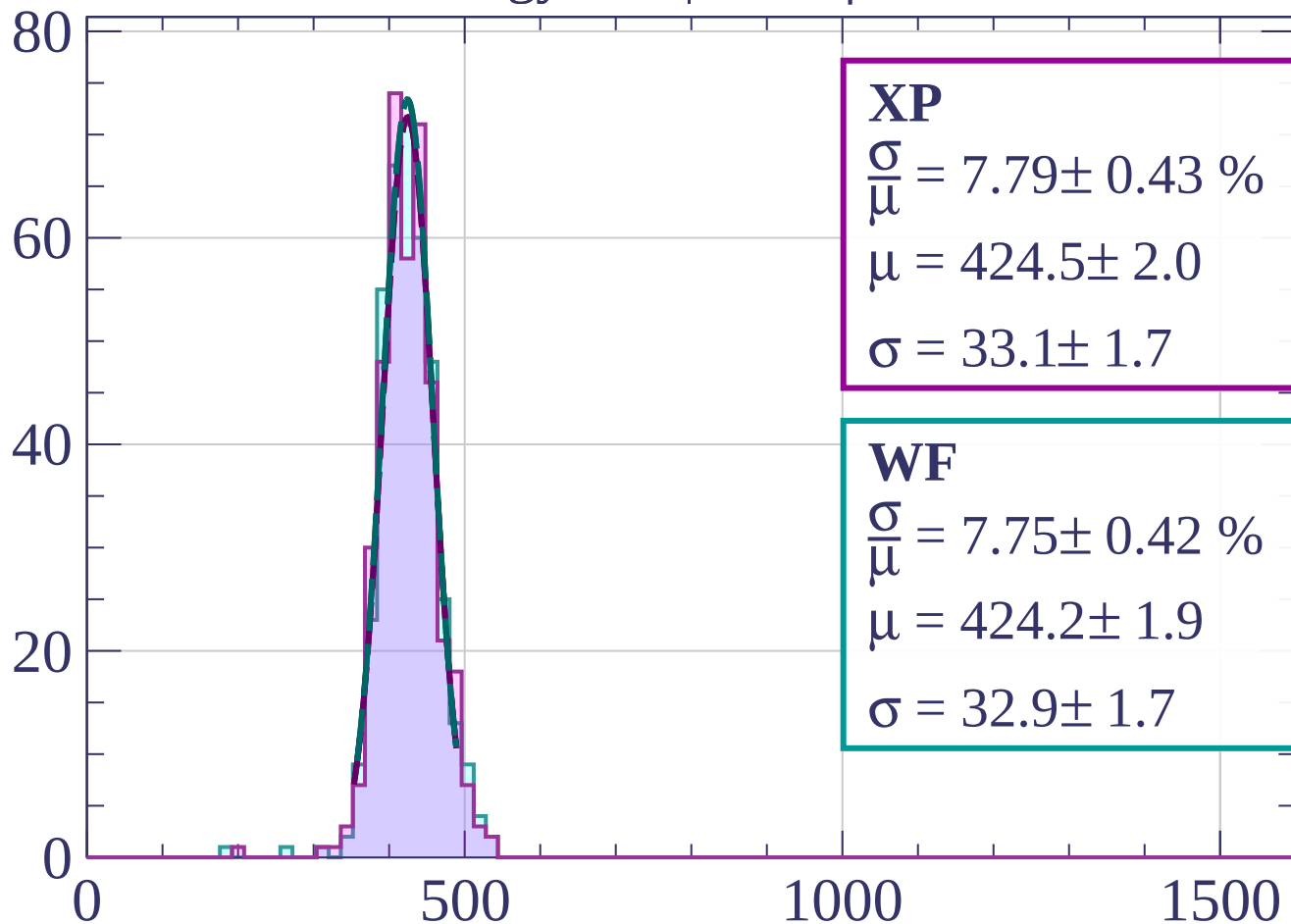
$$\mu = 421.7 \pm 1.9$$

$$\sigma = 29.6 \pm 1.9$$

dE/dx [ADC counts/cm]

Energy loss | $84 < \phi < 87$

Count



XP

$$\frac{\sigma}{\mu} = 7.79 \pm 0.43 \%$$

$$\mu = 424.5 \pm 2.0$$

$$\sigma = 33.1 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.75 \pm 0.42 \%$$

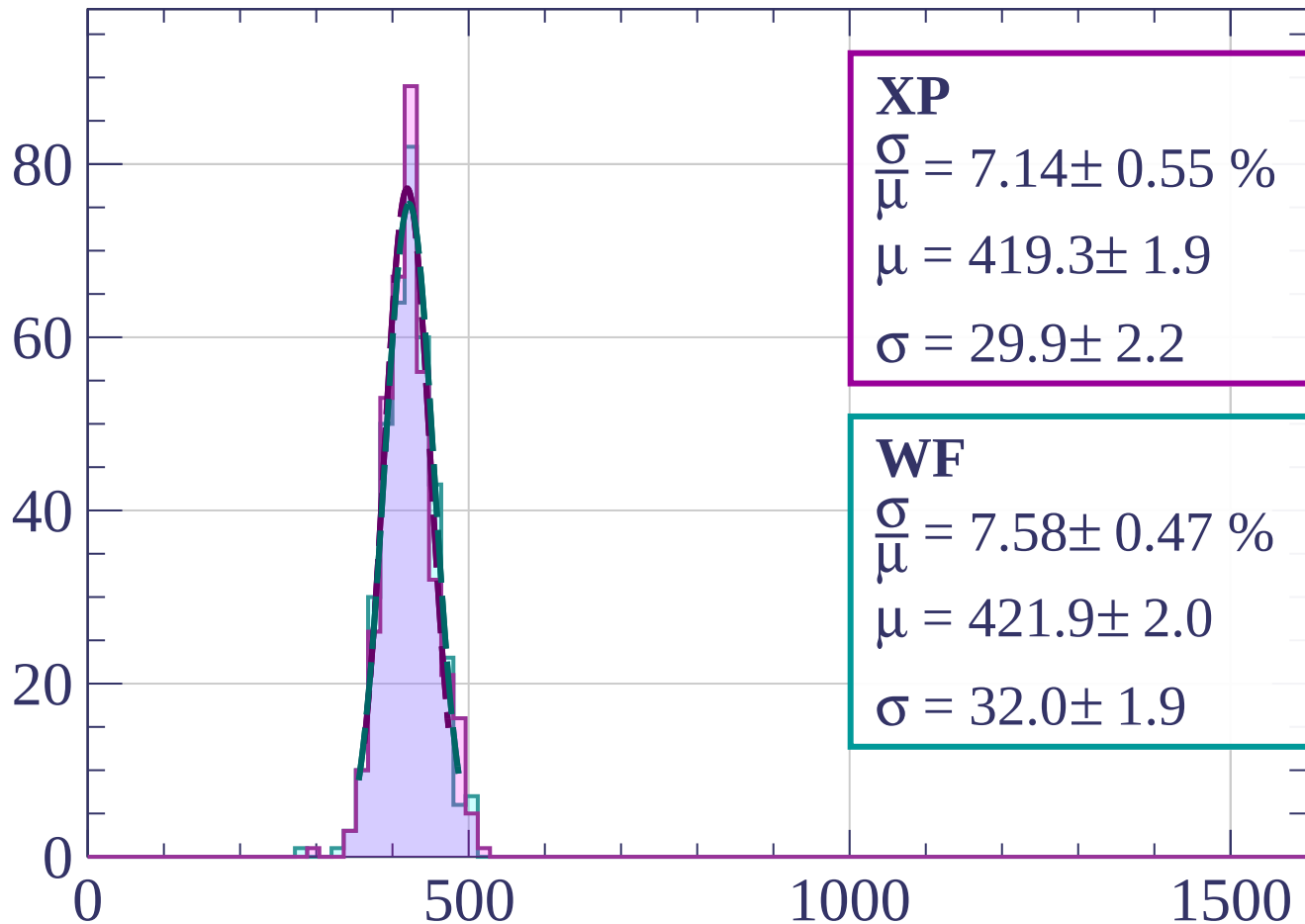
$$\mu = 424.2 \pm 1.9$$

$$\sigma = 32.9 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss | $87 < \phi < 90$

Count



XP

$$\frac{\sigma}{\mu} = 7.14 \pm 0.55 \%$$

$$\mu = 419.3 \pm 1.9$$

$$\sigma = 29.9 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 7.58 \pm 0.47 \%$$

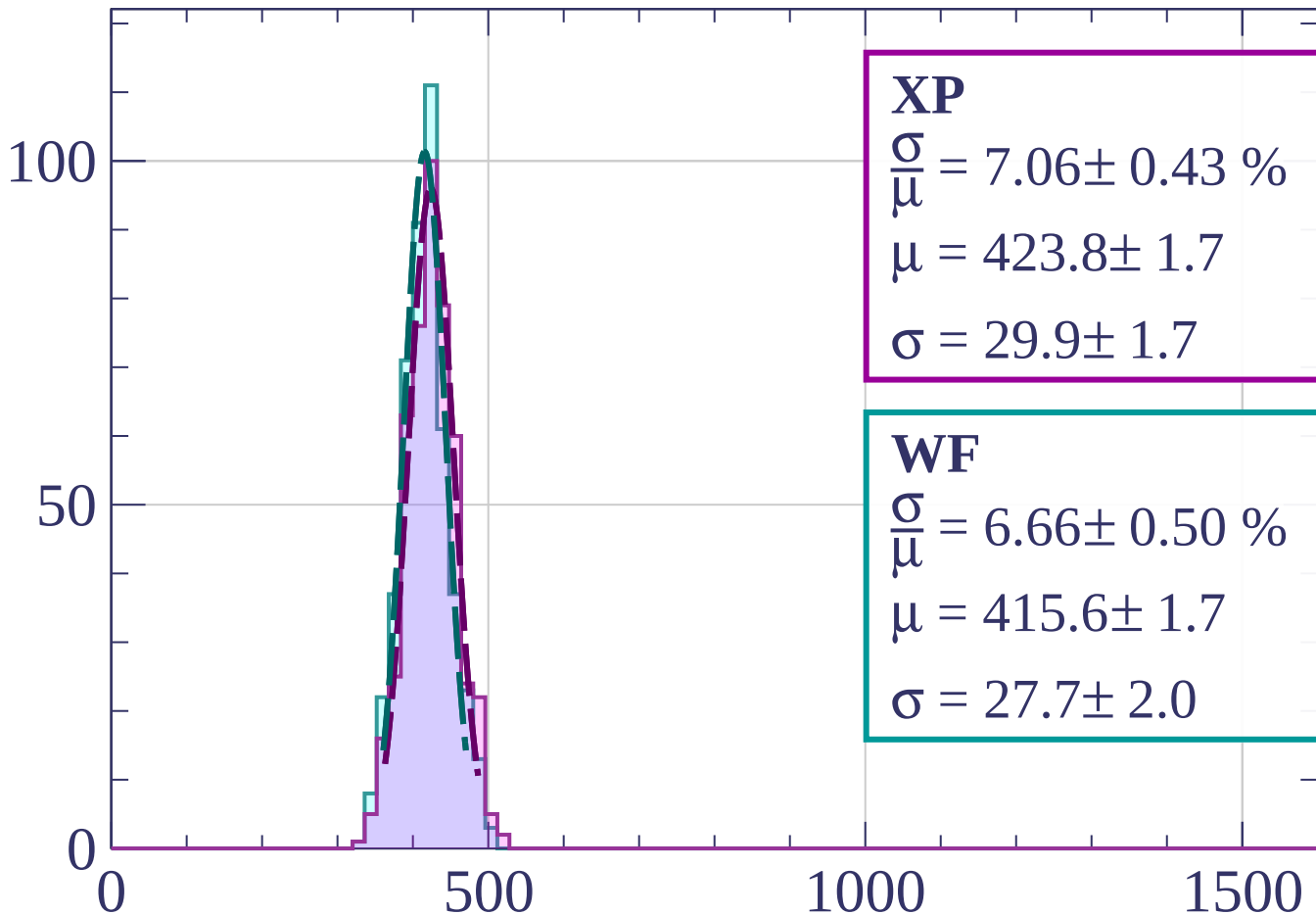
$$\mu = 421.9 \pm 2.0$$

$$\sigma = 32.0 \pm 1.9$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 7.06 \pm 0.43 \%$$

$$\mu = 423.8 \pm 1.7$$

$$\sigma = 29.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 6.66 \pm 0.50 \%$$

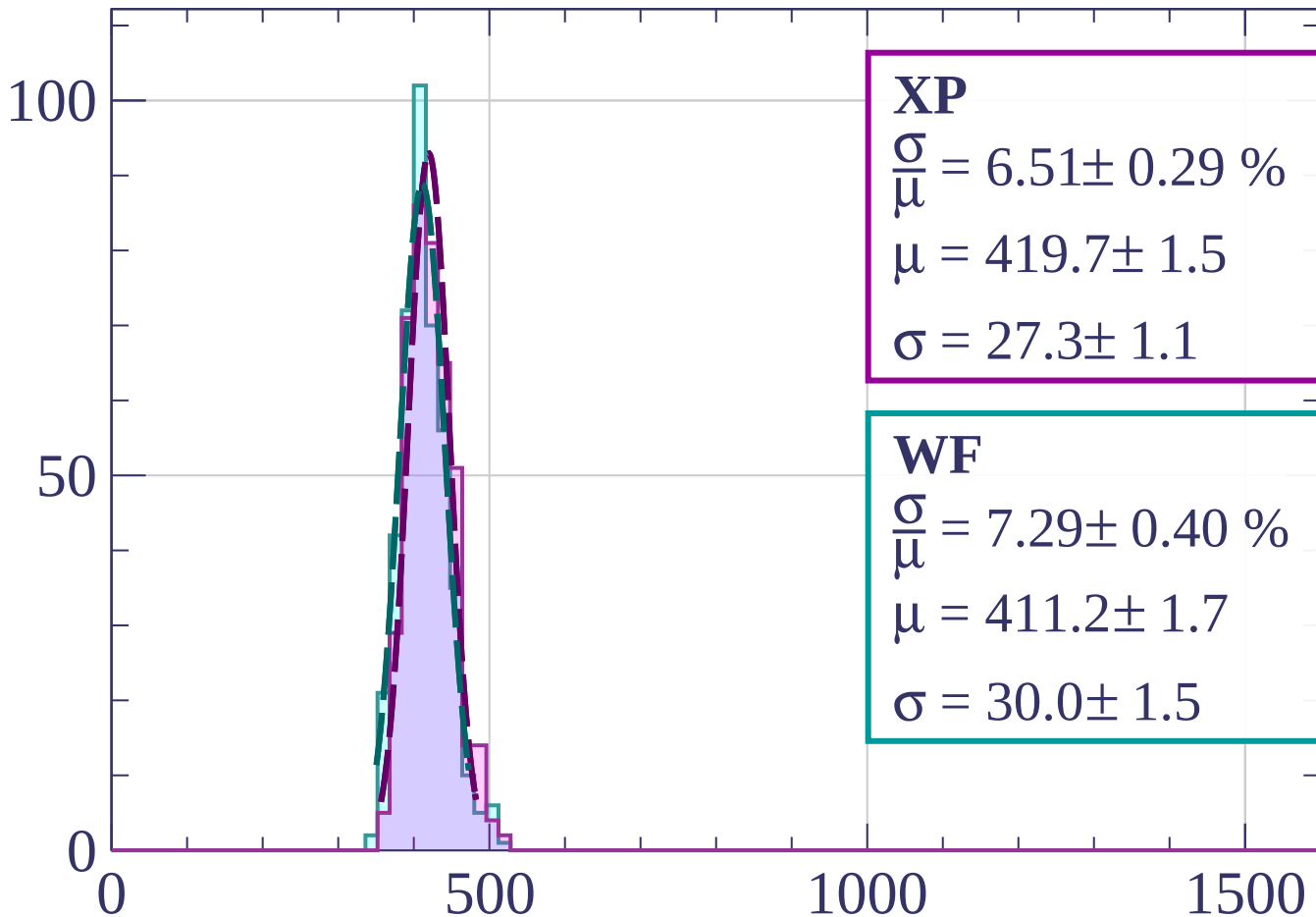
$$\mu = 415.6 \pm 1.7$$

$$\sigma = 27.7 \pm 2.0$$

dE/dx [ADC counts/cm]

Energy loss | $3 < \theta < 6$

Count



XP

$$\frac{\sigma}{\mu} = 6.51 \pm 0.29 \%$$

$$\mu = 419.7 \pm 1.5$$

$$\sigma = 27.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.29 \pm 0.40 \%$$

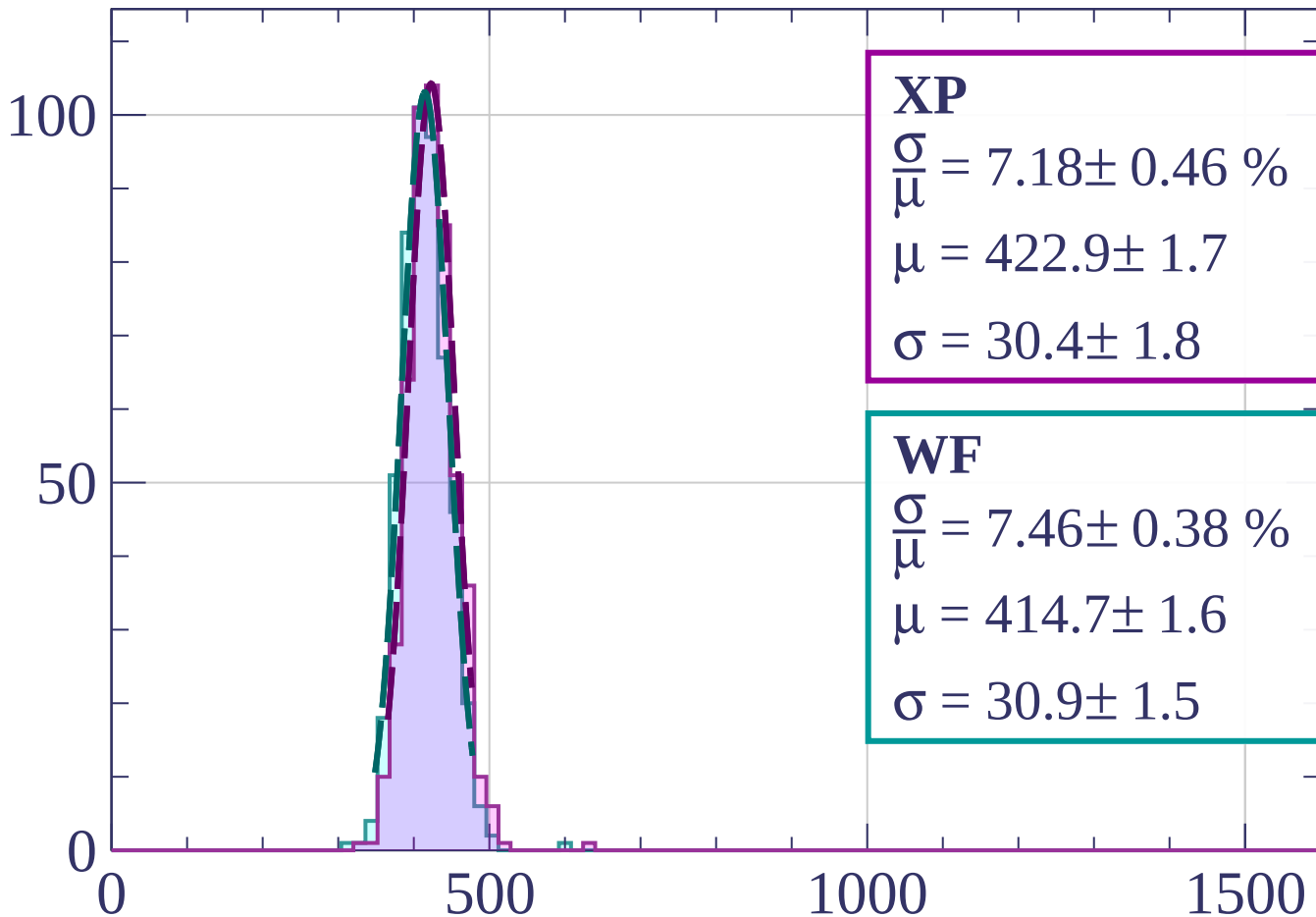
$$\mu = 411.2 \pm 1.7$$

$$\sigma = 30.0 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $6 < \theta < 9$

Count



XP

$$\frac{\sigma}{\mu} = 7.18 \pm 0.46 \%$$

$$\mu = 422.9 \pm 1.7$$

$$\sigma = 30.4 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 7.46 \pm 0.38 \%$$

$$\mu = 414.7 \pm 1.6$$

$$\sigma = 30.9 \pm 1.5$$

dE/dx [ADC counts/cm]

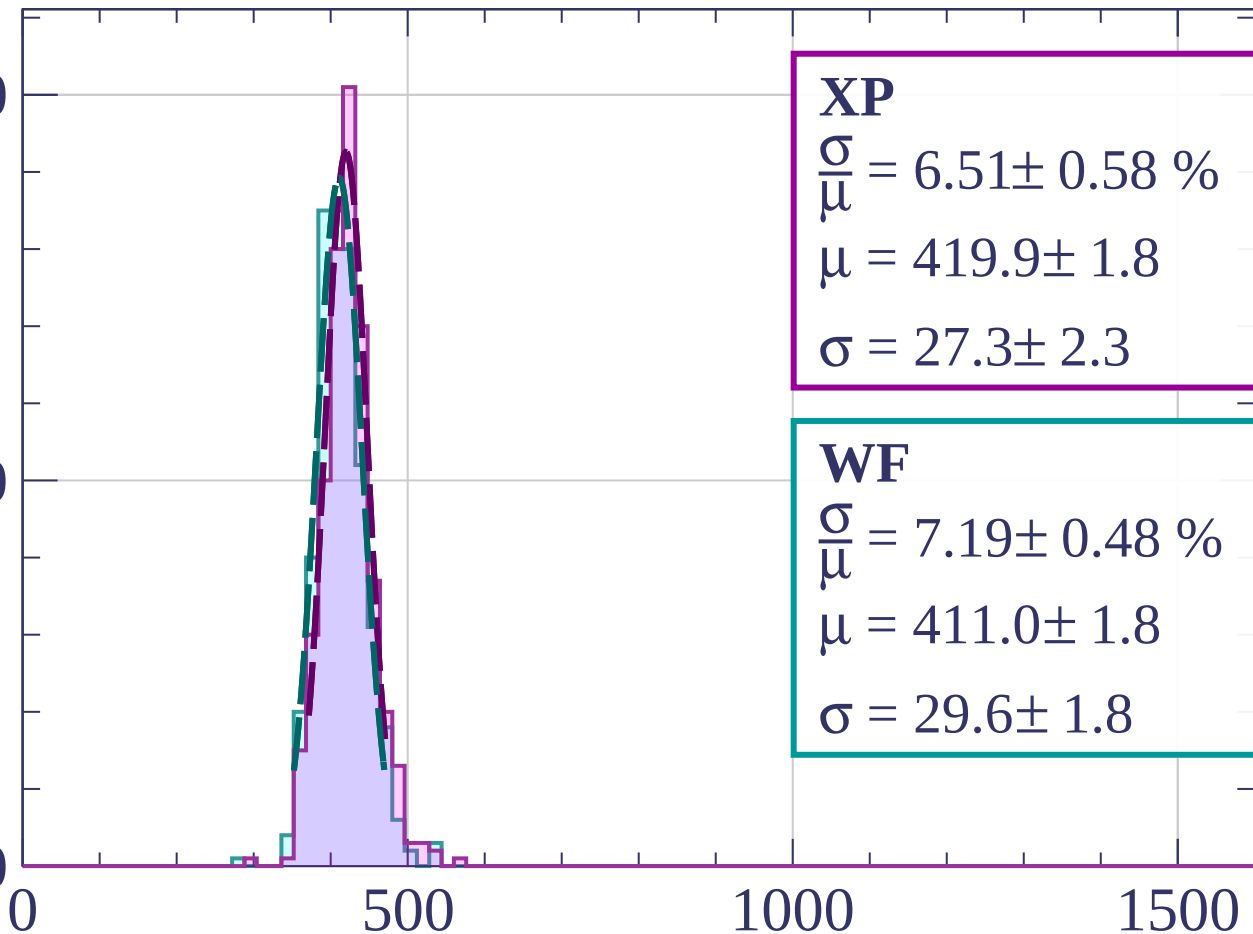
Energy loss | $9 < \theta < 12$

Count

100

50

0



XP

$$\frac{\sigma}{\mu} = 6.51 \pm 0.58 \%$$

$$\mu = 419.9 \pm 1.8$$

$$\sigma = 27.3 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 7.19 \pm 0.48 \%$$

$$\mu = 411.0 \pm 1.8$$

$$\sigma = 29.6 \pm 1.8$$

dE/dx [ADC counts/cm]

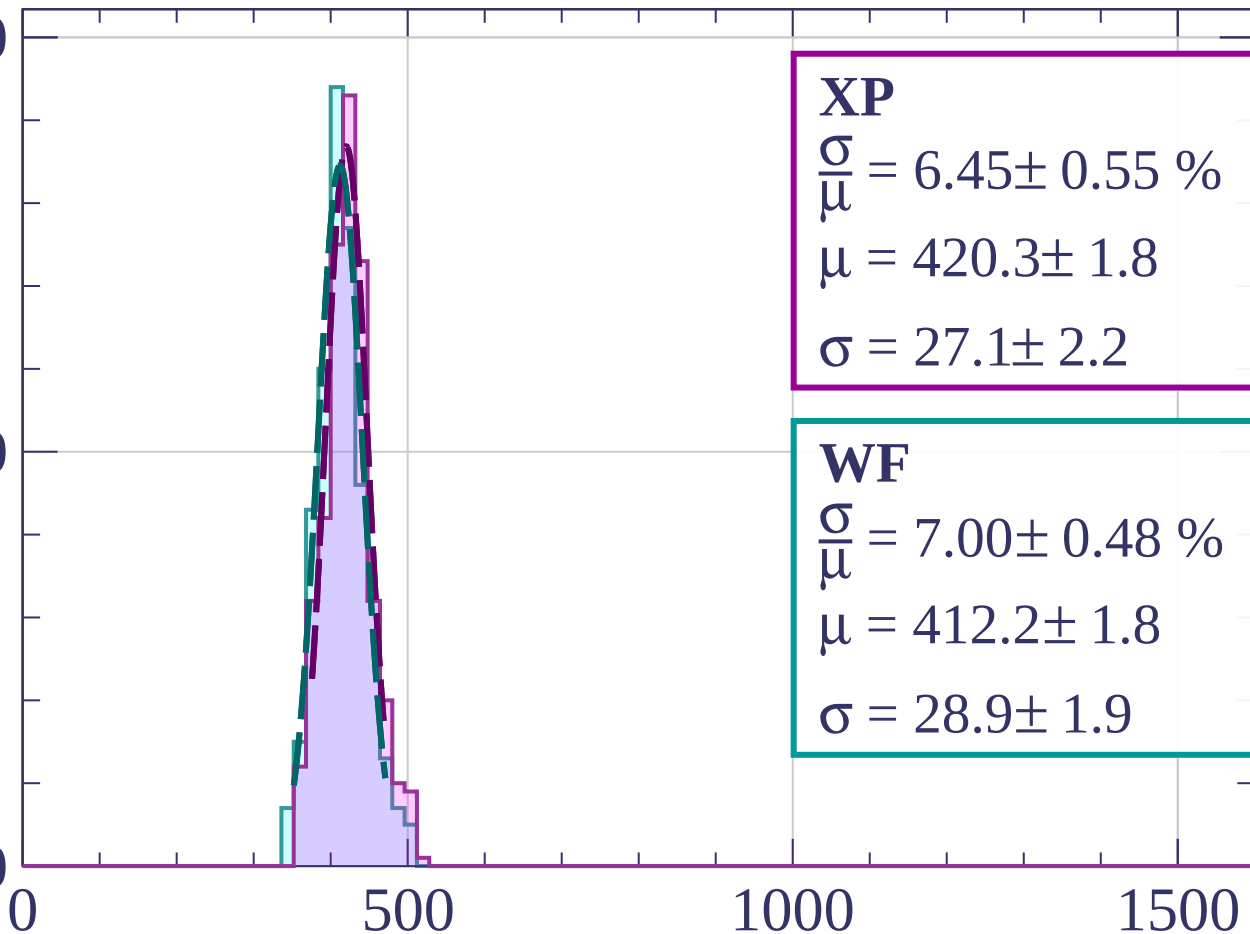
Energy loss | $12 < \theta < 15$

Count

100

50

0



XP

$$\frac{\sigma}{\mu} = 6.45 \pm 0.55 \%$$

$$\mu = 420.3 \pm 1.8$$

$$\sigma = 27.1 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 7.00 \pm 0.48 \%$$

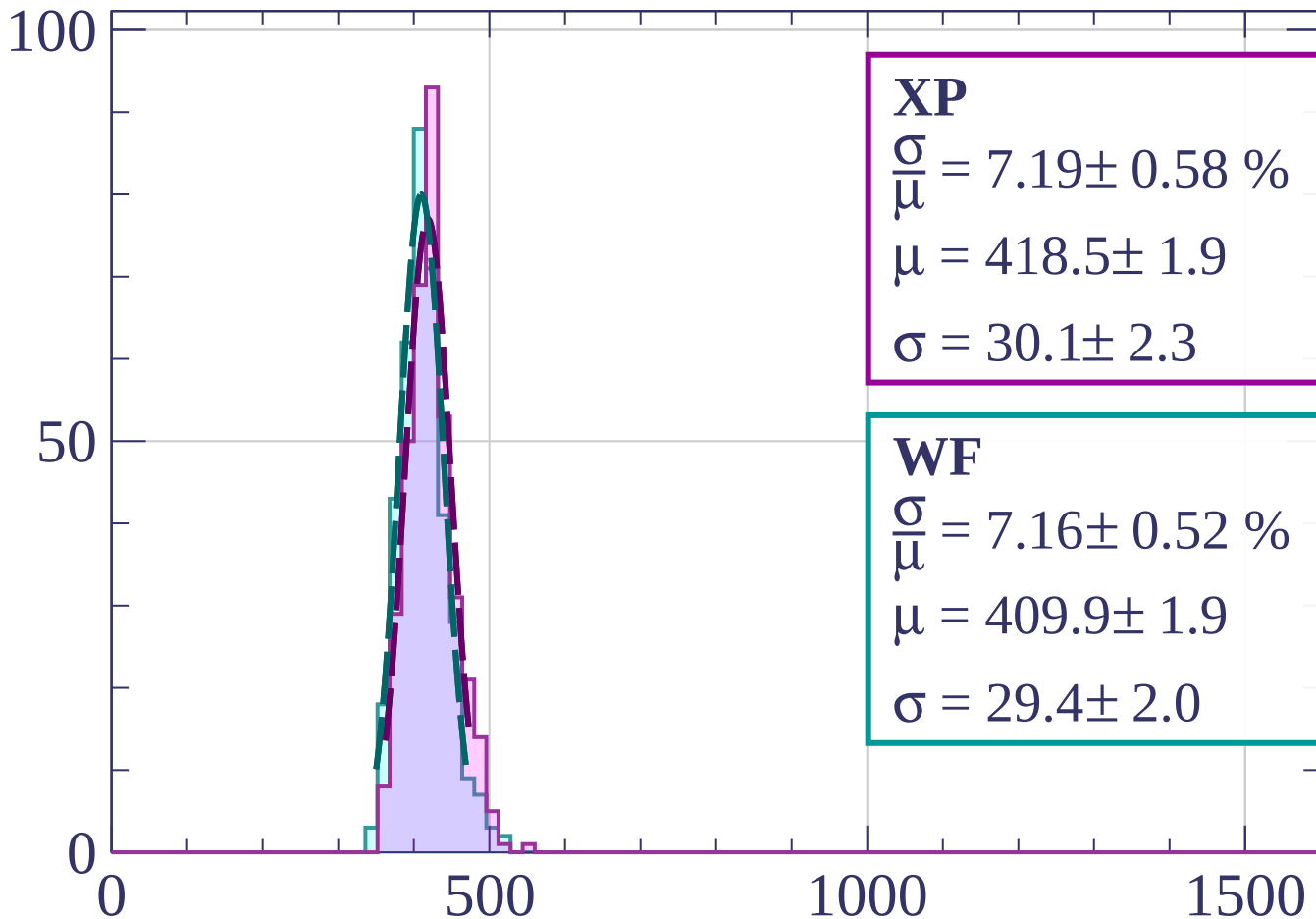
$$\mu = 412.2 \pm 1.8$$

$$\sigma = 28.9 \pm 1.9$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 7.19 \pm 0.58 \%$$

$$\mu = 418.5 \pm 1.9$$

$$\sigma = 30.1 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 7.16 \pm 0.52 \%$$

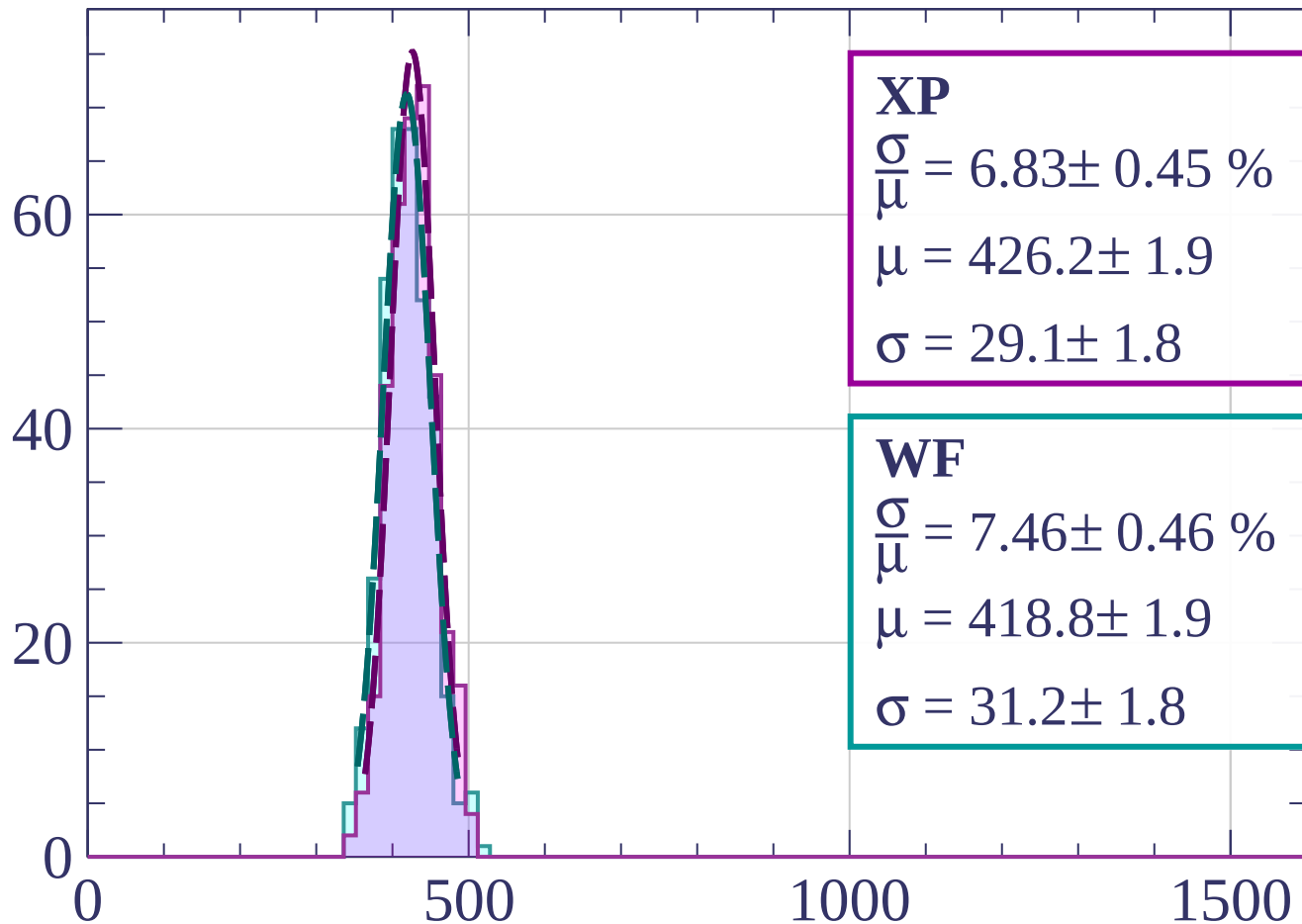
$$\mu = 409.9 \pm 1.9$$

$$\sigma = 29.4 \pm 2.0$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \theta < 21$

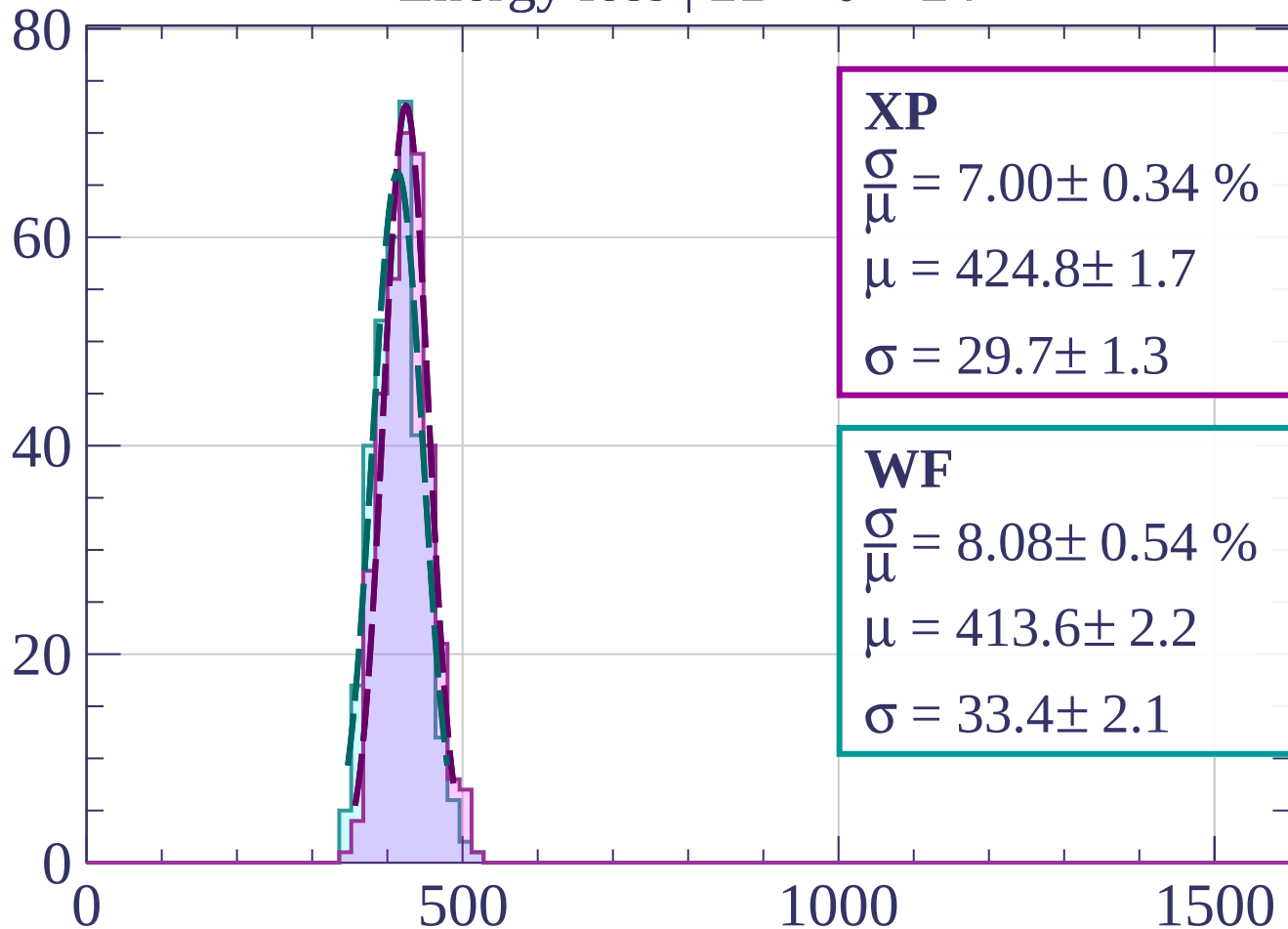
Count



dE/dx [ADC counts/cm]

Count

Energy loss | $21 < \theta < 24$



XP

$$\frac{\sigma}{\mu} = 7.00 \pm 0.34 \%$$

$$\mu = 424.8 \pm 1.7$$

$$\sigma = 29.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.08 \pm 0.54 \%$$

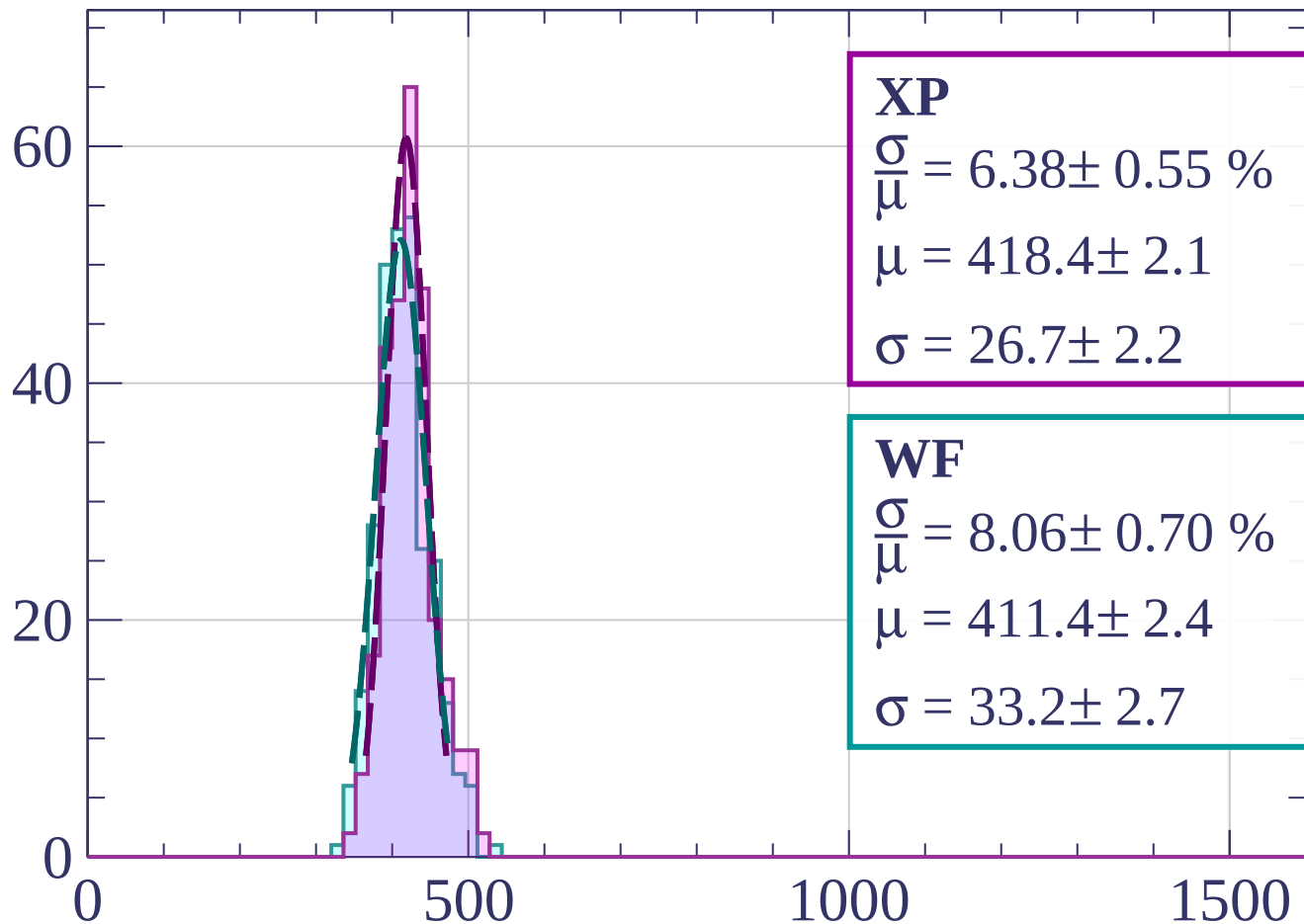
$$\mu = 413.6 \pm 2.2$$

$$\sigma = 33.4 \pm 2.1$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \theta < 27$

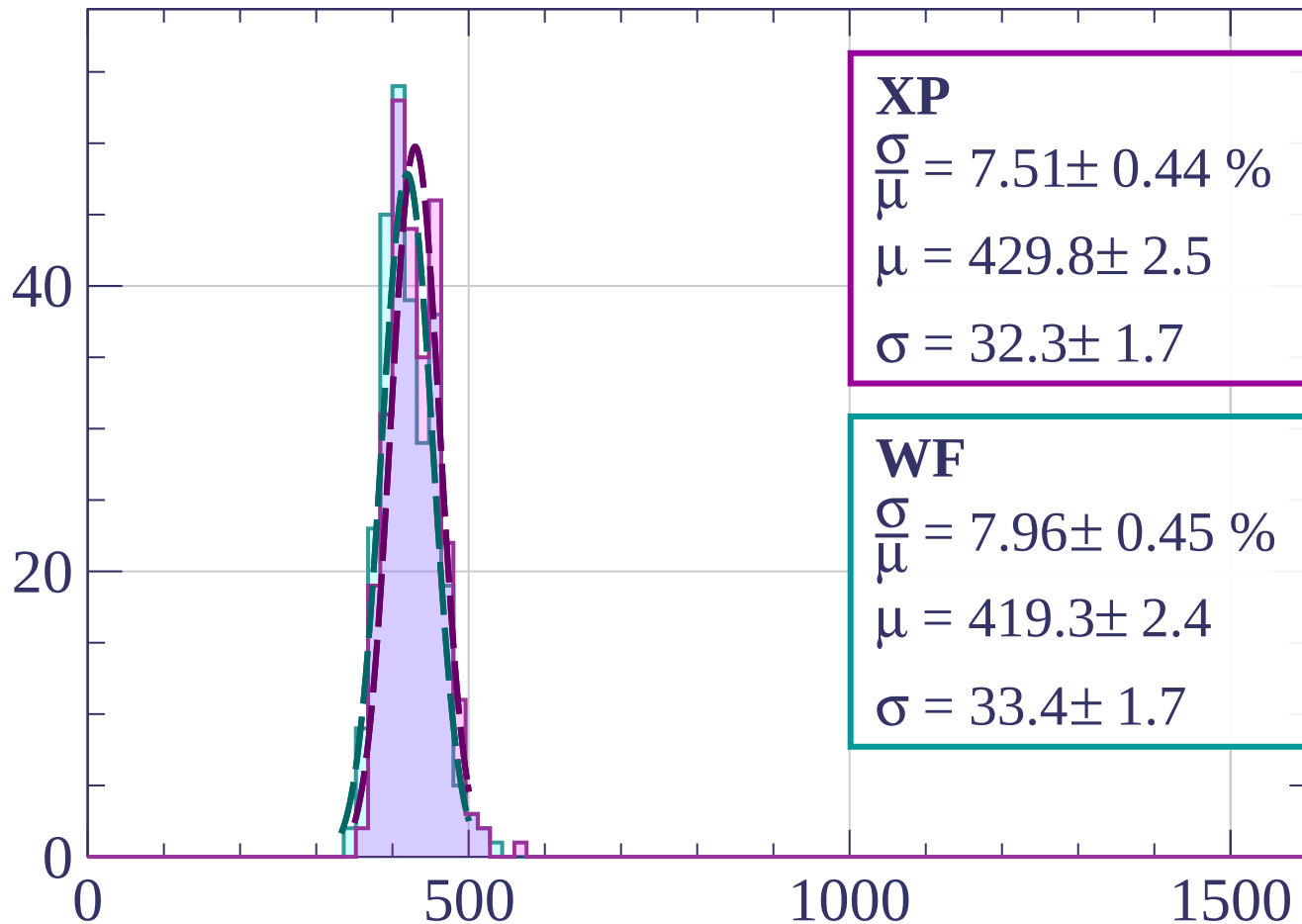
Count



dE/dx [ADC counts/cm]

Energy loss | $27 < \theta < 30$

Count



XP

$$\frac{\sigma}{\mu} = 7.51 \pm 0.44 \%$$

$$\mu = 429.8 \pm 2.5$$

$$\sigma = 32.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 7.96 \pm 0.45 \%$$

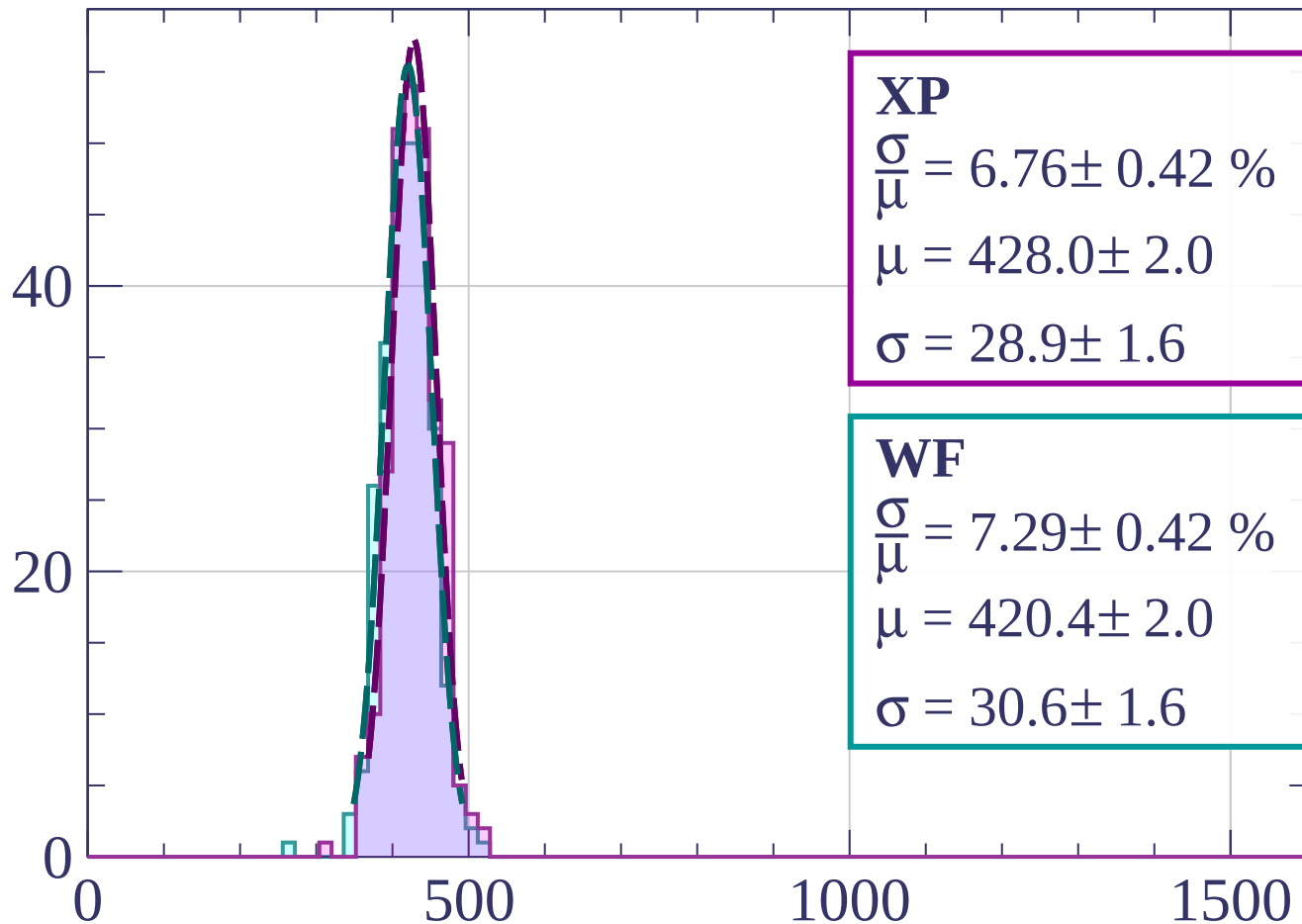
$$\mu = 419.3 \pm 2.4$$

$$\sigma = 33.4 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \theta < 33$

Count



XP

$$\frac{\sigma}{\mu} = 6.76 \pm 0.42 \%$$

$$\mu = 428.0 \pm 2.0$$

$$\sigma = 28.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.29 \pm 0.42 \%$$

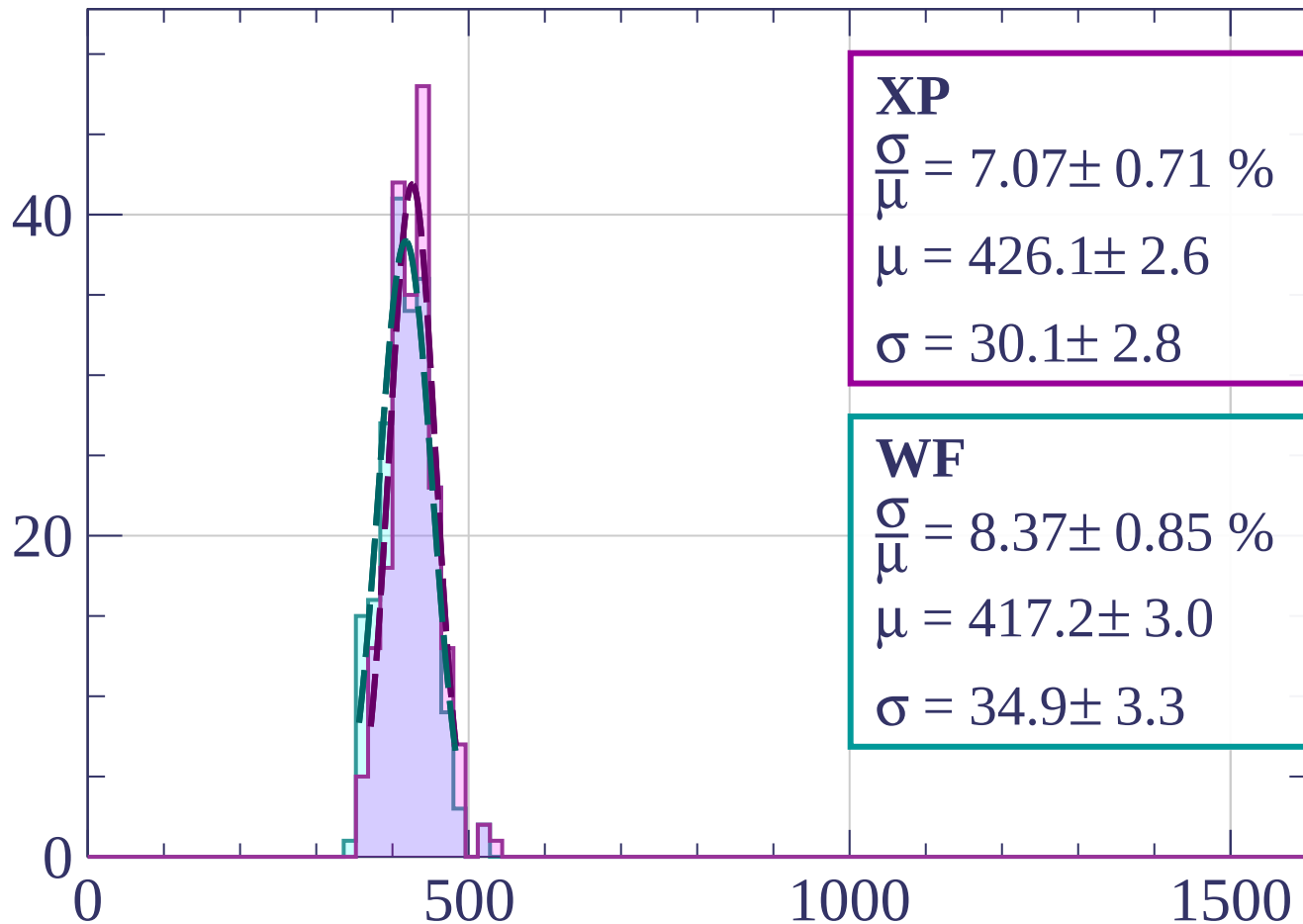
$$\mu = 420.4 \pm 2.0$$

$$\sigma = 30.6 \pm 1.6$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \theta < 36$

Count



XP

$$\frac{\sigma}{\mu} = 7.07 \pm 0.71 \%$$

$$\mu = 426.1 \pm 2.6$$

$$\sigma = 30.1 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 8.37 \pm 0.85 \%$$

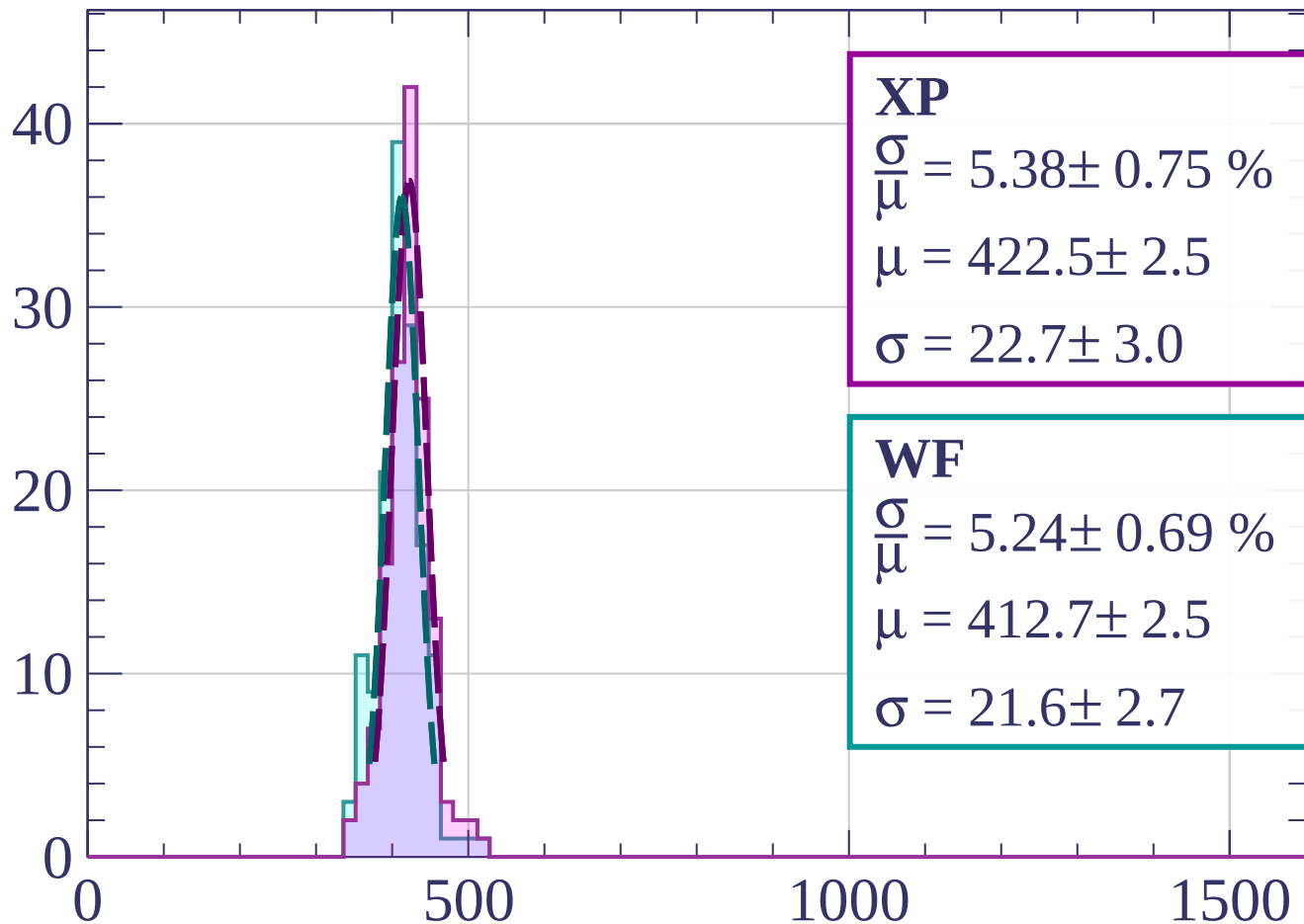
$$\mu = 417.2 \pm 3.0$$

$$\sigma = 34.9 \pm 3.3$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \theta < 39$

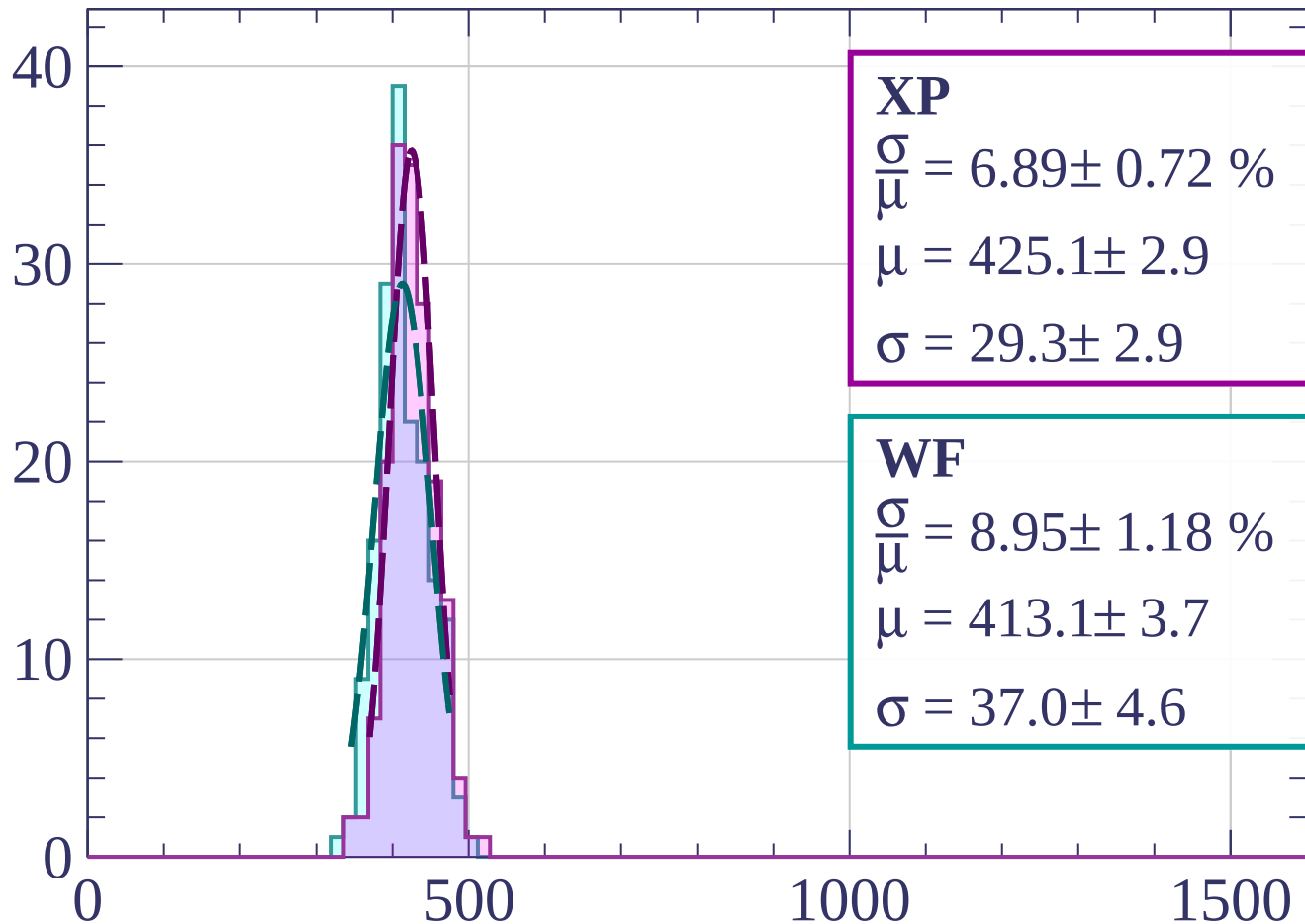
Count



dE/dx [ADC counts/cm]

Energy loss | $39 < \theta < 42$

Count



XP

$$\frac{\sigma}{\mu} = 6.89 \pm 0.72 \%$$

$$\mu = 425.1 \pm 2.9$$

$$\sigma = 29.3 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 8.95 \pm 1.18 \%$$

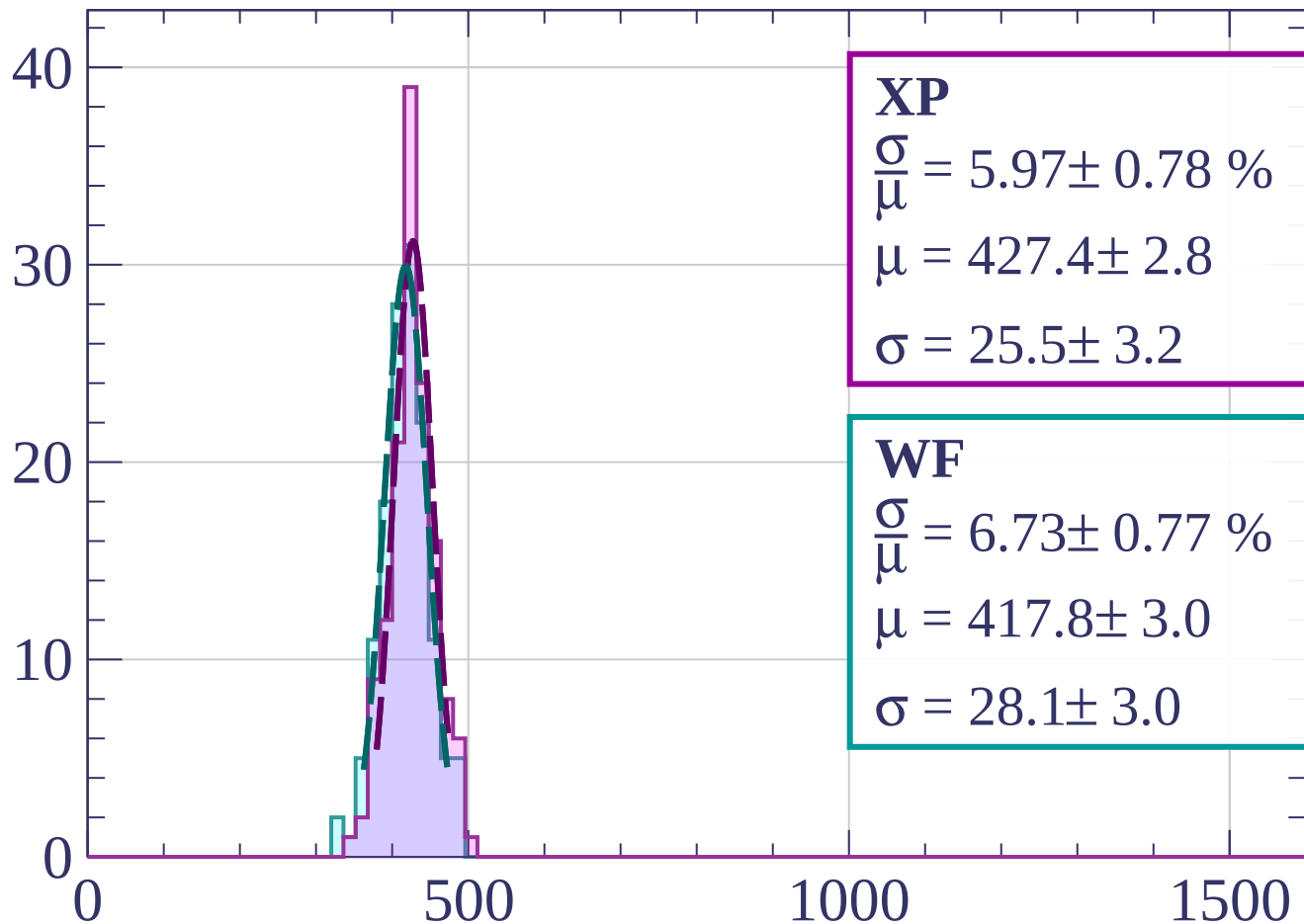
$$\mu = 413.1 \pm 3.7$$

$$\sigma = 37.0 \pm 4.6$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \theta < 45$

Count



XP

$$\frac{\sigma}{\mu} = 5.97 \pm 0.78 \%$$

$$\mu = 427.4 \pm 2.8$$

$$\sigma = 25.5 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 6.73 \pm 0.77 \%$$

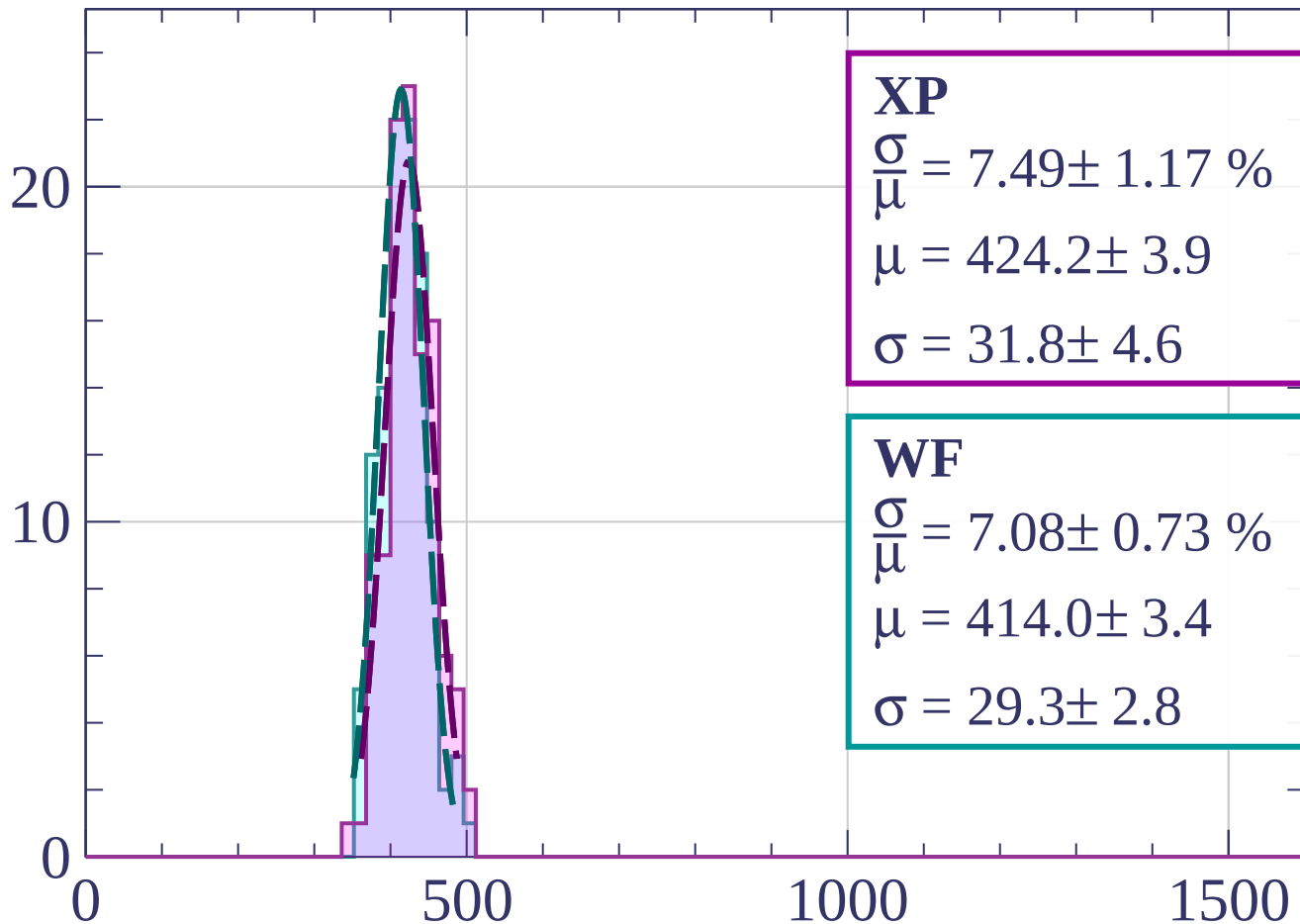
$$\mu = 417.8 \pm 3.0$$

$$\sigma = 28.1 \pm 3.0$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \theta < 48$

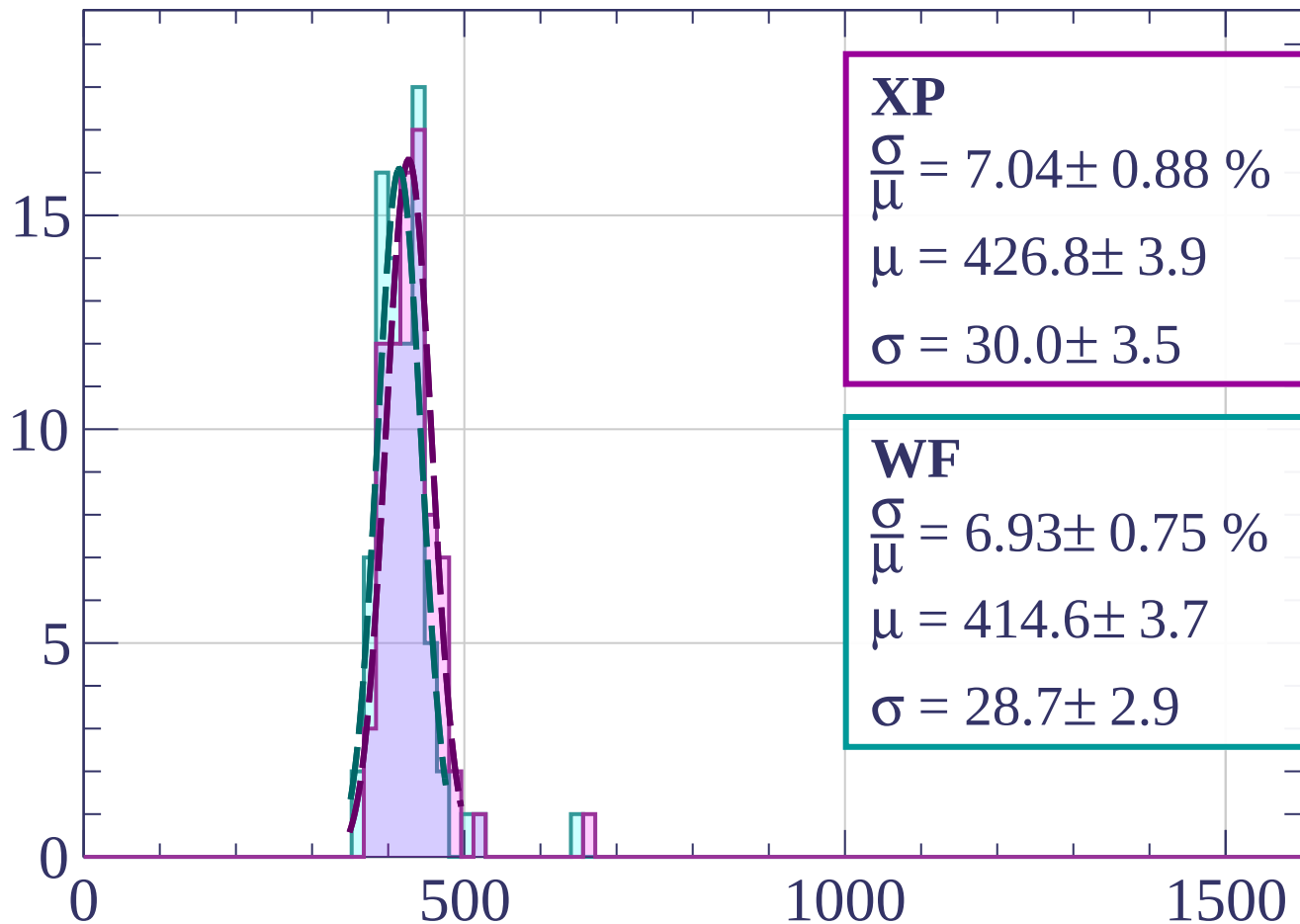
Count



dE/dx [ADC counts/cm]

Energy loss | $48 < \theta < 51$

Count



XP

$$\frac{\sigma}{\mu} = 7.04 \pm 0.88 \%$$

$$\mu = 426.8 \pm 3.9$$

$$\sigma = 30.0 \pm 3.5$$

WF

$$\frac{\sigma}{\mu} = 6.93 \pm 0.75 \%$$

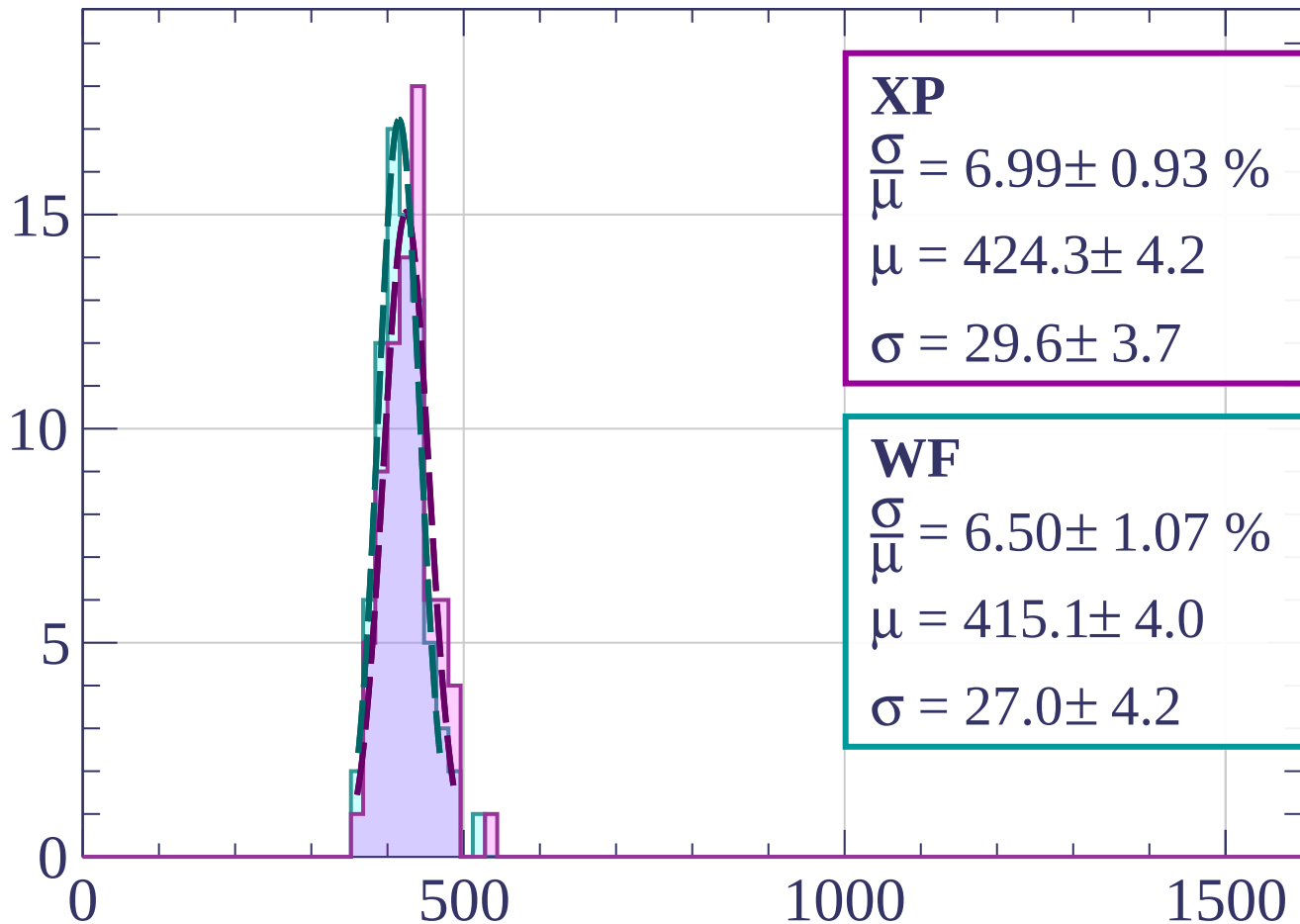
$$\mu = 414.6 \pm 3.7$$

$$\sigma = 28.7 \pm 2.9$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \theta < 54$

Count



XP

$$\frac{\sigma}{\mu} = 6.99 \pm 0.93 \%$$

$$\mu = 424.3 \pm 4.2$$

$$\sigma = 29.6 \pm 3.7$$

WF

$$\frac{\sigma}{\mu} = 6.50 \pm 1.07 \%$$

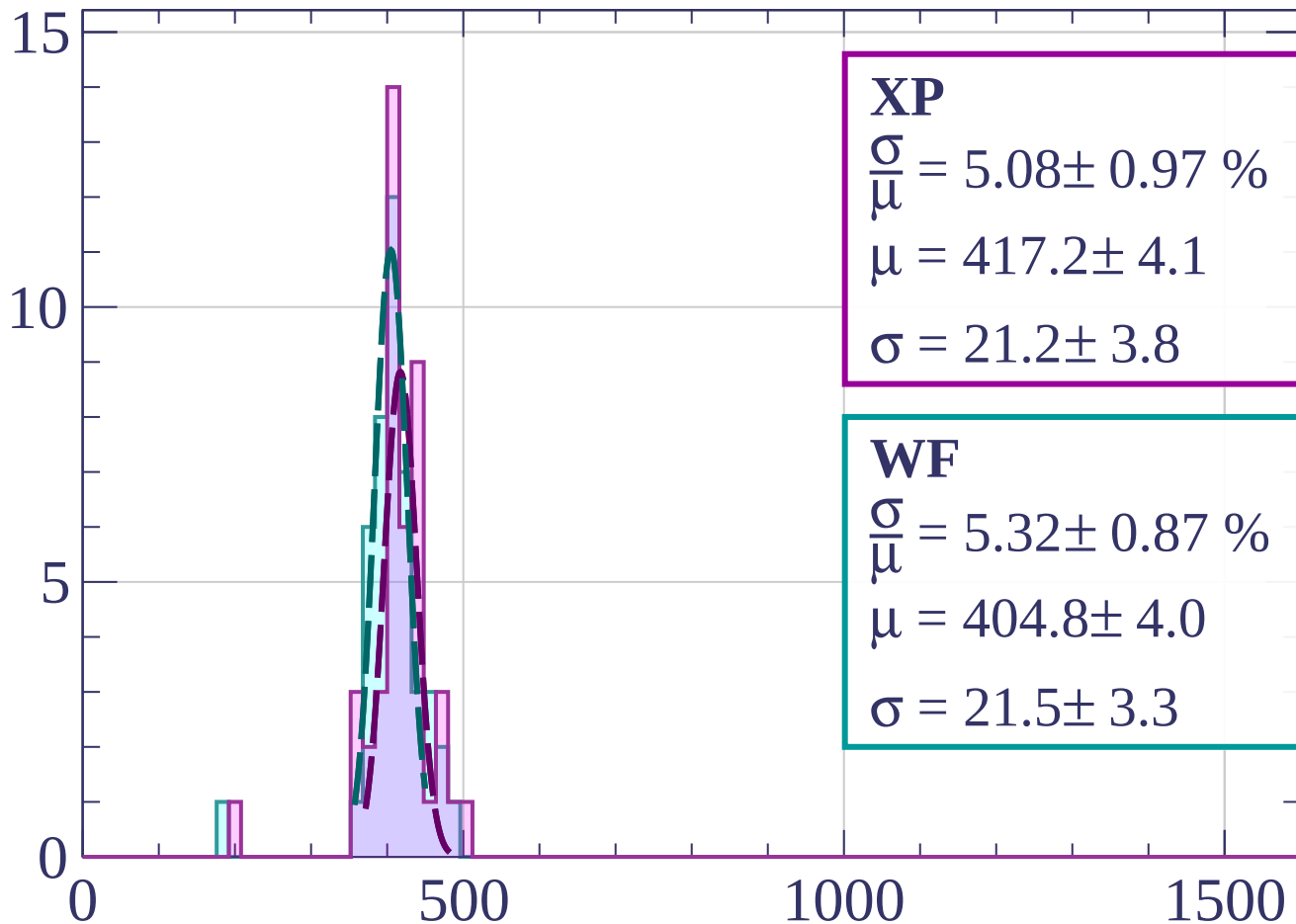
$$\mu = 415.1 \pm 4.0$$

$$\sigma = 27.0 \pm 4.2$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \theta < 57$

Count



XP

$$\frac{\sigma}{\mu} = 5.08 \pm 0.97 \%$$

$$\mu = 417.2 \pm 4.1$$

$$\sigma = 21.2 \pm 3.8$$

WF

$$\frac{\sigma}{\mu} = 5.32 \pm 0.87 \%$$

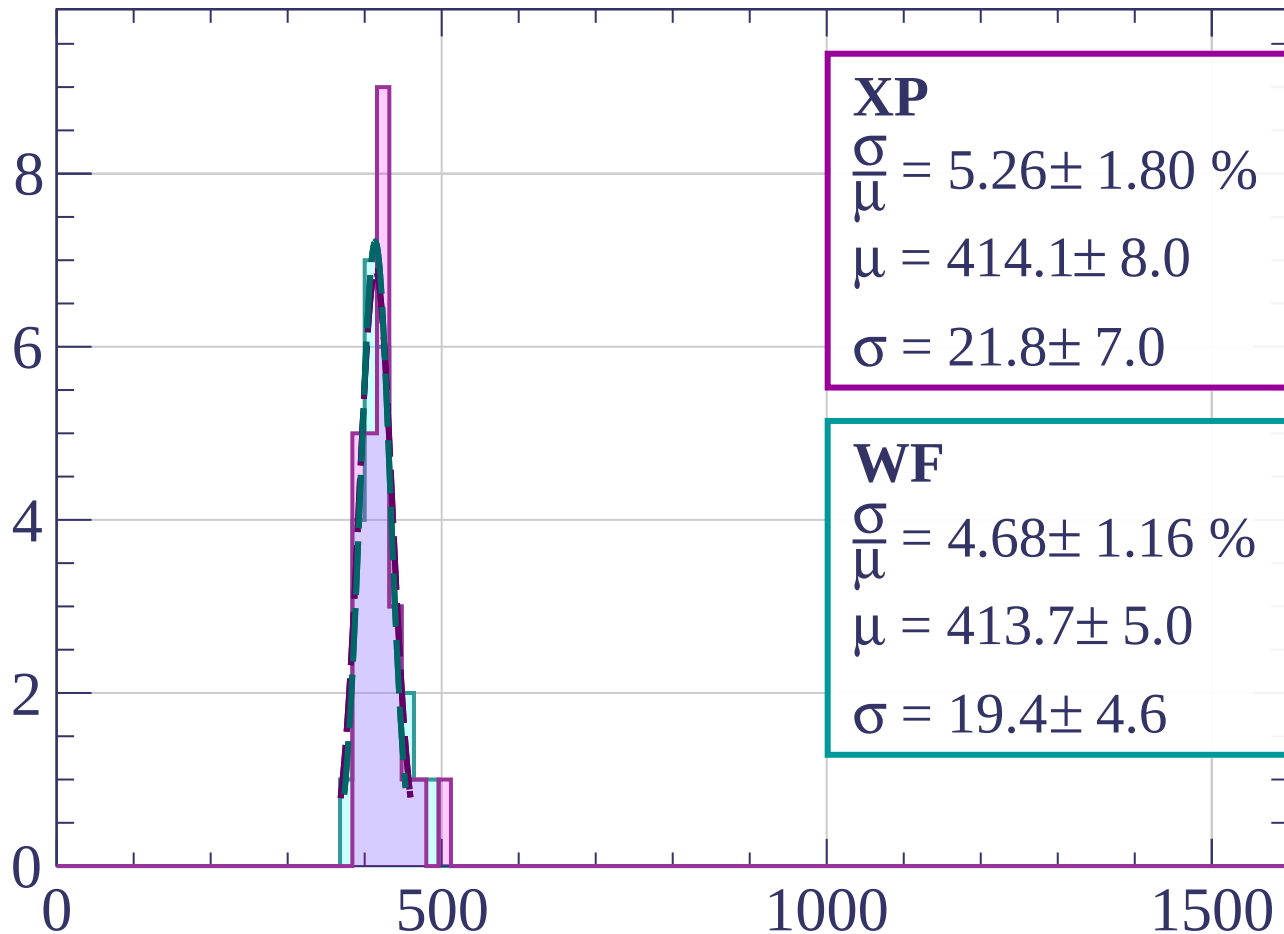
$$\mu = 404.8 \pm 4.0$$

$$\sigma = 21.5 \pm 3.3$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \theta < 60$

Count



XP

$$\frac{\sigma}{\mu} = 5.26 \pm 1.80 \%$$

$$\mu = 414.1 \pm 8.0$$

$$\sigma = 21.8 \pm 7.0$$

WF

$$\frac{\sigma}{\mu} = 4.68 \pm 1.16 \%$$

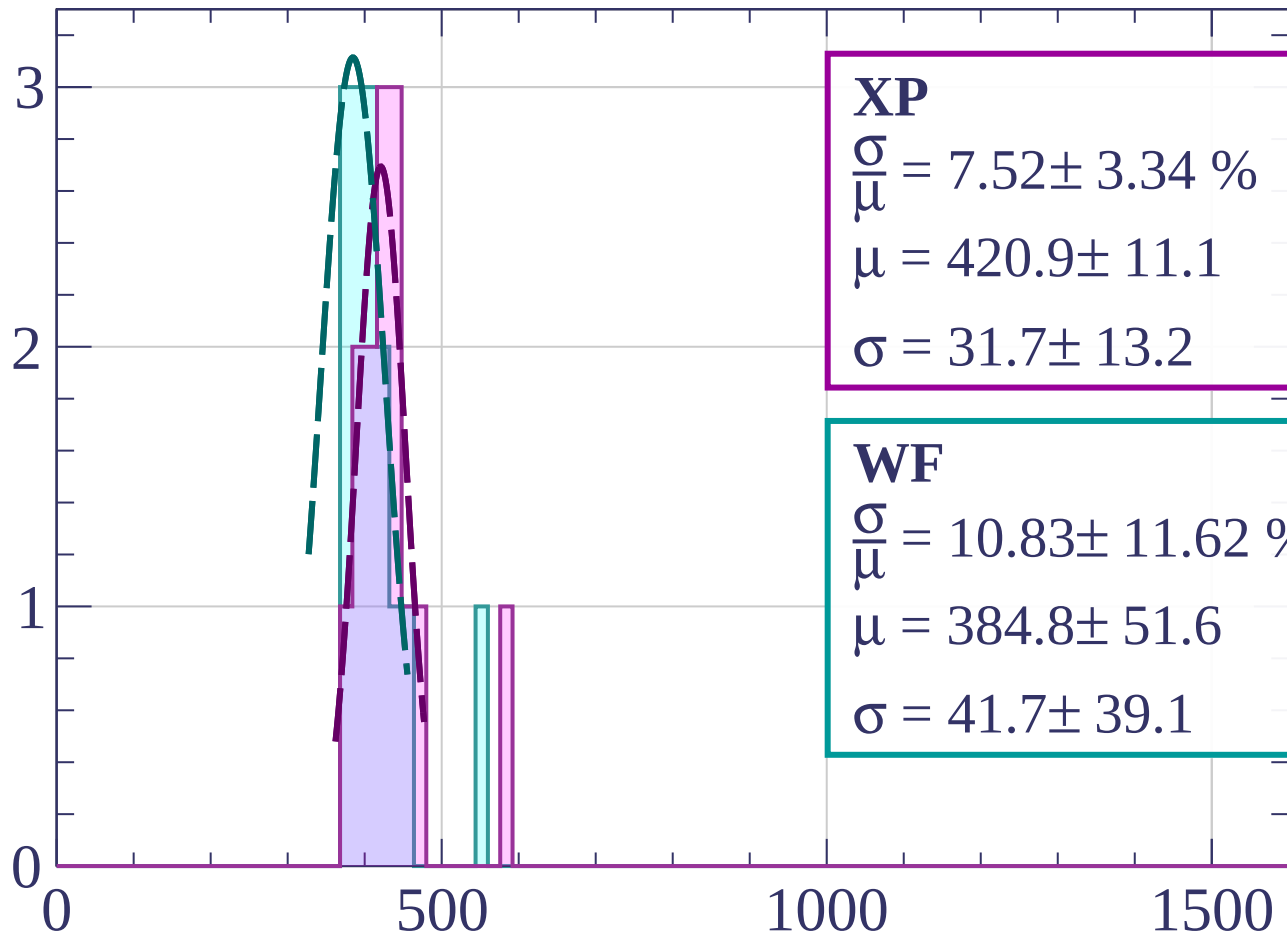
$$\mu = 413.7 \pm 5.0$$

$$\sigma = 19.4 \pm 4.6$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \theta < 63$

Count



XP

$$\frac{\sigma}{\mu} = 7.52 \pm 3.34 \%$$

$$\mu = 420.9 \pm 11.1$$

$$\sigma = 31.7 \pm 13.2$$

WF

$$\frac{\sigma}{\mu} = 10.83 \pm 11.62 \%$$

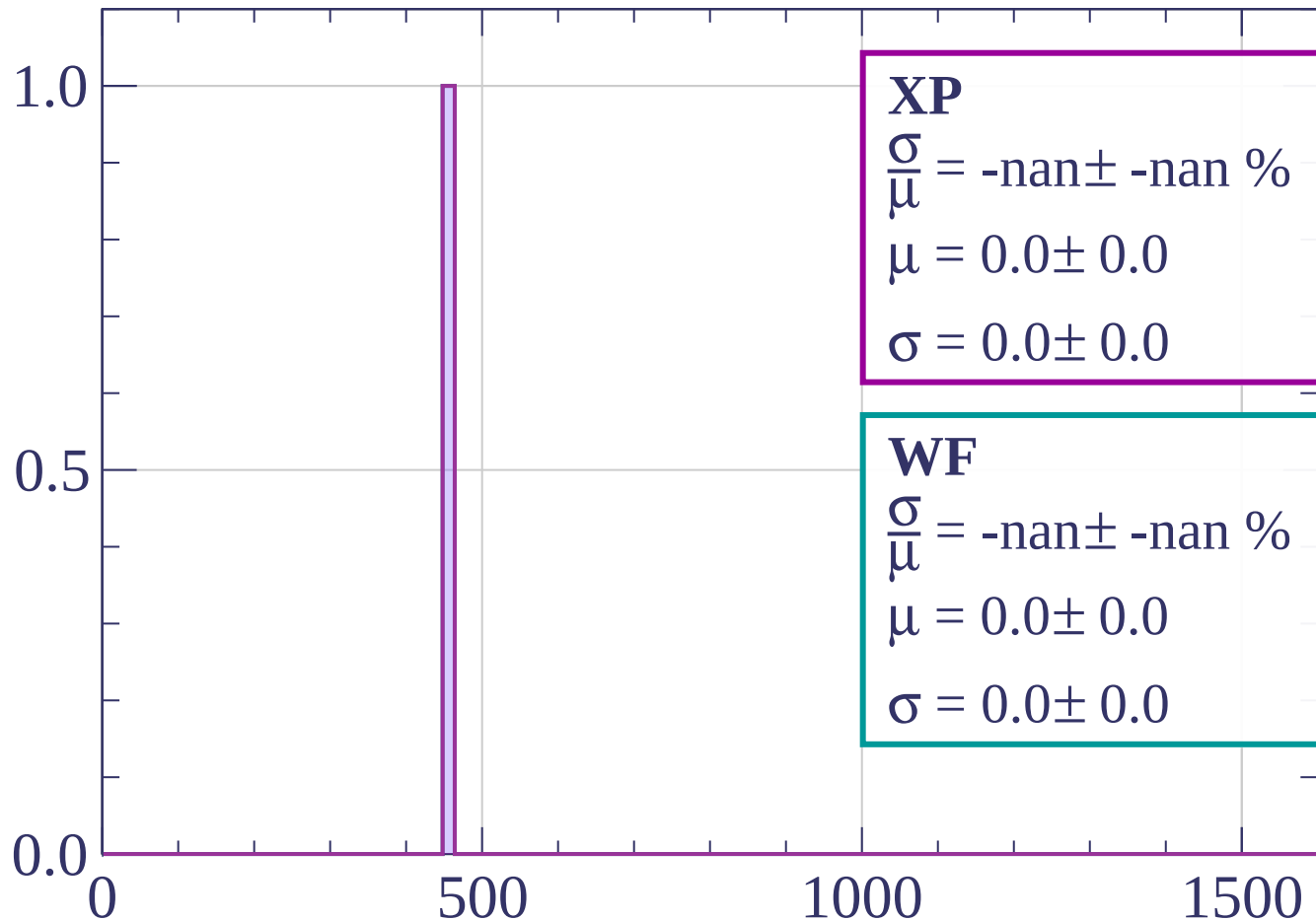
$$\mu = 384.8 \pm 51.6$$

$$\sigma = 41.7 \pm 39.1$$

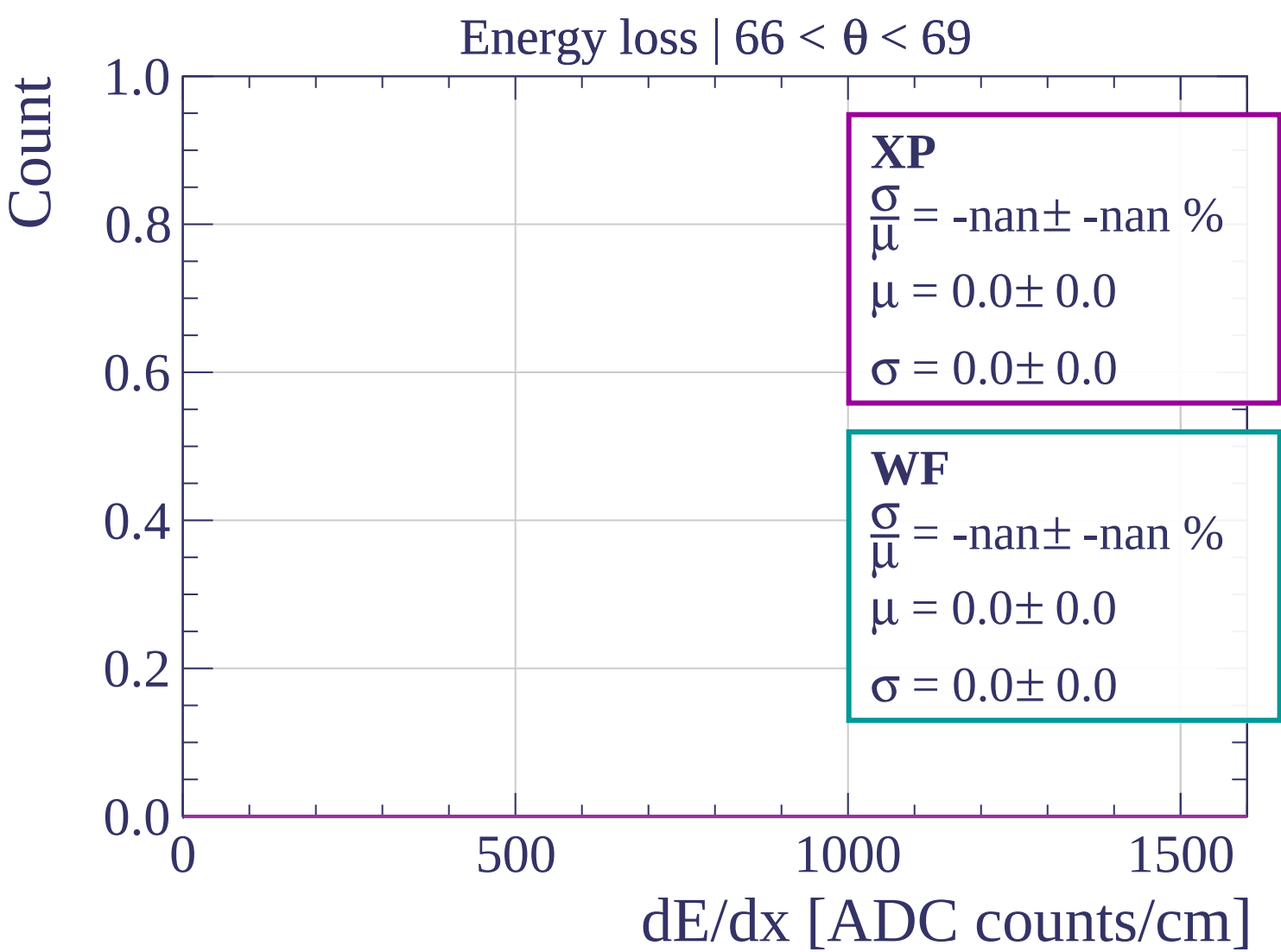
dE/dx [ADC counts/cm]

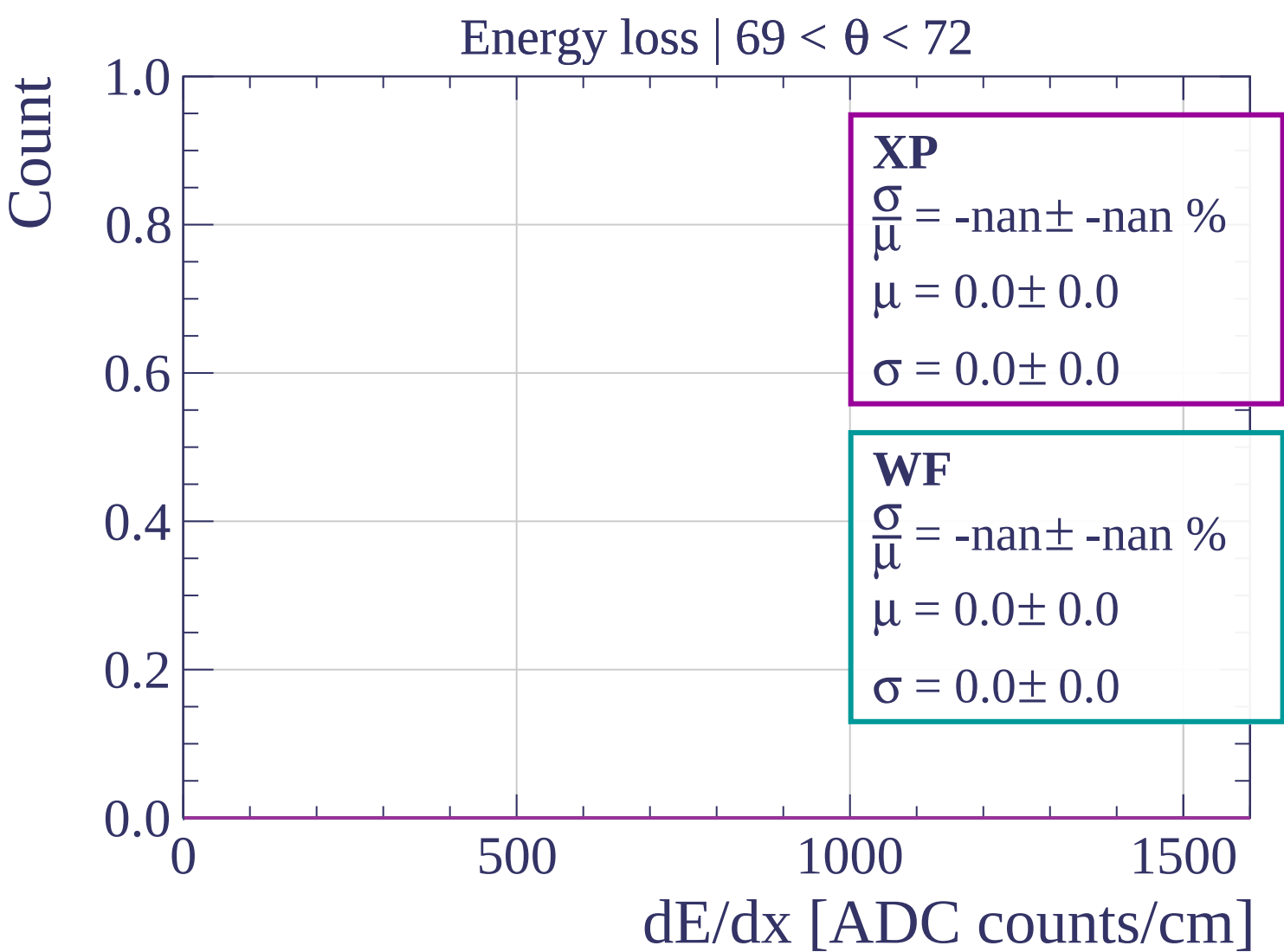
Energy loss | $63 < \theta < 66$

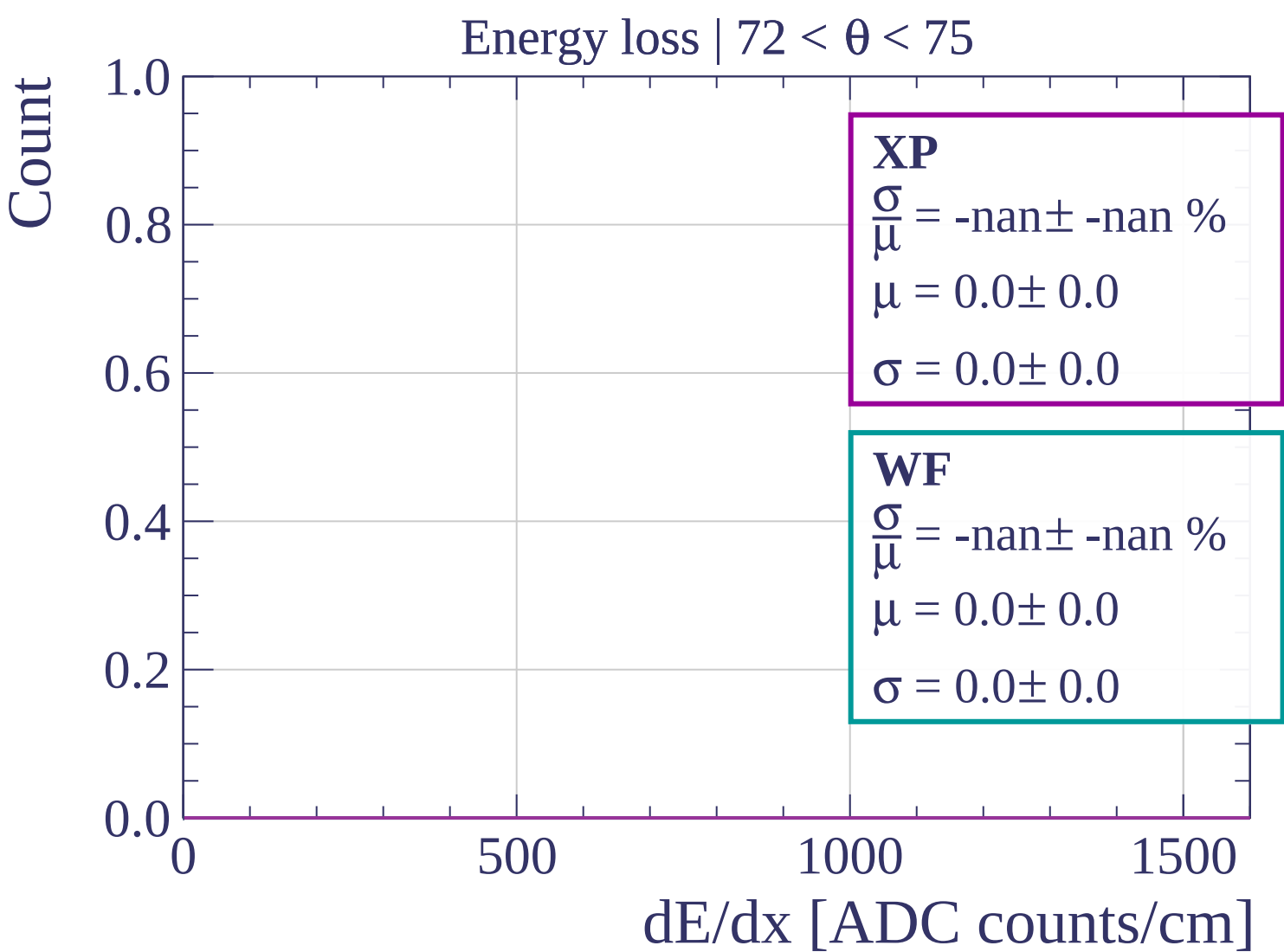
Count

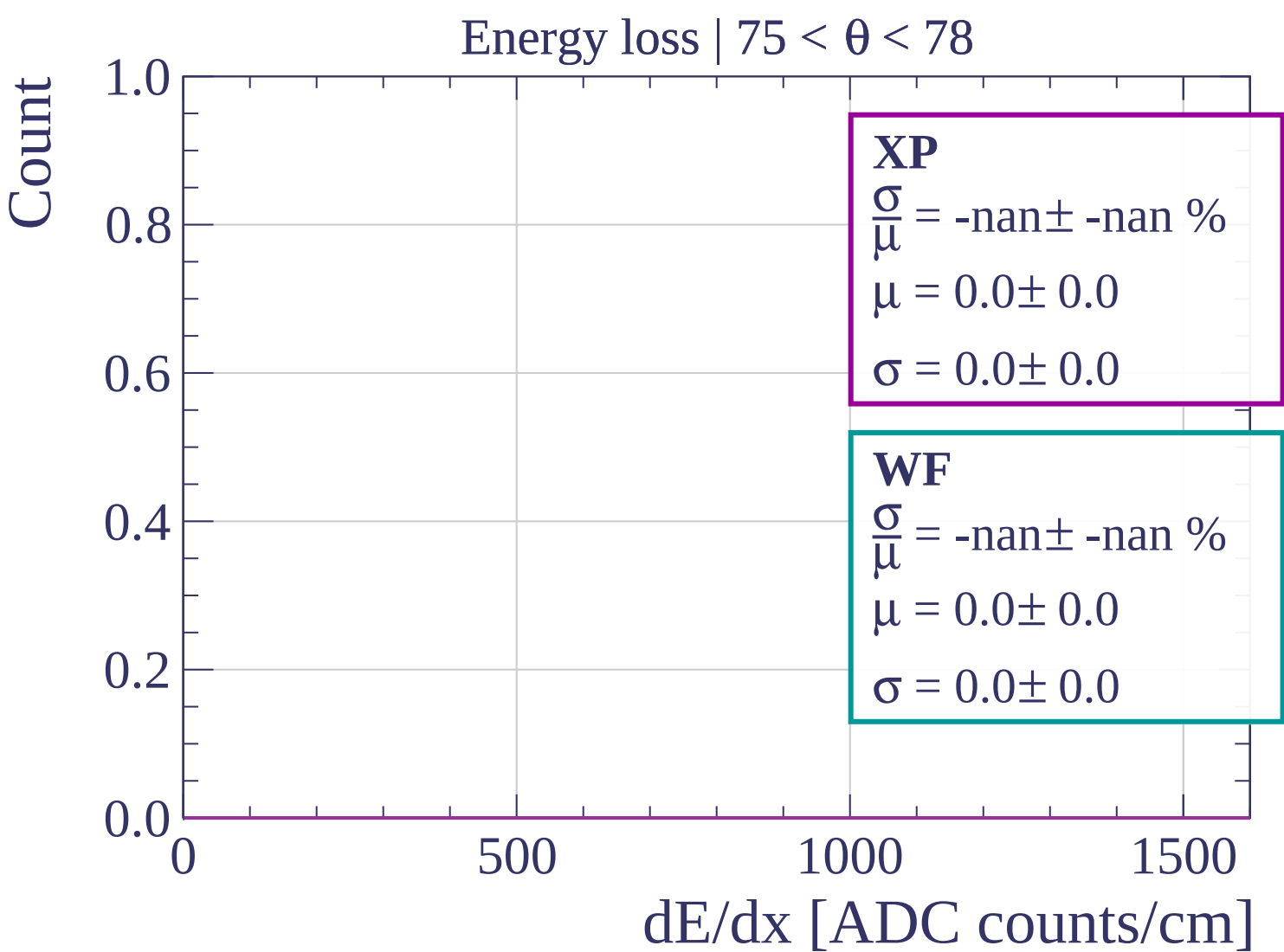


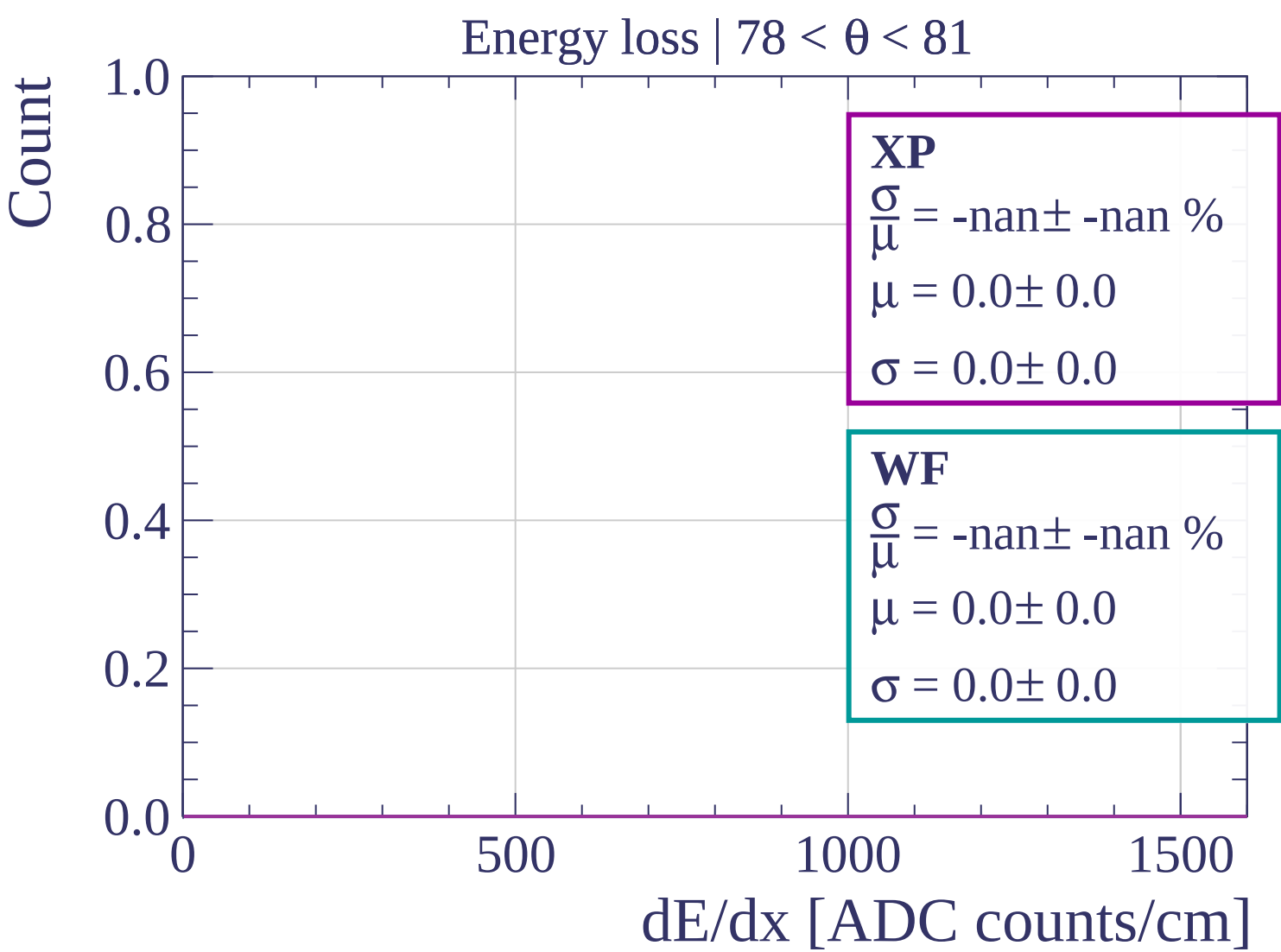
dE/dx [ADC counts/cm]











Count

Energy loss | $81 < \theta < 84$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

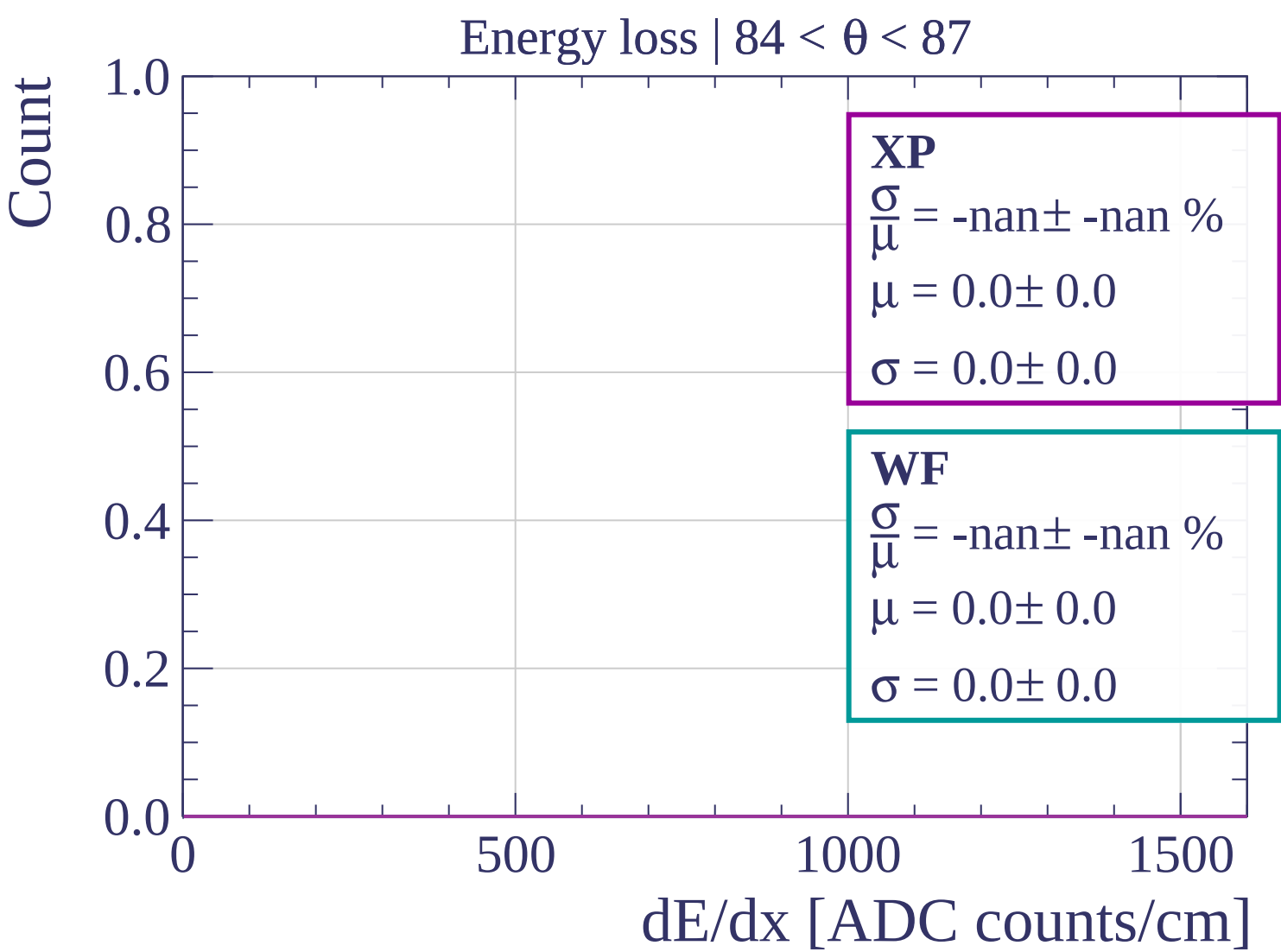
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

Energy loss | $87 < \theta < 90$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

0

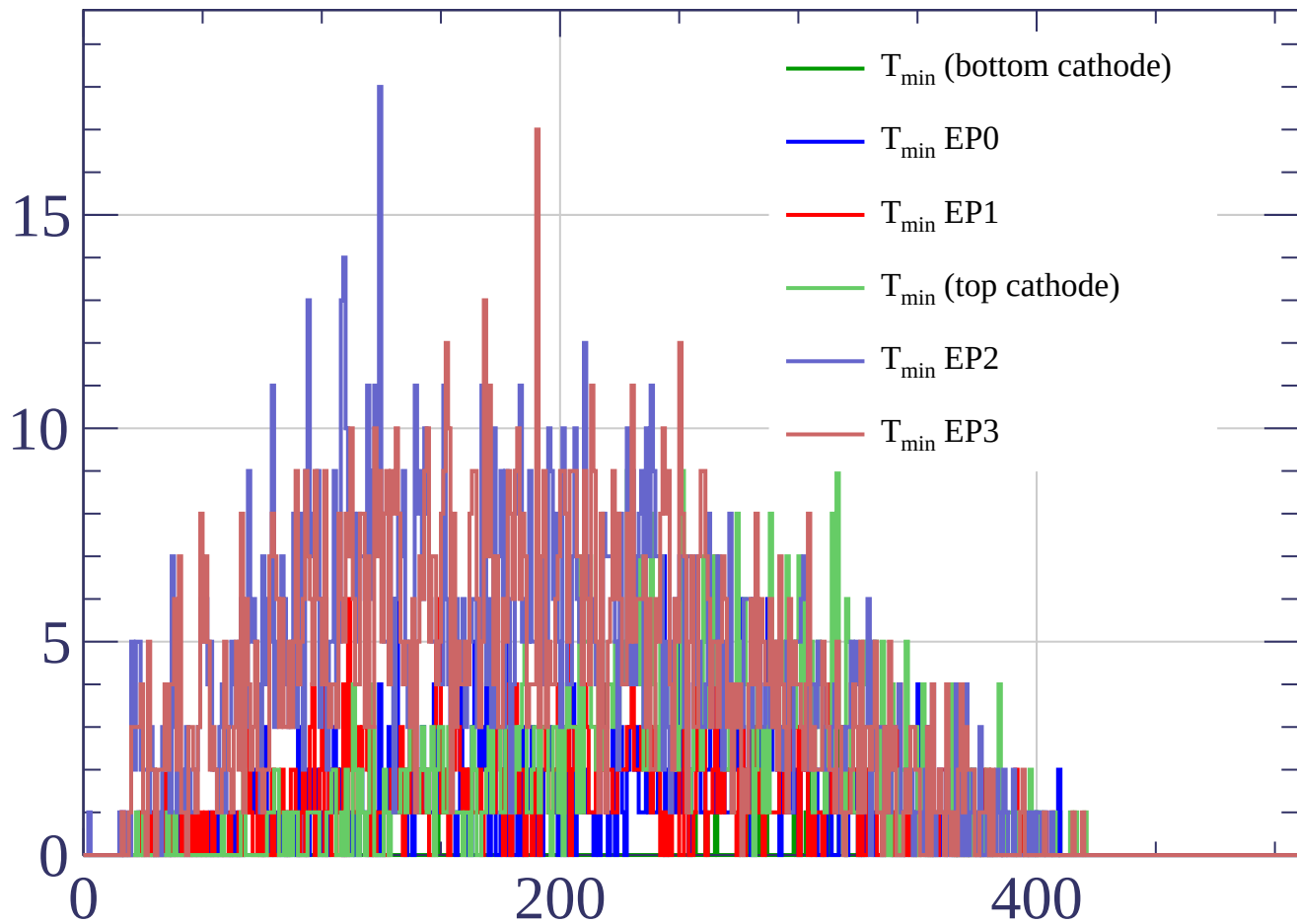
500

1000

1500

dE/dx [ADC counts/cm]

Count



Count

